



Lecture
Notes

2026

Analytical Elec- trodynamics

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Special Relativity

1 Lecture 01 — Introduction: Galilean Transformations and the Failure of Non-Relativistic Symmetry

Date: 22/03/2026

Overview

This lecture opens the course by motivating Analytical Electrodynamics as the field-theoretic analogue of Analytical Mechanics: where the latter reformulates finite-degree-of-freedom Newtonian dynamics via the action principle (Lagrange, Hamilton), this course does the same for fields — systems with *infinitely many* degrees of freedom. The central goal is to derive Maxwell's equations from an action principle, identify the symmetries of the theory via Noether's theorem, and ultimately understand why special relativity is not merely compatible with electrodynamics but is, in a precise sense, demanded by it. The lecture then begins the technical programme by reviewing the structure of Galilean spacetime, proving that the Galilean group is indeed a group, and demonstrating that Maxwell's equations are *not* invariant under Galilean transformations — a contradiction that forces us to abandon the Galilean picture in favour of Lorentzian spacetime and the Lorentz group.

Definitions

Physical Concept 1.1: Inertial reference frame

A reference frame in which a body subject to no external forces moves with constant velocity (including zero). Equivalently, Newton's second law holds in its standard form in such a frame. The existence of at least one inertial frame is taken as an axiom; all frames moving with constant velocity relative to a known inertial frame are also inertial.

Physical Concept 1.2: Homogeneity of spacetime

The laws of physics are invariant under arbitrary translations in both space and time: no spatial location and no instant of time is privileged.

Physical Concept 1.3: Isotropy of space

The laws of physics are invariant under arbitrary rotations of the spatial axes. There is no preferred direction in space. (Isotropy applies to space; homogeneity applies to spacetime.)

Physical Concept 1.4: Principle of Relativity

The *equations of motion* — i.e., the laws of physics — take identical form in every inertial reference frame. All inertial frames are equivalent; no preferred inertial frame exists.

Physical Concept 1.5: Galilean transformation

Given two inertial frames S and S' with S' moving at constant velocity $\mathbf{u}$ relative to S , with origins coinciding at $t = 0$, the Galilean transformation relating an event $(\mathbf{r}, t)$ in S to its coordinates $(\mathbf{r}', t')$ in S' is

$$\mathbf{r}' = \mathbf{r} - \mathbf{u} t, \quad (1)$$

$$t' = t. \quad (2)$$

Time is absolute (frame-independent) in the Galilean picture.

Physical Concept 1.6: Group

A set G together with a binary operation $\circ$ is a *group* if it satisfies:

1. **Closure.** For all $g_1, g_2 \in G$, $g_1 \circ g_2 \in G$.
2. **Identity.** There exists $e \in G$ such that $e \circ g = g \circ e = g$ for all $g \in G$.
3. **Inverse.** For every $g \in G$ there exists $g^{-1} \in G$ such that $g \circ g^{-1} = g^{-1} \circ g = e$.
4. **Associativity.** For all $g_1, g_2, g_3 \in G$, $(g_1 \circ g_2) \circ g_3 = g_1 \circ (g_2 \circ g_3)$.

Key Results

Physical Law 1.1 Galilean transformations form a group

The set of all Galilean boosts $\{G_{\mathbf{u}} : \mathbf{u} \in \mathbb{R}^3\}$, under the operation of composition, forms a group — the *Galilean group* (boost subgroup). The composition law is

$$G_{\mathbf{u}''} = G_{\mathbf{u}'} \circ G_{\mathbf{u}}, \quad \mathbf{u}'' = \mathbf{u} + \mathbf{u}'. \quad (3)$$

Physical Law 1.2 Newton's second law is Galilean-invariant

Under a Galilean transformation (1)–(2), the equation $m \ddot{\mathbf{r}} = -\nabla V(\mathbf{r})$ retains identical form:

$$m \frac{d^2 \mathbf{r}'}{dt'^2} = -\nabla' V(\mathbf{r}'). \quad (4)$$

Physical Law 1.3 Galilean field-transformation rules (from Lorentz force invariance)

Demanding that the Lorentz force law $m \ddot{\mathbf{r}} = q \mathbf{E} + \frac{q}{c} \mathbf{v} \times \mathbf{B}$ be invariant under Galilean transformations (working in Gaussian/CGS units) forces the electromagnetic fields to transform as

$$\mathbf{B}' = \mathbf{B}, \quad (5)$$

$$\mathbf{E}' = \mathbf{E} + \frac{1}{c} \mathbf{u} \times \mathbf{B}. \quad (6)$$

Physical Law 1.4 Maxwell's equations are *not* Galilean-invariant

The field-transformation rules (5)–(6), which follow from demanding Galilean invariance of the Lorentz force, are inconsistent with Gauss's law $\nabla' \cdot \mathbf{E}' = 0$ (in a source-free region). Specifically, computing $\nabla' \cdot \mathbf{E}'$ yields a term proportional to $\mathbf{u} \cdot \partial_t \mathbf{E}$, which is generically non-zero. Therefore the Galilean transformation law for spacetime is physically incorrect, and must be replaced by the

Lorentz transformation.

The single most important equation of the lecture is the incompatibility result:

$$\nabla' \cdot \mathbf{E}' = \frac{1}{c} \mathbf{u} \cdot \frac{\partial \mathbf{E}}{\partial t} \neq 0 \quad \text{in general,} \quad (7)$$

which signals that Galilean spacetime is incompatible with Maxwell's theory.

Derivations

Galilean boosts form a group: closure

Let $G_{\mathbf{u}}$ denote the boost by velocity $\mathbf{u}$:

$$\mathbf{r}' = \mathbf{r} - \mathbf{u} t, \quad t' = t. \quad (8)$$

Let $G_{\mathbf{u}'}$ act next on $(\mathbf{r}', t')$:

$$\mathbf{r}'' = \mathbf{r}' - \mathbf{u}' t' = (\mathbf{r} - \mathbf{u} t) - \mathbf{u}' t = \mathbf{r} - (\mathbf{u} + \mathbf{u}') t, \quad (9)$$

$$t'' = t' = t. \quad (10)$$

The result has the same form as a single Galilean boost with velocity $\mathbf{u}'' = \mathbf{u} + \mathbf{u}'$, proving closure. The identity element is G_0 ($\mathbf{r}' = \mathbf{r}$, $t' = t$); the inverse of $G_{\mathbf{u}}$ is $G_{-\mathbf{u}}$; associativity follows from function composition.

Newton's second law is Galilean-invariant

Consider a particle of mass m moving in a scalar potential V :

$$m \frac{d^2 \mathbf{r}}{dt^2} = -\nabla_{\mathbf{r}} V(\mathbf{r}). \quad (11)$$

Left-hand side. Since $t' = t$, time derivatives are unchanged. From $\mathbf{r}' = \mathbf{r} - \mathbf{u} t$,

$$\frac{d\mathbf{r}'}{dt'} = \frac{d\mathbf{r}}{dt} - \mathbf{u}, \quad (12)$$

the velocity shifts by $-\mathbf{u}$ (Galilean velocity addition). Taking a second derivative (and recalling $\mathbf{u}$ is constant):

$$\frac{d^2 \mathbf{r}'}{dt'^2} = \frac{d^2 \mathbf{r}}{dt^2}. \quad (13)$$

Acceleration is the same in both frames.

Right-hand side. The chain rule gives

$$\frac{\partial}{\partial x} V(\mathbf{r}) = \frac{\partial x'}{\partial x} \frac{\partial V}{\partial x'} + \frac{\partial y'}{\partial x} \frac{\partial V}{\partial y'} + \frac{\partial z'}{\partial x} \frac{\partial V}{\partial z'}. \quad (14)$$

From $x' = x - u_x t$, $y' = y - u_y t$, $z' = z - u_z t$, one finds $\partial x' / \partial x = 1$, $\partial y' / \partial x = 0$, $\partial z' / \partial x = 0$. Hence $\partial V / \partial x = \partial V / \partial x'$, and likewise for y and z , so

$$\nabla_{\mathbf{r}} V(\mathbf{r}) = \nabla_{\mathbf{r}'} V(\mathbf{r}'). \quad (15)$$

Equations (13) and the above together show that (11) becomes (4) in S' , with no change of form.

Galilean field transformation from Lorentz force invariance

The Lorentz force in S (CGS units) is

$$m \ddot{\mathbf{r}} = q\mathbf{E} + \frac{q}{c} \mathbf{v} \times \mathbf{B}, \quad (16)$$

where $\mathbf{E}$ and $\mathbf{B}$ are external fields. By the principle of relativity under Galilean transformations the same equation must hold in S' :

$$m \ddot{\mathbf{r}}' = q\mathbf{E}' + \frac{q}{c} \mathbf{v}' \times \mathbf{B}'. \quad (17)$$

Since $\ddot{\mathbf{r}}' = \ddot{\mathbf{r}}$ (shown above), the right-hand sides of (16) and (17) must be equal. The Galilean velocity transformation gives $\mathbf{v}' = \mathbf{v} - \mathbf{u}$. Substituting:

$$q\mathbf{E}' + \frac{q}{c}(\mathbf{v} - \mathbf{u}) \times \mathbf{B}' = q\mathbf{E} + \frac{q}{c} \mathbf{v} \times \mathbf{B}. \quad (18)$$

Rearranging, collecting terms with and without $\mathbf{v}$:

$$q\mathbf{E}' - \frac{q}{c} \mathbf{u} \times \mathbf{B}' + \frac{q}{c} \mathbf{v} \times (\mathbf{B}' - \mathbf{B}) = q\mathbf{E}. \quad (19)$$

The fields $\mathbf{E}$, $\mathbf{B}$, $\mathbf{E}'$, $\mathbf{B}'$ are generated by *external* sources and are independent of $\mathbf{v}$ (the velocity of the test particle). Because $\mathbf{v}$ is freely controllable, the coefficient of $\mathbf{v}$ in (19) must vanish identically:

$$\mathbf{B}' - \mathbf{B} = \mathbf{0} \implies \mathbf{B}' = \mathbf{B}. \quad (20)$$

Substituting this back into (19):

$$\mathbf{E}' = \mathbf{E} + \frac{1}{c} \mathbf{u} \times \mathbf{B}. \quad (21)$$

These are equations (5)–(6).

Galilean field transforms are inconsistent with Maxwell's equations

Work in a source-free region: $\nabla \cdot \mathbf{E} = 0$ in S , and by the principle of relativity $\nabla' \cdot \mathbf{E}' = 0$ must also hold in S' . Compute $\nabla' \cdot \mathbf{E}'$ using the derived transformation:

$$\nabla' \cdot \mathbf{E}' = \nabla' \cdot \mathbf{E} + \frac{1}{c} \nabla' \cdot (\mathbf{u} \times \mathbf{B}). \quad (22)$$

Under a Galilean transformation $\partial/\partial x'^i = \partial/\partial x^i$ (the Jacobian is the identity), so $\nabla' \cdot \mathbf{E} = \nabla \cdot \mathbf{E} = 0$. For the second term, using the vector identity $\nabla \cdot (\mathbf{A} \times \mathbf{C}) = \mathbf{C} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{C})$ with $\mathbf{u}$ constant:

$$\nabla \cdot (\mathbf{u} \times \mathbf{B}) = -\mathbf{u} \cdot (\nabla \times \mathbf{B}). \quad (23)$$

The Ampère–Maxwell law (in CGS, source-free) gives $\nabla \times \mathbf{B} \propto \partial\mathbf{E}/\partial t$, so:

$$\nabla' \cdot \mathbf{E}' = -\frac{1}{c} \mathbf{u} \cdot \frac{\partial\mathbf{E}}{\partial t}. \quad (24)$$

Because $\partial\mathbf{E}/\partial t \neq 0$ in general (e.g. in an electromagnetic wave), and $\mathbf{u}$ can be chosen with any direction, this is generically non-zero — in contradiction with the requirement $\nabla' \cdot \mathbf{E}' = 0$. The Galilean law of transformation of inertial frames is therefore physically untenable: it must be replaced by the Lorentz transformation.

Examples

Worked Example 1.1 (Closure of Galilean boosts: explicit computation)

Let $G_{\mathbf{u}}$ boost from S to S' and $G_{\mathbf{u}'}$ boost from S' to S'' . With S' moving at $\mathbf{u}$ and S'' moving at $\mathbf{u}'$ relative to S' :

$$\mathbf{r}'' = \mathbf{r}' - \mathbf{u}'t' = \mathbf{r} - \mathbf{u}t - \mathbf{u}'t = \mathbf{r} - (\mathbf{u} + \mathbf{u}')t, \quad t'' = t. \quad (25)$$

The composition equals a single boost $G_{\mathbf{u}+\mathbf{u}'}$, confirming the group law $\mathbf{u}'' = \mathbf{u} + \mathbf{u}'$ and group closure.

Worked Example 1.2 (Integers under addition as a group)

Smolkin used $(\mathbb{Z}, +)$ as a concrete illustration of the group axioms: the identity is 0, and the inverse of any integer n is $-n$. Closure ($\mathbb{Z}$ is closed under $+$) and associativity are immediate. This parallels the Galilean boosts, where 0 corresponds to the trivial boost and $-\mathbf{u}$ to its inverse.

Connections

- **Analytical Mechanics.** Smolkin explicitly frames this course as the field-theory analogue of Analytical Mechanics: the action principle and Noether's theorem for fields generalize what Lagrange and Hamilton did for finite-degree-of-freedom systems. Prior exposure to the Lagrangian formulation (including the Euler–Lagrange equations) is assumed.
- **Quantum Field Theory.** Smolkin noted that the formalism developed here — classical field theory with an action principle — is the direct precursor to QFT, which combines special relativity with quantum mechanics and underpins all of modern elementary-particle physics.
- **Special Relativity (from Mechanics course).** Students are expected to know the Lorentz transformation from their mechanics course. The next lectures will review and extend this knowledge; Lorentz transformations will replace Galilean ones as the correct symmetry of spacetime.
- **Landau & Lifshitz, Vol. 2 (Classical Theory of Fields).** Smolkin stated that the majority of the course follows this text closely. The title “Classical Theory of Fields” reflects that, when the book was written, there were exactly two classical field theories: electrodynamics and gravity. The course covers only the electrodynamics part.
- **Jackson, Classical Electrodynamics.** Recommended as an equally authoritative alternative reference, covering most topics in this course.

To Remember

- The Galilean transformation asserts absolute time ($t' = t$) and the position rule $\mathbf{r}' = \mathbf{r} - \mathbf{u}t$; it implies Galilean velocity addition $\mathbf{v}' = \mathbf{v} - \mathbf{u}$.
- Acceleration is an *absolute* quantity under Galilean transformations: $\ddot{\mathbf{r}}' = \ddot{\mathbf{r}}$. This makes Newton's second law (with velocity-independent forces) Galilean-invariant.
- The Galilean boosts $\{G_{\mathbf{u}}\}$ form a group under composition, with composition law $G_{\mathbf{u}'} \circ G_{\mathbf{u}} = G_{\mathbf{u}+\mathbf{u}'}$.
- Every symmetry of physics has the mathematical structure of a group; the symmetry group of non-relativistic mechanics is the Galilean group.
- Demanding Galilean invariance of the Lorentz force law forces $\mathbf{B}' = \mathbf{B}$ and $\mathbf{E}' = \mathbf{E} + \mathbf{u} \times \mathbf{B}/c$.

- These field-transformation rules are inconsistent with $\nabla \cdot \mathbf{E} = 0$ in S' (source-free Gauss's law) because $\nabla' \cdot (\mathbf{u} \times \mathbf{B}) = -\mathbf{u} \cdot (\nabla \times \mathbf{B}) \propto \mathbf{u} \cdot \partial_t \mathbf{E} \neq 0$ in general.
- **Conclusion:** Maxwell's equations are not Galilean-invariant; the correct transformation is the Lorentz transformation, to be studied in the next lectures.
- The course works exclusively in **Gaussian (CGS) units** because relativistic factors of c appear symmetrically; in SI units they are obscured.

Exam / HW Hints

- Smolkin strongly recommended treating homework seriously and working through it independently (or with understanding), as it directly prepares for the exam.
- He noted that AI tools and solutions from peers are permissible but warned explicitly: you must read and understand whatever you submit; copying without understanding is both ineffective for the exam and a poor learning strategy.
- Office hours are by appointment via email (michael.smolkin@mail.huji.ac.il); demand peaks in the week before the exam, so he advised scheduling early.
- Assessment: final exam 90%; HW bonus up to 15% (maximum grade 100%); submitting $\geq 70\%$ of assignments is required for the full bonus.

2 Lecture 02 — Invariance of the Spacetime Interval and the Lorentz Group

Date: 24/03/2026

Overview

This lecture establishes the central kinematic fact of special relativity: the *spacetime interval* ds^2 is invariant under passage between inertial frames. Starting from the specific result of Lecture 01 — that a signal propagating at the maximum speed c has $ds^2 = 0$ in every frame — Smolkin argues, using homogeneity, isotropy, and a three-frame consistency condition, that ds^2 is proportional to ds'^2 for *any* pair of events, and that the proportionality constant must equal unity. The lecture concludes by connecting the interval to the Lorentz group: the Lorentz transformations are precisely the linear maps that preserve ds^2 , in exact analogy with orthogonal transformations preserving the Euclidean distance $d\ell^2 = dx^2 + dy^2 + dz^2$. The next lecture will derive the explicit form of the Lorentz transformation from this invariance requirement.

Definitions

Physical Concept 2.1: Spacetime event

An *event* is a point in spacetime specified by a coordinate quadruplet $(t, \mathbf{r})$ or, in component form, (t, x, y, z) .

Physical Concept 2.2: Spacetime interval

Given two events $(t_1, \mathbf{r}_1)$ and $(t_2, \mathbf{r}_2)$ in an inertial frame S , the *spacetime interval* (squared) is

$$\Delta s^2 = c^2(t_2 - t_1)^2 - |\mathbf{r}_2 - \mathbf{r}_1|^2. \quad (26)$$

In the infinitesimal form:

$$ds^2 = c^2 dt^2 - d\mathbf{r}^2. \quad (27)$$

Physical Concept 2.3: Lorentz transformations

The *Lorentz transformations* are the set of linear maps between inertial frames that leave the spacetime interval (26) invariant. They form a group — the *Lorentz group* $O(3, 1)$ — under composition, in complete analogy with the orthogonal group $O(3)$ which preserves the Euclidean distance $d\ell^2 = dx^2 + dy^2 + dz^2$.

Key Results

Physical Law 2.1 Invariance of the spacetime interval

The spacetime interval between any two events is the same in all inertial frames:

$$\boxed{ds^2 = ds'^2} \quad (28)$$

Equivalently, the proportionality factor λ defined by $ds^2 = \lambda(|\mathbf{v}|/c) ds'^2$ satisfies $\lambda = 1$.

Physical Law 2.2 Lorentz group as the symmetry of the interval

Lorentz transformations are the unique linear transformations preserving ds^2 . Mathematically, they are defined as the group $O(3, 1)$, the set of matrices that preserve the indefinite quadratic form $\eta_{\mu\nu}$ with signature $(+, -, -, -)$. This is the relativistic analogue of $O(3)$ preserving δ_{ij} .

Derivations**Proof (physical argument) that $ds^2 = ds'^2$ for arbitrary events**

Setup. Let S and S' be two inertial frames, with S' moving at velocity $\mathbf{v}'$ relative to S . Consider two events that are infinitesimally close; denote their separation in S by ds and in S' by ds' . The goal is to determine the relationship between ds^2 and ds'^2 for *arbitrary* events, not merely for light-propagation events.

Step 1: most general ansatz. We write

$$ds^2 = \lambda(t_1, \mathbf{r}_1, \mathbf{v}', \hat{\mathbf{v}}') ds'^2, \quad (29)$$

where the proportionality factor λ is, a priori, a function of the location of the event $(t_1, \mathbf{r}_1)$, the magnitude of the relative velocity $|\mathbf{v}'|$, and its direction $\hat{\mathbf{v}}'$. (The infinitesimal separation ds'^2 itself already encodes the displacement dt , $d\mathbf{r}$, so no additional dependence on dt or $d\mathbf{r}$ is needed beyond what is already in ds'^2 .)

Step 2: eliminate dependence on location (homogeneity). Homogeneity of spacetime means no spacetime point is privileged: the laws of physics — and hence the transformation factor — cannot depend on where or when the events occur. Therefore

$$\lambda \text{ is independent of } t_1, \mathbf{r}_1. \quad (30)$$

Step 3: eliminate dependence on the direction of $\mathbf{v}'$ (isotropy). Isotropy of space means no spatial direction is preferred. The magnitude of ds^2 (a scalar) cannot depend on which direction the frame S' moves relative to S . Therefore

$$\lambda = \lambda\left(\frac{|\mathbf{v}'|}{c}\right). \quad (31)$$

The factor $1/c$ appears because λ must be dimensionless and c is the only invariant speed in the theory.

After steps 2 and 3 the ansatz reduces to

$$ds^2 = \lambda\left(\frac{|\mathbf{v}'|}{c}\right) ds'^2. \quad (32)$$

Step 4: three-frame consistency (key argument). Introduce a third inertial frame S'' moving at velocity $\mathbf{v}''$ relative to S . Applying (32) twice:

$$ds^2 = \lambda(|\mathbf{v}'|/c) ds'^2, \quad (33)$$

$$ds^2 = \lambda(|\mathbf{v}''|/c) ds''^2, \quad (34)$$

$$ds'^2 = \lambda(|\mathbf{v}|/c) ds''^2, \quad (35)$$

where $|\mathbf{v}|$ is the speed of S'' relative to S' . Dividing (33) by (34):

$$\frac{ds'^2}{ds''^2} = \frac{\lambda(|\mathbf{v}''|/c)}{\lambda(|\mathbf{v}|/c)}. \quad (36)$$

But from (35),

$$\frac{ds'^2}{ds''^2} = \lambda(|\mathbf{v}|/c). \quad (37)$$

Equating (36) and (37):

$$\lambda\left(\frac{|\mathbf{v}|}{c}\right) = \frac{\lambda(|\mathbf{v}''|/c)}{\lambda(|\mathbf{v}'|/c)}. \quad (38)$$

Step 5: the right-hand side cannot depend on the angle between $\mathbf{v}'$ and $\mathbf{v}''$. By isotropy, λ depends only on the *magnitude* of its argument, never on a direction. The right-hand side of (38) involves only the magnitudes $|\mathbf{v}'|$ and $|\mathbf{v}''|$, so it carries no angular information. The left-hand side, however, contains $|\mathbf{v}|$ — the speed of S'' relative to S' — which in general *does* depend on the angle θ between $\mathbf{v}'$ and $\mathbf{v}''$. (Concretely: if $\mathbf{v}' \parallel \mathbf{v}''$ then $|\mathbf{v}| = 0$; if $\mathbf{v}' \perp \mathbf{v}''$ then $|\mathbf{v}| \neq 0$.)

Equation (38) therefore requires the left-hand side to be independent of θ , which is possible only if λ is *constant* (independent of its argument):

$$\lambda = \text{const.} \quad (39)$$

Step 6: the constant must equal 1. Substitute $\lambda = \text{const}$ back into (38):

$$\text{const} = \frac{\text{const}}{\text{const}} \implies \text{const}^2 = \text{const} \implies \text{const} = 1. \quad (40)$$

(The value $\text{const} = 0$ is excluded because $ds^2 = 0$ for all events would be unphysical.)

Conclusion.

$$ds^2 = ds'^2 \quad \text{for any two infinitesimally separated events.} \quad (41)$$

By integration (summing infinitesimal intervals), the result extends to any finite pair of events:

$$\Delta s^2 = \Delta s'^2. \quad (42)$$

Remark on rigor. Smolkin emphasized explicitly that this is a *physical argument*, not a rigorous mathematical proof. The extension from the infinitesimal to the finite case rests on the fact that we showed the proportionality factor equals 1 *at every spacetime point*, so integrating up the infinitesimal intervals cannot introduce any discrepancy. Formally constructing the Lorentz group from the requirement of interval preservation (the mathematician's route) makes this rigorous but takes the interval invariance as an axiom rather than deriving it.

Recap: light signals satisfy $ds^2 = 0$ in every frame

For a signal propagating at the maximum speed c in frame S , $|d\mathbf{r}| = c|dt|$, hence

$$ds^2 = c^2 dt^2 - d\mathbf{r}^2 = 0. \quad (43)$$

The principle of relativity demands that the same signal satisfies $|d\mathbf{r}'| = c|dt'|$ in every frame S' (the maximum speed is a law of nature, hence frame-independent), so $ds'^2 = 0$. This is the special case $ds^2 = ds'^2 = 0$ that motivated the general invariance result.

Analogy: orthogonal group $O(3)$ preserves Euclidean distance

The Euclidean distance $d\ell^2 = dx^2 + dy^2 + dz^2$ is preserved by all *orthogonal* transformations $R \in O(3)$:

$$d\ell^2 = (R d\mathbf{r})^T (R d\mathbf{r}) = d\mathbf{r}^T R^T R d\mathbf{r} = d\mathbf{r}^T d\mathbf{r} = d\ell^2 \quad (R^T R = \mathbf{1}). \quad (44)$$

Lorentz transformations Λ play the identical role for the indefinite form $ds^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2 = \eta_{\mu\nu} dx^\mu dx^\nu$:

$$\Lambda^T \eta \Lambda = \eta, \quad \eta = \text{diag}(+1, -1, -1, -1). \quad (45)$$

The group of such Λ is $O(3, 1)$, the Lorentz group.

Examples**Worked Example 2.1** (Why the angle between $\mathbf{v}'$ and $\mathbf{v}''$ matters for $|\mathbf{v}|$)

Let $|\mathbf{v}'| = |\mathbf{v}''| = V$.

- If $\mathbf{v}'' \parallel \mathbf{v}'$ (same direction): S'' moves at speed V in the same direction as S' ; both frames co-move, so $|\mathbf{v}| = 0$.
- If $\mathbf{v}'' \perp \mathbf{v}'$: S'' moves orthogonally to S' ; the relative speed $|\mathbf{v}| \neq 0$.

Thus the left-hand side of (38) varies with the angle between $\mathbf{v}'$ and $\mathbf{v}''$, while the right-hand side does not — contradicting (38) unless λ is constant.

Worked Example 2.2 (Integers under addition and the Lorentz group)

Just as $(\mathbb{Z}, +)$ is a group with identity 0 and inverse $-n$ for each n , the Lorentz group is a set of transformations closed under composition, with the identity transformation (no boost, no rotation) and with an inverse for each element. The specific feature distinguishing the Lorentz group from $O(3)$ is the indefinite signature of the preserved quadratic form.

Connections

- **Lecture 01.** The starting point is the Lecture 01 result that Maxwell's equations are inconsistent with the Galilean group. The natural question — what group *does* leave Maxwell's equations invariant? — motivates the construction of the Lorentz group from the invariance of ds^2 .
- **Euclidean geometry / $O(3)$.** Smolkin used the orthogonal group $O(3)$ as the canonical example of a group defined by preservation of a quadratic form ($d\ell^2$). The Lorentz group $O(3, 1)$ is its spacetime analogue.
- **Next lecture.** Interval invariance will be taken as the axiom from which the explicit form of the Lorentz transformation is derived. Smolkin stated: “invariance of the interval is sufficient to uniquely determine the transformation.”
- **Landau & Lifshitz, Vol. 2, §2.** The argument presented here (homogeneity + isotropy + three-frame consistency $\Rightarrow \lambda = 1$) follows the L&L treatment closely, though Smolkin noted it should be read as physical motivation rather than a rigorous proof.

To Remember

- The spacetime interval is $ds^2 = c^2 dt^2 - d\mathbf{r}^2$; it is a Lorentz *scalar* — the same in every inertial frame.
- The argument for $\lambda = 1$ uses three ingredients: (1) homogeneity eliminates dependence on the location of the events; (2) isotropy eliminates dependence on the direction of $\mathbf{v}'$, leaving $\lambda = \lambda(|\mathbf{v}'|/c)$; (3) three-frame consistency forces λ to be constant, and the constant must be 1.
- The speed c enters as the unique invariant speed scale that makes λ dimensionless; its invariance is a consequence of the principle of relativity applied to the maximum-speed signal, not an independent postulate at this stage.
- Lorentz transformations are the group $O(3, 1)$, defined by $\Lambda^T \eta \Lambda = \eta$ with $\eta = \text{diag}(+1, -1, -1, -1)$. They play the same role for ds^2 as orthogonal matrices play for $d\ell^2$.
- The argument presented is physical motivation, not a mathematically rigorous proof. The mathematician's route takes interval invariance as an axiom and defines the Lorentz group directly as the symmetry group of the quadratic form.

3 Lecture 03 — Deriving the Lorentz Group from Interval Invariance

Date: 12/04/2026

Overview

Lecture 03 picks up directly from the result of Lecture 02 — that the spacetime interval $ds^2 = c^2 dt^2 - d\mathbf{x}^2$ is invariant under passage between inertial frames — and uses this single fact to *derive* the structure of the allowed coordinate transformations.

The strategy is the following. Write the interval in a compact index notation using the Minkowski metric $\eta_{\alpha\beta}$ (signature $+- - -$). Demand that $ds^2 = ds'^2$ for an *arbitrary* smooth, invertible map $x'^\alpha = x'^\alpha(x)$. Differentiating the invariance condition twice forces the Jacobian of the transformation to satisfy the defining relation of the Lorentz group, and a further differentiation argument shows that the Jacobian must be *constant* — i.e., the transformation is necessarily *linear*. The most general inertial-frame transition is therefore an affine map: a homogeneous Lorentz transformation $\Lambda^\alpha{}_\beta$ (which mixes all four coordinates) composed with a constant spacetime translation a^α (which merely synchronises origins). The lecture closes by introducing the proper orthochronous subgroup $SO^+(3, 1)$ — the physically relevant sector — characterised by $\det \Lambda = +1$ and $\Lambda^0{}_0 \geq +1$.

Notation and Definitions

Physical Concept 3.1: Four-coordinates

Choose a unified coordinate label

$$x^0 \equiv ct, \quad x^1 \equiv x, \quad x^2 \equiv y, \quad x^3 \equiv z, \quad (46)$$

so that all four components carry dimensions of length. Greek indices $\alpha, \beta, \dots$ run from 0 to 3. The four-displacement is $dx^\alpha = (c dt, dx, dy, dz)$.

Physical Concept 3.2: Minkowski metric

The *Minkowski metric* is the diagonal 4×4 matrix

$$\eta_{\alpha\beta} = \text{diag}(+1, -1, -1, -1), \quad (47)$$

with row and column indices running from 0 to 3. Explicitly: $\eta_{00} = +1$, $\eta_{11} = \eta_{22} = \eta_{33} = -1$, and $\eta_{\alpha\beta} = 0$ for $\alpha \neq \beta$.

Physical Concept 3.3: Interval in compact form

With the Einstein summation convention (repeated upper–lower index pairs are summed from 0 to 3), the infinitesimal squared interval is

$$ds^2 = \eta_{\alpha\beta} dx^\alpha dx^\beta = c^2 dt^2 - dx^2 - dy^2 - dz^2. \quad (48)$$

This is the relativistic generalisation of the Euclidean inner product $\delta_{ij} dx^i dx^j = dx^2 + dy^2 + dz^2$, but with an indefinite metric: ds^2 can be positive (time-like), negative (space-like), or zero (null/light-like).

Remark 1. The notation $\eta_{\alpha\beta}$ with both indices *down* is standard; upper-index coordinates x^α are *contravariant* vectors. The metric serves the role of lowering indices: $x_\alpha \equiv \eta_{\alpha\beta} x^\beta$, so $x_0 = +ct$ while $x_i = -x^i$ for $i = 1, 2, 3$. This distinction will become significant when covariant and contravariant components are compared in later lectures.

Derivations

Step 1 — Invariance condition in index notation

By Lecture 02's result, $ds^2 = ds'^2$ for any two inertial frames. In the notation of Eq. (48),

$$ds^2 = \eta_{\alpha\beta} dx^\alpha dx^\beta, \quad (49)$$

$$ds'^2 = \eta_{\alpha\beta} dx'^\alpha dx'^\beta, \quad (50)$$

where dx'^α are the coordinate differentials in the primed frame. Since the same metric tensor $\eta_{\alpha\beta}$ appears in both frames (the metric is part of the physical structure of flat spacetime, not a choice of coordinates), the invariance requirement is

$$\eta_{\alpha\beta} dx^\alpha dx^\beta = \eta_{\alpha\beta} dx'^\alpha dx'^\beta. \quad (51)$$

Step 2 — Jacobian substitution

Assume a smooth, invertible transformation $x'^\alpha = x'^\alpha(x)$. Its differential is (chain rule)

$$dx'^\alpha = \frac{\partial x'^\alpha}{\partial x^\gamma} dx^\gamma, \quad (52)$$

where $\frac{\partial x'^\alpha}{\partial x^\gamma}$ is the (4×4) *Jacobian matrix* of the transformation. Substituting into Eq. (50):

$$ds'^2 = \eta_{\alpha\beta} \frac{\partial x'^\alpha}{\partial x^\gamma} \frac{\partial x'^\beta}{\partial x^\delta} dx^\gamma dx^\delta. \quad (53)$$

Setting $ds^2 = ds'^2$ and writing both sides with the same dummy indices γ, δ :

$$\left(\eta_{\alpha\beta} \frac{\partial x'^\alpha}{\partial x^\gamma} \frac{\partial x'^\beta}{\partial x^\delta} - \eta_{\gamma\delta} \right) dx^\gamma dx^\delta = 0 \quad \forall dx^\alpha. \quad (54)$$

Define the symmetric 4×4 matrix

$$M_{\gamma\delta} \equiv \eta_{\alpha\beta} \frac{\partial x'^\alpha}{\partial x^\gamma} \frac{\partial x'^\beta}{\partial x^\delta} - \eta_{\gamma\delta}. \quad (55)$$

$M_{\gamma\delta}$ is symmetric. Under the exchange $\gamma \leftrightarrow \delta$: the first term transforms as $\eta_{\alpha\beta} \partial_\delta x'^\alpha \partial_\gamma x'^\beta$. Renaming the dummy indices $\alpha \leftrightarrow \beta$ (which is permissible since $\eta_{\alpha\beta} = \eta_{\beta\alpha}$) returns this to the original form. The second term $\eta_{\gamma\delta}$ is symmetric by definition. Hence $M_{\gamma\delta} = M_{\delta\gamma}$.

$M_{\gamma\delta} = 0$ everywhere. Because Eq. (54) holds for *all* dx^γ , and $M_{\gamma\delta}$ is symmetric, one may contract it with eigenvectors of M (which exist since M is a real symmetric matrix and therefore diagonalisable). For any eigenvector v^γ with eigenvalue λ :

$$M_{\gamma\delta} v^\gamma v^\delta = \lambda \underbrace{v^\gamma v_\gamma}_{>0} = 0 \Rightarrow \lambda = 0.$$

Since all eigenvalues vanish and M is symmetric, $M_{\gamma\delta} = 0$ identically. Therefore:

$$\boxed{\eta_{\alpha\beta} \frac{\partial x'^\alpha}{\partial x^\gamma} \frac{\partial x'^\beta}{\partial x^\delta} = \eta_{\gamma\delta}}. \quad (56)$$

This is sometimes called the *fundamental identity* $(\star)$ of the Lorentz group. It is the direct analogue of $R^T R = 1$ for orthogonal matrices.

Step 3 — Differentiating (*) to prove linearity

Take $\partial/\partial x^\varepsilon$ on both sides of Eq. (56). The right-hand side is the constant matrix $\eta_{\gamma\delta}$, so its derivative vanishes:

$$\begin{aligned} \partial_\varepsilon \left(\eta_{\alpha\beta} \partial_\gamma x'^\alpha \partial_\delta x'^\beta \right) &= 0, \\ \eta_{\alpha\beta} \frac{\partial^2 x'^\alpha}{\partial x^\varepsilon \partial x^\gamma} \frac{\partial x'^\beta}{\partial x^\delta} + \eta_{\alpha\beta} \frac{\partial x'^\alpha}{\partial x^\gamma} \frac{\partial^2 x'^\beta}{\partial x^\varepsilon \partial x^\delta} &= 0. \end{aligned} \quad (57)$$

Denote the second-derivative terms by $A_{\varepsilon\gamma}^{(\alpha)} \equiv \partial_\varepsilon \partial_\gamma x'^\alpha$. Equation (57) reads

$$\eta_{\alpha\beta} A_{\varepsilon\gamma}^\alpha \partial_\delta x'^\beta + \eta_{\alpha\beta} \partial_\gamma x'^\alpha A_{\varepsilon\delta}^\beta = 0. \quad (58)$$

Since $A_{\varepsilon\gamma}^\alpha = A_{\gamma\varepsilon}^\alpha$ (partial derivatives commute), this equation is symmetric under $\varepsilon \leftrightarrow \gamma$. Write two further permuted copies of the same equation:

$$(\text{original, indices } \varepsilon, \gamma, \delta) : \quad \eta_{\alpha\beta} A_{\varepsilon\gamma}^\alpha \partial_\delta x'^\beta + \eta_{\alpha\beta} \partial_\gamma x'^\alpha A_{\varepsilon\delta}^\beta = 0, \quad (59)$$

$$(\text{swap } \gamma \leftrightarrow \varepsilon) : \quad \eta_{\alpha\beta} A_{\gamma\varepsilon}^\alpha \partial_\delta x'^\beta + \eta_{\alpha\beta} \partial_\varepsilon x'^\alpha A_{\gamma\delta}^\beta = 0, \quad (60)$$

$$(\text{swap } \delta \leftrightarrow \varepsilon) : \quad \eta_{\alpha\beta} A_{\delta\varepsilon}^\alpha \partial_\gamma x'^\beta + \eta_{\alpha\beta} \partial_\varepsilon x'^\alpha A_{\delta\gamma}^\beta = 0. \quad (61)$$

Using $A_{\varepsilon\gamma}^\alpha = A_{\gamma\varepsilon}^\alpha$, add Eq. (59) and Eq. (60) and subtract Eq. (61):

$$2 \eta_{\alpha\beta} A_{\varepsilon\gamma}^\alpha \partial_\delta x'^\beta = 0. \quad (62)$$

The Jacobian matrix $\partial_\delta x'^\beta$ is invertible (the transformation is assumed bijective, so its Jacobian is non-singular). Multiplying both sides by the inverse Jacobian and using $\eta_{\alpha\beta}$ being non-degenerate:

$$A_{\varepsilon\gamma}^\alpha = \frac{\partial^2 x'^\alpha}{\partial x^\varepsilon \partial x^\gamma} = 0 \quad \forall \alpha, \varepsilon, \gamma. \quad (63)$$

All second (and hence all higher) partial derivatives of x'^α vanish. A smooth function with vanishing second derivative is *at most linear* (first-order polynomial) in its arguments.

Step 4 — Most general form of the transformation

The most general affine map consistent with Eq. (63) is

$$x'^\alpha = \Lambda^\alpha_\beta x^\beta + a^\alpha, \quad (64)$$

where Λ^α_β is a constant invertible 4×4 matrix (the *Lorentz matrix*) and a^α is a constant four-vector (a *spacetime translation*).

Physical meaning of a^α . The constant a^α encodes the offset between the coordinate origins of frames K and K' . By convention one synchronises the two frames so that their origins coincide at $t = 0$, i.e. $x'^\alpha = 0$ when $x^\alpha = 0$, which forces $a^\alpha = 0$. This is the standard choice; the general case with $a^\alpha \neq 0$ describes a spacetime translation (Poincaré transformation).

Setting $a^\alpha = 0$. With this convention,

$$x'^\alpha = \Lambda^\alpha_\beta x^\beta, \quad (65)$$

and the Jacobian reduces to the constant matrix Λ itself: $\partial x'^\alpha / \partial x^\gamma = \Lambda^\alpha_\gamma$.

Step 5 — The defining relation of the Lorentz group

Substituting $\partial x'^\alpha / \partial x^\gamma = \Lambda^\alpha_\gamma$ into the fundamental identity (56):

$$\eta_{\alpha\beta} \Lambda^\alpha_\gamma \Lambda^\beta_\delta = \eta_{\gamma\delta}. \quad (66)$$

In matrix notation (with Λ^T denoting the transpose of Λ):

$$\boxed{\Lambda^T \eta \Lambda = \eta}, \quad (67)$$

where $\eta = \text{diag}(+1, -1, -1, -1)$. This is the relativistic analogue of the orthogonality condition $R^T \mathbf{1} R = \mathbf{1}$ (i.e. $R^T R = \mathbf{1}$) for Euclidean rotations $R \in O(3)$.

The set of all 4×4 real matrices satisfying Eq. (67) is called the *Lorentz group* $O(3, 1)$. Closure under matrix multiplication, existence of an identity ($\Lambda = \mathbf{1}$), and existence of inverses all follow from Eq. (67).

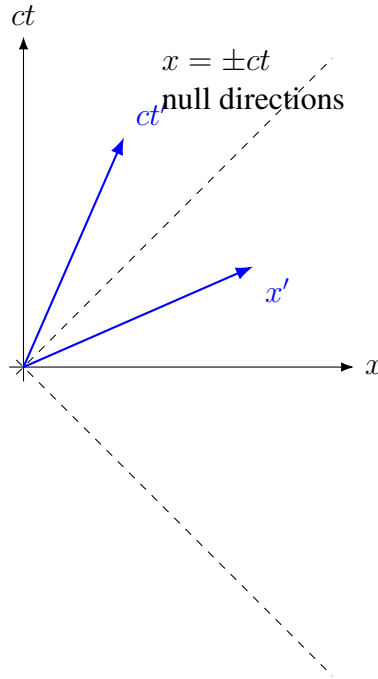


Figure 1: A 1 + 1-dimensional Lorentz boost mixes the ct and x axes while preserving the null lines $x = \pm ct$, which encode the invariant interval structure.

Step 6 — Determinant and time-orientation constraints

Taking the determinant of both sides of Eq. (67):

$$(\det \Lambda)^2 \det \eta = \det \eta \Rightarrow (\det \Lambda)^2 = 1 \Rightarrow \det \Lambda = \pm 1. \quad (68)$$

Setting $\gamma = \delta = 0$ in Eq. (66) and using $\eta_{00} = +1, \eta_{ii} = -1$:

$$(\Lambda^0_0)^2 - \sum_{i=1}^3 (\Lambda^i_0)^2 = 1, \quad (69)$$

so $(\Lambda^0_0)^2 \geq 1$, giving $\Lambda^0_0 \geq +1$ or $\Lambda^0_0 \leq -1$.

These two discrete conditions ($\det \Lambda = \pm 1, \Lambda^0_0 \gtrless \pm 1$) divide $O(3, 1)$ into four disconnected components. The physically realised component — continuously connected to the identity transformation — satisfies

$$\det \Lambda = +1 \quad \text{and} \quad \Lambda^0_0 \geq +1. \quad (70)$$

This subgroup is called the *proper orthochronous Lorentz group* $SO^+(3, 1)$ (also written $L_+^\uparrow$).

Excluded transformations.

- **Time reversal** (T): $\Lambda^0_0 = -1$, $\det \Lambda = +1$. Reverses the time coordinate; physically possible but not continuously connected to the identity.
- **Spatial parity** (P): flips all three spatial coordinates; $\det \Lambda = -1$, $\Lambda^0_0 = +1$. Equivalent to a rotation by π about a spatial axis combined with a single reflection; has $\det \Lambda = -1$.
- **Combined** PT : $\det \Lambda = -1$, $\Lambda^0_0 = -1$.

Remark 2. A reflection of only two spatial axes (e.g. $x \rightarrow -x$, $y \rightarrow -y$, $z \rightarrow z$) is equivalent to a rotation by 180° about the z -axis and therefore has $\det \Lambda = +1$; it belongs to $SO^+(3, 1)$ and is *not* a parity inversion.

Step 7 — The Poincaré group

The full set of symmetries of flat spacetime that preserve the interval is the *Poincaré group* (inhomogeneous Lorentz group), whose elements are pairs (Λ, a) with multiplication law

$$(\Lambda_2, a_2) \circ (\Lambda_1, a_1) = (\Lambda_2 \Lambda_1, \Lambda_2 a_1 + a_2). \quad (71)$$

The Poincaré group has 10 generators: 6 (from $O(3, 1)$: 3 boosts + 3 rotations) + 4 (spacetime translations a^α).

For the purpose of deriving the explicit form of Λ (next lecture), one restricts to the homogeneous case $a = 0$.

Examples

Worked Example 3.1 (Spatial rotations are Lorentz transformations)

A spatial rotation $R \in SO(3)$ embedded in 4×4 form as

$$\Lambda = \begin{pmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & R \end{pmatrix}$$

satisfies Eq. (67) because $R^T R = \mathbf{1}$ and the $(0, 0)$ block is unaffected. Moreover $\det \Lambda = \det R = +1$ and $\Lambda^0_0 = 1 \geq +1$, so rotations belong to $SO^+(3, 1)$.

Worked Example 3.2 (Anti-symmetric matrices cannot represent $M_{\gamma\delta}$)

An anti-symmetric 2×2 matrix $M = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ satisfies $v^T M v = 0$ for all $v \in \mathbb{R}^2$. This illustrates why the conclusion $M_{\gamma\delta} = 0$ requires the additional input that M is *symmetric* — without symmetry, the quadratic form vanishing identically does not force $M = 0$.

Worked Example 3.3 (Spatial parity has $\det \Lambda = -1$)

For $\Lambda = \text{diag}(1, -1, -1, -1)$: $\det \Lambda = 1 \cdot (-1)^3 = -1$, so parity is *not* in $SO^+(3, 1)$.

Connections

- **Lecture 02.** The starting point is the invariance $ds^2 = ds'^2$ proved (by physical argument) in Lecture 02. The present lecture extracts the full algebraic content of that one scalar equation.

- **Euclidean analogy.** The derivation is formally identical to showing that distance-preserving maps in $\mathbb{R}^3$ are affine with an orthogonal linear part ($R^T R = \mathbf{1}$). Replacing δ_{ij} by $\eta_{\mu\nu}$ promotes $O(3)$ to $O(3, 1)$.
- **Jacobi matrix / Jacobian.** The matrix $\partial x'^\alpha / \partial x^\gamma$ is the relativistic Jacobian. After proving linearity, it collapses to the constant matrix Λ^α_γ . This connection to Jacobians will reappear when transforming tensor components and when changing integration measures.
- **Lie group structure.** $SO^+(3, 1)$ is a Lie group; every element can be reached from the identity by continuous parameter deformation. The argument in Step 6 — that $\det \Lambda$ is a continuous function, so its value cannot jump between $+1$ and -1 along a continuous path — is a standard topological argument in Lie group theory.
- **Next lecture.** With the structure $x'^\alpha = \Lambda^\alpha_\beta x^\beta$ established, one can parametrise Λ explicitly by boost velocity and rotation angles, yielding the textbook Lorentz transformation formulas.
- **Quantum Field Theory.** The representation theory of $SO^+(3, 1)$ (finite-dimensional, non-unitary representations plus the infinite-dimensional unitary representations) determines the particle content of relativistic quantum field theories.

To Remember

- The four-coordinate notation $x^\alpha = (ct, x, y, z)$ makes all components carry dimensions of length and allows the interval to be written $ds^2 = \eta_{\alpha\beta} dx^\alpha dx^\beta$.
- The metric $\eta_{\alpha\beta} = \text{diag}(+1, -1, -1, -1)$ encodes the Minkowski indefinite inner product. Unlike the Euclidean metric δ_{ij} , it is *not* positive definite.
- Any smooth invertible transformation preserving ds^2 must satisfy the fundamental identity (56). Differentiating it once shows all second derivatives of the transformation vanish; hence the transformation is *linear* (plus a possible constant translation).
- The constant linear part Λ^α_β obeys $\Lambda^T \eta \Lambda = \eta$. The set of such matrices is the Lorentz group $O(3, 1)$.
- Physical transformations (continuously connected to the identity) form the proper orthochronous subgroup $SO^+(3, 1)$: $\det \Lambda = +1$, $\Lambda^0_0 \geq +1$. Reflections (parity, time reversal) lie in other disconnected components.
- Spatial rotations are a subgroup of $SO^+(3, 1)$; they act trivially on $x^0 = ct$ and satisfy $R^T R = \mathbf{1}$ in the spatial block.
- A general inertial-frame transformation is a *Poincaré transformation*: $x'^\alpha = \Lambda^\alpha_\beta x^\beta + a^\alpha$, with (Λ, a) parametrised by 10 real numbers.

4 Lecture 04 — Derivation of Lorentz Transformations and Causality

Date: 14/04/2026

Overview

In this lecture, we finalize the mathematical derivation of the Lorentz transformation matrix. By imposing the invariance of the spacetime interval under changes of inertial reference frames, we completely constrain the form of the coordinate transformation. We restrict our analysis to the *proper, orthochronous Lorentz group* (transformations that preserve the direction of time and the parity of space, explicitly excluding reflections). Following the derivation, we classify spacetime intervals into three causally distinct categories—timelike, spacelike, and lightlike—and analyze the relativistic notions of simultaneity and time-ordering.

Derivations: Constructing the Lorentz Transformation Matrix

We postulate that the transformation between inertial reference frames is linear and preserves the spacetime interval Δs^2 . Let the coordinates be given by the four-vector $x^\mu = (ct, x, y, z)$. The transformation from frame K to frame K' is:

$$x'^\mu = \Lambda^\mu_\nu x^\nu \quad (72)$$

The invariance of the spacetime interval, $ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu = ds'^2$, implies the fundamental constraint on the Lorentz transformation matrix Λ :

$$\eta_{\mu\nu} = \Lambda^\alpha_\mu \Lambda^\beta_\nu \eta_{\alpha\beta} \quad (73)$$

where $\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$ is the Minkowski metric. For simplicity, we restrict our analysis to 1+1 dimensions (ct, x) , so $\mu, \nu \in \{0, 1\}$.

Finding the Temporal Components

Let us evaluate the $\mu = 0, \nu = 0$ component of the metric identity. Expanding the summation over α, β :

$$\eta_{00} = \Lambda^\alpha_0 \Lambda^\beta_0 \eta_{\alpha\beta} \quad (74)$$

$$1 = (\Lambda^0_0)^2 \eta_{00} + (\Lambda^1_0)^2 \eta_{11} \quad (75)$$

$$1 = (\Lambda^0_0)^2 - (\Lambda^1_0)^2 \quad (76)$$

To relate these matrix elements to physical kinematics, consider a particle (or the origin of frame K') at rest in frame K' . Thus, in K' , $dx'^1 = 0$. Using the inverse transformation $dx^i = (\Lambda^{-1})^i_\mu dx'^\mu$, we can determine the relative velocity v between the frames. By carefully tracking the components, one identifies the physical meaning of the ratio of these matrix elements:

$$\frac{\Lambda^1_0}{\Lambda^0_0} = \frac{v}{c} \equiv \beta \quad (77)$$

Substituting this back into our metric equation:

$$1 = (\Lambda^0_0)^2 - \left(\frac{v}{c} \Lambda^0_0\right)^2 \quad (78)$$

$$1 = (\Lambda^0_0)^2 \left[1 - \left(\frac{v}{c}\right)^2\right] \quad (79)$$

Solving for Λ^0_0 , and selecting the positive root to ensure time flows forward (orthochronous transformation, $\Lambda^0_0 \geq 1$):

$$\Lambda^0_0 = \frac{1}{\sqrt{1 - v^2/c^2}} \equiv \gamma \quad (80)$$

Consequently, we also find:

$$\Lambda^1_0 = \gamma \frac{v}{c} = \gamma\beta \quad (81)$$

Solving the Full 2x2 Matrix

Our 2×2 transformation matrix now takes the form:

$$\Lambda = \begin{pmatrix} \gamma & \Lambda^0_1 \\ \gamma\beta & \Lambda^1_1 \end{pmatrix} \quad (82)$$

To find the remaining components, we employ the matrix form of the interval invariance: $\Lambda^T \eta \Lambda = \eta$.

$$\Lambda^T \eta \Lambda = \begin{pmatrix} \gamma & \gamma\beta \\ \Lambda^0_1 & \Lambda^1_1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \gamma & \Lambda^0_1 \\ \gamma\beta & \Lambda^1_1 \end{pmatrix} = \eta \quad (83)$$

Evaluating the off-diagonal terms yields:

$$\gamma\Lambda^0_1 - \gamma\beta\Lambda^1_1 = 0 \implies \Lambda^0_1 = \beta\Lambda^1_1 \quad (84)$$

Evaluating the bottom-right $(1, 1)$ component yields:

$$(\Lambda^0_1)^2 - (\Lambda^1_1)^2 = -1 \quad (85)$$

Substitute $\Lambda^0_1 = \beta\Lambda^1_1$ into the second equation:

$$\beta^2(\Lambda^1_1)^2 - (\Lambda^1_1)^2 = -1 \quad (86)$$

$$(\Lambda^1_1)^2(1 - \beta^2) = 1 \quad (87)$$

$$(\Lambda^1_1)^2 = \gamma^2 \implies \Lambda^1_1 = \pm\gamma \quad (88)$$

We must choose between $\pm\gamma$. The determinant of the transformation is $\det \Lambda = \gamma\Lambda^1_1 - \gamma\beta\Lambda^0_1$. To avoid spatial reflections, we demand $\det \Lambda = +1$ (a Proper Lorentz transformation). If we choose $\Lambda^1_1 = +\gamma$, then $\Lambda^0_1 = \gamma\beta$, and:

$$\det \Lambda = \gamma(\gamma) - \gamma\beta(\gamma\beta) = \gamma^2(1 - \beta^2) = 1 \quad (89)$$

Thus, the correct choice is $+$, yielding the final matrix.

$$\Lambda^\mu_\nu = \begin{pmatrix} \gamma & \gamma\beta \\ \gamma\beta & \gamma \end{pmatrix} \quad (90)$$

Note: The symmetry (positive off-diagonal elements) obtained here naturally represents a boost where the unprimed frame moves with a relative velocity in the negative x -direction compared to the primed frame. The standard inverse (or forward boost) involves swapping the sign of β . Written explicitly as coordinates:

$$\begin{cases} ct' = \gamma \left(ct + \frac{v}{c}x \right) \\ x' = \gamma(x + vt) \end{cases} \quad (91)$$

Spacetime Intervals and The Light Cone

Consider an event at the origin $(0, 0)$ and a second generic event at (ct, x) . We define the spacetime interval relative to the origin:

$$\Delta s^2 = c^2 t^2 - x^2 \quad (92)$$

Depending on which term dominates, the relative spacetime separation takes one of three forms:

1. **Lightlike (Null) Interval:** $\Delta s^2 = 0 \implies x = \pm ct$. These events lie exactly on the Light Cone. Only signals moving exactly at c (like photons) can connect them. Under a Lorentz boost, a lightlike interval remains lightlike.
2. **Spacelike Interval:** $\Delta s^2 < 0 \implies |x| > c|t|$. The spatial separation dominates. The event lies outside the light cone. No physical signal can travel between the origin and this event without exceeding c .
3. **Timelike Interval:** $\Delta s^2 > 0 \implies c|t| > |x|$. The temporal separation dominates. The event lies inside the light cone. A physical particle with $v < c$ can travel between the origin and this event.

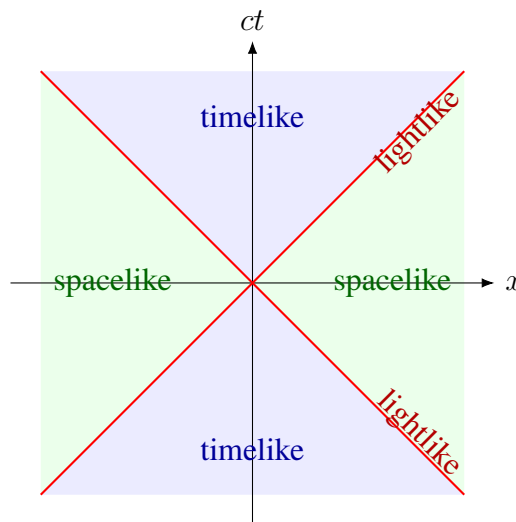


Figure 2: The light cone in $1 + 1$ dimensions. Timelike events lie inside the cone, spacelike events lie outside it, and null events lie on its boundary.

Connections: Causality and Simultaneity

Spacelike Intervals and Relative Simultaneity

If two events are spacelike separated ($\Delta s^2 < 0$), can we find an inertial frame K' where they occur *simultaneously*? We demand $t' = 0$:

$$ct' = \gamma(ct + \beta x) = 0 \quad (93)$$

$$\implies \beta = -\frac{ct}{x} \quad (94)$$

Because the interval is spacelike ($|x| > c|t|$), the required velocity magnitude is $|\beta| = \left|\frac{ct}{x}\right| < 1$. Thus, there exists a perfectly valid physical frame ($v < c$) where the two events occur at the exact same time. Because the time ordering can be reversed depending on the observer's frame, **spacelike events cannot be causally connected**. Saying "Event A caused Event B" is meaningless if another observer sees Event B occurring before Event A.

Timelike Intervals and Absolute Time-Ordering

If two events are timelike separated ($\Delta s^2 > 0$), can we find a frame K' where they occur at the *exact same spatial location*? We demand $x' = 0$:

$$x' = \gamma(x + \beta ct) = 0 \quad (95)$$

$$\implies \beta = -\frac{x}{ct} \quad (96)$$

Because the interval is timelike ($c|t| > |x|$), the required velocity magnitude is $|\beta| = \left|\frac{x}{ct}\right| < 1$. Thus, we can easily find a frame where both events occur at the same place (this is simply the rest frame of a particle traveling between the two events).

However, we *cannot* reverse the time ordering of timelike events. A Lorentz boost is continuous with respect to the velocity parameter v . To flip the sign of the time interval ($\Delta t' < 0$ when $\Delta t > 0$), $\Delta t'$ would have to pass continuously through 0. But setting $\Delta t' = 0$ requires $|\beta| > 1$, which is physically impossible. Therefore, the distinction between past and future is absolute for events inside the light cone, preserving **causality**.

Exam / HW Hints

- **Identifying Reference Frames:** If a problem asks you to find a frame where two events are simultaneous, first verify the interval $\Delta s^2 < 0$. Then use the condition $\Delta t' = 0 \implies \frac{v}{c} = -\frac{c\Delta t}{\Delta x}$.
- **Colocated Events:** If asked to find a frame where events happen at the same location, check that $\Delta s^2 > 0$ and use $\Delta x' = 0 \implies \frac{v}{c} = -\frac{\Delta x}{c\Delta t}$. Note that v represents the relative velocity required to "catch up" to the event.
- **Invariance is Key:** Never forget that $c^2\Delta t^2 - \Delta x^2 = c^2\Delta t'^2 - \Delta x'^2$. If Δs^2 is negative in one frame, it is negative in *all* valid inertial frames.
- **Length Measurement:** As briefly mentioned in the lecture, the relativistic definition of length requires measuring the endpoints of an object *simultaneously* in a given frame ($\Delta t = 0$). This directly leads to the phenomena of Lorentz contraction.

5 Lecture 05 — Length Contraction, Time Dilation, the Twin Paradox, and Four-Vectors

Date: 19/04/2026

Overview

This lecture begins by revisiting two foundational kinematic consequences of the Lorentz transformation—length contraction and time dilation—deriving each directly from the invariant interval and the definition of measurement. The discussion of time dilation naturally leads to a careful analysis of the *Twin Paradox*, resolving the apparent contradiction by identifying the asymmetry between an inertial and a non-inertial (accelerating) observer. The second half of the lecture introduces the covariant language required for the rest of the course: **scalars**, **four-vectors**, **proper time**, the **Minkowski metric** and its **inverse metric**, the notions of **covariant and contravariant** indices, and the **four-velocity**.

5.1 Length Contraction (Lorentz Contraction)

Setup and Definition of Length Measurement

Consider a ruler at rest in frame K' . Let the two endpoints of the ruler occupy positions x'_1 and x'_2 in K' , so the **rest length** (proper length) is:

$$L_0 = x'_2 - x'_1 \quad (97)$$

Frame K moves at velocity $-v$ relative to K' (equivalently, K' moves at velocity v relative to K). To measure the length of the ruler in frame K , we must locate *both endpoints at the same instant of time* t in K . This is the operational definition of measuring the length of a moving object.

Applying the Lorentz Transformation

The Lorentz transformation (boost in x -direction, K' moving at $+v$ relative to K) gives:

$$x' = \gamma(x - vt), \quad ct' = \gamma\left(ct - \frac{v}{c}x\right) \quad (98)$$

We consider two **events**: Event 1 is the observation of endpoint 1 at (t, x_1) in K , and Event 2 is the observation of endpoint 2 at (t, x_2) in K , both at *the same time* t in K .

Applying the Lorentz transformation to each event:

$$\begin{aligned} x'_1 &= \gamma(x_1 - vt) \\ x'_2 &= \gamma(x_2 - vt) \end{aligned}$$

Taking the difference:

$$x'_2 - x'_1 = \gamma(x_2 - x_1) \quad (99)$$

Therefore:

$$L_0 = \gamma L \quad \implies \quad \boxed{L = \frac{L_0}{\gamma} = L_0 \sqrt{1 - \frac{v^2}{c^2}}} \quad (100)$$

Since $\gamma \geq 1$, we have $L \leq L_0$: **a moving ruler appears shorter** than its proper (rest-frame) length. This is Lorentz contraction.

Key Remark: Simultaneity in the Rest Frame

A subtle but important point: in the rest frame K' , the two events (measuring the endpoints) occur at *different times* $t'_1 \neq t'_2$. However, because the ruler is stationary in K' , it does not matter at which times we observe the endpoints there — the ruler's length in K' is the same regardless. The distinction between simultaneous and non-simultaneous only matters in the frame where the ruler is *moving*.

Physical Consequence: No Rigid Bodies

Length contraction implies that there are no truly **rigid bodies** in special relativity. A cube at rest is a cube; a moving cube is contracted along the direction of motion. Elementary particles must be treated as **point-like**, or else they would change shape between reference frames, violating the principle that the laws of physics (and elementary constituents) are the same in all inertial frames.

5.2 Time Dilation

Setup

Consider a particle at rest in frame K' . The particle undergoes some process (e.g., radioactive decay) that takes a time $\Delta\tau$ as measured by a clock co-moving with the particle (i.e., at rest in K'). This is the **proper time** interval. Since the particle is at rest in K' , both the creation and decay events occur at the same spatial location in K' : $\Delta x' = 0$.

We observe this process from frame K (in which the particle moves at velocity v) and ask: how much time Δt passes in K ?

Derivation via the Invariant Interval

The spacetime interval is the same in both frames:

$$\Delta s^2 = c^2 \Delta t'^2 - \Delta x'^2 = c^2 \Delta t^2 - \Delta x^2 \quad (101)$$

In the rest frame K' : $\Delta x' = 0$, so $\Delta s^2 = c^2 \Delta\tau^2$.

In frame K : the particle travels a distance $\Delta x = v \Delta t$ during time Δt , so:

$$c^2 \Delta\tau^2 = c^2 \Delta t^2 - v^2 \Delta t^2 = \Delta t^2 (c^2 - v^2) \quad (102)$$

Dividing both sides by c^2 :

$$\Delta\tau^2 = \Delta t^2 \left(1 - \frac{v^2}{c^2} \right) = \frac{\Delta t^2}{\gamma^2} \quad (103)$$

Therefore:

$$\boxed{\Delta t = \gamma \Delta\tau} \quad (104)$$

Since $\gamma \geq 1$, we have $\Delta t \geq \Delta\tau$: **the time measured in the frame where the particle is moving is longer** than the proper time measured in the particle's rest frame. A moving clock runs slow.

Physical Example

A radioactive element placed at rest on a table decays with a mean lifetime τ_0 . If the same element is accelerated to a relativistic velocity v , an external observer measures a longer lifetime $\tau = \gamma\tau_0$. This is experimentally confirmed, for instance, in the observed lifetimes of cosmic-ray muons reaching the Earth's surface.

5.3 The Twin Paradox

The Setup

Two identical twins are born together at the same event (the origin). Twin A remains stationary in an inertial frame K throughout. Twin B leaves, travels through the universe at some trajectory (necessarily involving acceleration), and eventually returns to meet Twin A.

The paradox appears to arise because each twin can claim to be "at rest" and the other twin to be "moving." It seems symmetric: each should observe the other's clock as running slow. Yet upon reunion, both observers can look at each other and compare ages, a manifestly frame-independent (Lorentz-scalar) outcome.

Resolution: The Asymmetry of Acceleration

The key insight is that the two twins are **not** in equivalent situations:

- Twin A remains in a single inertial frame the entire time.
- Twin B *must* accelerate at some point to leave, turn around, and return. A frame that is accelerating is physically distinguishable from an inertial frame — there are fictitious (inertial) forces present, which can be measured locally (e.g., by a spring, or a freely released object that accelerates within that frame).

Therefore, the two reference frames are not symmetrically related. The Lorentz transformation laws we derived apply only between *inertial* frames.

Quantitative Argument via Proper Time

We can calculate the elapsed proper time for each twin.

Twin A (inertial): travels along a straight worldline from event P_0 to event P_f . The elapsed time is simply $t_A = T$ (the coordinate time in K).

Twin B (non-inertial): follows some curved worldline. We can partition the worldline into infinitesimal segments, on each of which B has an instantaneous velocity $v(t)$. At each instant, B is momentarily at rest in some instantaneous inertial frame K'_{inst} . The proper time elapsed in one infinitesimal segment is:

$$d\tau_B = \frac{dt}{\gamma(v(t))} = dt \sqrt{1 - \frac{v(t)^2}{c^2}} \quad (105)$$

The total elapsed proper time for Twin B is:

$$\tau_B = \int_0^T \frac{dt}{\gamma(v(t))} = \int_0^T \sqrt{1 - \frac{v(t)^2}{c^2}} dt \quad (106)$$

Since $\gamma(v) \geq 1$ for all v , we have $\frac{1}{\gamma} \leq 1$, and thus:

$$\boxed{\tau_B \leq T = \tau_A} \quad (107)$$

Twin A is always older upon reunion. The traveling twin B ages less, regardless of the specific trajectory taken. The more extreme the velocities reached, the greater the discrepancy.

Spacetime Diagram Interpretation

On a spacetime diagram (axes ct and x):

- Twin A's worldline is a straight vertical line.
- Twin B's worldline is a curved path that departs from and returns to the ct -axis.

This is the **geometrical content**: the straight worldline maximizes the proper time (the Minkowski "length" along the worldline). Any deviation from the straight (inertial) worldline reduces the accumulated proper time. This is the opposite of the Euclidean intuition (where the straight line gives the *minimum* distance), and reflects the indefinite signature of the Minkowski metric.

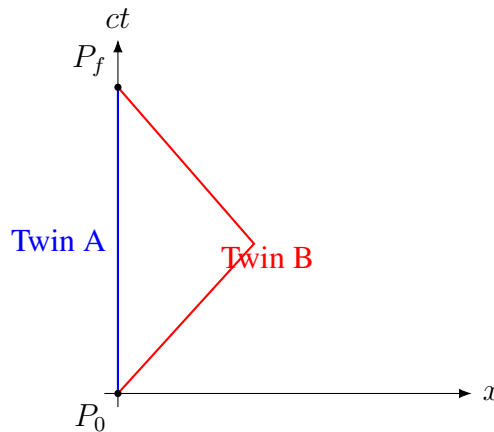


Figure 3: Twin A follows the inertial straight worldline, while Twin B departs and returns. Between the same endpoints P_0 and P_f , the straight timelike path accumulates the maximal proper time.

5.4 Proper Time

Definition

The **proper time** τ is a Lorentz scalar: it is the time measured on a clock that is co-moving with the object of interest (i.e., at rest in the object's instantaneous rest frame).

For two infinitesimally separated events on the worldline of a particle:

$$ds^2 = c^2 d\tau^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2 \quad (108)$$

In the rest frame of the particle ($d\vec{x} = 0$), this reduces to $ds^2 = c^2 dt_{\text{rest}}^2$, confirming that $d\tau$ is the time interval in the rest frame.

More explicitly, in any frame where the particle has velocity v :

$$d\tau = \frac{ds}{c} = dt \sqrt{1 - \frac{v^2}{c^2}} = \frac{dt}{\gamma} \quad (109)$$

For a finite path:

$$\tau = \int_{\text{path}} d\tau = \int \frac{dt}{\gamma(t)} \quad (110)$$

Geometric Meaning: Extremal Paths in Minkowski Space

Consider all timelike worldlines connecting two fixed events P_1 and P_2 . The integral $\int d\tau$ along each worldline is a Lorentz-scalar "length." The key result (analogous to but opposite from Euclidean geometry) is:

*The straight (inertial) worldline connecting two timelike-separated events has the **maximum** proper time.*

In Euclidean space, the straight line minimizes path length. In Minkowski space, the straight worldline *maximizes* proper time. This is a direct consequence of the indefinite metric signature $(+, -, -, -)$.

5.5 Four-Vectors and Scalars in Minkowski Space

Lorentz Scalars

A **Lorentz scalar** is a quantity whose numerical value is the same in all inertial reference frames (i.e., it is invariant under Lorentz transformations). Examples include:

- The spacetime interval: $ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu$
- The proper time: $d\tau$
- The rest mass (invariant mass): m (also called the **rest mass** or **invariant mass**; the old concept of "relativistic mass" γm is deprecated in modern physics)
- The electric charge: q

Four-Vectors: Definition

A **four-vector** (or contravariant 4-vector) is a set of four components $A^\mu = (A^0, A^1, A^2, A^3)$ that transforms under a Lorentz transformation exactly as the coordinate differential dx^μ does:

$$A'^\mu = \Lambda^\mu_\nu A^\nu \quad (111)$$

where Λ^μ_ν is the Lorentz transformation matrix. The superscript index (μ) is called a **contravariant index**.

Analogy: In 3D, a vector $\vec{V} = (V_x, V_y, V_z)$ is defined by how it transforms under rotations ($\vec{V}' = R\vec{V}$). A four-vector is defined by how it transforms under Lorentz boosts (and spatial rotations). Not every set of four numbers forms a four-vector; the transformation law must be satisfied.

The prototype four-vector is the spacetime **position four-vector**:

$$x^\mu = (ct, x, y, z) \quad (112)$$

Since we know dx^μ transforms as $dx'^\mu = \Lambda^\mu_\nu dx^\nu$ (this is the definition of the Lorentz transformation), dx^μ is indeed a four-vector. Any quantity built from dx^μ by multiplying by a scalar also transforms correctly.

The Minkowski Metric

The **Minkowski metric** $\eta_{\mu\nu}$ (with lower indices, the **covariant** metric) is:

$$\eta_{\mu\nu} = \text{diag}(+1, -1, -1, -1) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \quad (113)$$

It is used to compute the invariant interval: $ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu$.

The Inverse Metric (Contravariant Metric)

The **inverse metric** $\eta^{\mu\nu}$ (with upper indices) is defined so that:

$$\eta^{\mu\rho}\eta_{\rho\nu} = \delta^\mu_\nu \quad (114)$$

where δ^μ_ν is the Kronecker delta (the identity matrix). For the Minkowski metric, one finds:

$$\eta^{\mu\nu} = \text{diag}(+1, -1, -1, -1) = \eta_{\mu\nu} \quad (115)$$

The inverse metric is numerically identical to the metric itself in flat Minkowski space. (In curved spacetime, this is generally not the case.) The notation with upper indices is kept to distinguish the role of the matrix in index-raising operations.

Covariant and Contravariant Indices: Raising and Lowering

Given a contravariant four-vector A^μ , we define its associated **covariant four-vector** (or **covector**, or **1-form**) A_μ by lowering the index with the metric:

$$A_\mu \equiv \eta_{\mu\nu} A^\nu \quad (116)$$

Explicitly:

$$A_\mu = (A^0, -A^1, -A^2, -A^3) \quad (117)$$

Similarly, we raise an index with the inverse metric:

$$A^\mu = \eta^{\mu\nu} A_\nu \quad (118)$$

The **Lorentz-invariant inner product** (a scalar) of two four-vectors A^μ and B^μ is:

$$A \cdot B \equiv \eta_{\mu\nu} A^\mu B^\nu = A^\mu B_\mu = A_\mu B^\mu = A^0 B^0 - A^1 B^1 - A^2 B^2 - A^3 B^3 \quad (119)$$

Consistency of the Inverse Metric: Derivation

Starting from the Lorentz transformation constraint $\eta_{\mu\nu} = \Lambda^\alpha_\mu \Lambda^\beta_\nu \eta_{\alpha\beta}$ (in matrix form: $\Lambda^T \eta \Lambda = \eta$), one can show that the inverse metric $\eta^{\mu\nu}$ satisfies the analogous relation:

$$\eta^{\mu\nu} = \Lambda^\mu_\alpha \Lambda^\nu_\beta \eta^{\alpha\beta} \quad (120)$$

This is derived by multiplying the original constraint by $\eta^{\mu\rho}$ and $\eta^{\nu\sigma}$, using the invertibility of both η and Λ , and simplifying via the identity $\eta^{\mu\rho}\eta_{\rho\nu} = \delta^\mu_\nu$. The result confirms that the inverse metric transforms correctly and is consistently defined.

5.6 Four-Velocity

Definition

The **four-velocity** u^μ of a particle is defined as the derivative of the particle's spacetime position with respect to its own proper time τ :

$$u^\mu \equiv \frac{dx^\mu}{d\tau} \quad (121)$$

Since dx^μ is a four-vector and $d\tau$ is a Lorentz scalar (proper time is invariant), the ratio $u^\mu = dx^\mu/d\tau$ is automatically a four-vector.

Explicit Components

Using $d\tau = dt/\gamma$ (so $dt/d\tau = \gamma$), we can express the components in terms of the ordinary 3-velocity $\vec{v} = d\vec{x}/dt$:

$$u^0 = \frac{d(ct)}{d\tau} = c \frac{dt}{d\tau} = \gamma c, \quad (122)$$

$$u^i = \frac{dx^i}{d\tau} = \frac{dx^i}{dt} \frac{dt}{d\tau} = \gamma v^i, \quad i = 1, 2, 3 \quad (123)$$

Therefore:

$$\boxed{u^\mu = \gamma(c, v_x, v_y, v_z) = (\gamma c, \gamma \vec{v})} \quad (124)$$

The Magnitude of the Four-Velocity is Constant

A fundamental property: the Lorentz-invariant "magnitude squared" of the four-velocity is a constant:

$$\begin{aligned} u^\mu u_\mu &= \eta_{\mu\nu} u^\mu u^\nu = (u^0)^2 - (u^1)^2 - (u^2)^2 - (u^3)^2 \\ &= \gamma^2 c^2 - \gamma^2 v^2 = \gamma^2 (c^2 - v^2) = \gamma^2 c^2 \left(1 - \frac{v^2}{c^2}\right) = c^2 \end{aligned} \quad (125)$$

$$\boxed{u^\mu u_\mu = c^2} \quad (126)$$

This is consistent with the definition: $u^\mu u_\mu = \frac{dx^\mu}{d\tau} \frac{dx_\mu}{d\tau} = \frac{ds^2}{d\tau^2} = \frac{c^2 d\tau^2}{d\tau^2} = c^2$.

In the particle's rest frame ($\vec{v} = 0$, $\gamma = 1$): $u^\mu = (c, 0, 0, 0)$, and trivially $u^\mu u_\mu = c^2$. Since this is a scalar (Lorentz-invariant), it holds in all frames. This is a powerful consistency check.

Remark: Particles vs. Fields (Photons)

The four-velocity u^μ is defined only for massive particles ($v < c$). For massless particles (photons), $d\tau = 0$ along the worldline (lightlike interval), so u^μ diverges. In the classical (non-quantum) field theory framework of this course, light is described as an *electromagnetic field*, not as a stream of photons. Four-velocity therefore applies exclusively to massive particles.

Summary and Looking Ahead

1. **Length Contraction:** $L = L_0/\gamma$. Measuring the length of a moving object requires recording both endpoints at the *same coordinate time*.
2. **Time Dilation:** $\Delta t = \gamma \Delta\tau$. A moving clock runs slow; the proper time is the shortest elapsed time.
3. **Twin Paradox:** Resolved by the asymmetry between an inertial frame (Twin A) and an accelerated frame (Twin B). The traveling twin always ages less: $\tau_B \leq \tau_A$.
4. **Proper Time:** A Lorentz scalar, $d\tau = ds/c$. Geometrically, it is the Minkowski "length" of a worldline segment; a straight (inertial) worldline *maximizes* $\int d\tau$.
5. **Scalars and Four-Vectors:** A Lorentz scalar is invariant under Λ ; a four-vector transforms as $A'^\mu = \Lambda^\mu_\nu A^\nu$.
6. **Minkowski Metric and Inverse:** $\eta_{\mu\nu} = \eta^{\mu\nu} = \text{diag}(+1, -1, -1, -1)$; used to raise/lower indices and compute invariant inner products.
7. **Four-Velocity:** $u^\mu = dx^\mu/d\tau = \gamma(c, \vec{v})$; its norm is always $u^\mu u_\mu = c^2$.

In the next lecture we will combine this covariant language with the physics of electromagnetism, showing how the electric and magnetic fields unify into the **electromagnetic field tensor** $F^{\mu\nu}$.

6 Lecture 06 — Covariant and Contravariant Tensors; Relativistic Mechanics

Date: 21/04/2026

Overview

This lecture has two parts. In the first part we complete the tensor formalism begun in Lecture 05. We introduce the distinction between **contravariant** (upper-index) and **covariant** (lower-index) objects, derive the transformation law for covariant four-vectors from first principles, discuss mixed-rank tensors and why the position of each index matters, and prove that the Minkowski metric is itself a Lorentz-invariant rank-2 tensor. We also give a self-contained proof that any full contraction of tensor indices yields a Lorentz scalar.

In the second part we build **relativistic mechanics** from the action principle. Following the modern approach—identify the symmetries first, then write the most general action consistent with them—we show that the symmetries of special relativity essentially fix the free-particle action uniquely. Matching the non-relativistic limit fixes the remaining constant, and the Euler–Lagrange equations then deliver the relativistic momentum, energy, and dispersion relation. We close by adding the minimal electromagnetic coupling and recovering the covariant Lorentz force law.

6.1 Tensors in Special Relativity (Concluded)

Recap: Contravariant Four-Vectors

In the previous lecture we defined a **four-vector** as any set of four components V^μ ($\mu = 0, 1, 2, 3$) that transforms under a Lorentz transformation in exactly the same way as the coordinate differential dx^μ :

$$V'^\mu = \Lambda^\mu_\nu V^\nu. \quad (127)$$

An object satisfying this law is called a **contravariant four-vector**; its single index is written *upstairs*. Not every collection of four numbers is a four-vector: the transformation law must be verified.

Standard examples: the position four-vector $x^\mu = (ct, \mathbf{x})$, the four-velocity $u^\mu = \gamma(c, \mathbf{v})$, the four-momentum p^μ , and the four-potential A^μ .

Covariant Four-Vectors: Lowering an Index

Given a contravariant four-vector V^μ , we define a companion object with a *lower* index by contracting with the Minkowski metric $\eta_{\mu\nu} = \text{diag}(+1, -1, -1, -1)$:

$$V_\mu \equiv \eta_{\mu\nu} V^\nu. \quad (128)$$

Because $\eta_{\mu\nu}$ is diagonal with entries $(+1, -1, -1, -1)$, the explicit components are

$$V_\mu = (V^0, -V^1, -V^2, -V^3). \quad (129)$$

This object V_μ is called a **covariant four-vector** (or **covector**). It is a genuinely new object: $V_\mu \neq V^\mu$ because the Minkowski metric is not the identity. We keep the same letter but use a *downstairs* index to distinguish the two.

Transformation Law of the Covariant Four-Vector

We derive how V_μ transforms. Starting from the definition in the primed frame:

$$V'_\mu = \eta_{\mu\nu} V'^\nu = \eta_{\mu\nu} \Lambda^\nu_\rho V^\rho. \quad (130)$$

We want the right-hand side expressed in terms of V_σ (lower index), not V^ρ (upper index). Insert $V^\rho = \eta^{\rho\sigma} V_\sigma$ (raising with the inverse metric $\eta^{\mu\nu}$, satisfying $\eta_{\mu\rho}\eta^{\rho\nu} = \delta_\mu^\nu$):

$$V'_\mu = \underbrace{\eta_{\mu\nu} \Lambda^\nu_\rho \eta^{\rho\sigma}}_{\equiv \tilde{\Lambda}_\mu^\sigma} V_\sigma. \quad (131)$$

We define the **covariant Lorentz matrix**

$$\tilde{\Lambda}_\mu^\sigma \equiv \eta_{\mu\nu} \Lambda^\nu_\rho \eta^{\rho\sigma}. \quad (132)$$

Claim: $\tilde{\Lambda}_\mu^\sigma = (\Lambda^{-1})^\sigma_\mu$.

Proof. Lorentz transformations are defined by the condition that they preserve the Minkowski metric:

$$\eta_{\alpha\beta} \Lambda^\alpha_\mu \Lambda^\beta_\nu = \eta_{\mu\nu} \iff \Lambda^T \eta \Lambda = \eta. \quad (133)$$

Rearranging: $\Lambda^{-1} = \eta^{-1} \Lambda^T \eta = \eta \Lambda^T \eta$ (since $\eta^{-1} = \eta$ numerically in flat space). In components:

$$(\Lambda^{-1})^\sigma_\mu = \eta^{\sigma\rho} \Lambda^\nu_\rho \eta_{\nu\mu} = \eta_{\mu\nu} \Lambda^\nu_\rho \eta^{\rho\sigma} = \tilde{\Lambda}_\mu^\sigma. \quad \square \quad (134)$$

Therefore the transformation law of a covariant four-vector is

$$\boxed{V'_\mu = (\Lambda^{-1})^\nu_\mu V_\nu.} \quad (135)$$

A contravariant four-vector transforms with Λ ; a covariant four-vector transforms with the inverse Λ^{-1} . Both are legitimate “four-vectors” — they are distinguished only by the position of the index and by their respective transformation matrices.

The Inner Product is a Lorentz Scalar

Contracting one contravariant and one covariant four-vector:

$$V^\mu W_\mu = V^0 W^0 - V^1 W^1 - V^2 W^2 - V^3 W^3, \quad (136)$$

is automatically a Lorentz scalar. Under a Lorentz transformation:

$$V'^\mu W'_\mu = (\Lambda^\mu_\alpha V^\alpha) ((\Lambda^{-1})^\beta_\mu W_\beta) = (\Lambda^{-1})^\beta_\mu \Lambda^\mu_\alpha V^\alpha W_\beta = \delta^\beta_\alpha V^\alpha W_\beta = V^\alpha W_\alpha. \quad (137)$$

The product of the two Lorentz matrices is the identity by definition. This illustrates the core book-keeping rule: *any fully contracted expression — with every free index appearing once upstairs and once downstairs — is Lorentz invariant.*

Classification of Tensor Objects

- **Scalar (rank 0):** no free indices; invariant under all Lorentz transformations. Examples: ds^2 , m , $d\tau$, $V^\mu W_\mu$.
- **Contravariant four-vector (rank 1, upper):** $V'^\mu = \Lambda^\mu_\nu V^\nu$.
- **Covariant four-vector (rank 1, lower):** $V'_\mu = (\Lambda^{-1})^\nu_\mu V_\nu$.
- **Rank-2 contravariant tensor:** $T'^{\mu\nu} = \Lambda^\mu_\rho \Lambda^\nu_\sigma T^{\rho\sigma}$.
- **Rank-2 covariant tensor:** $T'_{\mu\nu} = (\Lambda^{-1})^\rho_\mu (\Lambda^{-1})^\sigma_\nu T_{\rho\sigma}$.
- **Mixed rank-2 tensor:** $T'^\mu_\nu = \Lambda^\mu_\rho (\Lambda^{-1})^\sigma_\nu T^\rho_\sigma$.
- **Rank- n tensor:** each upper index transforms with a factor Λ ; each lower index transforms with a factor Λ^{-1} .

Index Position Matters in Mixed Tensors

For a tensor with at least one upper and one lower index, the *relative position* of the indices matters. In general:

$$T^\mu_\nu \neq T^\mu_\nu, \quad (138)$$

because these represent different index-lowering choices applied to a tensor that may not be symmetric. When writing mixed tensors, always specify which index was raised or lowered and preserve its left-right order relative to the other indices. For example, T^μ_ν has the upper index to the left, while T^μ_ν has the upper index to the right; unless T is symmetric these are different components.

Raising and Lowering Indices

For any tensor, any index can be raised or lowered independently using the metric or its inverse:

$$V_\mu = \eta_{\mu\nu} V^\nu, \quad V^\mu = \eta^{\mu\nu} V_\nu, \quad (139)$$

$$T_{\mu\nu} = \eta_{\mu\rho} T^\rho_\nu = \eta_{\mu\rho} \eta_{\nu\sigma} T^{\rho\sigma}, \quad (140)$$

and analogously for higher-rank tensors. Applying η followed by η^{-1} (or vice versa) returns the original tensor.

With the $(+, -, -, -)$ metric, lowering a time-component index leaves it unchanged while lowering a spatial-component index introduces a minus sign:

$$V_0 = V^0, \quad V_i = -V^i \quad (i = 1, 2, 3). \quad (141)$$

Contraction and the Trace

Contraction means setting one upper and one lower index equal and summing (Einstein convention). Each contraction reduces the tensor rank by 2:

- A rank-2 tensor $T^{\mu\nu}$ contracted to a scalar:

$$T \equiv T^\mu_\mu = \eta_{\mu\nu} T^{\mu\nu}. \quad (142)$$

- A rank-3 tensor $T^{\mu\nu\rho}$ contracted on the first and third indices to a four-vector:

$$S^\nu = T^\mu_{\nu\mu} = \eta_{\mu\rho} T^{\mu\nu\rho}. \quad (143)$$

- More generally, contracting k pairs of indices in a rank- n tensor yields a rank- $(n - 2k)$ tensor.

Why Every Full Contraction is a Lorentz Scalar

We verify directly for the trace of a rank-2 tensor. In the primed frame:

$$\begin{aligned} T' &= \eta_{\mu\nu} T'^{\mu\nu} \\ &= \eta_{\mu\nu} \Lambda^\mu_\rho \Lambda^\nu_\sigma T^{\rho\sigma} \\ &= T^{\rho\sigma} (\eta_{\mu\nu} \Lambda^\mu_\rho \Lambda^\nu_\sigma) \\ &= T^{\rho\sigma} \eta_{\rho\sigma} = T. \end{aligned} \quad (144)$$

In the third step we rearranged numerical factors (permitted since sums of products of numbers commute). In the fourth step we used the defining property (133) of Lorentz transformations. The same argument extends to any rank: each pair of contracted indices produces a factor $(\Lambda^{-1})\Lambda = \mathbf{1}$.

The Minkowski Metric is a Lorentz-Invariant Tensor

We now show that $\eta_{\mu\nu}$ is itself a rank-2 covariant tensor — and a *constant* one: it is numerically the same in every inertial frame.

Applying the covariant rank-2 transformation law to $\eta_{\mu\nu}$:

$$\eta'_{\mu\nu} = (\Lambda^{-1})^\rho{}_\mu (\Lambda^{-1})^\sigma{}_\nu \eta_{\rho\sigma}. \quad (145)$$

Inserting $(\Lambda^{-1})^\rho{}_\mu = \eta_{\mu\alpha} \Lambda^\alpha{}_\beta \eta^{\beta\rho}$ and collecting, the right-hand side reduces to $\eta_{\mu\nu}$ by the Lorentz condition (133):

$$\boxed{\eta'_{\mu\nu} = \eta_{\mu\nu}} \quad (146)$$

Equation (146) is not a coincidence: it is exactly the defining property of Lorentz transformations. The Minkowski metric is therefore a **constant tensor**: it looks the same to every inertial observer.

This completes the tensor vocabulary required for the rest of the course.

6.2 Relativistic Mechanics

The Modern Paradigm: Symmetries Determine the Action

The Lagrangian/action formulation of mechanics says that a physical system is specified by its action $S = \int L dt$. Rather than guessing L , the modern approach proceeds in three steps:

1. **Identify the symmetries.** For a relativistic point particle in flat spacetime these are: Lorentz boosts, spatial rotations, spatial-translation invariance (homogeneity of space), and time-translation invariance (homogeneity of time).
2. **Write the most general action consistent with those symmetries.** Any term that is not a Lorentz scalar violates Lorentz symmetry and is forbidden. Spatial/temporal homogeneity forbids explicit dependence on x^μ itself.
3. **Fix any remaining undetermined constants** by demanding agreement with the known non-relativistic limit.

This symmetry-first approach is how all of modern field theory is built. For a free particle the constraints are so tight that the action is essentially unique.

The Only Available Lorentz Scalar for a Free Particle

For a free point particle the dynamical degrees of freedom are its worldline coordinates $x^\mu(\lambda)$ and the only dimensional constant is c . We must construct a Lorentz scalar that:

- depends only on dx^μ (translational invariance forbids x^μ itself),
- is reparameterisation-invariant (independent of the labelling parameter λ).

The unique (up to a constant factor) object satisfying all requirements is the spacetime line element:

$$ds^2 \equiv \eta_{\mu\nu} dx^\mu dx^\nu = c^2 dt^2 - d\vec{x}^2 > 0 \quad (\text{timelike worldline}), \quad (147)$$

giving the invariant arc length $ds = c dt \sqrt{1 - v^2/c^2} = c d\tau$. Any polynomial in x^μ would violate translational invariance; higher-order differential invariants reduce to powers of ds^2 and can be absorbed into the constant. The action must therefore take the form

$$S = -\alpha \int ds \quad (148)$$

for some constant α to be determined.

Why the Minus Sign?

In Minkowski space, a straight (inertial) worldline *maximises* the proper time $\tau = \int d\tau = \int ds/c$ between two fixed events. (This is the geometrical content of the twin paradox: any deviation from a straight worldline shortens the elapsed proper time.) Therefore $\int ds$ is *maximised* along the classical trajectory, not minimised.

The principle of **least action** demands $\delta S = 0$ with S stationary at a *minimum*. Inserting the minus sign converts the maximum of $\int ds$ into a minimum of $S = -\alpha \int ds$ (for $\alpha > 0$), restoring compatibility with the variational principle.

Fixing the Constant α : the Non-Relativistic Limit

Substituting $ds = c dt \sqrt{1 - \beta^2}$ (where $\beta = v/c$):

$$S = -\alpha c \int \sqrt{1 - \beta^2} dt. \quad (149)$$

Expanding for $v \ll c$:

$$\sqrt{1 - \beta^2} \approx 1 - \frac{v^2}{2c^2} + O(v^4/c^4), \quad (150)$$

so

$$S \approx -\alpha c \int dt + \frac{\alpha}{2c} \int v^2 dt. \quad (151)$$

The first term is a constant added to L (it does not affect the equations of motion). Matching the second term to the classical kinetic action $\frac{1}{2}m \int v^2 dt$ requires

$$\frac{\alpha}{2c} = \frac{m}{2} \implies \alpha = mc. \quad (152)$$

The free relativistic particle action and its Lagrangian are therefore:

$$S = -mc \int ds = -mc^2 \int \sqrt{1 - \frac{v^2}{c^2}} dt, \quad (153)$$

$$L = -mc^2 \sqrt{1 - \frac{v^2}{c^2}} = -\frac{mc^2}{\gamma}. \quad (154)$$

Non-Relativistic Limit

Expanding (153):

$$S \approx -mc^2 \int dt + \frac{1}{2}m \int v^2 dt. \quad (155)$$

The constant $-mc^2 \int dt$ shifts L by a constant and is irrelevant for the dynamics. The remaining term $\frac{1}{2}mv^2$ is exactly the classical free-particle Lagrangian, confirming internal consistency.

Conservation Laws from Noether's Theorem

The Lagrangian (154) depends on $v = |\dot{x}|$ but neither on the coordinates x^i nor on t explicitly:

$$\frac{\partial L}{\partial x^i} = 0, \quad \frac{\partial L}{\partial t} = 0. \quad (156)$$

By Noether's theorem:

- **Spatial homogeneity** ($\partial L / \partial x^i = 0$) $\Rightarrow$ the canonical momentum $p_i = \partial L / \partial \dot{x}^i$ is a constant of the motion (conserved for all time).
- **Temporal homogeneity** ($\partial L / \partial t = 0$) $\Rightarrow$ the energy $E = \dot{x}^i (\partial L / \partial \dot{x}^i) - L$ is conserved.

We now compute both.

Relativistic Momentum

$$p_i = \frac{\partial L}{\partial v^i} = \frac{\partial}{\partial v^i} \left(-mc^2 \sqrt{1 - \frac{v^2}{c^2}} \right) = \frac{-mc^2 \cdot (-v^i/c^2)}{\sqrt{1 - v^2/c^2}} = \frac{mv^i}{\sqrt{1 - v^2/c^2}} = \gamma mv^i. \quad (157)$$

In vector form:

$$\boxed{\vec{p} = \gamma m \vec{v}}. \quad (158)$$

For $v \ll c$, $\gamma \rightarrow 1$ and $\vec{p} \rightarrow m\vec{v}$, recovering Newton. For $v \rightarrow c$, $\gamma \rightarrow \infty$ and no finite force can accelerate the particle to the speed of light.

Relativistic Energy

$$\begin{aligned} E &= \vec{v} \cdot \frac{\partial L}{\partial \vec{v}} - L = \vec{v} \cdot \vec{p} - L \\ &= \gamma mv^2 + mc^2 \sqrt{1 - \beta^2} = \gamma mv^2 + \frac{mc^2}{\gamma}. \end{aligned} \quad (159)$$

Using $v^2 = c^2(1 - \gamma^{-2})$:

$$E = \gamma mc^2 \left(1 - \frac{1}{\gamma^2} \right) + \frac{mc^2}{\gamma} = \gamma mc^2 - \frac{mc^2}{\gamma} + \frac{mc^2}{\gamma} = \gamma mc^2. \quad (160)$$

Therefore:

$$\boxed{E = \gamma mc^2}. \quad (161)$$

Expanding for $v \ll c$:

$$E = mc^2 \left(1 + \frac{v^2}{2c^2} + \dots \right) = mc^2 + \frac{1}{2}mv^2 + \dots \quad (162)$$

The energy contains the **rest energy** mc^2 plus the classical kinetic energy as the leading correction. The rest energy is unobservable in purely mechanical processes but becomes physically relevant whenever mass is converted into radiation or kinetic energy.

Energy-Momentum Dispersion Relation

From (158) and (161) one can eliminate γ and v to obtain a frame-independent relation between E and $|\vec{p}| \equiv p$:

$$E^2 - p^2 c^2 = (\gamma mc^2)^2 - (\gamma mv)^2 c^2 = \gamma^2 m^2 c^2 (c^2 - v^2) = \gamma^2 m^2 c^4 \left(1 - \frac{v^2}{c^2} \right) = m^2 c^4. \quad (163)$$

$$\boxed{E^2 = p^2 c^2 + m^2 c^4}. \quad (164)$$

This relation is more fundamental than the individual formulas for E and $\vec{p}$: the combination $E^2 - p^2 c^2$ is a Lorentz scalar equal to $m^2 c^4$. The four-momentum $p^\mu = (E/c, \vec{p})$ is a contravariant four-vector, and (164) is the statement $\eta_{\mu\nu} p^\mu p^\nu = p^\mu p_\mu = m^2 c^2$ (in the $(+, -, -, -)$ convention).

Massless Particles Must Travel at the Speed of Light

Setting $m = 0$ in (164):

$$E = pc. \quad (165)$$

From $\vec{p} = \gamma m \vec{v}$ and $E = \gamma m c^2$, the ratio gives $v = pc^2/E = c$ as long as $m \neq 0$. Taking $m \rightarrow 0$ while keeping $E = pc$ finite yields $v = c$. Therefore:

$$\boxed{m = 0 \implies v = c.} \quad (166)$$

A massless particle has no rest frame; the only velocity consistent with Lorentz symmetry is the invariant speed c . In Lecture 07 we will derive electromagnetic waves from the covariant Maxwell equations and confirm that they propagate at exactly c .

Coupling to the Electromagnetic Field

To describe a particle of charge e interacting with the electromagnetic field, we add to (153) the only Lorentz-scalar coupling linear in the **electromagnetic four-potential** $A_\mu = (\phi, -\mathbf{A})$:

$$S_{\text{int}} = -\frac{e}{c} \int A_\mu dx^\mu. \quad (167)$$

This is a Lorentz scalar because A_μ is a covariant four-vector and dx^μ is a contravariant four-vector; their inner product $A_\mu dx^\mu$ is invariant. The total action is

$$S = -mc \int ds - \frac{e}{c} \int A_\mu dx^\mu. \quad (168)$$

Writing explicitly in terms of coordinate time ($\dot{x}^\mu = (c, \mathbf{v})$):

$$L = -mc^2 \sqrt{1 - \beta^2} - \underbrace{\frac{e}{c} A_\mu \dot{x}^\mu}_{= -e\phi + \frac{e}{c} \mathbf{A} \cdot \mathbf{v}} = -mc^2 \sqrt{1 - \beta^2} - e\phi + \frac{e}{c} \mathbf{A} \cdot \mathbf{v}, \quad (169)$$

which is the standard **minimal-coupling Lagrangian** for a charged relativistic particle.

Covariant Equation of Motion: the Relativistic Lorentz Force

Varying the total action (168) with respect to the worldline $x^\mu(\lambda)$, the free-particle part yields the inertial term $m du^\mu/d\tau$. The variation of the interaction term generates the antisymmetric combination

$$F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu, \quad (170)$$

the **electromagnetic field tensor**. (F_{0i} encodes $\mathbf{E}$; F_{ij} encodes $\mathbf{B}$.) Combining both contributions, the Euler–Lagrange equations take the manifestly covariant form:

$$\boxed{m \frac{du^\mu}{d\tau} = \frac{e}{c} F^\mu{}_\nu u^\nu} \quad (171)$$

where $u^\mu = dx^\mu/d\tau$ is the four-velocity. This is the **relativistic Lorentz force law**. In the non-relativistic limit it reduces to the familiar $m d\mathbf{v}/dt = e(\mathbf{E} + \mathbf{v}/c \times \mathbf{B})$.

Summary and Looking Ahead

1. **Contravariant vs. covariant:** V^μ transforms with Λ ; $V_\mu = \eta_{\mu\nu}V^\nu$ transforms with Λ^{-1} . Index position is the sole distinguishing notation.
2. **Covariant transformation matrix:** $\tilde{\Lambda}_\mu^\sigma = \eta_{\mu\nu}\Lambda^\nu_\rho\eta^{\rho\sigma} = (\Lambda^{-1})^\sigma_\mu$, a direct consequence of $\Lambda^T\eta\Lambda = \eta$.
3. **Mixed tensors:** index position matters; $T^\mu_\nu \neq T^\mu_\nu$ unless the tensor is symmetric.
4. **Minkowski metric:** $\eta'_{\mu\nu} = \eta_{\mu\nu}$ in every inertial frame — a constant rank-2 covariant tensor.
5. **Contraction:** summing one upper and one lower index reduces rank by 2 and yields a Lorentz-invariant object.
6. **Modern approach:** symmetries (Lorentz + translations) fix the action of a free particle essentially uniquely as $S = -mc \int ds$. The minus sign converts the maximum of $\int ds$ to a least-action minimum.
7. **Relativistic momentum and energy:** $\vec{p} = \gamma m \vec{v}$, $E = \gamma mc^2$, $E^2 = p^2c^2 + m^2c^4$.
8. **Massless limit:** $m = 0 \Rightarrow E = pc \Rightarrow v = c$.
9. **Electromagnetic coupling:** $S_{\text{int}} = -(e/c) \int A_\mu dx^\mu$; equation of motion is $m du^\mu/d\tau = (e/c)F^\mu_\nu u^\nu$.

In the next lecture we will write Maxwell's equations in manifestly covariant form using $F^{\mu\nu}$, introduce the **energy-momentum tensor** $T^{\mu\nu}$ of the electromagnetic field, and connect Noether's theorem to the conservation of energy and momentum in the field.

7 Lecture 07 — Electromagnetic Lagrangian, Lorentz Force, and Gauge Invariance

Date: 28/04/2026

Overview

This lecture has three conceptual parts. In the first part we pick up directly from the covariant electromagnetic action of Lecture 06, choose coordinate time as the worldline parameter, and extract the explicit three-dimensional Lagrangian. We identify the **scalar potential** ϕ and the **vector potential** $\mathbf{A}$ as the temporal and spatial parts of the covariant four-potential A_μ , and carefully distinguish the **canonical momentum** from the **mechanical momentum**.

In the second part we apply the Euler–Lagrange equations to that Lagrangian. The key technical step is evaluating the total time derivative of the potential along the particle’s worldline via the chain rule; a careful bookkeeping of the Levi-Civita identity then causes two gradient terms to cancel and the familiar **Lorentz force law** drops out.

In the third part we identify a profound symmetry of the entire formulation: **gauge invariance**. Shifting A_μ by the four-gradient of any smooth scalar function f changes the action by a pure boundary term, leaving the equations of motion and the physical fields $\mathbf{E}$ and $\mathbf{B}$ unchanged. We close with a manifestly covariant re-derivation of the equations of motion that uses the **proper time** τ as the parameter; the calculation keeps Lorentz invariance explicit at every step and delivers the electromagnetic **field tensor** $F^{\mu\nu}$.

7.1 From the Covariant Action to the Explicit Lagrangian

Physical Motivation

In Lecture 06 we derived the total action for a relativistic charged particle in an external electromagnetic field:

$$S = -mc \int ds - \frac{e}{c} \int A_\mu dx^\mu. \quad (172)$$

This expression is manifestly Lorentz invariant, but it is written without committing to any particular parameter along the worldline. To derive explicit equations of motion we must *choose a parameter*. The natural choice is coordinate time t : this breaks 4-covariance at the level of the Lagrangian but is perfectly consistent and reproduces Newton’s equation in the non-relativistic limit. The translation into $L(t)$ is a straightforward but instructive exercise in decomposing four-vectors.

Decomposing the Four-Potential

The electromagnetic four-potential with a raised (contravariant) index is defined as

$$A^\mu = (\phi, \mathbf{A}), \quad (173)$$

where ϕ is the **scalar potential** and $\mathbf{A}$ the **three-vector potential**. Lowering with the Minkowski metric $\eta_{\mu\nu} = \text{diag}(+1, -1, -1, -1)$:

$$A_\mu = \eta_{\mu\nu} A^\nu = (\phi, -\mathbf{A}). \quad (174)$$

The time component is preserved ($A_0 = A^0 = \phi$), while the spatial components pick up a minus sign: $A_i = -A^i$. This sign is a direct consequence of the $(+, -, -, -)$ signature; it appears in every contraction involving spatial indices.

Expanding the Interaction Term

With $x^\mu(t) = (ct, \mathbf{x}(t))$ as the parameterisation, the coordinate differentials are $dx^\mu = (c dt, d\mathbf{x})$. Contracting with A_μ :

$$\begin{aligned} A_\mu dx^\mu &= A_0 dx^0 + A_i dx^i = \phi \cdot c dt + (-A^i) dx^i = \phi c dt - \mathbf{A} \cdot d\mathbf{x} \\ &= (c\phi - \mathbf{A} \cdot \mathbf{v}) dt, \end{aligned} \quad (175)$$

where $\mathbf{v} = d\mathbf{x}/dt$ is the three-velocity. The interaction term in the action becomes:

$$-\frac{e}{c} \int A_\mu dx^\mu = -\frac{e}{c} \int (c\phi - \mathbf{A} \cdot \mathbf{v}) dt = \int \left(-e\phi + \frac{e}{c} \mathbf{A} \cdot \mathbf{v} \right) dt. \quad (176)$$

Adding the free-particle Lagrangian derived in Lecture 06 ($L_{\text{free}} = -mc^2\sqrt{1 - v^2/c^2}$), the complete Lagrangian is:

$$L = -mc^2\sqrt{1 - \frac{v^2}{c^2}} - e\phi + \frac{e}{c} \mathbf{A} \cdot \mathbf{v}. \quad (177)$$

Physical significance: The three terms encode, in order: the relativistic inertia (whose non-relativistic expansion yields $\frac{1}{2}mv^2$ plus the irrelevant constant $-mc^2$); the electrostatic potential energy $e\phi$; and a *velocity-dependent* magnetic coupling $\frac{e}{c} \mathbf{A} \cdot \mathbf{v}$. The magnetic coupling is the origin of the $\mathbf{v} \times \mathbf{B}$ term in the Lorentz force: it is the price paid for the field carrying four-momentum.

7.2 Equations of Motion: Deriving the Lorentz Force

Physical Motivation

Because ϕ and $\mathbf{A}$ depend on position, the action is not invariant under spatial translations, and the canonical momentum is not conserved. The task is to apply the Euler–Lagrange equations

$$\frac{d}{dt} \left(\frac{\partial L}{\partial v^i} \right) - \frac{\partial L}{\partial x^i} = 0 \quad (178)$$

and work out both sides explicitly. The key subtlety is the total time derivative of the potential along the moving worldline, which requires the chain rule. A vector identity (proved via Levi-Civita symbols) then causes two gradient terms to cancel, leaving a clean result.

The Canonical Momentum

Differentiating L with respect to v^i :

$$p_i^{\text{can}} \equiv \frac{\partial L}{\partial v^i} = \underbrace{\frac{mv^i}{\sqrt{1 - v^2/c^2}}}_{p_i^{\text{mech}}} + \frac{e}{c} A^i. \quad (179)$$

Two contributions appear. The **mechanical momentum** $p_i^{\text{mech}} = \gamma mv^i$ is the same quantity derived from the free Lagrangian in Lecture 06. The field-dependent correction $\frac{e}{c} A^i$ is new — it arises from the velocity-dependent coupling and makes the canonical momentum **gauge dependent** (since A^i changes under a gauge transformation, as we will see).

Remark 3. In quantum mechanics (and quantum field theory), canonical commutation relations are imposed between position x^i and the *canonical* momentum p_j^{can} . The minimal-coupling substitution $\mathbf{p}^{\text{can}} \rightarrow -i\hbar\nabla$ therefore replaces the *mechanical* momentum by $-i\hbar\nabla - \frac{e}{c}\mathbf{A}$. This is the origin of the covariant derivative in quantum electrodynamics and is a non-trivial physical consequence of the distinction between the two momenta.

The Generalised Force: A Vector Identity via Levi-Civita Symbols

The right-hand side of the Euler–Lagrange equation requires $\partial L / \partial x^i$ with $\mathbf{v}$ held fixed:

$$\frac{\partial L}{\partial x^i} = -e \partial_i \phi + \frac{e}{c} \frac{\partial(\mathbf{A} \cdot \mathbf{v})}{\partial x^i} = -e \partial_i \phi + \frac{e}{c} v^j \partial_i A^j. \quad (180)$$

(Since $\mathbf{v}$ does not depend on $\mathbf{x}$, only the spatial components of $\mathbf{A}$ are differentiated.)

We now rewrite $v^j \partial_i A^j$ using the Levi-Civita symbol. Recall the identity

$$\epsilon_{ijk} \epsilon_{klm} = \delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}. \quad (181)$$

The i -th component of $\mathbf{v} \times (\nabla \times \mathbf{A})$ is

$$\begin{aligned} [\mathbf{v} \times (\nabla \times \mathbf{A})]^i &= \epsilon^{ijk} v^j (\nabla \times \mathbf{A})^k = \epsilon^{ijk} v^j \epsilon_{klm} \partial^l A^m \\ &= (\delta_l^i \delta_m^j - \delta_m^i \delta_l^j) v^j \partial^l A^m = v^j \partial^i A^j - v^j \partial^j A^i. \end{aligned} \quad (182)$$

With $\mathbf{v}$ treated as spatially constant: $v^j \partial^i A^j = \partial^i (v^j A^j) = \partial_i (\mathbf{v} \cdot \mathbf{A})$ (using $\partial^i = -\partial_i$ for spatial indices with our metric, but $\partial_i (\mathbf{v} \cdot \mathbf{A})$ suffices here since $\mathbf{v}$ is constant). Therefore:

$$v^j \partial_i A^j = [\mathbf{v} \times (\nabla \times \mathbf{A})]^i + v^j \partial_j A^i = [\mathbf{v} \times \mathbf{B}]^i + (\mathbf{v} \cdot \nabla) A^i, \quad (183)$$

where $\mathbf{B} \equiv \nabla \times \mathbf{A}$. Substituting back:

$$\frac{\partial L}{\partial x^i} = -e \partial_i \phi + \frac{e}{c} (\mathbf{v} \cdot \nabla) A^i + \frac{e}{c} [\mathbf{v} \times \mathbf{B}]^i. \quad (184)$$

Total Time Derivative Along the Worldline

The left-hand side of the Euler–Lagrange equation involves $\frac{d}{dt}(\gamma m v^i + \frac{e}{c} A^i)$. The potential $\mathbf{A} = \mathbf{A}(\mathbf{x}(t), t)$ depends on both position and time; as the particle moves, both change. Applying the chain rule:

$$\frac{dA^i}{dt} = \underbrace{\frac{\partial A^i}{\partial t}}_{\text{field changes at fixed point}} + \underbrace{v^j \frac{\partial A^i}{\partial x^j}}_{\text{particle moves to new point}} = \frac{\partial A^i}{\partial t} + (\mathbf{v} \cdot \nabla) A^i. \quad (185)$$

The first term is the explicit time dependence of the field; the second arises because the particle drifts to a location where the field has a different value.

Assembling the Euler–Lagrange Equation

Inserting (185) and (184) into the Euler–Lagrange equation:

$$\frac{d(\gamma m v^i)}{dt} + \frac{e}{c} \left(\frac{\partial A^i}{\partial t} + (\mathbf{v} \cdot \nabla) A^i \right) = -e \partial_i \phi + \frac{e}{c} (\mathbf{v} \cdot \nabla) A^i + \frac{e}{c} [\mathbf{v} \times \mathbf{B}]^i. \quad (186)$$

The two $(\mathbf{v} \cdot \nabla) A^i$ terms cancel identically. Rearranging:

$$\frac{d(\gamma m v^i)}{dt} = -e \partial_i \phi - \frac{e}{c} \frac{\partial A^i}{\partial t} + \frac{e}{c} [\mathbf{v} \times \mathbf{B}]^i. \quad (187)$$

Definitions of the Physical Fields and the Lorentz Force

Recognising the two combinations that appear, we define:

$$\mathbf{E} \equiv -\nabla\phi - \frac{1}{c} \frac{\partial \mathbf{A}}{\partial t}, \quad (188)$$

$$\mathbf{B} \equiv \nabla \times \mathbf{A}. \quad (189)$$

Substituting into (187):

$$\boxed{\frac{d\mathbf{p}^{\text{mech}}}{dt} = e \left(\mathbf{E} + \frac{\mathbf{v}}{c} \times \mathbf{B} \right)}. \quad (190)$$

This is the **Lorentz force law**.

Physical significance: We have derived—purely from symmetry (Lorentz invariance) and the single coupling $-\frac{e}{c} \int A_\mu dx^\mu$ —the complete electromagnetic force on a charged relativistic particle. The form was not assumed; it followed from the Euler–Lagrange equations applied to the action. The cancellation of the two $(\mathbf{v} \cdot \nabla)A^i$ terms is not accidental: it is mandated by gauge invariance, as will become clear in Section 7.3.

Why the Equation of Motion Is Lorentz Invariant

The equation (190) looks correct but its Lorentz invariance is not manifest: $\mathbf{v}$ is not a four-vector, $\mathbf{E}$ and $\mathbf{B}$ transform into each other under boosts, and it is laborious to verify explicitly that both sides transform the same way.

The key insight is: because we *derived* the equation from the Lorentz-invariant action (172), it *must* be Lorentz invariant. Any Lorentz transformation merely relabels which inertial frame we call “home”; since the action is the same in all frames, so are its Euler–Lagrange equations, up to a change of variables. This is a powerful application of the symmetry-first approach: we enforce invariance at the level of the action (one scalar object) and the equations of motion inherit it automatically, even if their form looks frame-dependent.

7.3 Gauge Invariance

Physical Motivation

The Lagrangian (177) is written in terms of ϕ and $\mathbf{A}$ rather than the directly measurable fields $\mathbf{E}$ and $\mathbf{B}$. This raises a natural question: how much freedom do we have in choosing A_μ while describing the *same* physical situation? Answering this question reveals a deep structure: the electromagnetic field is not uniquely described by a single A_μ , but by an entire *equivalence class* of potentials. The freedom to choose among them is called **gauge freedom**, and it is one of the defining features of every fundamental force in nature.

The Gauge Transformation

A **gauge transformation** replaces the four-potential by

$$\boxed{A_\mu \longrightarrow A'_\mu = A_\mu + \partial_\mu f}, \quad (191)$$

where $f = f(x^\alpha)$ is any smooth scalar function of spacetime. In components, using $\partial_\mu f = (\frac{1}{c}\partial_t f, \nabla f)$:

$$\phi \longrightarrow \phi + \frac{1}{c} \frac{\partial f}{\partial t}, \quad (192)$$

$$\mathbf{A} \longrightarrow \mathbf{A} + \nabla f. \quad (193)$$

The function f is completely arbitrary: there are infinitely many gauge-related descriptions of any given electromagnetic field.

The Action Changes by a Boundary Term Only

Under (191), the action changes by

$$\delta S = -\frac{e}{c} \int (\partial_\mu f) dx^\mu = -\frac{e}{c} \int \frac{\partial f}{\partial x^\mu} dx^\mu = -\frac{e}{c} \int df = -\frac{e}{c} [f(x_{\text{final}}) - f(x_{\text{initial}})]. \quad (194)$$

The integral of an exact differential df telescopes to the endpoints. Since the variational principle holds the endpoints fixed ($\delta x^\mu = 0$ there), δS is a constant that depends only on the fixed boundary values of f and is therefore **irrelevant for the equations of motion**.

Physical significance: The gauge freedom means that A_μ is not a physical observable. Any two potentials related by $A'_\mu = A_\mu + \partial_\mu f$ represent the *same* physical state. The actual physical content is encoded in the equivalence class of all such potentials. Singling out one representative — a **gauge choice** — is analogous to choosing a coordinate system: it simplifies calculations but carries no physical meaning. Different gauges that are useful in different contexts include the *Lorenz gauge* ($\partial^\mu A_\mu = 0$, useful for manifestly covariant calculations) and the *Coulomb gauge* ($\nabla \cdot \mathbf{A} = 0$, useful for non-relativistic problems).

Uniqueness of the Action up to Boundary Terms

This observation also answers a deeper question: *how unique is the action?* Two actions that yield identical equations of motion must have identical functional derivatives (“functional gradients”). By the same argument that shows two ordinary functions with equal gradients differ by at most a constant, two actions with identical Euler–Lagrange equations differ by at most a functional of the boundary values — i.e. a total time derivative. Adding a total derivative to L (equivalently, a boundary term to S) never changes the equations of motion. The gauge transformation is precisely an instance of this: it adds $-\frac{e}{c}df$ to the Lagrangian, which is a total derivative.

The Physical Fields Are Gauge Invariant

We verify directly that $\mathbf{E}$ and $\mathbf{B}$ are unchanged under (192)–(193):

$$\mathbf{E}' = -\nabla \left(\phi + \frac{\partial_t f}{c} \right) - \frac{1}{c} \frac{\partial}{\partial t} (\mathbf{A} + \nabla f) = \mathbf{E} - \frac{1}{c} \nabla \partial_t f + \frac{1}{c} \partial_t \nabla f = \mathbf{E}, \quad (195)$$

$$\mathbf{B}' = \nabla \times (\mathbf{A} + \nabla f) = \mathbf{B} + \underbrace{\nabla \times \nabla f}_{=0} = \mathbf{B}. \quad (196)$$

In (195) the two extra terms cancel because partial derivatives commute; in (196) the curl of a gradient is identically zero.

Both physical fields are gauge invariant: $\mathbf{E}' = \mathbf{E}$ and $\mathbf{B}' = \mathbf{B}$. This is consistent with the Lorentz force law (190), which is written entirely in terms of $\mathbf{E}$ and $\mathbf{B}$, not ϕ or $\mathbf{A}$.

7.4 Manifestly Covariant Derivation of the Equations of Motion

Physical Motivation

In Section 7.2 we derived the Lorentz force using coordinate time t , which breaks the democratic treatment of space and time. The result is correct, but its Lorentz invariance is hidden. There is a conceptually cleaner approach: work with the **proper time** τ as the parameter, never commit to a specific frame, and vary the covariant action (172) directly. The payoff is a single, manifestly covariant equation of motion in which the Lorentz symmetry is self-evident, and it naturally introduces the **electromagnetic field tensor** $F^{\mu\nu}$ — a single, gauge-invariant, Lorentz-covariant object that encodes both $\mathbf{E}$ and $\mathbf{B}$.

Setup: Proper Time as Parameter

Since $ds = c d\tau$, the action becomes

$$S = -mc^2 \int d\tau - \frac{e}{c} \int A_\mu u^\mu d\tau, \quad (197)$$

where the **four-velocity** $u^\mu = dx^\mu/d\tau = \gamma(c, \mathbf{v})$ satisfies $u^\mu u_\mu = c^2$.

Variation of the Worldline

We shift the worldline: $x^\mu(\tau) \rightarrow x^\mu(\tau) + \delta x^\mu(\tau)$, with $\delta x^\mu = 0$ at both endpoints. The resulting changes are:

$$\delta u^\mu = \frac{d(\delta x^\mu)}{d\tau}, \quad (198)$$

$$\delta A_\mu = (\partial_\nu A_\mu) \delta x^\nu \quad (\text{Taylor expansion to first order}). \quad (199)$$

First Variation of the Free-Particle Term

From $c^2 d\tau^2 = \eta_{\mu\nu} dx^\mu dx^\nu$ one finds $c^2 \delta(d\tau) = u_\nu d(\delta x^\nu)$, so:

$$-mc^2 \delta \int d\tau = -m \int u_\nu \frac{d(\delta x^\nu)}{d\tau} d\tau \stackrel{\text{IBP}}{=} +m \int \frac{du_\nu}{d\tau} \delta x^\nu d\tau, \quad (200)$$

where integration by parts was used and the boundary terms vanish because $\delta x^\nu = 0$ at the endpoints.

First Variation of the Interaction Term

$$-\frac{e}{c} \delta \int A_\mu u^\mu d\tau = -\frac{e}{c} \delta \int \left[(\partial_\nu A_\mu) \delta x^\nu u^\mu + A_\mu \frac{d(\delta x^\mu)}{d\tau} \right] d\tau. \quad (201)$$

Integrating the second piece by parts (boundary terms vanish):

$$-\frac{e}{c} \delta \int A_\mu \frac{d(\delta x^\mu)}{d\tau} d\tau = +\frac{e}{c} \delta \int \frac{dA_\mu}{d\tau} \delta x^\mu d\tau = +\frac{e}{c} \delta \int (\partial_\nu A_\mu) u^\nu \delta x^\mu d\tau. \quad (202)$$

Combining (201) and (202), and renaming the dummy index in the first term ($\mu \leftrightarrow \nu$) so that δx^μ factors out:

$$-\frac{e}{c} \delta \int A_\mu u^\mu d\tau = \frac{e}{c} \delta \int [(\partial_\mu A_\nu - \partial_\nu A_\mu) u^\nu] \delta x^\mu d\tau. \quad (203)$$

The Field Tensor and the Covariant Equation of Motion

Adding (200) and (203) and setting $\delta S = 0$ for arbitrary δx^μ :

$$\int \left[m \frac{du_\mu}{d\tau} - \frac{e}{c} (\partial_\mu A_\nu - \partial_\nu A_\mu) u^\nu \right] \delta x^\mu d\tau = 0. \quad (204)$$

Since δx^μ is an arbitrary function, the integrand must vanish at every point. We define the **electromagnetic field tensor**:

$$\boxed{F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu}, \quad (205)$$

an antisymmetric rank-2 covariant tensor. Raising both indices (or, equivalently, using $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$), the equation of motion takes the manifestly covariant form:

$$\boxed{m \frac{du^\mu}{d\tau} = \frac{e}{c} F^\mu{}_\nu u^\nu}. \quad (206)$$

Physical significance: This is the **relativistic Lorentz force law** in fully covariant form. It is a single 4-component equation: the spatial components reproduce (190), while the time component gives the power equation $dE/dt = e\mathbf{E} \cdot \mathbf{v}$ (the rate at which the field does work on the particle). Lorentz invariance is manifest: F^μ_ν is a tensor and u^ν is a four-vector, so both sides transform identically under any Lorentz transformation. There is no need to verify invariance term by term.

Structure of the Field Tensor: Encoding $\mathbf{E}$ and $\mathbf{B}$

The tensor $F^{\mu\nu} = -F^{\nu\mu}$ has $\binom{4}{2} = 6$ independent components. Comparing the definition (205) with the definitions (188)–(189):

- **Time-space components:** F^{0i} encodes the electric field.
- **Space-space components:** F^{ij} encodes the magnetic field via $F^{ij} = -\epsilon^{ijk} B_k$.

All six physical field components ($\mathbf{E}$ and $\mathbf{B}$) are packaged into a single Lorentz tensor. Under a Lorentz boost, $F^{\mu\nu}$ transforms as a tensor and mixes the electric and magnetic components — which is precisely the well-known fact that a purely electric field in one frame contains a magnetic component in a boosted frame.

Gauge Invariance of the Field Tensor

Under $A_\mu \rightarrow A_\mu + \partial_\mu f$:

$$F_{\mu\nu} \rightarrow F_{\mu\nu} + \underbrace{(\partial_\mu \partial_\nu f - \partial_\nu \partial_\mu f)}_{=0} = F_{\mu\nu}. \quad (207)$$

Partial derivatives commute, so $F_{\mu\nu}$ is **identically gauge invariant**. This is the deeper reason why $\mathbf{E}$ and $\mathbf{B}$ — which are the components of $F^{\mu\nu}$ — are gauge invariant, as we verified directly in Section 7.3.

Summary and Looking Ahead

1. **Explicit Lagrangian:** $L = -mc^2 \sqrt{1 - v^2/c^2} - e\phi + \frac{e}{c} \mathbf{A} \cdot \mathbf{v}$. The three terms are inertia, electrostatic coupling, and magnetic coupling.
2. **Canonical vs. mechanical momentum:** $\mathbf{p}^{\text{can}} = \gamma m \mathbf{v} + \frac{e}{c} \mathbf{A}$. The canonical momentum enters Poisson brackets and canonical quantisation; the mechanical momentum appears in the force law.
3. **Lorentz force derived from symmetry:** The Euler–Lagrange equations, combined with the chain rule for dA^i/dt along the worldline and a Levi-Civita identity, yield $d\mathbf{p}^{\text{mech}}/dt = e(\mathbf{E} + \mathbf{v}/c \times \mathbf{B})$. The two $(\mathbf{v} \cdot \nabla)A^i$ terms cancel.
4. **Gauge invariance:** $A_\mu \rightarrow A_\mu + \partial_\mu f$ changes the action by a boundary term only. Physical states correspond to equivalence classes of potentials. The fields $\mathbf{E}$ and $\mathbf{B}$ are gauge invariant. Different gauge choices simplify different calculations.
5. **Uniqueness of the action:** Two Lagrangians with identical equations of motion differ by at most a total time derivative. The gauge transformation generates precisely such a difference.
6. **Covariant equation of motion:** $m du^\mu/d\tau = \frac{e}{c} F^\mu_\nu u^\nu$, derived by varying the covariant action with proper time as parameter and integrating by parts.
7. **The field tensor:** $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is antisymmetric, gauge invariant, and packs all six electromagnetic field components ($\mathbf{E}$ and $\mathbf{B}$) into a single Lorentz tensor.

In Lecture 08 we will give the field tensor $F^{\mu\nu}$ a life of its own: we will write the **action for the electromagnetic field itself** (not just for a particle in the field) and show that its Euler–Lagrange equations reproduce **Maxwell’s equations** in covariant form. We will also meet the **energy-momentum tensor** $T^{\mu\nu}$ of the electromagnetic field and connect it to conservation laws via Noether’s theorem.

Classical Field Theory

8 Lecture 08 — The Electromagnetic Field Tensor and the Field Action

Date: 03/05/2026

Overview

In the previous lecture the action for a charged particle in an electromagnetic field,

$$S = -mc \int ds - \frac{e}{c} \int A_\mu dx^\mu, \quad (208)$$

was varied to obtain the covariant equation of motion

$$\frac{dp^\mu}{d\tau} = \frac{e}{c} F^\mu{}_\nu u^\nu, \quad u^\nu = \frac{dx^\nu}{d\tau}, \quad (209)$$

where the electromagnetic field tensor is defined as

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \quad (210)$$

This lecture completes the treatment of $F_{\mu\nu}$ in three stages: (1) prove rigorously that $F_{\mu\nu}$ is a genuine rank-2 Lorentz tensor; (2) identify its six independent components with the physical electric and magnetic fields; (3) derive the Lorentz-invariant action that governs the free electromagnetic field, applying the three guiding principles of Lorentz invariance, gauge invariance, and the superposition principle.

8.1 $F_{\mu\nu}$ as a Rank-2 Lorentz Tensor

Physical Motivation

The equation of motion (209) is manifestly covariant only if $F^\mu{}_\nu$ truly transforms as a tensor. Since the left-hand side is a four-vector and u^ν is a four-vector, the right-hand side must also be a four-vector for any choice of inertial frame. We therefore prove that $F_{\mu\nu}$ inherits the correct transformation law from its definition as a derivative of a four-vector.

Transformation of the four-gradient

Let $f(x)$ be a Lorentz scalar, i.e. $f'(x) = f(x')$ where $x'^\mu = \Lambda^\mu{}_\nu x^\nu$. Using the chain rule,

$$\partial_\mu f'(x) = \frac{\partial f'}{\partial x^\mu} = \frac{\partial f'}{\partial x'^\nu} \frac{\partial x'^\nu}{\partial x^\mu} = \Lambda^\nu{}_\mu \partial_\nu f(x). \quad (211)$$

Here $\partial_\mu \equiv \partial/\partial x^\mu$ is the covariant four-gradient (lower index), and $\Lambda^\nu{}_\mu$ is the Lorentz transformation matrix. The result shows:

$$\boxed{\partial'_\mu f'(x) = \Lambda^\nu{}_\mu \partial_\nu f(x).} \quad (212)$$

Physical significance: Applying a single derivative to a Lorentz scalar yields a covariant four-vector (rank-1 tensor). More generally, each additional derivative raises the tensor rank by one.

$F_{\mu\nu}$ transforms as a rank-2 tensor

The four-potential A_μ is a four-vector: $A'_\mu(x) = \Lambda^\nu_\mu A_\nu(x')$. Applying two derivatives and antisymmetrizing,

$$\begin{aligned} F'_{\mu\nu}(x) &= \partial'_\mu A'_\nu(x) - \partial'_\nu A'_\mu(x) \\ &= \Lambda^\alpha_\mu \partial_\alpha (\Lambda^\beta_\nu A_\beta(x')) - \Lambda^\beta_\nu \partial_\beta (\Lambda^\alpha_\mu A_\alpha(x')) \\ &= \Lambda^\alpha_\mu \Lambda^\beta_\nu (\partial_\alpha A_\beta(x') - \partial_\beta A_\alpha(x')). \end{aligned} \quad (213)$$

Hence

$$\boxed{F'_{\mu\nu}(x) = \Lambda^\alpha_\mu \Lambda^\beta_\nu F_{\alpha\beta}(x')}. \quad (214)$$

Physical significance: $F_{\mu\nu}$ transforms with two copies of the Lorentz matrix, exactly the definition of a covariant rank-2 tensor. The equation of motion is therefore genuinely Lorentz covariant, taking the same form in every inertial frame.

8.2 Electric and Magnetic Fields from $F_{\mu\nu}$ **Physical Motivation**

$F_{\mu\nu}$ is an antisymmetric 4×4 matrix ($F_{\mu\nu} = -F_{\nu\mu}$, so $F_{\mu\mu} = 0$), with $\frac{1}{2} \cdot 4 \cdot 3 = 6$ independent components. Since the physical fields $\mathbf{E}$ and $\mathbf{B}$ together carry $3 + 3 = 6$ real components, there is a one-to-one correspondence.

Identifying the components

Let spatial indices $i, j, k \in \{1, 2, 3\}$. Recall $\partial_0 = (1/c)\partial_t$, $\partial_i = \partial/\partial x^i$, and in the convention with signature $(+, -, -, -)$: $A_0 = \phi/c$, $A_i = -A^i$.

Electric field.

$$\begin{aligned} F_{0i} &= \partial_0 A_i - \partial_i A_0 \\ &= \frac{1}{c} \frac{\partial A_i}{\partial t} - \partial_i \left(\frac{\phi}{c} \right) \\ &= -\frac{1}{c} \frac{\partial A^i}{\partial t} - \frac{\partial \phi}{\partial x^i}. \end{aligned} \quad (215)$$

The right-hand side is the i -th component of $\mathbf{E} = -\nabla\phi - \frac{1}{c}\partial_t\mathbf{A}$, so

$$\boxed{E_i = F_{0i}}. \quad (216)$$

Magnetic field. For the spatial-spatial components, $F_{jk} = \partial_j A_k - \partial_k A_j$. Since $\mathbf{B} = \nabla \times \mathbf{A}$,

$$B_i = (\nabla \times \mathbf{A})_i = \epsilon_{ijk} \partial_j A_k = \frac{1}{2} \epsilon_{ijk} (\partial_j A_k - \partial_k A_j) = \frac{1}{2} \epsilon_{ijk} F_{jk}, \quad (217)$$

where ϵ_{ijk} is the three-dimensional Levi-Civita symbol ($\epsilon_{123} = +1$). Therefore

$$\boxed{B_i = \frac{1}{2} \epsilon_{ijk} F_{jk}}. \quad (218)$$

Physical significance: All six physical field components are encoded in the single antisymmetric tensor $F_{\mu\nu}$. The explicit matrix (rows/columns labeled 0, 1, 2, 3) is:

$$F_{\mu\nu} = \begin{pmatrix} 0 & E_1 & E_2 & E_3 \\ -E_1 & 0 & -B_3 & B_2 \\ -E_2 & B_3 & 0 & -B_1 \\ -E_3 & -B_2 & B_1 & 0 \end{pmatrix}. \quad (219)$$

Under a Lorentz boost, $\mathbf{E}$ and $\mathbf{B}$ mix with each other — they are two aspects of a single unified field.

8.3 Gauge Invariance of $F_{\mu\nu}$

Physical Motivation

Only $\mathbf{E}$ and $\mathbf{B}$ are directly measurable; the four-potential A_μ is auxiliary. Different choices of A_μ describing the same physical fields are related by gauge transformations.

Under the gauge transformation

$$A_\mu \longrightarrow A_\mu + \partial_\mu f, \quad (220)$$

where $f(x)$ is an arbitrary smooth scalar function,

$$\begin{aligned} F_{\mu\nu} &\longrightarrow \partial_\mu(A_\nu + \partial_\nu f) - \partial_\nu(A_\mu + \partial_\mu f) \\ &= F_{\mu\nu} + \underbrace{\partial_\mu \partial_\nu f - \partial_\nu \partial_\mu f}_{=0}. \end{aligned} \quad (221)$$

Hence

$$\boxed{F_{\mu\nu} \text{ is invariant under gauge transformations.}} \quad (222)$$

Physical significance: Because $\mathbf{E}$ and $\mathbf{B}$ are components of $F_{\mu\nu}$, the physical fields are gauge-invariant. Any valid action for electromagnetism must be expressed solely in terms of $F_{\mu\nu}$.

8.4 The $\mu = 0$ Component: Work–Energy Theorem

Physical Motivation

The covariant equation of motion (209) contains four equations. The spatial components ($\mu = 1, 2, 3$) give the relativistic Lorentz force. The $\mu = 0$ component encodes the rate at which the field does work on the particle — it is the relativistic power equation.

Derivation

The four-momentum is $p^\mu = (\mathcal{E}/c, \mathbf{p})$ and the four-velocity is $u^\nu = \gamma(c, \mathbf{v})$. Setting $\mu = 0$ in the equation of motion,

$$\frac{dp^0}{d\tau} = \frac{e}{c} F^0{}_\nu u^\nu = \frac{e}{c} (F^0{}_0 u^0 + F^0{}_i u^i). \quad (223)$$

Antisymmetry gives $F^0{}_0 = 0$. Using $F^\mu{}_\nu = \eta^{\mu\alpha} F_{\alpha\nu}$ and $\eta^{00} = +1$, we get $F^0{}_i = \eta^{00} F_{0i} = E_i$. Then,

$$\frac{dp^0}{d\tau} = \frac{e}{c} E_i u^i = \frac{e}{c} \mathbf{E} \cdot (\gamma \mathbf{v}). \quad (224)$$

Converting to coordinate time ($d\tau = dt/\gamma$) and using $p^0 = \mathcal{E}/c$:

$$\frac{1}{c} \frac{d\mathcal{E}}{d\tau} = \frac{e}{c} \gamma \mathbf{E} \cdot \mathbf{v} \implies \frac{d\mathcal{E}}{dt} = e \mathbf{E} \cdot \mathbf{v}. \quad (225)$$

$$\boxed{\frac{d\mathcal{E}}{dt} = e \mathbf{E} \cdot \mathbf{v}.} \quad (226)$$

Physical significance: Only the electric field does work on the particle. The magnetic force $e\mathbf{v} \times \mathbf{B}$ is always perpendicular to $\mathbf{v}$ and performs no work; it only redirects the particle. This is the relativistic generalisation of the classical power equation.

8.5 Transition to Classical Field Theory

Physical Motivation

So far the electromagnetic field has entered only as an external background. But a moving charged particle itself generates a field. To obtain a self-consistent description we must give the field its own dynamical degrees of freedom and derive its equations of motion from a variational principle.

From particles to fields

In mechanics, N particles carry a finite set of coordinates $q_i(t)$. The electromagnetic field $A_\mu(t, \mathbf{x})$ is a function over all of spacetime, carrying *infinitely many* degrees of freedom — one four-component value per spatial point $\mathbf{x} \in \mathbb{R}^3$.

The dynamics is encoded in a **Lagrangian density** $\mathcal{L}(A_\mu, \partial_\nu A_\mu)$, and the action is

$$S_{\text{field}} = \int \mathcal{L}(A_\mu, \partial_\nu A_\mu) d^4x. \quad (227)$$

The correspondence between mechanics and field theory is:

Classical mechanics	Classical field theory
D.O.F.: $q_i(t)$, $1 \leq i \leq N$	D.O.F.: $A_\mu(t, \mathbf{x})$, $\mathbf{x} \in \mathbb{R}^3$
Lagrangian: $L(q_i, \dot{q}_i)$	Lagrangian density: $\mathcal{L}(A_\mu, \partial_\nu A_\mu)$
$S = \int L dt$	$S = \int \mathcal{L} d^4x$
E-L: $\frac{\partial L}{\partial q_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} = 0$	E-L: $\frac{\partial \mathcal{L}}{\partial A_\mu} - \partial_\nu \frac{\partial \mathcal{L}}{\partial (\partial_\nu A_\mu)} = 0$

Field Euler–Lagrange equations

Varying $A_\mu \rightarrow A_\mu + \delta A_\mu$ with $\delta A_\mu = 0$ on the boundary and integrating by parts, $\delta S = 0$ yields:

$$\boxed{\frac{\partial \mathcal{L}}{\partial A_\mu} - \partial_\nu \frac{\partial \mathcal{L}}{\partial (\partial_\nu A_\mu)} = 0.} \quad (228)$$

Physical significance: These field equations of motion will yield Maxwell's equations when the electromagnetic Lagrangian density is substituted — which is the goal of the next lecture.

8.6 Determining the Electromagnetic Field Lagrangian

Physical Motivation

We want to write $\mathcal{L}_{\text{field}}$ without arbitrary assumptions, relying only on three symmetry principles that are established by experiment.

Three constraints

1. **Lorentz invariance.** $\mathcal{L}$ must be a Lorentz scalar — no free four-indices.
2. **Gauge invariance.** $\mathcal{L}$ must be invariant under $A_\mu \rightarrow A_\mu + \partial_\mu f$. Since A_μ transforms non-trivially but $F_{\mu\nu}$ does not, $\mathcal{L}$ must depend on A_μ only through $F_{\mu\nu}$. For example, a term $\alpha A_\mu A^\mu$ would transform as

$$A_\mu A^\mu \rightarrow A_\mu A^\mu + 2A^\mu \partial_\mu f + (\partial_\mu f)(\partial^\mu f),$$

which is not gauge invariant; hence such terms are forbidden.

3. **Superposition principle.** The electromagnetic field obeys linear superposition (experimentally verified). Linearity of the field equations requires the Lagrangian to be *quadratic* in the fields. A term $\sim (F_{\mu\nu}F^{\mu\nu})^2$ would give equations of motion cubic in F , violating linearity.

Unique Lagrangian

The most general Lorentz scalar quadratic in $F_{\mu\nu}$ is a linear combination of

$$I_1 = F_{\mu\nu}F^{\mu\nu}, \quad I_2 = \epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}, \quad (229)$$

where $\epsilon^{\mu\nu\rho\sigma}$ is the four-dimensional Levi-Civita tensor (totally antisymmetric, $\epsilon^{0123} = +1$) and $I_2 = F_{\mu\nu}\tilde{F}^{\mu\nu}$ with $\tilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}$ the dual tensor.

The combination I_2 is a total derivative:

$$\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma} = 4\partial_\mu(\epsilon^{\mu\nu\rho\sigma}A_\nu\partial_\rho A_\sigma).$$

A total derivative does not affect the equations of motion and can be discarded.

The unique candidate is therefore $I_1 = F_{\mu\nu}F^{\mu\nu}$, and the Lagrangian density is fixed up to an overall constant:

$$\mathcal{L}_{\text{field}} = -\frac{1}{16\pi}F_{\mu\nu}F^{\mu\nu}. \quad (230)$$

Physical significance: The coefficient $-1/(16\pi)$ is chosen in Gaussian units so that the resulting Maxwell equations take their standard form. The negative sign ensures a positive field energy density.

Relation to E and B

Using the matrix (219) and the metric ($\eta^{00} = +1$, $\eta^{ii} = -1$),

$$\begin{aligned} F_{\mu\nu}F^{\mu\nu} &= 2(F_{0i}F^{0i} + \tfrac{1}{2}F_{ij}F^{ij}) \\ &= 2(-E_i^2 + B_i^2) = 2(\mathbf{B}^2 - \mathbf{E}^2), \end{aligned} \quad (231)$$

so that

$$\mathcal{L}_{\text{field}} = -\frac{1}{16\pi} \cdot 2(\mathbf{B}^2 - \mathbf{E}^2) = \frac{\mathbf{E}^2 - \mathbf{B}^2}{8\pi}. \quad (232)$$

Physical significance: $\mathbf{E}^2 - \mathbf{B}^2$ is a Lorentz invariant; all inertial observers agree on its value at any event in spacetime.

Complete action

Combining the free-particle, interaction, and free-field terms,

$$S = -mc \int ds - \frac{e}{c} \int A_\mu dx^\mu - \frac{1}{16\pi} \int F_{\mu\nu}F^{\mu\nu} d^4x. \quad (233)$$

Physical significance: This is the complete action of classical electrodynamics. Varying with respect to the particle trajectory yields the Lorentz force equation; varying with respect to A_μ (next lecture) will yield Maxwell's equations.

Summary

- The four-gradient of a scalar transforms as a covariant four-vector; $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ transforms as a covariant rank-2 tensor.
- Electric field: $E_i = F_{0i}$; magnetic field: $B_i = \frac{1}{2}\epsilon_{ijk}F_{jk}$. Together they fill all 6 independent components of the antisymmetric 4×4 matrix $F_{\mu\nu}$.
- $F_{\mu\nu}$ is gauge-invariant; the action must be written in terms of $F_{\mu\nu}$, not A_μ directly.
- The $\mu = 0$ equation of motion gives $d\mathcal{E}/dt = e\mathbf{E} \cdot \mathbf{v}$: only the electric field does work.
- Promoting the field to a dynamical entity requires a Lagrangian density $\mathcal{L}(A_\mu, \partial_\nu A_\mu)$ and the field Euler–Lagrange equations.
- Three experimental principles (Lorentz invariance, gauge invariance, superposition) uniquely fix the free-field Lagrangian to $\mathcal{L}_{\text{field}} = -F_{\mu\nu}F^{\mu\nu}/(16\pi) = (\mathbf{E}^2 - \mathbf{B}^2)/(8\pi)$.

9 Recitation 04 – Four-current and Fields of a Moving Charge

Date: 05/05/2026

Overview

This recitation uses Lorentz covariance as a practical calculation tool. The first example asks what happens to a neutral electric dipole distribution when it is observed from a frame in which it moves. The answer is physically important: a purely electrostatic configuration in its rest frame carries a current in the lab frame, and therefore acquires a magnetic dipole moment. The second example derives the electric and magnetic fields of a uniformly moving point charge by solving the easy problem in the charge rest frame and then transforming the fields back to the lab frame.

The central lesson is that electric and magnetic phenomena are not separate objects. Charge density and current density form a four-vector,

$$J^\mu = (c\rho, \mathbf{J}), \quad (234)$$

while $\mathbf{E}$ and $\mathbf{B}$ are components of the electromagnetic field tensor. A Lorentz boost reshuffles the components of these objects. What one observer calls a charge density, another observer may call both charge and current; what one observer calls a purely electric field, another observer may call both electric and magnetic fields.

9.1 Moving Electric Dipole Distribution

Physical Motivation

Imagine a localized distribution of positive and negative charge. In its own rest frame it is static: charges sit in place and there is no current. If the total charge is zero but the electric dipole moment is not zero, the object looks like an ordinary neutral electric dipole. Now observe the same object from a lab frame in which it moves. Since the charge density is moving, it must create a current density. A current distribution generally has a magnetic dipole moment, so special relativity predicts that a moving electric dipole looks partly magnetic.

Rest-frame data

Let O' be the rest frame of the charge distribution. The charge density is $\rho'(\mathbf{r}', t')$, and the current density vanishes:

$$\mathbf{J}'(\mathbf{r}', t') = 0. \quad (235)$$

The total charge is assumed to vanish,

$$q' = \int \rho'(\mathbf{r}', t') d^3r' = 0, \quad (236)$$

but the electric dipole moment is nonzero,

$$\mathbf{p}' = \int \mathbf{r}' \rho'(\mathbf{r}', t') d^3r' \neq 0. \quad (237)$$

The lab frame O sees O' moving with velocity

$$\mathbf{v} = v\hat{\mathbf{x}} = \beta c\hat{\mathbf{x}}, \quad \beta \equiv \frac{v}{c}, \quad \gamma \equiv \frac{1}{\sqrt{1 - \beta^2}}. \quad (238)$$

Transforming the four-current

In the rest frame,

$$J'^{\mu} = \begin{pmatrix} c\rho' \\ 0 \\ 0 \\ 0 \end{pmatrix}. \quad (239)$$

Since J^{μ} is a four-vector, a boost along the x -direction gives

$$J^{\mu} = \begin{pmatrix} \gamma & \beta\gamma & 0 & 0 \\ \beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} c\rho' \\ 0 \\ 0 \\ 0 \end{pmatrix} = \gamma \begin{pmatrix} c\rho' \\ c\beta\rho' \\ 0 \\ 0 \end{pmatrix}. \quad (240)$$

Identifying $J^{\mu} = (c\rho, J_x, J_y, J_z)$, we obtain

$$\boxed{\rho(\mathbf{r}, t) = \gamma\rho'(\mathbf{r}', t'), \quad \mathbf{J}(\mathbf{r}, t) = \gamma\rho'(\mathbf{r}', t') \mathbf{v}.} \quad (241)$$

Physical significance: The current is not an added physical assumption. It appears automatically because a boosted charge density is a moving charge density.

The result in (241) is not yet a complete lab-frame answer unless we also express $(\mathbf{r}', t')$ in terms of $(\mathbf{r}, t)$. For a boost along x ,

$$ct = \gamma ct' + \gamma\beta x', \quad (242)$$

$$x = \gamma\beta ct' + \gamma x', \quad (243)$$

$$y = y', \quad (244)$$

$$z = z'. \quad (245)$$

Equivalently, the inverse relation used inside ρ' is

$$ct' = \gamma(ct - \beta x), \quad (246)$$

$$x' = \gamma(x - \beta ct), \quad (247)$$

$$y' = y, \quad (248)$$

$$z' = z. \quad (249)$$

Thus the full statement is: evaluate the rest-frame density at the spacetime point corresponding to the lab event, and then multiply by the Lorentz factors shown in (241).

Magnetic dipole moment generated by motion

In Gaussian-CGS units the magnetic dipole moment of a localized current distribution is

$$\mathbf{m} = \frac{1}{2c} \int \mathbf{r} \times \mathbf{J}(\mathbf{r}, t) d^3r. \quad (250)$$

Substituting (241),

$$\mathbf{m} = \frac{1}{2c} \int \mathbf{r} \times (\gamma\rho'(\mathbf{r}', t') \mathbf{v}) d^3r. \quad (251)$$

This expression is already first order in v/c , because the current itself is proportional to $\mathbf{v}$. Therefore, to find the leading nonzero term, all other factors may be taken at zeroth order in β :

$$\gamma \simeq 1, \quad d^3r \simeq d^3r', \quad \mathbf{r} \simeq \mathbf{r}' + \mathbf{a}(t), \quad (252)$$

where $\mathbf{a}(t)$ is a time-dependent spatial shift between the two origins. The shift does not contribute because the total charge is zero:

$$\int \mathbf{a}(t) \rho'(\mathbf{r}', t') d^3 r' = \mathbf{a}(t) \int \rho'(\mathbf{r}', t') d^3 r' = 0. \quad (253)$$

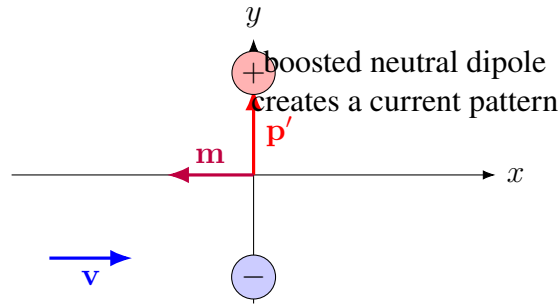
Hence

$$\begin{aligned} \mathbf{m} &\simeq \frac{1}{2c} \int \mathbf{r}' \times (\rho'(\mathbf{r}', t') \mathbf{v}) d^3 r' \\ &= \frac{1}{2c} \left(\int \mathbf{r}' \rho'(\mathbf{r}', t') d^3 r' \right) \times \mathbf{v}. \end{aligned} \quad (254)$$

Using the definition of $\mathbf{p}'$,

$$\boxed{\mathbf{m} \simeq \frac{\mathbf{p}' \times \mathbf{v}}{2c} \quad (v \ll c).} \quad (255)$$

Physical significance: A purely electric dipole in its own rest frame is not purely electric in every frame. The factor $1/(2c)$ is the Gaussian-CGS normalization of magnetic moment, and the cross product says that the induced magnetic moment is perpendicular both to the dipole direction and to the motion.



9.2 Field Transformations

Physical Motivation

The easiest way to find the field of a uniformly moving charge is not to solve Maxwell's equations directly in the lab frame. Instead, go to the frame where the charge is at rest. There the field is just Coulomb's field and $\mathbf{B}' = 0$. Then use the Lorentz transformation laws for the fields. This is the same strategy used for four-current: solve in the simple frame, then transform the covariant object.

For a boost with dimensionless velocity vector $\beta = \mathbf{v}/c$, the field components parallel and perpendicular to the boost direction satisfy

$$\mathbf{E}_{\parallel} = \mathbf{E}'_{\parallel}, \quad (256)$$

$$\mathbf{E}_{\perp} = \gamma (\mathbf{E}'_{\perp} + \beta \times \mathbf{B}'), \quad (257)$$

$$\mathbf{B}_{\parallel} = \mathbf{B}'_{\parallel}, \quad (258)$$

$$\mathbf{B}_{\perp} = \gamma (\mathbf{B}'_{\perp} - \beta \times \mathbf{E}'). \quad (259)$$

The sign convention here corresponds to transforming from O' to O with the relative velocity specified above. In any concrete problem, the safest method is to define the two frames carefully and then check one simple direction by the right-hand rule.

9.3 Fields of a Uniformly Moving Point Charge

Physical Motivation

A point charge at rest produces a radial electric field and no magnetic field. If the charge moves uniformly, observers in the lab frame must see a magnetic field, because a moving charge is a current. Less obvious is the exact shape of the electric field. The relativistic result is not an isotropic Coulomb field centered on where the charge appears optically; instead the electric field is compressed into directions transverse to the motion and points toward the charge's instantaneous position.

Geometry and choice of frames

Let O' be the rest frame of a point charge q , with the charge at the origin. Let O be the lab frame in which the charge moves with velocity

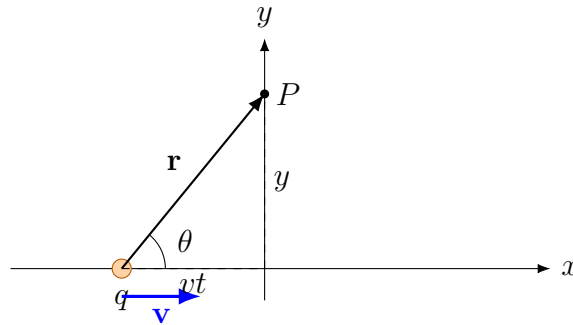
$$\mathbf{v} = v\hat{\mathbf{x}}. \quad (260)$$

To keep the algebra transparent, evaluate the lab field at the event

$$(ct, x, y, z) = (ct, 0, y, 0). \quad (261)$$

This is not as restrictive as it first appears. Rotational symmetry around the x -axis lets us set $z = 0$, and a uniformly moving charge passes every longitudinal separation from the observer at some time. The Lorentz transformation to the charge rest frame gives

$$(ct', x', y', z') = (\gamma ct, -\gamma vt, y, 0). \quad (262)$$



Fields in the charge rest frame

In O' the charge is static, so

$$\mathbf{B}' = 0, \quad (263)$$

and the electric field is Coulomb's field:

$$E'_x = \frac{qx'}{(x'^2 + y'^2 + z'^2)^{3/2}}, \quad (264)$$

$$E'_y = \frac{qy'}{(x'^2 + y'^2 + z'^2)^{3/2}}, \quad (265)$$

$$E'_z = 0. \quad (266)$$

At the event (262),

$$E'_x = -\frac{q\gamma vt}{[(\gamma vt)^2 + y^2]^{3/2}}, \quad (267)$$

$$E'_y = \frac{qy}{[(\gamma vt)^2 + y^2]^{3/2}}, \quad (268)$$

$$E'_z = 0. \quad (269)$$

Transforming to the lab frame

The x -direction is parallel to the boost, so $E_x = E'_x$ and $B_x = B'_x = 0$. The transverse electric field gains a factor of γ , because $\mathbf{B}' = 0$:

$$E_x = -\frac{q\gamma vt}{[(\gamma vt)^2 + y^2]^{3/2}}, \quad (270)$$

$$E_y = \frac{q\gamma y}{[(\gamma vt)^2 + y^2]^{3/2}}, \quad (271)$$

$$E_z = 0. \quad (272)$$

For the magnetic field, the only nonzero component is along $\hat{\mathbf{z}}$:

$$B_x = 0, \quad (273)$$

$$B_y = 0, \quad (274)$$

$$B_z = \frac{q\gamma\beta y}{[(\gamma vt)^2 + y^2]^{3/2}}. \quad (275)$$

Therefore

$$\boxed{\mathbf{E}(0, y, 0, t) = \frac{q\gamma}{[(\gamma vt)^2 + y^2]^{3/2}} (-vt \hat{\mathbf{x}} + y \hat{\mathbf{y}}),} \quad (276)$$

$$\boxed{\mathbf{B}(0, y, 0, t) = \frac{q\gamma\beta y}{[(\gamma vt)^2 + y^2]^{3/2}} \hat{\mathbf{z}}.} \quad (277)$$

Physical significance: The magnetic field is azimuthal around the direction of motion. It is exactly what one expects from a moving charge, but with the same relativistic angular weighting as the electric field.

Rewriting the result in spherical geometry

The formulas above are correct but still tied to the special coordinates $(0, y, 0)$. Define $\mathbf{r}$ as the vector from the instantaneous position of the charge to the observation point, and let θ be the angle between $\mathbf{v}$ and $\mathbf{r}$. In the lab frame,

$$y = r \sin \theta, \quad (278)$$

$$vt = -r \cos \theta. \quad (279)$$

The denominator becomes

$$\begin{aligned} (\gamma vt)^2 + y^2 &= r^2 (\gamma^2 \cos^2 \theta + \sin^2 \theta) \\ &= r^2 [1 + (\gamma^2 - 1) \cos^2 \theta]. \end{aligned} \quad (280)$$

Since $(-vt \hat{\mathbf{x}} + y \hat{\mathbf{y}})$ points along $\hat{\mathbf{r}}$ with magnitude r , the electric field becomes

$$\boxed{\mathbf{E}(\mathbf{r}, t) = \frac{q\gamma}{r^2 [1 + (\gamma^2 - 1) \cos^2 \theta]^{3/2}} \hat{\mathbf{r}}.} \quad (281)$$

The magnetic field can be written as

$$\boxed{\mathbf{B}(\mathbf{r}, t) = \boldsymbol{\beta} \times \mathbf{E}(\mathbf{r}, t).} \quad (282)$$

Physical significance: The electric field points radially away from the instantaneous position of the charge, while the magnetic field circles the line of motion. The compact relation $\mathbf{B} = \boldsymbol{\beta} \times \mathbf{E}$ is the field-level version of the statement that magnetism appears when electric sources move.

It is often useful to rewrite the angular factor using

$$1 + (\gamma^2 - 1) \cos^2 \theta = \frac{1 - \beta^2 \sin^2 \theta}{1 - \beta^2}. \quad (283)$$

Since $\gamma^{-2} = 1 - \beta^2$, (281) is equivalent to the standard form

$$\boxed{\mathbf{E}(\mathbf{r}, t) = \frac{q(1 - \beta^2)}{r^2 (1 - \beta^2 \sin^2 \theta)^{3/2}} \hat{\mathbf{r}}, \quad \mathbf{B} = \boldsymbol{\beta} \times \mathbf{E}.} \quad (284)$$

Physical significance: For $\beta \rightarrow 0$, the factor tends to 1 and the usual Coulomb field is recovered. For relativistic motion, the field is enhanced near $\theta = \pi/2$ and suppressed near $\theta = 0, \pi$, producing a “pancake” angular pattern transverse to the motion.

Limiting cases and intuition

At closest approach, $\theta = \pi/2$, so $\sin \theta = 1$. Equation (284) gives

$$|\mathbf{E}|_{\theta=\pi/2} = \frac{q\gamma}{r^2}. \quad (285)$$

In front of or behind the charge, $\theta = 0$ or $\theta = \pi$, so

$$|\mathbf{E}|_{\theta=0,\pi} = \frac{q}{\gamma^2 r^2}. \quad (286)$$

Thus the field is stronger sideways and weaker along the direction of motion. The total flux is still consistent with Gauss’s law; the field lines have been redistributed angularly by the Lorentz boost.

A subtle point is the direction of $\mathbf{E}$. The field points toward the charge’s instantaneous position, not toward the earlier optical position from which light emitted by the charge would now arrive. This does not violate causality. For uniform motion, the retardation effects and the Lorentz transformation combine so that the final field direction is radial from the instantaneous position. Acceleration would change this story and produce radiation fields.

9.4 Handout-Only Example: A Frame Where the Fields Are Parallel

Physical Motivation

The handout includes one additional example that was not developed in the recorded recitation. It illustrates how field transformations can be used in reverse: instead of finding the lab fields from simple rest-frame fields, we search for a moving frame with a simpler relation between the fields. Here the goal is to find a frame in which $\mathbf{E}'$ and $\mathbf{B}'$ are parallel.

Suppose in frame O the fields obey

$$\mathbf{E} \cdot \mathbf{B} \neq 0. \quad (287)$$

Choose the boost direction perpendicular to both fields, i.e. along $\mathbf{E} \times \mathbf{B}$. Then the fields have only transverse components with respect to the boost direction. Let that boost direction be $\hat{\mathbf{x}}$. The transformed fields are parallel when

$$\mathbf{E}' \times \mathbf{B}' = 0. \quad (288)$$

Using the transformation laws, this condition reduces to

$$(1 + \beta^2)(\mathbf{E} \times \mathbf{B})_x = \beta(\mathbf{E}^2 + \mathbf{B}^2). \quad (289)$$

Vectorially,

$$\frac{\boldsymbol{\beta}}{1 + \beta^2} = \frac{\mathbf{E} \times \mathbf{B}}{\mathbf{E}^2 + \mathbf{B}^2}. \quad (290)$$

Solving the quadratic equation and choosing the subluminal branch gives

$$\beta = \frac{\mathbf{E}^2 + \mathbf{B}^2 - \sqrt{(\mathbf{E}^2 + \mathbf{B}^2)^2 - 4|\mathbf{E} \times \mathbf{B}|^2}}{2|\mathbf{E} \times \mathbf{B}|^2} (\mathbf{E} \times \mathbf{B}). \quad (291)$$

Physical significance: The direction $\mathbf{E} \times \mathbf{B}$ is the natural direction of electromagnetic energy flow. Boosting along that direction can remove the transverse mismatch between the fields, leaving a frame where the electric and magnetic vectors point along the same line.

10 Lecture 09 – Field Variation and Four-Current

Date: 05/05/2026

Overview

The previous lecture constructed the electromagnetic field tensor

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad (292)$$

and argued that Lorentz invariance, gauge invariance, and the superposition principle almost uniquely determine the free-field action. This lecture begins the variational derivation of Maxwell's equations. The main idea is simple: the complete system has two kinds of degrees of freedom. Varying the particle worldlines gives the Lorentz force law; varying the electromagnetic field $A_\mu(x)$ gives the field equations.

The live lecture reached the following points:

- the total action for point charges plus the electromagnetic field;
- the first variation of the free-field action;
- the definition of charge density, current density, and four-current;
- the proof that the four-current obeys the covariant continuity equation.

The derivation of Maxwell's equations is therefore almost complete: one still has to vary the interaction term and combine it with the free-field variation. That combination will give the inhomogeneous Maxwell equations $\partial_\mu F^{\mu\nu} = (4\pi/c)J^\nu$.

10.1 The Total Action

Physical Motivation

A charged particle and the electromagnetic field are not separate mechanical problems. The particle moves in the field, and the field is sourced by the particle. The action must therefore contain three pieces: a free-particle piece, an interaction piece, and a free-field piece. Each term has a clear role: inertia, coupling, and field dynamics.

For N point charges, labelled by $n = 1, \dots, N$, the total action is

$$S = - \sum_{n=1}^N m_n c \int ds_n - \frac{1}{c} \sum_{n=1}^N e_n \int_{\Gamma_n} A_\mu dx_n^\mu - \frac{1}{16\pi c} \int F_{\mu\nu} F^{\mu\nu} d^4x. \quad (293)$$

Physical significance: The first term says that a free relativistic particle extremizes its proper time. The second term couples the particle worldline Γ_n to the four-potential. The third term gives the electromagnetic field its own dynamics.

Here $x^\mu = (ct, \mathbf{x})$, $d^4x = dx^0 d^3x = c dt d^3x$, and the metric convention is $(+, -, -, -)$. The factor of $1/c$ in the field action is the standard Gaussian-units normalization when the four-volume element is written as d^4x .

Why the field action is quadratic

The free-field part is not guessed arbitrarily. It is forced by three physical requirements:

1. **Lorentz invariance:** the action must be a scalar.
2. **Gauge invariance:** the action may depend on A_μ only through gauge-invariant combinations, especially $F_{\mu\nu}$.
3. **Superposition:** the field equations must be linear, so the action must be quadratic in the field.

The derivative matrix $\partial_\mu A_\nu$ can be decomposed into a symmetric and an antisymmetric part:

$$\partial_\mu A_\nu = \frac{1}{2} (\partial_\mu A_\nu + \partial_\nu A_\mu) + \frac{1}{2} (\partial_\mu A_\nu - \partial_\nu A_\mu). \quad (294)$$

The antisymmetric part is precisely $F_{\mu\nu}/2$. Under a gauge transformation,

$$A_\mu \longrightarrow A_\mu + \partial_\mu f, \quad (295)$$

the antisymmetric combination is unchanged:

$$\begin{aligned} F_{\mu\nu} &\longrightarrow \partial_\mu (A_\nu + \partial_\nu f) - \partial_\nu (A_\mu + \partial_\mu f) \\ &= F_{\mu\nu} + \partial_\mu \partial_\nu f - \partial_\nu \partial_\mu f = F_{\mu\nu}. \end{aligned} \quad (296)$$

Thus the allowed quadratic Lorentz scalar is

$$S_{\text{field}} = -\frac{1}{16\pi c} \int F_{\mu\nu} F^{\mu\nu} d^4x. \quad (297)$$

Physical significance: Gauge symmetry throws away the symmetric part of $\partial_\mu A_\nu$ and keeps only the curl-like object $F_{\mu\nu}$. This is the covariant version of the familiar fact that physical electromagnetic fields are built from derivatives of potentials.

Why d^4x is Lorentz invariant

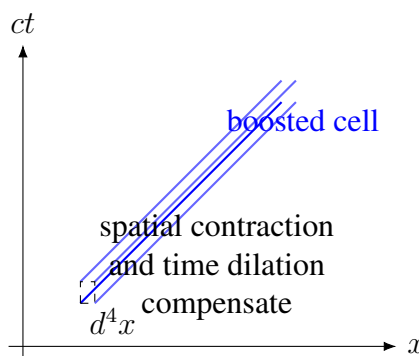
The four-volume element behaves like an ordinary volume element under a change of variables:

$$d^4x' = |\det \Lambda| d^4x. \quad (298)$$

For a proper Lorentz transformation, $\det \Lambda = 1$, so

$$d^4x' = d^4x. \quad (299)$$

Physical significance: A Lorentz boost squeezes spatial volume and stretches time in just the compensating way needed to leave the four-dimensional spacetime volume unchanged.



10.2 Varying the Free Electromagnetic Field

Physical Motivation

In mechanics, one varies the path $q(t) \rightarrow q(t) + \delta q(t)$ and demands that the first-order change in the action vanish. For a field, the same idea is applied at every spacetime point:

$$A_\mu(x) \longrightarrow A_\mu(x) + \delta A_\mu(x). \quad (300)$$

The variation δA_μ is assumed to vanish on the boundary of the spacetime integration region, just as $\delta q(t_1) = \delta q(t_2) = 0$ in ordinary mechanics.

First variation of $F_{\mu\nu}$

Since $F_{\mu\nu}$ is linear in A_μ ,

$$\begin{aligned} F_{\mu\nu}[A + \delta A] &= \partial_\mu(A_\nu + \delta A_\nu) - \partial_\nu(A_\mu + \delta A_\mu) \\ &= F_{\mu\nu}[A] + \partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu. \end{aligned} \quad (301)$$

Therefore

$$\boxed{\delta F_{\mu\nu} = \partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu.} \quad (302)$$

Physical significance: A small change in the potential changes the field tensor only through derivatives of that change. This is why integration by parts will move derivatives from δA_μ onto $F^{\mu\nu}$ and produce field equations.

First variation of the free-field action

Starting from

$$S_{\text{field}} = -\frac{1}{16\pi c} \int F_{\mu\nu} F^{\mu\nu} d^4x, \quad (303)$$

the first-order variation is

$$\delta S_{\text{field}} = -\frac{1}{16\pi c} \int [\delta F_{\mu\nu} F^{\mu\nu} + F_{\mu\nu} \delta F^{\mu\nu}] d^4x. \quad (304)$$

The two terms are equal. Indeed,

$$F_{\mu\nu} \delta F^{\mu\nu} = F_{\mu\nu} \eta^{\mu\alpha} \eta^{\nu\beta} \delta F_{\alpha\beta} = F^{\alpha\beta} \delta F_{\alpha\beta}. \quad (305)$$

Renaming the dummy indices α, β back to μ, ν gives

$$F_{\mu\nu} \delta F^{\mu\nu} = F^{\mu\nu} \delta F_{\mu\nu}. \quad (306)$$

Thus

$$\delta S_{\text{field}} = -\frac{1}{8\pi c} \int F^{\mu\nu} \delta F_{\mu\nu} d^4x. \quad (307)$$

Substituting (302),

$$\delta S_{\text{field}} = -\frac{1}{8\pi c} \int F^{\mu\nu} (\partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu) d^4x. \quad (308)$$

The two terms again combine. In the second term, exchange the dummy indices $\mu \leftrightarrow \nu$:

$$-F^{\mu\nu} \partial_\nu \delta A_\mu \longrightarrow -F^{\nu\mu} \partial_\mu \delta A_\nu = +F^{\mu\nu} \partial_\mu \delta A_\nu, \quad (309)$$

where $F^{\nu\mu} = -F^{\mu\nu}$. Therefore

$$\delta S_{\text{field}} = -\frac{1}{4\pi c} \int F^{\mu\nu} \partial_\mu \delta A_\nu d^4x. \quad (310)$$

Integration by parts in spacetime

Use the product rule

$$F^{\mu\nu} \partial_\mu \delta A_\nu = \partial_\mu (F^{\mu\nu} \delta A_\nu) - (\partial_\mu F^{\mu\nu}) \delta A_\nu. \quad (311)$$

Then

$$\delta S_{\text{field}} = -\frac{1}{4\pi c} \int \partial_\mu (F^{\mu\nu} \delta A_\nu) d^4x + \frac{1}{4\pi c} \int (\partial_\mu F^{\mu\nu}) \delta A_\nu d^4x. \quad (312)$$

The first term is a four-dimensional boundary term. Since $\delta A_\nu = 0$ on the boundary, it vanishes. Hence

$$\delta S_{\text{field}} = \frac{1}{4\pi c} \int (\partial_\mu F^{\mu\nu}) \delta A_\nu d^4x. \quad (313)$$

Physical significance: The quantity $\partial_\mu F^{\mu\nu}$ is the field-theory analogue of “force equals derivative of the action”. Once the source term is included, it will be set equal to the charge-current source.

10.3 Charge Density and Current Density for Point Charges

Physical Motivation

To vary the interaction term with respect to the field, we want to express the point particles as a spacetime source distribution. A point charge is not a smooth lump of charge; all of its charge is concentrated at one position. Dirac delta functions are the correct language for that concentration.

For point charges e_n moving along spatial trajectories $\mathbf{x}_n(t)$, the charge density is

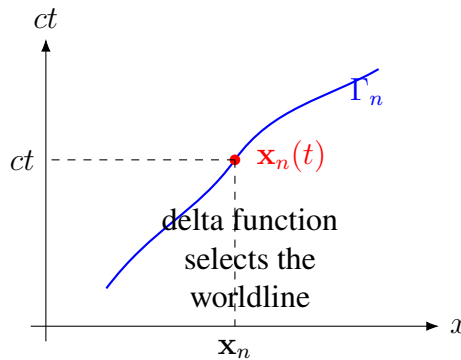
$$\rho(t, \mathbf{x}) = \sum_{n=1}^N e_n \delta^{(3)}(\mathbf{x} - \mathbf{x}_n(t)). \quad (314)$$

Physical significance: Integrating ρ over a region returns the total charge of the particles inside that region. If a particle is outside the region, its delta function contributes zero.

The ordinary current density is charge density times velocity:

$$\mathbf{J}(t, \mathbf{x}) = \sum_{n=1}^N e_n \dot{\mathbf{x}}_n(t) \delta^{(3)}(\mathbf{x} - \mathbf{x}_n(t)). \quad (315)$$

Physical significance: $\mathbf{J}$ measures how much charge crosses a unit area per unit time. A static charge contributes to ρ but not to $\mathbf{J}$.



10.4 The Four-Current

Physical Motivation

The electromagnetic field is written covariantly using four-vectors and tensors. The source must also be written covariantly. The natural object is the four-current: its time component is charge density, and its spatial components are ordinary current density.

Define

$$J^\mu = (c\rho, \mathbf{J}). \quad (316)$$

Physical significance: Charge density and current density are not separate relativistic objects. They are frame-dependent components of one four-vector, just as energy and momentum are components of one four-momentum.

Using (314) and (315), this can be written componentwise as

$$J^\mu(x) = \sum_{n=1}^N e_n \frac{dx_n^\mu}{dt} \delta^{(3)}(\mathbf{x} - \mathbf{x}_n(t)). \quad (317)$$

Indeed, for $\mu = 0$,

$$J^0 = \sum_{n=1}^N e_n \frac{d(ct)}{dt} \delta^{(3)}(\mathbf{x} - \mathbf{x}_n(t)) = c\rho, \quad (318)$$

and for $\mu = i$,

$$J^i = \sum_{n=1}^N e_n \frac{dx_n^i}{dt} \delta^{(3)}(\mathbf{x} - \mathbf{x}_n(t)) = J_{\text{spatial}}^i. \quad (319)$$

Why this is really a four-vector

At first glance (317) is not manifestly covariant because it contains dt and a three-dimensional delta function. The more covariant way to see the structure is to insert a delta function in time as well, so that the source is supported on the entire worldline:

$$J^\mu(x) = c \sum_{n=1}^N e_n \int d\tau_n u_n^\mu(\tau_n) \delta^{(4)}(x - x_n(\tau_n)), \quad (320)$$

where

$$u_n^\mu = \frac{dx_n^\mu}{d\tau_n} \quad (321)$$

is the four-velocity of particle n . The four-dimensional delta function is defined by

$$\int d^4x \delta^{(4)}(x - a) = 1. \quad (322)$$

Since d^4x is Lorentz invariant, $\delta^{(4)}(x - a)$ transforms as a scalar under proper Lorentz transformations. The charge e_n is a scalar, $d\tau_n$ is a scalar, and u_n^μ is a four-vector. Therefore J^μ is a four-vector.

$$J^\mu \text{ is a Lorentz four-vector.} \quad (323)$$

Physical significance: Different observers disagree on how much source is called “charge density” and how much is called “current”, but they agree that both are components of the same spacetime current.

10.5 Charge Conservation as a Continuity Equation

Physical Motivation

Charge cannot disappear from one place without flowing somewhere else. Locally, this means that a decrease of charge density inside a small volume must be balanced by current flowing through its boundary. The continuity equation is the differential statement of this local conservation law.

Start with the ordinary current density:

$$J^i(t, \mathbf{x}) = \sum_{n=1}^N e_n \dot{x}_n^i(t) \delta^{(3)}(\mathbf{x} - \mathbf{x}_n(t)). \quad (324)$$

Its spatial divergence is

$$\partial_i J^i = \sum_{n=1}^N e_n \dot{x}_n^i(t) \partial_i \delta^{(3)}(\mathbf{x} - \mathbf{x}_n(t)). \quad (325)$$

Use the delta-function identity

$$\partial_i \delta^{(3)}(\mathbf{x} - \mathbf{x}_n(t)) = -\frac{\partial}{\partial x_n^i} \delta^{(3)}(\mathbf{x} - \mathbf{x}_n(t)). \quad (326)$$

Then

$$\begin{aligned} \partial_i J^i &= -\sum_{n=1}^N e_n \dot{x}_n^i(t) \frac{\partial}{\partial x_n^i} \delta^{(3)}(\mathbf{x} - \mathbf{x}_n(t)) \\ &= -\frac{\partial}{\partial t} \sum_{n=1}^N e_n \delta^{(3)}(\mathbf{x} - \mathbf{x}_n(t)) \\ &= -\frac{\partial \rho}{\partial t}. \end{aligned} \quad (327)$$

Thus

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0.} \quad (328)$$

Physical significance: Charge conservation is local. If ρ changes inside a region, the change is accounted for by current crossing the boundary, not by charge being created or destroyed.

Covariant form

Since $x^0 = ct$ and $J^0 = c\rho$,

$$\partial_0 J^0 = \frac{\partial}{\partial x^0}(c\rho) = \frac{1}{c} \frac{\partial}{\partial t}(c\rho) = \frac{\partial \rho}{\partial t}. \quad (329)$$

Therefore the ordinary continuity equation (328) becomes

$$\boxed{\partial_\mu J^\mu = 0.} \quad (330)$$

Physical significance: Charge conservation is Lorentz covariant. It is not a statement true only in one preferred frame; it is the vanishing four-divergence of the source four-vector.

10.6 How This Leads to Maxwell's Equations

Physical Motivation

The free-field variation knows how the electromagnetic field wants to evolve in empty space. The interaction term tells the field where its sources are. Maxwell's equations are obtained by demanding that the total variation vanish for every allowed δA_μ .

Using the four-current, the interaction term can be rewritten as a spacetime integral:

$$S_{\text{int}} = -\frac{1}{c^2} \int J^\mu A_\mu d^4x. \quad (331)$$

Physical significance: The compact contraction $J^\mu A_\mu$ is the covariant version of “sources couple to potentials”. The charge density couples to the scalar potential, and the current density couples to the vector potential.

Varying this term with respect to A_μ gives

$$\delta S_{\text{int}} = -\frac{1}{c^2} \int J^\nu \delta A_\nu d^4x. \quad (332)$$

Combining with (313) and imposing $\delta S_{\text{field}} + \delta S_{\text{int}} = 0$ for arbitrary δA_ν gives

$$\partial_\mu F^{\mu\nu} = \frac{4\pi}{c} J^\nu. \quad (333)$$

Physical significance: This is the covariant form of the inhomogeneous Maxwell equations. The $\nu = 0$ component is Gauss's law, and the spatial components are the Ampere–Maxwell law.

Although the live lecture stopped just before completing this last combination, all ingredients were prepared: the field variation (313), the source J^μ , and the conservation law $\partial_\mu J^\mu = 0$.

Summary

- The complete action contains free-particle, interaction, and free-field terms.
- Gauge invariance selects the antisymmetric field tensor $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.
- The first variation of the free-field action is $\delta S_{\text{field}} = (4\pi c)^{-1} \int (\partial_\mu F^{\mu\nu}) \delta A_\nu d^4x$.
- Point charges are represented by delta-function charge and current densities.
- The four-current $J^\mu = (c\rho, \mathbf{J})$ is a Lorentz four-vector.
- Local charge conservation is the covariant equation $\partial_\mu J^\mu = 0$.
- Combining the field variation with the interaction variation gives $\partial_\mu F^{\mu\nu} = (4\pi/c)J^\nu$, the inhomogeneous Maxwell equations.

11 Lecture 10 – Maxwell's Equations, Discrete Symmetries, and Lorentz Invariants of the EM Field

Date: 10/05/2026

Overview

The previous lecture prepared all ingredients needed to derive Maxwell's equations from the action principle. The total action

$$S = - \sum_n m_n c \int ds_n - \frac{1}{c} \sum_n e_n \int A_\mu dx_n^\mu - \frac{1}{16\pi c} \int F_{\mu\nu} F^{\mu\nu} d^4x \quad (334)$$

was varied with respect to the field A_μ , giving $\delta S_{\text{field}} = (4\pi c)^{-1} \int (\partial_\mu F^{\mu\nu}) \delta A_\nu d^4x$. The four-current $J^\mu = (c\rho, \mathbf{J})$ was introduced and shown to be a four-vector obeying $\partial_\mu J^\mu = 0$.

This lecture:

- completes the variational derivation of the inhomogeneous Maxwell equations;
- shows how the covariant form implies the continuity equation automatically;
- unpacks the covariant equation into the familiar 3-vector Maxwell equations;
- studies the behaviour of $\mathbf{E}$ and $\mathbf{B}$ under time-reversal ($t \rightarrow -t$) and spatial inversion ($\mathbf{r} \rightarrow -\mathbf{r}$), introducing polar and axial vectors;
- constructs the two independent Lorentz scalars built from $F_{\mu\nu}$ and extracts physical consequences about which configurations can be related by a boost.

11.1 Completing the Derivation of Maxwell's Equations

Physical Motivation

The action has two types of degrees of freedom: particle worldlines and the field A_μ . We already varied with respect to the particles and obtained the covariant Lorentz force. Now we impose stationarity under variations of the field. Because S_{particle} does not depend on A_μ , only the last two terms contribute.

Variation of the interaction term

Writing the interaction in terms of the four-current (derived in Lecture 09),

$$S_{\text{int}} = -\frac{1}{c^2} \int J^\mu A_\mu d^4x, \quad (335)$$

a shift $A_\mu \rightarrow A_\mu + \delta A_\mu$ (keeping particle trajectories fixed, so J^μ is unchanged) gives

$$\delta S_{\text{int}} = -\frac{1}{c^2} \int J^\nu \delta A_\nu d^4x. \quad (336)$$

Note: J^μ does not vary because it depends only on the particle degrees of freedom, which are held fixed.

Combining the two variations

Adding (336) to the field variation derived in Lecture 09,

$$\delta S_{\text{field}} = \frac{1}{4\pi c} \int (\partial_\mu F^{\mu\nu}) \delta A_\nu d^4x, \quad (337)$$

the total variation is

$$\delta S = \int \left[\frac{1}{4\pi c} \partial_\mu F^{\mu\nu} - \frac{1}{c^2} J^\nu \right] \delta A_\nu d^4x. \quad (338)$$

By the principle of least action, $\delta S = 0$ for every admissible δA_ν . Since δA_ν is arbitrary, the bracket must vanish at every spacetime point:

$$\boxed{\partial_\mu F^{\mu\nu} = \frac{4\pi}{c} J^\nu.} \quad (339)$$

Physical significance: These are the *inhomogeneous Maxwell equations* in compact covariant form. The left-hand side is determined entirely by the geometry of A_μ ; the right-hand side is the four-current source. The equation is manifestly Lorentz covariant: both sides are four-vectors.

The full set of Maxwell equations

Together with the particle equation of motion (from varying the worldlines),

$$\boxed{m \frac{du^\mu}{d\tau} = \frac{e}{c} F^\mu{}_\nu u^\nu,} \quad (340)$$

equations (339) and (340) encode *all of classical electrodynamics* in just two compact lines. This economy is possible because we understood the system's symmetries well enough to write a natural action from first principles.

11.2 Continuity Equation as an Automatic Consequence

Physical Motivation

Charge conservation says that charge cannot be created or destroyed. In a field theory derived from an action, conservation laws should emerge automatically, either from symmetries (Noether) or from algebraic identities. Maxwell's equations contain charge conservation as an identity.

Apply ∂_ν to both sides of (339):

$$\partial_\nu \partial_\mu F^{\mu\nu} = \frac{4\pi}{c} \partial_\nu J^\nu. \quad (341)$$

The left-hand side vanishes identically. The operator $\partial_\nu \partial_\mu$ is symmetric in $\mu \leftrightarrow \nu$, while $F^{\mu\nu}$ is antisymmetric. Contracting a symmetric operator with an antisymmetric tensor gives zero:

$$\partial_\nu \partial_\mu F^{\mu\nu} = -\partial_\nu \partial_\mu F^{\nu\mu} = -\partial_\mu \partial_\nu F^{\mu\nu} = 0. \quad (342)$$

(The second step uses $F^{\mu\nu} = -F^{\nu\mu}$; the third relabels dummy indices.) Therefore

$$\boxed{\partial_\mu J^\mu = 0.} \quad (343)$$

Physical significance: Local charge conservation is not an *additional* assumption; it follows from Maxwell's equations automatically, as a consequence of the antisymmetry of $F^{\mu\nu}$. This is an internal consistency check: the field equations can only be self-consistent if their source is conserved.

Standard 3-vector derivation (cross-check)

The same result follows from the familiar vector equations, providing a useful consistency check. Take the divergence of the Ampère–Maxwell law (351):

$$\nabla \cdot (\nabla \times \mathbf{B}) - \frac{1}{c} \frac{\partial}{\partial t} (\nabla \cdot \mathbf{E}) = \frac{4\pi}{c} \nabla \cdot J_{vec}. \quad (344)$$

The first term vanishes because the divergence of any curl is zero. Substituting Gauss's law (349), $\nabla \cdot \mathbf{E} = 4\pi\rho$:

$$0 - \frac{4\pi}{c} \frac{\partial \rho}{\partial t} = \frac{4\pi}{c} \nabla \cdot J_{vec}, \quad (345)$$

which rearranges to

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla \cdot J_{vec} = 0.} \quad (346)$$

Physical significance: Maxwell's equations are only mathematically self-consistent if charge is conserved. The displacement current term $\partial_t \mathbf{E}/c$ that Maxwell added to Ampère's law is precisely what ensures this consistency.

11.3 Explicit Component Form of Maxwell's Equations

Physical Motivation

The covariant equation (339) contains four equations (one for each value of ν). Let us unpack them. Recall (from Lecture 08) the explicit matrix of $F^{\mu\nu}$ in Gaussian CGS:

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & -B_z & B_y \\ E_y & B_z & 0 & -B_x \\ E_z & -B_y & B_x & 0 \end{pmatrix}, \quad (347)$$

so $F^{0i} = -E_i$ (and $F^{i0} = E_i$), and $F^{ij} = -\epsilon^{ijk} B_k$.

The $\nu = 0$ component: Gauss's law

$$\partial_\mu F^{\mu 0} = \partial_i F^{i0} = \partial_i E_i = \nabla \cdot \mathbf{E}, \quad (348)$$

where we used $F^{00} = 0$. The right-hand side is $(4\pi/c) J^0 = (4\pi/c)(c\rho) = 4\pi\rho$. Therefore

$$\boxed{\nabla \cdot \mathbf{E} = 4\pi\rho.} \quad (349)$$

Physical significance: This is Gauss's law. Electric field lines originate and terminate on charge.

The $\nu = 1$ component: Ampère–Maxwell law (x -direction)

$$\begin{aligned} \partial_\mu F^{\mu 1} &= \partial_0 F^{01} + \partial_2 F^{21} + \partial_3 F^{31} \\ &= -\frac{1}{c} \partial_t E_x + \partial_y B_z - \partial_z B_y \\ &= (\nabla \times \mathbf{B})_x - \frac{1}{c} \frac{\partial E_x}{\partial t}, \end{aligned} \quad (350)$$

where we computed: $F^{01} = -E_x$ so $\partial_0 F^{01} = -(1/c)\partial_t E_x$; $F^{21} = -F^{12} = B_z$ so $\partial_2 F^{21} = \partial_y B_z$; $F^{31} = -F^{13} = -B_y$ so $\partial_3 F^{31} = -\partial_z B_y$. The right-hand side is $(4\pi/c)J^1 = (4\pi/c)J_x$. Repeating for $\nu = 2, 3$ gives the full equation

$$\boxed{\nabla \times \mathbf{B} - \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} = \frac{4\pi}{c} \mathbf{J}.} \quad (351)$$

Physical significance: Currents and changing electric fields source the magnetic field. The term $(1/c)\partial_t \mathbf{E}$ is Maxwell's displacement current; without it, $\nabla \times \mathbf{B}$ would not be divergence-free, violating $\partial_\mu J^\mu = 0$.

The homogeneous equations: direct derivation from definitions

Physical Motivation

The fundamental object in our theory is the four-potential $A^\mu = (\phi, \mathbf{A})$. All physical fields are built from it. The two definitions

$$\mathbf{E} = -\nabla\phi - \frac{1}{c} \frac{\partial \mathbf{A}}{\partial t}, \quad (352)$$

$$\mathbf{B} = \nabla \times \mathbf{A}, \quad (353)$$

are *not* dynamical equations; they are kinematic identities that encode the choice to describe electromagnetism via a potential. The homogeneous Maxwell equations are then *automatic consequences of these definitions*, requiring no additional input from the action.

Faraday's law. Take the curl of both sides of (352):

$$\nabla \times \mathbf{E} = -\nabla \times (\nabla\phi) - \frac{1}{c} \frac{\partial}{\partial t} (\nabla \times \mathbf{A}). \quad (354)$$

The first term vanishes identically: the curl of any gradient is zero ($\nabla \times \nabla\phi = 0$). In the second term, partial derivatives with respect to space and time commute, and $\nabla \times \mathbf{A} = \mathbf{B}$ by definition (353). Therefore

$$\boxed{\nabla \times \mathbf{E} = -\frac{1}{c} \frac{\partial \mathbf{B}}{\partial t}.} \quad (355)$$

Physical significance: This is Faraday–Lenz's law. A time-varying magnetic field induces a curling electric field. It is a pure consequence of the potential structure, not of any dynamics.

No magnetic monopoles ($\nabla \cdot \mathbf{B} = 0$). Take the divergence of (353):

$$\nabla \cdot \mathbf{B} = \nabla \cdot (\nabla \times \mathbf{A}) = 0, \quad (356)$$

since the divergence of any curl vanishes identically. Therefore

$$\boxed{\nabla \cdot \mathbf{B} = 0.} \quad (357)$$

Physical significance: There are no magnetic charges (monopoles) in this theory. The result is not an experimental statement but a *definition*: we chose $\mathbf{B} = \nabla \times \mathbf{A}$, and for any vector field that is a curl, the divergence is zero.

Covariant statement (Bianchi identity). Both results are unified in the covariant identity

$$\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} = 0, \quad (358)$$

which holds for any $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. Taking $\lambda = 0$ recovers Faraday's law; contracting over the spatial indices recovers $\nabla \cdot \mathbf{B} = 0$.

The E–B asymmetry in the definitions

Before asking about monopoles, notice that the two definitions (352)–(353) are *not symmetric* in $\mathbf{E}$ and $\mathbf{B}$:

$$\begin{aligned}\mathbf{E} &= -\nabla\phi - \frac{1}{c} \frac{\partial \mathbf{A}}{\partial t} \quad (\text{gradient term} + \text{time derivative of } \mathbf{A}), \\ \mathbf{B} &= \nabla \times \mathbf{A} \quad (\text{curl of } \mathbf{A} \text{ only}).\end{aligned}$$

There is no symmetry between the electric and magnetic sectors at the level of the definitions, because the action contains only *one* four-potential A^μ and only *electric* charges. This asymmetry is a direct reflection of the experimental fact that isolated magnetic charges have never been observed.

Why $\nabla \cdot \mathbf{B} = 0$ is not a fundamental law of nature

The statement $\nabla \cdot \mathbf{B} = 0$ is a consequence of our choice to describe the field by the potential A^μ . If magnetic monopoles existed, the force law $\mathbf{F} = e(\mathbf{E} + \mathbf{v}/c \times \mathbf{B})$ would need an additional magnetic-charge term. We have no such term in our action because we have never observed magnetic monopoles in the laboratory.

If monopoles were discovered, one would introduce a *second* four-potential $A_{\text{mag}}^\mu = (\phi_M, \mathbf{A}_M)$, and the definitions would be extended to

$$\mathbf{E} = -\nabla\phi - \frac{1}{c} \dot{\mathbf{A}} + \nabla \times \mathbf{A}_M, \quad (359)$$

$$\mathbf{B} = \nabla \times \mathbf{A} - \nabla\phi_M - \frac{1}{c} \dot{\mathbf{A}}_M, \quad (360)$$

making the electric and magnetic sectors fully symmetric: $\mathbf{E}$ acquires a curl of $\mathbf{A}_M$ and $\mathbf{B}$ acquires the same gradient + time-derivative structure as $\mathbf{E}$. The equation $\nabla \cdot \mathbf{B} = 0$ would be replaced by

$$\nabla \cdot \mathbf{B} = 4\pi\rho_M. \quad (361)$$

Why we do not simply add monopoles speculatively

Introducing a monopole potential immediately predicts a new magnetic force. If that force were comparable in strength to the electric force, it would have been detected long ago. Any consistent monopole theory must therefore contain a new coupling constant that suppresses the monopole interaction to below current experimental sensitivity.

An analogy: general relativity predicts gravitational waves, but they are extraordinarily weak because Newton's constant G is very small. If one had never detected gravity directly, one would still have to include G as a free parameter that the theory cannot fix on symmetry grounds alone. Similarly, a monopole theory would require a new coupling constant to explain the absence of observed magnetic-charge effects.

Physical significance: The framework of action + symmetry + potential does not forbid monopoles on logical grounds. Their absence is an experimental input, encoded in the choice of which four-potential we include in the action.

11.4 The Methodology of Analytical Field Theory

Physical Motivation

The derivation of Maxwell's equations from the action is not just a mathematical trick. It reflects a deep methodological principle that underlies all of modern theoretical physics. The professor describes it as *analytical mechanics of field theory*: the same philosophy that Lagrange and Hamilton applied to mechanics, now applied to fields.

Analytical mechanics as the prototype

In Newtonian mechanics one writes forces and integrates $F = ma$. Analytical mechanics (Lagrangian/Hamiltonian) produces *exactly the same equations* but arrives at them from two ingredients:

1. **Symmetry principle:** the action must be invariant under the symmetries of the system (Galilean invariance for non-relativistic mechanics).
2. **Principle of least action:** the physical trajectory extremises $S = \int L dt$.

The Lagrangian itself is not derived from first principles; one must know from experiment what interactions are present (e.g. a $1/r$ potential for the Kepler problem). The symmetry + action framework then generates the equations of motion unambiguously.

The same logic applied to electrodynamics

This course performs exactly the same programme for the electromagnetic field:

1. **Symmetry:** Lorentz invariance (special relativity), gauge invariance, and the superposition principle.
2. **Experimental inputs:** The observed force on a charge is $\mathbf{F} = e(\mathbf{E} + \mathbf{v}/c \times \mathbf{B})$ (Lorentz force), not a magnetic-monopole force. The superposition principle is observed.
3. **Principle of least action:** the physical fields and trajectories extremise (334).

Together these three inputs *uniquely* determine the action and hence all of Maxwell's equations. As the professor put it in Lecture 01: "this is analytical mechanics of field theory."

Why experiment is irreplaceable

Symmetry constraints are necessary but not sufficient to fix a theory. There are in general many actions compatible with a given symmetry group. Experimental observations select which terms appear and fix the coupling constants.

- In the Kepler problem, symmetry allows any central force; only experiment (Kepler's laws) selects $F \propto 1/r^2$.
- In electrodynamics, Lorentz + gauge invariance allows both $F_{\mu\nu}F^{\mu\nu}$ and $\epsilon^{\mu\nu\alpha\beta}F_{\mu\nu}F_{\alpha\beta}$ in the Lagrangian; the latter is a total derivative and is discarded because it contributes nothing to the equations of motion. The coupling e/c in the interaction term is fixed by measuring the electron charge.

The Standard Model as the same programme applied further

The same methodology underlies the Standard Model of particle physics, built roughly 50–60 years ago:

1. Identify the symmetry group (Lorentz invariance + gauge symmetries $SU(3) \times SU(2) \times U(1)$).
2. Write all actions (Lagrangian terms) consistent with those symmetries.
3. Use experimental data (particle masses, decay rates, scattering cross-sections) to fix the free parameters and eliminate terms not seen in nature.

The result is a compact set of equations that encode all known non-gravitational physics. The guiding principle throughout is: *action + symmetry, then constrain with experiment*.

Physical significance: Electrodynamics, studied in this course, is both the simplest and the historical prototype of this programme. Mastering it is the prerequisite for understanding how all fundamental interactions are formulated.

11.5 Aside: Feynman's Path-Integral Formulation

Physical Motivation

The discussion of the action principle prompted a brief digression on Feynman's alternative formulation of quantum mechanics, which generalises the least-action idea from classical to quantum physics.

In classical mechanics, a particle follows the single trajectory that *extremises* the action S : of all conceivable paths from $(\mathbf{x}_1, t_1)$ to $(\mathbf{x}_2, t_2)$, only the one satisfying the Euler–Lagrange equations is realised.

Feynman proposed that in quantum mechanics every path is realised *simultaneously*. The probability amplitude for propagation from point 1 to point 2 is the sum (integral) over all paths, each weighted by the phase factor

$$\mathcal{A}_{1 \rightarrow 2} \propto \int_{\text{all paths}} \mathcal{D}[x(t)] e^{iS[x(t)]/\hbar}, \quad (362)$$

where $S[x(t)]$ is the classical action evaluated along a given path and $\mathcal{D}[x(t)]$ denotes integration over the space of all trajectories.

Physical significance: In the classical limit $\hbar \rightarrow 0$, the phase $e^{iS/\hbar}$ oscillates infinitely rapidly for paths far from the classical one, so they cancel by destructive interference. Only the neighbourhood of the *stationary-phase path* (i.e. the classical trajectory) contributes, recovering the principle of least action. Quantum mechanics is therefore the deeper framework: classical mechanics is its stationary-phase approximation.

11.6 Discrete Symmetries: Time Reversal and Spatial Inversion

Physical Motivation

The proper orthochronous Lorentz group $SO^+(3, 1)$ is continuously connected to the identity. But there are also discrete transformations: time reversal $T : t \rightarrow -t$ and spatial parity $P : \mathbf{r} \rightarrow -\mathbf{r}$ (in three dimensions). Electromagnetism *is* symmetric under both, but in nature the weak force violates T and P (and even CP). Here we determine how $\mathbf{E}$ and $\mathbf{B}$ transform under these discrete symmetries by using Maxwell's equations as constraints.

Time reversal $t \rightarrow -t$

Charge density: $\rho(\mathbf{x}, t) \rightarrow \rho(\mathbf{x}, -t)$. A static charge is unaffected; the value of ρ does not change sign:

$$\rho \longrightarrow \rho. \quad (363)$$

Current density: $\mathbf{J} = \rho \mathbf{v}$. The velocity $\mathbf{v} = d\mathbf{x}/dt$ changes sign under $t \rightarrow -t$, so

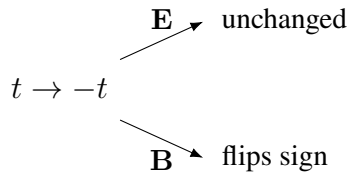
$$\mathbf{J} \longrightarrow -\mathbf{J}. \quad (364)$$

Electric field: Gauss's law $\nabla \cdot \mathbf{E} = 4\pi\rho$ relates $\mathbf{E}$ to ρ . The right-hand side does not change sign. The spatial derivatives ∂_i are unaffected by $t \rightarrow -t$. Therefore

$$\mathbf{E} \longrightarrow \mathbf{E}. \quad (365)$$

Magnetic field: Consider Faraday's law $\nabla \times \mathbf{E} = -(1/c)\partial_t \mathbf{B}$. The left-hand side transforms as $\mathbf{E} \rightarrow \mathbf{E}$ (purely spatial curl), so it is unchanged. The right-hand side acquires a sign from $\partial_t \rightarrow -\partial_t$ and another from $\mathbf{B} \rightarrow ?$. For the equation to remain satisfied, the two signs must cancel:

$$\mathbf{B} \longrightarrow -\mathbf{B}. \quad (366)$$



Spatial inversion $\mathbf{r} \rightarrow -\mathbf{r}$

Charge and current: ρ is a scalar density; it is unchanged. $\mathbf{J} = \rho\mathbf{v}$ changes sign because $\mathbf{v} = d\mathbf{x}/dt$ changes sign:

$$\rho \longrightarrow \rho, \quad \mathbf{J} \longrightarrow -\mathbf{J}. \quad (367)$$

Electric field: Under $\mathbf{r} \rightarrow -\mathbf{r}$, the spatial gradient $\nabla_i \rightarrow -\nabla_i$. Gauss's law $\nabla \cdot \mathbf{E} = 4\pi\rho$ then requires $\mathbf{E} \rightarrow -\mathbf{E}$ (to cancel the sign of $\nabla \rightarrow -\nabla$):

$$\mathbf{E} \longrightarrow -\mathbf{E}. \quad (368)$$

Magnetic field: In Faraday's law, the curl $\nabla \times \mathbf{E} \rightarrow -(-\mathbf{E}) = +(\nabla \times \mathbf{E})$ (two sign flips cancel). The time derivative ∂_t is unaffected. Therefore $\mathbf{B}$ must be unchanged to keep the equation satisfied:

$$\mathbf{B} \longrightarrow +\mathbf{B}. \quad (369)$$

	Time reversal ($t \rightarrow -t$)	Spatial inversion ($\mathbf{r} \rightarrow -\mathbf{r}$)
ρ	$+\rho$	$+\rho$
$\mathbf{J}$	$-\mathbf{J}$	$-\mathbf{J}$
$\mathbf{E}$	$+\mathbf{E}$	$-\mathbf{E}$
$\mathbf{B}$	$-\mathbf{B}$	$+\mathbf{B}$

Polar and axial vectors

A **polar vector** (ordinary vector) changes sign under spatial inversion, just like $\mathbf{r}$ itself. Examples: $\mathbf{r}$, $\mathbf{E}$, $\mathbf{J}$.

An **axial vector** (pseudovector) does *not* change sign under spatial inversion. It behaves like a rotation axis. Examples: $\mathbf{B}$, angular momentum $\mathbf{L} = \mathbf{r} \times \mathbf{p}$.

Physical significance: $\mathbf{B}$ is axial because it is built as a curl (a cross product), which inherits an extra sign flip that compensates the inversion. This distinction matters when writing Lorentz-invariant combinations: only contractions that respect the parity of each vector produce true scalars.

Caveat: parity in two dimensions

A common error: in two spatial dimensions, $\mathbf{r} \rightarrow -\mathbf{r}$ is a rotation by 180, not a parity transformation, because $\det(-\mathbf{1}_{2 \times 2}) = +1$. True parity requires an odd number of sign flips, i.e. $\det(\text{transformation}) = -1$. In three dimensions, $\mathbf{r} \rightarrow -\mathbf{r}$ has $\det = -1$ and is a genuine parity.

11.7 Lorentz Invariants of the Electromagnetic Field

Physical Motivation

The components $\mathbf{E}$ and $\mathbf{B}$ mix under Lorentz boosts (they are entries of the rank-2 tensor $F^{\mu\nu}$, not separate four-vectors). Yet certain combinations must be frame-independent. These Lorentz scalars encode absolute properties of the electromagnetic field: which configurations are related by a boost, and which are not.

First invariant: $F_{\mu\nu}F^{\mu\nu}$

The free-field Lagrangian density $\mathcal{L} = -F_{\mu\nu}F^{\mu\nu}/(16\pi)$ is a Lorentz scalar (it enters the Lorentz-scalar action). Computing $F_{\mu\nu}F^{\mu\nu}$ explicitly:

The matrix $F_{\mu\nu}$ (indices both down) differs from $F^{\mu\nu}$ by raising both indices, i.e. multiplying by $\eta^{\mu\alpha}\eta^{\nu\beta}$. Raising a timelike index (0) costs no sign; raising a spacelike index costs a -1 factor per index. Therefore $F^{i0} = -F_{i0}$ and $F^{ij} = F_{ij}$ (both indices spatial, two minus signs cancel). The time-space components thus pick up a relative sign:

$$\begin{aligned} F_{\mu\nu}F^{\mu\nu} &= 2[F_{01}F^{01} + F_{02}F^{02} + F_{03}F^{03}] + 2[F_{12}F^{12} + F_{13}F^{13} + F_{23}F^{23}] \\ &= 2(-E_x^2 - E_y^2 - E_z^2) + 2(B_z^2 + B_y^2 + B_x^2) \\ &= 2(B^2 - E^2). \end{aligned} \quad (370)$$

(Each off-diagonal pair contributes twice because of the symmetry of the sum.)

$$\boxed{F_{\mu\nu}F^{\mu\nu} = 2(B^2 - E^2).} \quad (371)$$

Physical significance: The sign of $B^2 - E^2$ is a Lorentz-invariant property. An electromagnetic configuration with $B^2 > E^2$ remains so in every frame; a pure-electric configuration ($B = 0, E \neq 0$) has $B^2 - E^2 < 0$ and stays that way. In particular, it is impossible to boost to a frame where $B = 0$ if $B^2 - E^2 > 0$ in the original frame.

Second invariant: the Levi-Civita pseudoscalar

There is a second quadratic combination,

$$\epsilon^{\alpha\beta\mu\nu} F_{\alpha\beta} F_{\mu\nu}, \quad (372)$$

where $\epsilon^{\alpha\beta\mu\nu}$ is the **four-dimensional Levi-Civita symbol**:

$$\epsilon^{0123} = +1, \quad \text{completely antisymmetric in all four indices.} \quad (373)$$

This symbol is a *pseudotensor* (it picks up a $\det \Lambda$ factor under Lorentz transformations), so (372) is a pseudoscalar.

A direct calculation (summing over all index permutations) yields contributions of the form $F_{0i}F_{jk}$, which combine to give $E_x B_x + E_y B_y + E_z B_z$:

$$\boxed{\epsilon^{\alpha\beta\mu\nu} F_{\alpha\beta} F_{\mu\nu} = -8 \mathbf{E} \cdot \mathbf{B}.} \quad (374)$$

Physical significance: The angle between $\mathbf{E}$ and $\mathbf{B}$ (or rather, the sign of $\mathbf{E} \cdot \mathbf{B}$) is a Lorentz-invariant property. If $\mathbf{E} \perp \mathbf{B}$ in one frame, they are orthogonal in all frames.

Why this term does not affect the field equations. At first sight, one might add $\alpha \epsilon^{\alpha\beta\mu\nu} F_{\alpha\beta} F_{\mu\nu}$ to the Lagrangian as a second quadratic invariant. However, one can show that it is a *total spacetime derivative*:

$$\epsilon^{\alpha\beta\mu\nu} F_{\alpha\beta} F_{\mu\nu} = \partial_\mu (\epsilon^{\mu\nu\alpha\beta} A_\nu F_{\alpha\beta}). \quad (375)$$

Adding a total derivative to the Lagrangian density changes the action by a boundary term, which does not affect the Euler–Lagrange equations. This is why we discarded it when constructing the action in Lecture 08.

11.8 Physical Consequences of the Two Invariants

Physical Motivation

The two invariants $B^2 - E^2$ and $\mathbf{E} \cdot \mathbf{B}$ are the complete set of independent Lorentz scalars quadratic in F . They classify electromagnetic field configurations and determine what can be transformed into what by a Lorentz boost.

Case 1: Radiation ($B = E$, $\mathbf{E} \perp \mathbf{B}$)

For a plane electromagnetic wave (radiation), $|\mathbf{E}| = |\mathbf{B}|$ and $\mathbf{E} \perp \mathbf{B}$. Therefore both invariants vanish: $B^2 - E^2 = 0$ and $\mathbf{E} \cdot \mathbf{B} = 0$. Since both invariants are zero, they remain zero in every frame. *It is impossible to eliminate electromagnetic radiation by switching to a different inertial frame.* This is physically obvious: if you are “burned” by radiation in one frame, you are burned in all frames.

Case 2: $B^2 > E^2$

One can always boost to a frame where $\mathbf{E}' = 0$ (purely magnetic field). The invariant $B^2 - E^2 > 0$ is preserved and now reads $B'^2 > 0$, consistent. But one *cannot* boost to a frame with $\mathbf{B}' = 0$, because that would require $B'^2 - E'^2 = -E'^2 \leq 0$, contradicting $B^2 - E^2 > 0$.

Case 3: $E^2 > B^2$

One can always boost to a frame where $\mathbf{B}' = 0$ (purely electric field). Conversely, one cannot boost to a frame with $\mathbf{E}' = 0$.

Case 4: $F^{\mu\nu} = 0$

If all components of $F^{\mu\nu}$ vanish in one frame, both invariants are zero and the tensor remains zero in every frame (it transforms tensorially; $\Lambda F \Lambda^T = 0$).

Condition (any frame)	Can boost to	Cannot boost to
$B^2 - E^2 > 0$	$\mathbf{E}' = 0$	$\mathbf{B}' = 0$
$B^2 - E^2 < 0$	$\mathbf{B}' = 0$	$\mathbf{E}' = 0$
$B^2 - E^2 = 0$, $\mathbf{E} \cdot \mathbf{B} = 0$ (radiation)	neither	neither
$\mathbf{E} \cdot \mathbf{B} = 0$	$\mathbf{E}' \perp \mathbf{B}'$	$\mathbf{E}' \parallel \mathbf{B}' \neq 0$

Physical significance: The two invariants are the complete set of information that survives a boost. All other properties of $\mathbf{E}$ and $\mathbf{B}$ (individual magnitudes, directions relative to coordinate axes) can be changed by choosing a different frame.

Summary

- Varying the total action with respect to A_μ gives the inhomogeneous Maxwell equations $\partial_\mu F^{\mu\nu} = (4\pi/c)J^\nu$.
- The covariant Lorentz force law $m du^\mu/d\tau = (e/c)F^\mu{}_\nu u^\nu$ and the field equation together encode all of classical electrodynamics.
- Applying ∂_ν to the field equation and using antisymmetry of $F^{\mu\nu}$ immediately yields charge conservation $\partial_\mu J^\mu = 0$. The same result follows from the 3-vector equations by taking $\nabla \cdot$ of the Ampère–Maxwell law and substituting Gauss’s law; Maxwell’s displacement current is essential for consistency.

- The $\nu = 0$ component gives Gauss's law; the spatial components give the Ampère–Maxwell law.
- The homogeneous pair ($\nabla \times \mathbf{E} = -(1/c)\dot{\mathbf{B}}$ and $\nabla \cdot \mathbf{B} = 0$) follows directly from the definitions $\mathbf{E} = -\nabla\phi - (1/c)\dot{\mathbf{A}}$ and $\mathbf{B} = \nabla \times \mathbf{A}$, via $\nabla \times \nabla f = 0$ and $\nabla \cdot (\nabla \times \mathbf{F}) = 0$. These are kinematic identities, not dynamical equations. The absence of monopoles is an experimental input, not a logical necessity.
- The definitions of $\mathbf{E}$ and $\mathbf{B}$ in terms of A^μ are *not* E–B symmetric; this asymmetry reflects the absence of magnetic charges. Adding a second four-potential A_{mag}^μ restores full symmetry and gives $\nabla \cdot \mathbf{B} = 4\pi\rho_M$. Any monopole theory requires a new suppression constant to explain non-observation.
- The derivation exemplifies *analytical field theory*: symmetry (Lorentz + gauge + superposition) + principle of least action + experimental inputs (Lorentz force, no monopoles) uniquely fix the action and hence all of Maxwell's equations. This is the same programme used to construct the Standard Model.
- Under time reversal: $\mathbf{E} \rightarrow +\mathbf{E}$, $\mathbf{B} \rightarrow -\mathbf{B}$. Under spatial inversion: $\mathbf{E} \rightarrow -\mathbf{E}$ (polar vector), $\mathbf{B} \rightarrow +\mathbf{B}$ (axial vector).
- The first Lorentz invariant $F_{\mu\nu}F^{\mu\nu} = 2(B^2 - E^2)$ determines which of $\mathbf{E}$, $\mathbf{B}$ can be boosted away.
- The second Lorentz invariant $\epsilon^{\alpha\beta\mu\nu}F_{\alpha\beta}F_{\mu\nu} = -8\mathbf{E} \cdot \mathbf{B}$ determines whether $\mathbf{E}$ and $\mathbf{B}$ are orthogonal in all frames.
- Radiation has both invariants zero; it cannot be eliminated by any boost.

Electrostatics

12 Lecture 11 – Field Transformations and Electrostatics

Date: 12/05/2026

Overview

This lecture closes one conceptual loop from Lecture 10 and opens the next topic. The central message is that the electric and magnetic fields are not two independent relativistic objects. They are different components of one antisymmetric tensor $F_{\mu\nu}$, so a Lorentz boost generally mixes them.

The lecture has four parts:

- a clarification of the invariant $E^2 - B^2$: it is compared at the same spacetime event, not at unrelated spatial points;
- a proof that the four-dimensional Levi-Civita symbol $\epsilon_{\mu\nu\rho\sigma}$ behaves as a tensor under proper Lorentz transformations, and as a pseudotensor if reflections are included;
- a derivation of the Lorentz transformation laws for $\mathbf{E}$ and $\mathbf{B}$;
- the beginning of electrostatics, defined as the time-reversal symmetric sector of Maxwell theory.

The physical bridge is simple: after Maxwell's equations have been written in covariant form, the next useful question is how familiar non-covariant language, such as “electric field” and “magnetic field”, is seen by different inertial observers.

12.1 What Does an Electromagnetic Invariant Compare?

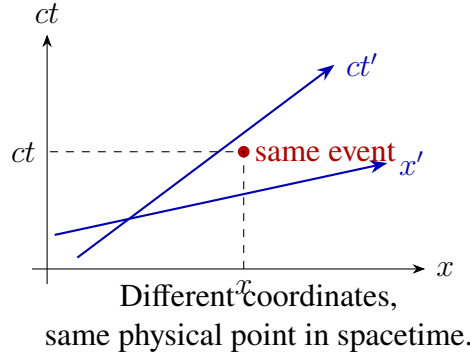
Physical Motivation

An invariant is not a statement that a field has the same value everywhere in space. Fields can vary from point to point. The invariant statement compares what two observers assign to the *same event* in spacetime. This distinction matters because a Lorentz transformation changes the coordinates of the event, not the event itself.

Recall from Lecture 10 that the scalar

$$\boxed{\frac{1}{2}F_{\mu\nu}F^{\mu\nu} = B^2 - E^2} \quad (376)$$

is Lorentz invariant, equivalently $E^2 - B^2$ is invariant up to the conventional overall sign used in writing the contraction. *Physical significance:* If two observers describe the same spacetime event $x^\mu = (ct, \mathbf{r})$ and $x'^\mu = (ct', \mathbf{r}')$, they may disagree about the separate values of $\mathbf{E}$ and $\mathbf{B}$, but they agree on the scalar combination $E^2 - B^2$ at that event.



12.2 The Levi-Civita Symbol as a Tensor and Pseudotensor

Physical Motivation

In three dimensions, cross products and oriented volumes rely on the Levi-Civita symbol. In four-dimensional electrodynamics the analogous object $\epsilon_{\mu\nu\rho\sigma}$ is needed to build the second electromagnetic invariant and the dual field tensor. Before using it in relativistic equations, we must know how it transforms.

Define the completely antisymmetric object by

$$\epsilon_{0123} = +1, \quad \epsilon_{\mu\nu\rho\sigma} = 0 \quad \text{if any two indices are equal,} \quad (377)$$

with all other nonzero components fixed by antisymmetry. *Physical significance:* $\epsilon_{\mu\nu\rho\sigma}$ is the algebraic device that remembers orientation in spacetime, just as the determinant remembers the oriented volume scaling of a matrix.

To test whether it is a tensor, apply the tensor transformation rule:

$$X_{\mu_1\mu_2\mu_3\mu_4} = \Lambda^{\nu_1}_{\mu_1} \Lambda^{\nu_2}_{\mu_2} \Lambda^{\nu_3}_{\mu_3} \Lambda^{\nu_4}_{\mu_4} \epsilon_{\nu_1\nu_2\nu_3\nu_4}. \quad (378)$$

The object $X_{\mu_1\mu_2\mu_3\mu_4}$ is completely antisymmetric. Indeed, if two free indices are interchanged, the corresponding two dummy indices in $\epsilon_{\nu_1\nu_2\nu_3\nu_4}$ can be interchanged, and the antisymmetry of ϵ gives a minus sign. In four dimensions, every completely antisymmetric rank-four object is proportional to $\epsilon_{\mu_1\mu_2\mu_3\mu_4}$, so

$$X_{\mu_1\mu_2\mu_3\mu_4} = \alpha \epsilon_{\mu_1\mu_2\mu_3\mu_4}. \quad (379)$$

To find α , choose $(\mu_1, \mu_2, \mu_3, \mu_4) = (0, 1, 2, 3)$:

$$\begin{aligned} \alpha &= \Lambda^{\nu_1}_0 \Lambda^{\nu_2}_1 \Lambda^{\nu_3}_2 \Lambda^{\nu_4}_3 \epsilon_{\nu_1\nu_2\nu_3\nu_4} \\ &= \det \Lambda. \end{aligned} \quad (380)$$

Therefore

$$\Lambda^{\nu_1}_{\mu_1} \Lambda^{\nu_2}_{\mu_2} \Lambda^{\nu_3}_{\mu_3} \Lambda^{\nu_4}_{\mu_4} \epsilon_{\nu_1\nu_2\nu_3\nu_4} = (\det \Lambda) \epsilon_{\mu_1\mu_2\mu_3\mu_4}. \quad (381)$$

Physical significance: For proper Lorentz transformations, $\det \Lambda = +1$, so $\epsilon_{\mu\nu\rho\sigma}$ behaves as a tensor. Under a reflection, such as parity, $\det \Lambda = -1$, so it gains an extra sign. That is why it is often called a pseudotensor when improper transformations are allowed.

This also explains why angular momentum in relativistic notation is naturally an antisymmetric rank-two tensor rather than an ordinary four-vector: in more than three dimensions, a rotation is best associated with a plane, not only with an axis.

12.3 Lorentz Boost Setup

Physical Motivation

The cleanest way to see the mixing of $\mathbf{E}$ and $\mathbf{B}$ is to compare two inertial frames related by a boost along the x -axis. This loses no conceptual content: any boost can be decomposed into a rotation, a boost along one axis, and another rotation.

Let K' move with speed v along the positive x direction as seen from K . Write

$$\boxed{\beta = \frac{v}{c}, \quad \gamma = \frac{1}{\sqrt{1 - \beta^2}}.} \quad (382)$$

Physical significance: β measures the boost speed in units of the maximum signal speed c , while γ is the relativistic stretching factor that controls time dilation, length contraction, and field mixing.

The coordinate transformation from primed coordinates to unprimed coordinates is

$$x = \gamma(x' + vt'), \quad ct = \gamma(ct' + \beta x'), \quad y = y', \quad z = z'. \quad (383)$$

Equivalently, for the four-potential

$$A^\mu = (\phi, \mathbf{A}), \quad (384)$$

the same boost gives

$$\phi = \gamma(\phi' + \beta A'_x), \quad A_x = \gamma(A'_x + \beta \phi'), \quad A_y = A'_y, \quad A_z = A'_z. \quad (385)$$

The fields are defined by

$$\boxed{\mathbf{E} = -\nabla\phi - \frac{1}{c} \frac{\partial \mathbf{A}}{\partial t}, \quad \mathbf{B} = \nabla \times \mathbf{A}.} \quad (386)$$

Physical significance: The potentials transform simply as a four-vector, but the observed fields involve spacetime derivatives of the potentials. This is why the transformation of $\mathbf{E}$ and $\mathbf{B}$ contains both the boost and the curl-like structure of electromagnetism.

12.4 Transforming the Electric Field

Physical Motivation

We first transform $\mathbf{E}$ directly from the potentials. This derivation is algebraic, but its physical content is important: a purely magnetic part of the field tensor in one frame can contribute to the electric field measured in another frame.

The y component in K is

$$E_y = -\frac{\partial \phi}{\partial y} - \frac{1}{c} \frac{\partial A_y}{\partial t}. \quad (387)$$

Since $y = y'$, the first derivative is

$$\frac{\partial}{\partial y} = \frac{\partial}{\partial y'}. \quad (388)$$

For the time derivative, the inverse boost gives

$$\frac{\partial}{\partial t} = \frac{\partial t'}{\partial t} \frac{\partial}{\partial t'} + \frac{\partial x'}{\partial t} \frac{\partial}{\partial x'} = \gamma \frac{\partial}{\partial t'} - \gamma v \frac{\partial}{\partial x'}. \quad (389)$$

Substituting (385) into (387),

$$\begin{aligned}
 E_y &= -\frac{\partial}{\partial y'}[\gamma(\phi' + \beta A'_x)] - \frac{1}{c} \left(\gamma \frac{\partial}{\partial t'} - \gamma v \frac{\partial}{\partial x'} \right) A'_y \\
 &= \gamma \left(-\frac{\partial \phi'}{\partial y'} - \frac{1}{c} \frac{\partial A'_y}{\partial t'} \right) + \gamma \beta \left(\frac{\partial A'_y}{\partial x'} - \frac{\partial A'_x}{\partial y'} \right) \\
 &= \gamma (E'_y + \beta B'_z).
 \end{aligned} \tag{390}$$

In the last line we used

$$B'_z = (\nabla' \times \mathbf{A}')_z = \frac{\partial A'_y}{\partial x'} - \frac{\partial A'_x}{\partial y'}.$$
(391)

Similarly,

$$E_x = E'_x, \tag{392}$$

$$E_z = \gamma(E'_z - \beta B'_y). \tag{393}$$

Thus, for a boost along the x direction,

$$\boxed{E_x = E'_x, \quad E_y = \gamma(E'_y + \beta B'_z), \quad E_z = \gamma(E'_z - \beta B'_y).} \tag{394}$$

Physical significance: The component parallel to the boost is unchanged, while the perpendicular electric field mixes with the perpendicular magnetic field.

It is useful to write this in vector form. With the sign convention of (383), let $\beta_{K|K'}$ denote the velocity of K as seen from K' , divided by c . Then

$$\boxed{\mathbf{E}_{\parallel} = \mathbf{E}'_{\parallel}, \quad \mathbf{E}_{\perp} = \gamma (\mathbf{E}'_{\perp} + \beta_{K|K'} \times \mathbf{B}')} \tag{395}$$

Physical significance: The cross product says that only fields transverse to the relative motion are mixed; a field already pointing along the boost direction has no transverse area element with which to mix.

12.5 Transforming the Magnetic Field

Physical Motivation

The magnetic field can be transformed by repeating the potential calculation, but the tensor method shows the deeper reason immediately. The magnetic components are spatial components of $F^{\mu\nu}$, and a boost mixes spatial and temporal indices. Since temporal-spatial components are electric-field components, $\mathbf{B}$ must mix with $\mathbf{E}$.

For the electromagnetic tensor,

$$F^{\mu\nu} = \Lambda^{\mu}_{\alpha} \Lambda^{\nu}_{\beta} F'^{\alpha\beta}. \tag{396}$$

Using the convention from Lecture 10,

$$F^{0i} = -E_i, \quad F^{i0} = E_i, \quad F^{ij} = -\epsilon^{ijk} B_k. \tag{397}$$

Take the component $F^{12} = -B_z$. Since the boost only mixes the 0 and 1 indices, while the 2 direction is untouched,

$$\begin{aligned}
 F^{12} &= \Lambda^1_{\alpha} \Lambda^2_{\beta} F'^{\alpha\beta} \\
 &= \Lambda^1_0 F'^{02} + \Lambda^1_1 F'^{12} \\
 &= \gamma \beta (-E'_y) + \gamma (-B'_z).
 \end{aligned} \tag{398}$$

Therefore

$$B_z = \gamma(B'_z + \beta E'_y). \tag{399}$$

The other components are obtained in the same way:

$$\boxed{B_x = B'_x, \quad B_y = \gamma(B'_y - \beta E'_z), \quad B_z = \gamma(B'_z + \beta E'_y).} \quad (400)$$

Physical significance: The magnetic field transforms in the same pattern as the electric field, but with the opposite sign in the mixing term. This is the tensor statement that electric and magnetic fields are two faces of one object.

In vector form,

$$\boxed{\mathbf{B}_{\parallel} = \mathbf{B}'_{\parallel}, \quad \mathbf{B}_{\perp} = \gamma (\mathbf{B}'_{\perp} - \beta_{K|K'} \times \mathbf{E}').} \quad (401)$$

Physical significance: If an observer moves relative to a purely electric configuration, they generally measure a magnetic field. Magnetism is therefore not an optional extra phenomenon; it is required by relativity once electric fields exist.

The inverse transformations are obtained by replacing $\beta_{K|K'}$ with $-\beta_{K|K'}$. The component formulas are often the safest way to check signs.

12.6 Why Use Both Languages?

Physical Motivation

Tensor notation and three-vector notation answer different practical questions. Tensor notation exposes symmetry; three-vector notation is often better for solving ordinary boundary-value problems.

The tensor $F_{\mu\nu}$ makes it clear that:

$$\boxed{\text{there is one electromagnetic field, not two unrelated fields.}} \quad (402)$$

Physical significance: What one observer calls “electric” and “magnetic” depends on their state of motion. The invariant object is the tensor, not the split into $\mathbf{E}$ and $\mathbf{B}$.

However, in electrostatics there is no need to carry all tensor components if only $\mathbf{E}$ is present in the chosen frame. In that setting, the scalar potential ϕ and the equation for $\mathbf{E}$ are the efficient language.

12.7 The Covariant Form of the Homogeneous Maxwell Equations

Physical Motivation

Lecture 10 wrote the inhomogeneous Maxwell equations as $\partial_{\mu} F^{\mu\nu} = (4\pi/c)J^{\nu}$. The other two Maxwell equations also have a covariant form. They do not come from sources; they come from the definition of $F_{\mu\nu}$ in terms of the potential.

The homogeneous Maxwell equations are

$$\boxed{\partial_{\alpha} F_{\beta\gamma} + \partial_{\beta} F_{\gamma\alpha} + \partial_{\gamma} F_{\alpha\beta} = 0.} \quad (403)$$

Physical significance: This equation is the covariant form of $\nabla \cdot \mathbf{B} = 0$ and $\nabla \times \mathbf{E} = -(1/c)\partial_t \mathbf{B}$. It says that the field tensor is built from a potential, so the antisymmetrized second derivative vanishes.

Indeed, with

$$F_{\mu\nu} = \partial_{\mu} A_{\nu} - \partial_{\nu} A_{\mu}, \quad (404)$$

substitution into (403) gives pairs of second derivatives such as $\partial_{\alpha}\partial_{\beta}A_{\gamma}$ and $-\partial_{\beta}\partial_{\alpha}A_{\gamma}$. Since partial derivatives commute, each pair cancels:

$$\partial_{\alpha}\partial_{\beta}A_{\gamma} - \partial_{\beta}\partial_{\alpha}A_{\gamma} = 0. \quad (405)$$

12.8 Electrostatics as a Symmetry Sector

Physical Motivation

Electrostatics is often introduced as “nothing depends on time”. That is true, but it is not the most structural definition. The lecture defines electrostatics as the sector of Maxwell theory that respects time reversal $t \rightarrow -t$. This distinguishes it from magnetostatics, where currents can be steady but reverse under time reversal.

The Maxwell equations in Gaussian units are

$$\nabla \cdot \mathbf{B} = 0, \quad (406)$$

$$\nabla \times \mathbf{E} = -\frac{1}{c} \frac{\partial \mathbf{B}}{\partial t}, \quad (407)$$

$$\nabla \cdot \mathbf{E} = 4\pi\rho, \quad (408)$$

$$\nabla \times \mathbf{B} = \frac{4\pi}{c} \mathbf{J} + \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t}. \quad (409)$$

The sources must also obey

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0.} \quad (410)$$

Physical significance: Charge conservation is the consistency condition that allows the sources to appear on the right-hand side of Maxwell’s equations.

Under time reversal, velocities reverse, so currents reverse:

$$\mathbf{J} \longrightarrow -\mathbf{J}. \quad (411)$$

The electric field is even under time reversal, while the magnetic field is odd:

$$\mathbf{E} \longrightarrow \mathbf{E}, \quad \mathbf{B} \longrightarrow -\mathbf{B}. \quad (412)$$

If a solution is invariant under time reversal for any choice of the time origin, then there can be no genuine time dependence, no current, and no magnetic field:

$$\boxed{\frac{\partial}{\partial t} = 0, \quad \mathbf{J} = 0, \quad \mathbf{B} = 0.} \quad (413)$$

Physical significance: Electrostatics is not merely “slow dynamics”. It is the time-reversal-even part of the theory: charges sit as prescribed sources, and the magnetic, current-carrying sector is absent.

With (413), Maxwell’s equations reduce to

$$\boxed{\nabla \times \mathbf{E} = 0, \quad \nabla \cdot \mathbf{E} = 4\pi\rho.} \quad (414)$$

Physical significance: The first equation says the electric field is conservative; the second says charge density is the source of electric flux.

Since $\nabla \times \mathbf{E} = 0$, one can write

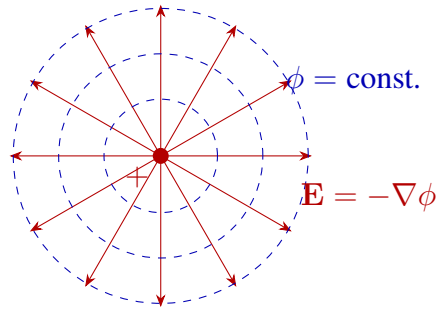
$$\boxed{\mathbf{E} = -\nabla\phi.} \quad (415)$$

Physical significance: In electrostatics the field can be described by one scalar function. Equipotential surfaces organize the problem, and the field points in the direction of steepest decrease of ϕ .

Substituting (415) into Gauss’s law gives Poisson’s equation:

$$\boxed{\nabla^2\phi = -4\pi\rho.} \quad (416)$$

Physical significance: Electrostatics becomes a boundary-value problem: given charges and boundary conditions, solve one elliptic equation for the scalar potential.



The lecture stops at the reduction to the electrostatic sector. The next step is to solve (416) for concrete charge distributions and boundary conditions.

12.9 Summary

- Electromagnetic invariants compare fields at the same spacetime event as described by different inertial observers.
- The four-dimensional Levi-Civita symbol satisfies $\epsilon \mapsto (\det \Lambda)\epsilon$; it is a tensor for proper Lorentz transformations and a pseudotensor if reflections are included.
- A boost mixes perpendicular electric and magnetic fields: $\mathbf{E}_\perp$ receives a contribution from $\mathbf{B}'$, and $\mathbf{B}_\perp$ receives a contribution from $\mathbf{E}'$.
- The tensor $F_{\mu\nu}$ is the invariant electromagnetic object; the split into $\mathbf{E}$ and $\mathbf{B}$ is observer-dependent.
- Electrostatics is the time-reversal-even sector of Maxwell theory, where $\mathbf{J} = 0$, $\mathbf{B} = 0$, and $\mathbf{E} = -\nabla\phi$ with $\nabla^2\phi = -4\pi\rho$.

13 Lecture 12 – Poisson’s Equation, Green’s Identities, and Uniqueness

Date: 17/05/2026

Overview

This lecture develops the mathematical machinery for solving electrostatics boundary-value problems. Starting from Poisson’s equation (derived last lecture as the electrostatic limit of Maxwell’s equations), we:

- construct the Green’s function as the inverse of the Laplacian operator, recovering Coulomb’s law from an operator-inversion argument;
- formulate the general boundary-value problem and discuss why the free-space solution is insufficient when boundaries are present;
- prove the two Green’s identities from Gauss’s theorem;
- use the second identity to prove uniqueness of solutions under Dirichlet or Neumann boundary conditions;
- derive a general representation formula for ϕ inside a domain in terms of boundary data (left incomplete — the full story comes next lecture);
- state and physically interpret Earnshaw’s theorem.

13.1 Poisson’s Equation as an Operator Inversion Problem

Mathematical Setup

Electrostatics reduces Maxwell’s equations to

$$\nabla^2 \phi = -4\pi \rho. \quad (417)$$

Significance: One scalar equation for one unknown function $\phi(\mathbf{r})$; the full electromagnetic problem has collapsed to an elliptic PDE.

Think of (417) in the language of linear algebra. The Laplacian ∇^2 is a linear operator acting on the infinite-dimensional space of sufficiently smooth functions. Poisson’s equation is

$$\hat{L} \phi = -4\pi \rho, \quad \hat{L} \equiv \nabla^2. \quad (418)$$

To find ϕ we need the *inverse operator* $\hat{L}^{-1}$, which in matrix language has “matrix elements” $G(\mathbf{r}, \mathbf{r}')$ defined by

$$\nabla_{\mathbf{r}}^2 G(\mathbf{r}, \mathbf{r}') = -4\pi \delta^{(3)}(\mathbf{r} - \mathbf{r}'). \quad (419)$$

Significance: $G(\mathbf{r}, \mathbf{r}')$ is the potential of a unit point charge at $\mathbf{r}'$; knowing it lets us reconstruct ϕ for any source ρ by superposition.

Remark 4. The Laplacian does have a non-trivial kernel in all of $\mathbb{R}^3$: harmonic functions that grow at infinity (e.g. linear functions) satisfy $\nabla^2 f = 0$. Restricting to functions that decay at infinity removes this kernel, making $\hat{L}$ invertible on that restricted space. This is why the free-space Green’s function exists uniquely.

13.2 Computing the Free-Space Green's Function

Mathematical Setup

We seek $G(\mathbf{r}, \mathbf{r}')$ satisfying (419) that decays at infinity. By translational invariance, G can only depend on $\mathbf{r} - \mathbf{r}'$, so place $\mathbf{r}' = \mathbf{0}$ and write $G = G(r)$ with $r = |\mathbf{r}|$.

Apply a Fourier transform: $\nabla^2 \rightarrow -k^2$ in momentum space. The equation becomes

$$-k^2 \tilde{G}(\mathbf{k}) = -4\pi \implies \tilde{G}(\mathbf{k}) = \frac{4\pi}{k^2}. \quad (420)$$

Inverting the Fourier transform in spherical coordinates (integrating over the polar angle trivially, then using $\int_0^\infty \frac{\sin y}{y} dy = \frac{\pi}{2}$):

$$G(\mathbf{r}, \mathbf{r}') = \frac{1}{|\mathbf{r} - \mathbf{r}'|}. \quad (421)$$

Significance: This is precisely the Coulomb potential of a unit point charge. The Green's function is not a new object; it is the familiar $1/r$ Coulomb kernel reinterpreted as the inverse of the Laplacian.

The general solution for charge distributed in infinite space follows immediately:

$$\phi(\mathbf{r}) = \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3r'. \quad (422)$$

Significance: This is the superposition principle for the potential — the textbook Year-1 result — derived here as an operator-inversion identity.

13.3 Why Free Space is Not Enough: Boundary-Value Problems

Physical Motivation

In practice, electrostatics problems live inside a *bounded domain* Ω with boundary $\partial\Omega = \Sigma$, and the experimenter fixes conditions on Σ (e.g. holding conductors at prescribed potentials). The free-space formula (422) only accounts for the known charges ρ inside Ω ; it ignores whatever surface charges accumulate on Σ to enforce the boundary condition.

The proper problem is:

$$\nabla^2 \phi = -4\pi\rho \quad \text{in } \Omega, \quad \text{with boundary condition on } \Sigma. \quad (423)$$

Two standard types of boundary condition:

- **Dirichlet:** $\phi|_{\Sigma} = f(\mathbf{r})$ (potential prescribed on the boundary).
- **Neumann:** $\partial_n \phi|_{\Sigma} = g(\mathbf{r})$ (normal component of $\mathbf{E}$ prescribed on the boundary, since $E_n = -\partial_n \phi$).

It is over-determined to impose *both* simultaneously; one condition already uniquely fixes ϕ (as we prove below). This is analogous to classical mechanics: position and velocity at one time uniquely determine the trajectory — you cannot additionally prescribe the acceleration.

13.4 Green's Identities

Mathematical Setup

The identities follow from Gauss's theorem applied to carefully chosen vector fields. Let ϕ, ψ be sufficiently smooth functions on Ω .

First Green's Identity. Apply Gauss's theorem to $\mathbf{F} = \phi \nabla \psi$:

$$\int_{\Omega} \nabla \cdot (\phi \nabla \psi) d^3r = \oint_{\Sigma} \phi \nabla \psi \cdot \hat{n} dA = \oint_{\Sigma} \phi \partial_n \psi dA. \quad (424)$$

Expanding the left-hand side, $\nabla \cdot (\phi \nabla \psi) = \nabla \phi \cdot \nabla \psi + \phi \nabla^2 \psi$, gives

$$\boxed{\int_{\Omega} (\phi \nabla^2 \psi + \nabla \phi \cdot \nabla \psi) d^3r = \oint_{\Sigma} \phi \partial_n \psi dA.} \quad (425)$$

Significance: Green's first identity is the integration-by-parts rule for the Laplacian: the Laplacian can be “transferred” from ψ to ϕ at the cost of a boundary term.

Second Green's Identity. Write (425) once with (ϕ, ψ) and once with (ψ, ϕ) , then subtract. The symmetric term $\nabla \phi \cdot \nabla \psi$ cancels:

$$\boxed{\int_{\Omega} (\phi \nabla^2 \psi - \psi \nabla^2 \phi) d^3r = \oint_{\Sigma} (\phi \partial_n \psi - \psi \partial_n \phi) dA.} \quad (426)$$

Significance: The Laplacian is self-adjoint up to a boundary term; when both functions satisfy homogeneous boundary conditions the Laplacian is truly self-adjoint on Ω . Equation (426) is the key tool for both uniqueness and the representation formula.

13.5 Uniqueness of Solutions

Motivation

We show that the boundary-value problem (423) has *at most one* solution under either Dirichlet or Neumann conditions (existence is guaranteed physically — nature produces a unique field).

Proof. Suppose ϕ_1 and ϕ_2 both solve (423) with the same source ρ and the same boundary data. Define

$$u \equiv \phi_1 - \phi_2. \quad (427)$$

Then u satisfies:

$$\nabla^2 u = 0 \quad \text{in } \Omega, \quad u|_{\Sigma} = 0 \text{ (Dirichlet)} \quad \text{or} \quad \partial_n u|_{\Sigma} = 0 \text{ (Neumann)}. \quad (428)$$

Apply the first Green's identity (425) with $\phi = \psi = u$:

$$\int_{\Omega} (u \nabla^2 u + |\nabla u|^2) d^3r = \oint_{\Sigma} u \partial_n u dA. \quad (429)$$

The right-hand side vanishes under either boundary condition (for Dirichlet: $u|_{\Sigma} = 0$; for Neumann: $\partial_n u|_{\Sigma} = 0$). The first term on the left also vanishes because $\nabla^2 u = 0$. Hence

$$\int_{\Omega} |\nabla u|^2 d^3r = 0. \quad (430)$$

Since $|\nabla u|^2 \geq 0$, the integrand must vanish identically:

$$\boxed{\nabla u = \mathbf{0} \implies u = \text{const.}} \quad (431)$$

For Dirichlet conditions, $u|_{\Sigma} = 0$ forces the constant to be zero, so $\phi_1 = \phi_2$ exactly. For Neumann conditions, u is an arbitrary constant, meaning the potential is unique *up to an overall additive constant* — which has no physical content. *Significance:* One cannot impose both Dirichlet and Neumann conditions on the same boundary; either one already pins the solution (up to an irrelevant constant).

13.6 Representation Formula via the Second Green's Identity

Mathematical Setup

We now use the second identity (426) to write $\phi(\mathbf{y})$ at any interior point $\mathbf{y} \in \Omega$ in terms of the source and boundary data.

Choose $\phi = \phi(\mathbf{r})$ (the solution we want) and $\psi = G(\mathbf{r}, \mathbf{y})$ where

$$G(\mathbf{r}, \mathbf{y}) = \frac{1}{|\mathbf{r} - \mathbf{y}|}, \quad \nabla_{\mathbf{r}}^2 G(\mathbf{r}, \mathbf{y}) = -4\pi \delta^{(3)}(\mathbf{r} - \mathbf{y}). \quad (432)$$

Substituting into (426):

$$\begin{aligned} & \int_{\Omega} [\phi(\mathbf{r}) (-4\pi \delta^{(3)}(\mathbf{r} - \mathbf{y})) - G(\mathbf{r}, \mathbf{y}) (-4\pi \rho(\mathbf{r}))] d^3r \\ &= \oint_{\Sigma} [\phi \partial_n G - G \partial_n \phi] dA. \end{aligned} \quad (433)$$

For $\mathbf{y} \in \Omega$ the delta function gives $-4\pi \phi(\mathbf{y})$; dividing through by -4π :

$$\phi(\mathbf{y}) = \int_{\Omega} \frac{\rho(\mathbf{r})}{|\mathbf{r} - \mathbf{y}|} d^3r + \frac{1}{4\pi} \oint_{\Sigma} [G(\mathbf{r}, \mathbf{y}) \partial_n \phi - \phi \partial_n G(\mathbf{r}, \mathbf{y})] dA. \quad (434)$$

Significance: The potential is expressed as a volume integral over sources plus a surface integral over boundary data. The first term is the familiar free-space Coulomb contribution. The second term captures the effect of the boundary.

Remark 5. Equation (434) is *not yet* a solution because the surface integrand contains both $\phi|_{\Sigma}$ and $\partial_n \phi|_{\Sigma}$. In a Dirichlet problem only $\phi|_{\Sigma}$ is given; in a Neumann problem only $\partial_n \phi|_{\Sigma}$ is given. The resolution — replacing G with a modified Green's function G_D or G_N that satisfies an additional boundary condition so that the unwanted term vanishes — will be carried out in the next lecture.

13.7 Earnshaw's Theorem

Physical Motivation

A harmonic function (one satisfying $\nabla^2 \phi = 0$) cannot have a local maximum or minimum *inside* a domain; extrema can occur only on the boundary. This is the maximum principle for Laplace's equation.

Physical consequence. Consider a system of static point charges interacting only through electrostatic forces. A charge q at position $\mathbf{r}_0$ is in stable equilibrium only if $\mathbf{r}_0$ is a local minimum of the potential $\Phi(\mathbf{r})$ created by all *other* charges. But Φ satisfies Laplace's equation in the neighbourhood of $\mathbf{r}_0$ (no source there). By the maximum principle, Φ has no local minimum at $\mathbf{r}_0$. Therefore:

$$\boxed{\text{No configuration of static charges can be held in stable equilibrium by electrostatic forces alone.}} \quad (435)$$

Significance: Electrostatics is an *effective* description only — real matter requires quantum mechanics, nuclear forces, or external constraints to remain stable. A “local” saddle point does exist (forces cancel), but any small perturbation drives the charge away. True stable equilibrium requires a local minimum, which Laplace's equation forbids in the interior.

Remark 6. The theorem does not forbid a charge sitting at a point where $\nabla \Phi = \mathbf{0}$ (a saddle); it only forbids a local minimum. A saddle gives unstable equilibrium: perturbations in some directions restore, in others they diverge.

13.8 Summary

- Poisson's equation $\nabla^2\phi = -4\pi\rho$ is solved in free space by convolving with the Green's function $G(\mathbf{r}, \mathbf{r}') = |\mathbf{r} - \mathbf{r}'|^{-1}$.
- In a bounded domain, boundary conditions (Dirichlet or Neumann) are needed; imposing *both* is over-determined.
- **Green's first identity:** $\int_{\Omega} (\phi \nabla^2 \psi + \nabla \phi \cdot \nabla \psi) d^3r = \oint_{\Sigma} \phi \partial_n \psi dA$.
- **Green's second identity:** $\int_{\Omega} (\phi \nabla^2 \psi - \psi \nabla^2 \phi) d^3r = \oint_{\Sigma} (\phi \partial_n \psi - \psi \partial_n \phi) dA$.
- Uniqueness: the Dirichlet or Neumann problem has a unique solution (up to an additive constant for Neumann).
- Representation formula: $\phi(\mathbf{y}) = \int_{\Omega} \frac{\rho}{|\mathbf{r}-\mathbf{y}|} d^3r + \frac{1}{4\pi} \oint_{\Sigma} (G \partial_n \phi - \phi \partial_n G) dA$; completing this into a true solution requires a modified Green's function (next lecture).
- Earnshaw's theorem: no stable electrostatic equilibrium exists from purely electrostatic forces — harmonic functions have no interior extrema.

14 Lecture 13 – Green’s Functions for Boundary-Value Problems

Date: 19/05/2026

Overview

This lecture completes the programme begun in Lecture 12. We have a representation formula for ϕ in terms of boundary data, but it is *not* a solution because it contains both $\phi|_{\Sigma}$ and $\partial_n \phi|_{\Sigma}$ simultaneously. We:

- prove the **mean value property** of harmonic functions from the representation formula, and use it to establish Earnshaw’s theorem rigorously;
- generalize from the bare $1/|\mathbf{r} - \mathbf{y}|$ to an arbitrary Green’s function G , deriving the general representation formula;
- define the **Dirichlet** and **Neumann** Green’s functions and show how each eliminates the unwanted boundary term;
- expose the inconsistency of the naive Neumann condition $\partial_n G_N = 0$ and derive the correct condition;
- connect the whole framework to the **method of images** studied in earlier courses;
- state the symmetry property $G_D(\mathbf{r}, \mathbf{y}) = G_D(\mathbf{y}, \mathbf{r})$ (proof deferred to next lecture).

14.1 Proof of the Mean Value Property

Motivation

Last lecture we proved (but did not use) Earnshaw’s theorem by appealing to the maximum principle. Here we derive it rigorously from the representation formula. The key tool is the *mean value property*: for a harmonic function u (one satisfying $\nabla^2 u = 0$), the value at any interior point equals the average over any sphere centred there.

Setup

Let u satisfy Laplace’s equation in a domain Ω :

$$\nabla^2 u = 0 \quad \text{in } \Omega. \quad (436)$$

Pick any interior point $\mathbf{y} \in \Omega$ and any sphere $B_R(\mathbf{y})$ of radius R that lies entirely inside Ω . Apply the representation formula (derived last lecture) to this smaller domain $B_R(\mathbf{y})$, with $\psi(\mathbf{r}; \mathbf{y}) = 1/|\mathbf{r} - \mathbf{y}|$:

$$u(\mathbf{y}) = -\frac{1}{4\pi} \oint_{\partial B_R} \left[u(\mathbf{r}) \partial_n \left(\frac{1}{|\mathbf{r} - \mathbf{y}|} \right) - \frac{1}{|\mathbf{r} - \mathbf{y}|} \partial_n u(\mathbf{r}) \right] dA. \quad (437)$$

On the sphere $|\mathbf{r} - \mathbf{y}| = R$, so $\psi = 1/R$ is constant. The normal $\hat{n}$ points radially outward, so

$$\partial_n \left(\frac{1}{|\mathbf{r} - \mathbf{y}|} \right) \Big|_{|\mathbf{r} - \mathbf{y}|=R} = -\frac{1}{R^2}. \quad (438)$$

Substituting into (437):

$$u(\mathbf{y}) = -\frac{1}{4\pi} \oint_{\partial B_R} \left[u(\mathbf{r}) \cdot \left(-\frac{1}{R^2} \right) - \frac{1}{R} \partial_n u(\mathbf{r}) \right] dA = \frac{1}{4\pi R^2} \oint_{\partial B_R} u dA + \frac{1}{4\pi R} \oint_{\partial B_R} \partial_n u dA. \quad (439)$$

The second term is $\frac{1}{4\pi R} \oint \nabla u \cdot \hat{n} dA$. By Gauss's theorem this equals $\frac{1}{4\pi R} \int_{B_R} \nabla^2 u d^3r = 0$ (since u is harmonic). Hence:

$$u(\mathbf{y}) = \frac{1}{4\pi R^2} \oint_{|\mathbf{r}-\mathbf{y}|=R} u(\mathbf{r}) dA = \langle u \rangle_{\partial B_R(\mathbf{y})}. \quad (440)$$

Significance: The value of a harmonic function at any point equals its spherical average over any concentric sphere inside the domain. This is a global constraint that a harmonic function must satisfy at every interior point.

Remark 7. The mean value (440) holds for every radius R such that $B_R(\mathbf{y}) \subset \Omega$. It is therefore a powerful intrinsic property of harmonic functions, independent of the boundary geometry.

14.2 Earnshaw's Theorem from the Mean Value Property

Motivation

A local minimum of u at $\mathbf{y}$ means $u(\mathbf{y}) < u(\mathbf{r})$ for all $\mathbf{r}$ in some punctured neighbourhood. But then the spherical average over a small enough sphere is *strictly greater* than $u(\mathbf{y})$, contradicting (440).

Physical Law 14.1 Earnshaw / Maximum Principle

A harmonic function on a bounded domain Ω (satisfying $\nabla^2 u = 0$) cannot attain a local interior minimum or maximum unless u is constant. Extrema occur only on the boundary $\partial\Omega$.

Physical consequence: No static arrangement of point charges can be in stable equilibrium under purely electrostatic forces. The potential Φ created by all other charges satisfies $\nabla^2 \Phi = 0$ near any charge-free point, so Φ has no interior minimum there — stable equilibrium is impossible.

14.3 Generalising the Green's Function

Conceptual Background

In the representation formula (437) we used $\psi = 1/|\mathbf{r} - \mathbf{y}|$, which satisfies $\nabla_{\mathbf{r}}^2 \psi = -4\pi \delta^{(3)}(\mathbf{r} - \mathbf{y})$. *Only this property was used.* Any function with the same singularity structure yields an equally valid formula. This gives us a free function to play with — and we will use it to eliminate one of the two boundary terms.

Physical Concept 14.1: Green's Function of the Laplacian

A function $G(\mathbf{r}, \mathbf{y})$ is called a *Green's function for the Laplacian* if it satisfies

$$\nabla_{\mathbf{r}}^2 G(\mathbf{r}, \mathbf{y}) = -4\pi \delta^{(3)}(\mathbf{r} - \mathbf{y}). \quad (441)$$

The most general such function is

$$G(\mathbf{r}, \mathbf{y}) = \frac{1}{|\mathbf{r} - \mathbf{y}|} + F(\mathbf{r}, \mathbf{y}), \quad (442)$$

where $F(\mathbf{r}, \mathbf{y})$ is *harmonic* in $\mathbf{r}$: $\nabla_{\mathbf{r}}^2 F = 0$ inside Ω .

General Representation Formula

Substituting G (instead of $\psi = 1/|\mathbf{r} - \mathbf{y}|$) into Green's second identity and using $\nabla^2 \phi = -4\pi\rho$ (Poisson's equation) gives:

$$\phi(\mathbf{y}) = \int_{\Omega} \rho(\mathbf{r}) G(\mathbf{r}, \mathbf{y}) d^3r - \frac{1}{4\pi} \oint_{\Sigma} [\phi(\mathbf{r}) \partial_n G(\mathbf{r}, \mathbf{y}) - G(\mathbf{r}, \mathbf{y}) \partial_n \phi(\mathbf{r})] dA. \quad (443)$$

Significance: This formula is valid for *any* choice of G satisfying (441). The freedom in F lets us impose one additional boundary condition on G so that one of the surface terms vanishes.

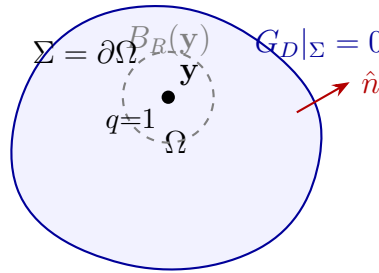


Figure 4: Domain Ω with outward normal $\hat{n}$ on boundary Σ . The Dirichlet Green's function $G_D(\mathbf{r}, \mathbf{y})$ is the potential of a unit point charge at $\mathbf{y}$ with the boundary held at zero — the method-of-images setup. The dashed sphere $B_R(\mathbf{y})$ is used in the mean value proof.

14.4 The Dirichlet Green's Function

Motivation

In the Dirichlet problem we *know* $\phi|_{\Sigma} = f(\mathbf{r})$ but *do not know* $\partial_n \phi|_{\Sigma}$. We want the term $G \partial_n \phi$ in (443) to vanish. The simplest way: require G itself to vanish on Σ .

Physical Concept 14.2: Dirichlet Green's Function G_D

$G_D(\mathbf{r}, \mathbf{y})$ is the unique solution of

$$\nabla_{\mathbf{r}}^2 G_D(\mathbf{r}, \mathbf{y}) = -4\pi \delta^{(3)}(\mathbf{r} - \mathbf{y}), \quad G_D(\mathbf{r}, \mathbf{y})|_{\mathbf{r} \in \Sigma} = 0. \quad (444)$$

With $G = G_D$, the surface term $G_D \partial_n \phi$ vanishes identically, and (443) becomes the **Dirichlet solution formula**:

$$\phi(\mathbf{y}) = \int_{\Omega} \rho(\mathbf{r}) G_D(\mathbf{r}, \mathbf{y}) d^3r - \frac{1}{4\pi} \oint_{\Sigma} f(\mathbf{r}) \partial_n G_D(\mathbf{r}, \mathbf{y}) dA, \quad (445)$$

where $f = \phi|_{\Sigma}$ is the prescribed Dirichlet data. Both integrands are now *known*, so (445) is a genuine solution.

Significance: The entire Dirichlet problem reduces to finding G_D . Once G_D is known for a given geometry, the solution for *any* source ρ and boundary datum f follows by a single integration.

Physical Interpretation of the G_D Problem

Equations (444) say: place a unit point charge at $\mathbf{y}$, remove all other sources ρ , and hold the boundary Σ at zero potential (grounded conductor). Solve for the resulting potential. That potential is $G_D(\cdot, \mathbf{y})$.

This is precisely the **method of images** problem studied in earlier courses:

- **Half-space:** a charge q at height d above a grounded infinite plane. Place an image charge $-q$ at $-d$; their combined potential vanishes on the plane and has the correct source singularity. The result is G_D for the half-space.
- **Sphere:** a charge q at distance r_0 from the centre of a grounded conducting sphere of radius a . The image charge is $-q a/r_0$ at distance a^2/r_0 . The resulting G_D gives the solution for any source inside the sphere with any potential on the boundary.

Remark 8. You used the method of images in year-1/2 courses without knowing you were computing Green's functions. The new insight is that *one* image calculation gives an *infinite family* of solutions: plug different ρ and f into (445) and read off the answer.

14.5 The Neumann Green's Function

Motivation

For the Neumann problem we know $\partial_n \phi|_\Sigma = g(\mathbf{r})$ but not $\phi|_\Sigma$. We want to eliminate the term $\phi \partial_n G$ from (443). The naive attempt is to demand $\partial_n G_N|_\Sigma = 0$.

Why the Naive Neumann Condition Fails

Suppose G_N satisfies (441) and $\partial_n G_N|_\Sigma = 0$. Integrate (441) over the domain Ω (with $\mathbf{y}$ inside):

$$\int_{\Omega} \nabla_{\mathbf{r}}^2 G_N d^3 r = -4\pi \xrightarrow{\text{Gauss}} \oint_{\Sigma} \partial_n G_N dA = -4\pi. \quad (446)$$

But if $\partial_n G_N|_\Sigma = 0$ then the left side is 0. This is a contradiction:

$$0 = -4\pi. \quad (\text{inconsistent!}) \quad (447)$$

The naive condition is therefore **incompatible** with the defining equation.

The Correct Neumann Condition

The flux integral must equal -4π . The simplest consistent condition is to distribute this equally over the boundary:

$$\partial_n G_N(\mathbf{r}, \mathbf{y})|_{\mathbf{r} \in \Sigma} = -\frac{4\pi}{S}, \quad (448)$$

where $S = \oint_{\Sigma} dA$ is the total area of the boundary.

Physical Concept 14.3: Neumann Green's Function G_N

$G_N(\mathbf{r}, \mathbf{y})$ satisfies (441) together with the boundary condition (448).

The Neumann Solution Formula

Substituting G_N into (443) and using (448):

$$\phi(\mathbf{y}) = \int_{\Omega} \rho(\mathbf{r}) G_N(\mathbf{r}, \mathbf{y}) d^3 r + \frac{1}{4\pi} \oint_{\Sigma} G_N(\mathbf{r}, \mathbf{y}) g(\mathbf{r}) dA + \langle \phi \rangle_{\Sigma}, \quad (449)$$

where $g = \partial_n \phi|_\Sigma$ is the prescribed Neumann data and $\langle \phi \rangle_{\Sigma} = \frac{1}{S} \oint_{\Sigma} \phi dA$ is the (unknown) average value of ϕ on the boundary.

Significance: The additive constant $\langle \phi \rangle_{\Sigma}$ is exactly the ambiguity expected from uniqueness theory: the Neumann problem determines ϕ only up to a global constant (physically, the potential reference level). This constant can be fixed by any convenient convention (e.g. setting $\langle \phi \rangle_{\Sigma} = 0$).

14.6 Comparison: Dirichlet vs. Neumann Problems

	Dirichlet	Neumann
Known data	$\phi _{\Sigma} = f$	$\partial_n \phi _{\Sigma} = g$
Condition on G	$G_D _{\Sigma} = 0$	$\partial_n G_N _{\Sigma} = -4\pi/S$
Eliminated term	$G \partial_n \phi$	$\phi \partial_n G$
Residual ambiguity	none	additive constant
Image problem	unit charge at $\mathbf{y}$, grounded Σ	unit charge at $\mathbf{y}$, $E_n = -4\pi/cS$ on Σ

14.7 Method of Images as Green's Function Computation

Conceptual Background

Finding G_D for a specific geometry is itself a boundary-value problem (Poisson's equation with a point source, homogeneous Dirichlet condition), and it is in general hard. For highly symmetric geometries the method of images provides an elegant exact answer.

Half-space ($z > 0$, grounded plane at $z = 0$). Place a unit charge at $\mathbf{y} = (0, 0, d)$. The image is -1 at $(0, 0, -d)$. Then

$$G_D(\mathbf{r}, \mathbf{y}) = \frac{1}{|\mathbf{r} - \mathbf{y}|} - \frac{1}{|\mathbf{r} - \mathbf{y}^*|}, \quad \mathbf{y}^* = (0, 0, -d). \quad (450)$$

One verifies $G_D|_{z=0} = 0$ (equal and opposite contributions cancel on the plane).

Sphere ($|\mathbf{r}| < a$, grounded at $|\mathbf{r}| = a$). For $\mathbf{y}$ at distance $r_0 < a$ from the centre, the image charge $-a/r_0$ is placed at $\mathbf{y}^* = (a^2/r_0)\hat{\mathbf{y}}$ (outside the sphere):

$$G_D(\mathbf{r}, \mathbf{y}) = \frac{1}{|\mathbf{r} - \mathbf{y}|} - \frac{a/r_0}{|\mathbf{r} - \mathbf{y}^*|}. \quad (451)$$

One verifies $G_D|_{|\mathbf{r}|=a} = 0$. This G_D immediately gives the solution for *any* charge distribution inside the sphere with any Dirichlet boundary condition on the sphere, via (445).

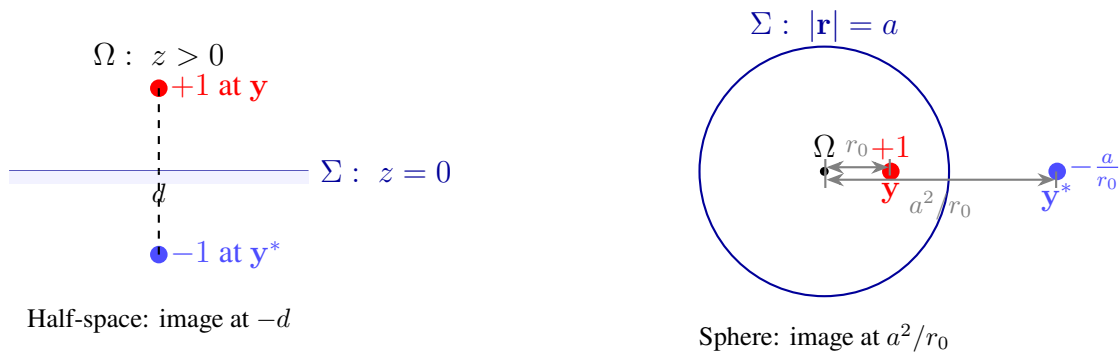


Figure 5: Method of images as computation of the Dirichlet Green's function. *Left*: grounded half-space; image charge -1 at $\mathbf{y}^* = (0, 0, -d)$. *Right*: grounded sphere of radius a ; image charge $-a/r_0$ at distance a^2/r_0 . In both cases the image ensures $G_D = 0$ on Σ .

14.8 Symmetry of the Dirichlet Green's Function

Mathematical Setup

It is not obvious from the definition (444) that $G_D(\mathbf{r}, \mathbf{y}) = G_D(\mathbf{y}, \mathbf{r})$. The free-space piece $1/|\mathbf{r} - \mathbf{y}|$ is manifestly symmetric. The harmonic part F that enforces the boundary condition is not obviously so.

For the half-space and sphere examples above one can check symmetry by direct calculation. In general:

Claim 14.1 Symmetry of G_D

The Dirichlet Green's function satisfies

$$G_D(\mathbf{r}, \mathbf{y}) = G_D(\mathbf{y}, \mathbf{r}) \quad (452)$$

for any $\mathbf{r}, \mathbf{y} \in \Omega$.

Proof (sketch, to be completed next lecture): Apply Green's second identity with $u = G_D(\cdot, \mathbf{y})$ and $v = G_D(\cdot, \mathbf{z})$. The boundary terms vanish (since $G_D = 0$ on Σ); the volume integrals yield $G_D(\mathbf{z}, \mathbf{y}) - G_D(\mathbf{y}, \mathbf{z}) = 0$.

Physical meaning: The potential at $\mathbf{y}$ due to a unit charge at $\mathbf{r}$ (with grounded boundary) equals the potential at $\mathbf{r}$ due to a unit charge at $\mathbf{y}$. This is a reciprocity theorem for electrostatics.

14.9 Summary

- **Mean value property:** $u(\mathbf{y}) = \frac{1}{4\pi R^2} \oint_{|\mathbf{r}-\mathbf{y}|=R} u \, dA$ for any harmonic u and any R with $B_R(\mathbf{y}) \subset \Omega$.
- **Earnshaw:** harmonic functions have no interior extrema (follows immediately from mean value property).
- **General Green's function:** $G = 1/|\mathbf{r} - \mathbf{y}| + F$ where F is harmonic; G satisfies $\nabla_{\mathbf{r}}^2 G = -4\pi\delta(\mathbf{r} - \mathbf{y})$.
- **Dirichlet:** $G_D|_{\Sigma} = 0$; gives $\phi(\mathbf{y}) = \int_{\Omega} \rho G_D \, d^3r - \frac{1}{4\pi} \oint_{\Sigma} f \, \partial_n G_D \, dA$. Physical interpretation: G_D is the potential of a unit charge at $\mathbf{y}$ in a grounded domain — precisely the method-of-images problem.
- **Neumann:** $\partial_n G_N|_{\Sigma} = -4\pi/S$ (not zero — that is inconsistent); gives ϕ up to an additive constant $\langle \phi \rangle_{\Sigma}$.
- **Symmetry:** $G_D(\mathbf{r}, \mathbf{y}) = G_D(\mathbf{y}, \mathbf{r})$ (proof next lecture).

Magnetostatics & Magnetic Matter

15 Lecture 14 – Magnetostatics, Biot-Savart Law, and Magnetic Forces

Date: 24/05/2026

Overview

This lecture starts a new static sector of Maxwell theory. In electrostatics we isolated the time-reversal invariant sector: no currents, no magnetic field, and only the scalar potential ϕ remained. Magnetostatics is different. The fields are still time independent, but the sources are steady currents. The main results are:

- magnetostatics describes neutral systems with steady currents, not static point charges sitting at rest;
- the equations reduce to $\nabla \cdot \mathbf{B} = 0$ and $\nabla \times \mathbf{B} = \frac{4\pi}{c} \mathbf{J}$;
- writing $\mathbf{B} = \nabla \times \mathbf{A}$ and choosing the Coulomb gauge $\nabla \cdot \mathbf{A} = 0$ turns the problem into three Poisson equations;
- solving those Poisson equations gives the vector potential $\mathbf{A}(\mathbf{r}) = \frac{1}{c} \int \mathbf{J}(\mathbf{r}')/|\mathbf{r} - \mathbf{r}'| d^3r'$;
- taking a curl gives the Biot-Savart law;
- for thin wires the Biot-Savart law reduces to the familiar line integral;
- the Lorentz force between two current loops can be written as a double line integral, and in the static limit Newton's third law is recovered.

The guiding idea is the same as in electrostatics: choose variables and gauge so that Maxwell's equations become Poisson equations. The difference is that the source is now a vector field $\mathbf{J}$, so the natural potential is the vector potential $\mathbf{A}$.

15.1 What Magnetostatics Means

Physical Motivation

The word “static” can be misleading. In magnetostatics charges are moving: otherwise there would be no current. What is static is the *macroscopic configuration*. A wire carrying a steady current contains moving electrons, but the positive ionic lattice makes the wire approximately neutral, and the current distribution does not change with time.

This is why magnetostatics is not obtained by demanding invariance under time reversal. Under time reversal,

$$t \mapsto -t, \quad \mathbf{J} \mapsto -\mathbf{J}, \quad \mathbf{B} \mapsto -\mathbf{B}, \quad \mathbf{E} \mapsto \mathbf{E}. \quad (453)$$

A current-carrying solution is therefore generally *not* time-reversal invariant. It is invariant only under continuous time translations $t \mapsto t + a$.

Physical Concept 15.1: Magnetostatic Configuration

A magnetostatic configuration is a time-independent current distribution $\mathbf{J}(\mathbf{r})$ satisfying charge conservation, with no time-dependent charge accumulation. In the simplest neutral sector used in this lecture,

$$\frac{\partial}{\partial t} = 0, \quad \rho = 0, \quad \mathbf{E} = 0, \quad \mathbf{J} \neq 0. \quad (454)$$

Significance: Magnetostatics is a sector of solutions, not a symmetry of the full theory. A concrete current pattern usually breaks time reversal even though Maxwell's equations themselves have the appropriate symmetry structure.

Remark 9. A physical wire is neutral overall: the moving electron charge is balanced by the positive lattice. The current is the charge flow through a cross-section per unit time, but the total charge density can still vanish macroscopically.

15.2 Maxwell Equations in the Magnetostatic Limit

Mathematical Setup

In Gaussian units, Maxwell's equations are

$$\nabla \cdot \mathbf{B} = 0, \quad (455)$$

$$\nabla \times \mathbf{E} + \frac{1}{c} \frac{\partial \mathbf{B}}{\partial t} = 0, \quad (456)$$

$$\nabla \cdot \mathbf{E} = 4\pi\rho, \quad (457)$$

$$\nabla \times \mathbf{B} = \frac{4\pi}{c} \mathbf{J} + \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t}. \quad (458)$$

With the assumptions (454), Faraday's law and Gauss's law for $\mathbf{E}$ become trivial in the neutral sector, and Ampere-Maxwell loses the displacement-current term. We are left with

$$\boxed{\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{B} = \frac{4\pi}{c} \mathbf{J}.} \quad (459)$$

Significance: Magnetostatics is the magnetic analogue of Poisson theory: given a steady current $\mathbf{J}$, solve for the magnetic field $\mathbf{B}$.

Charge conservation also simplifies. The continuity equation is

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0. \quad (460)$$

For time-independent sources, $\partial_t \rho = 0$, hence

$$\boxed{\nabla \cdot \mathbf{J} = 0.} \quad (461)$$

Significance: Steady currents cannot begin or end inside the domain; current lines either close on themselves or enter and leave through boundaries in a balanced way.

15.3 Introducing the Vector Potential

Physical Motivation

The equation $\nabla \cdot \mathbf{B} = 0$ says that magnetic field lines have no sources or sinks. Mathematically, a divergence-free vector field can be represented locally as the curl of another vector field. We therefore write

$$\boxed{\mathbf{B} = \nabla \times \mathbf{A}.} \quad (462)$$

Significance: This automatically solves $\nabla \cdot \mathbf{B} = 0$, because the divergence of a curl vanishes identically. Substituting (462) into Ampere's law gives

$$\nabla \times (\nabla \times \mathbf{A}) = \frac{4\pi}{c} \mathbf{J}. \quad (463)$$

To simplify the left-hand side, use the standard vector identity

$$\boxed{\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}.} \quad (464)$$

Significance: The only obstruction to a clean Poisson equation is the gradient term $\nabla(\nabla \cdot \mathbf{A})$.

Index Derivation of the Identity

Let ϵ_{ijk} be the three-dimensional Levi-Civita symbol, and use repeated-index summation. The i th component of $\nabla \times (\nabla \times \mathbf{A})$ is

$$\begin{aligned} [\nabla \times (\nabla \times \mathbf{A})]_i &= \epsilon_{ijk} \partial_j (\epsilon_{klm} \partial_\ell A_m) \\ &= \epsilon_{ijk} \epsilon_{klm} \partial_j \partial_\ell A_m. \end{aligned} \quad (465)$$

Using

$$\epsilon_{ijk} \epsilon_{klm} = \delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}, \quad (466)$$

we get

$$\begin{aligned} [\nabla \times (\nabla \times \mathbf{A})]_i &= (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) \partial_j \partial_\ell A_m \\ &= \partial_i (\partial_j A_j) - \partial_j \partial_j A_i \\ &= \partial_i (\partial_j A_j) - \nabla^2 A_i. \end{aligned} \quad (467)$$

This is exactly (464).

15.4 Coulomb Gauge and the Poisson Equation for $\mathbf{A}$

Conceptual Background

The vector potential is not unique. If $f(\mathbf{r})$ is any smooth scalar function, then

$$\mathbf{A} \mapsto \mathbf{A}' = \mathbf{A} + \nabla f \quad (468)$$

does not change the magnetic field:

$$\nabla \times \mathbf{A}' = \nabla \times \mathbf{A} + \nabla \times (\nabla f) = \nabla \times \mathbf{A}. \quad (469)$$

This freedom is not a nuisance; it is a tool. We use it to choose

$$\boxed{\nabla \cdot \mathbf{A} = 0.} \quad (470)$$

Significance: The Coulomb gauge removes the gradient term in (464), reducing Ampere's law to a vector Poisson equation.

Why is the gauge choice allowed? Suppose a preliminary potential $\mathbf{A}$ does not obey (470). We want $\mathbf{A}' = \mathbf{A} + \nabla f$ to obey it:

$$0 = \nabla \cdot \mathbf{A}' = \nabla \cdot \mathbf{A} + \nabla^2 f. \quad (471)$$

Therefore f must solve

$$\boxed{\nabla^2 f = -\nabla \cdot \mathbf{A}.} \quad (472)$$

Significance: Choosing the Coulomb gauge requires solving one scalar Poisson equation, which is exactly the type of problem already solved in electrostatics.

In Coulomb gauge, (463) becomes

$$\boxed{-\nabla^2 \mathbf{A} = \frac{4\pi}{c} \mathbf{J}.} \quad (473)$$

Significance: Magnetostatics has been reduced to three copies of Poisson's equation, one for each Cartesian component of $\mathbf{A}$.

For localized currents in free space, the solution is

$$\boxed{\mathbf{A}(\mathbf{r}) = \frac{1}{c} \int \frac{\mathbf{J}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3 r'.} \quad (474)$$

Significance: The vector potential of a steady current distribution is the current-density analogue of the electrostatic potential $\phi(\mathbf{r}) = \int \rho(\mathbf{r}')/|\mathbf{r} - \mathbf{r}'| d^3 r'$.

Consistency Check: $\nabla \cdot \mathbf{A} = 0$

Although (474) was obtained in Coulomb gauge, it is worth checking explicitly that it satisfies the gauge condition. Taking the divergence with respect to $\mathbf{r}$,

$$\nabla \cdot \mathbf{A}(\mathbf{r}) = \frac{1}{c} \int \mathbf{J}(\mathbf{r}') \cdot \nabla_{\mathbf{r}} \frac{1}{|\mathbf{r} - \mathbf{r}'|} d^3 r'. \quad (475)$$

Since

$$\nabla_{\mathbf{r}} \frac{1}{|\mathbf{r} - \mathbf{r}'|} = -\nabla_{\mathbf{r}'} \frac{1}{|\mathbf{r} - \mathbf{r}'|}, \quad (476)$$

we can write

$$\begin{aligned} \nabla \cdot \mathbf{A}(\mathbf{r}) &= -\frac{1}{c} \int \mathbf{J}(\mathbf{r}') \cdot \nabla_{\mathbf{r}'} \frac{1}{|\mathbf{r} - \mathbf{r}'|} d^3 r' \\ &= -\frac{1}{c} \int \nabla_{\mathbf{r}'} \cdot \left(\frac{\mathbf{J}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \right) d^3 r' + \frac{1}{c} \int \frac{\nabla_{\mathbf{r}'} \cdot \mathbf{J}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3 r'. \end{aligned} \quad (477)$$

The second integral vanishes by $\nabla \cdot \mathbf{J} = 0$. The first integral becomes a surface flux at infinity. For localized currents, $\mathbf{J} = 0$ on a sufficiently large surface, so the flux vanishes. Hence

$$\boxed{\nabla \cdot \mathbf{A} = 0.} \quad (478)$$

Significance: The current-conservation condition is exactly what makes the Coulomb-gauge solution self-consistent.

15.5 The Biot-Savart Law

Mathematical Setup

The magnetic field is the curl of (474):

$$\begin{aligned} \mathbf{B}(\mathbf{r}) &= \nabla_{\mathbf{r}} \times \mathbf{A}(\mathbf{r}) \\ &= \frac{1}{c} \int \nabla_{\mathbf{r}} \times \left(\frac{\mathbf{J}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \right) d^3 r'. \end{aligned} \quad (479)$$

Inside the curl, $\mathbf{J}(\mathbf{r}')$ is independent of $\mathbf{r}$, so

$$\nabla_{\mathbf{r}} \times (f(\mathbf{r})\mathbf{J}(\mathbf{r}')) = \nabla_{\mathbf{r}} f(\mathbf{r}) \times \mathbf{J}(\mathbf{r}'). \quad (480)$$

With

$$\nabla_{\mathbf{r}} \frac{1}{|\mathbf{r} - \mathbf{r}'|} = -\frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|^3}, \quad (481)$$

we obtain

$$\mathbf{B}(\mathbf{r}) = \frac{1}{c} \int \frac{\mathbf{J}(\mathbf{r}') \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3} d^3 r'. \quad (482)$$

Significance: The magnetic field at $\mathbf{r}$ is built by summing the transverse contribution of every current element; only the part of the current perpendicular to the line of sight contributes.

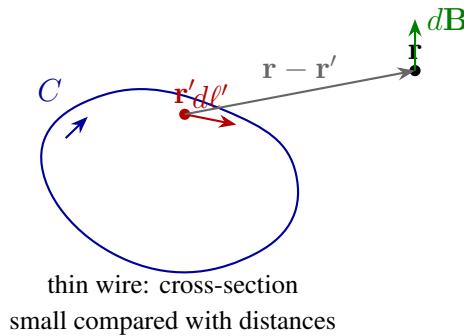


Figure 6: Geometry of the thin-wire Biot-Savart law. A current element $d\ell'$ at source point $\mathbf{r}'$ contributes a field at observation point $\mathbf{r}$ in the direction $d\ell' \times (\mathbf{r} - \mathbf{r}')$.

Thin-Wire Limit

For a wire whose cross-section is small compared with all distances of interest, write the volume element as $d^3 r' = dS' d\ell'$ and use

$$\mathbf{J}(\mathbf{r}') dS' d\ell' = I d\ell'. \quad (483)$$

Then (482) becomes

$$\mathbf{B}(\mathbf{r}) = \frac{I}{c} \oint_C \frac{d\ell' \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3}. \quad (484)$$

Significance: The familiar Biot-Savart line integral is a controlled approximation to the volume-current formula, valid when the wire is thin on the geometric scales being probed.

Remark 10. A truly open steady wire is impossible in an isolated magnetostatic problem: if current entered an endpoint and did not leave, charge would accumulate there. That is why the ideal line current is naturally a closed loop, or part of a larger closed circuit.

15.6 Force Between Two Current Loops

Physical Motivation

Once $\mathbf{B}$ is known, the force on moving charges is the magnetic part of the Lorentz force. For two current loops, the field of loop 2 exerts a force on the charges moving in loop 1. The self-field of loop 1 gives internal forces and is not part of the external force between the loops.

Let loop 1 carry current I_1 and loop 2 carry current I_2 . For source points $\mathbf{r}_1 \in C_1$ and $\mathbf{r}_2 \in C_2$, define

$$\mathbf{r}_{12} = \mathbf{r}_1 - \mathbf{r}_2, \quad r_{12} = |\mathbf{r}_{12}|. \quad (485)$$

The force on a small element $d\ell_1$ of loop 1 due to the magnetic field created by loop 2 is

$$d\mathbf{F}_{2 \rightarrow 1} = \frac{I_1}{c} d\ell_1 \times \mathbf{B}_2(\mathbf{r}_1). \quad (486)$$

Substituting Biot-Savart for $\mathbf{B}_2$,

$$\mathbf{F}_{2 \rightarrow 1} = \frac{I_1 I_2}{c^2} \oint_{C_1} \oint_{C_2} \frac{d\ell_1 \times (d\ell_2 \times \mathbf{r}_{12})}{r_{12}^3}. \quad (487)$$

This is correct, but the action-reaction symmetry is not yet transparent.

Simplifying the Double Integral

Use the vector triple-product identity

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b}). \quad (488)$$

With $\mathbf{a} = d\ell_1$, $\mathbf{b} = d\ell_2$, and $\mathbf{c} = \mathbf{r}_{12}$,

$$\frac{d\ell_1 \times (d\ell_2 \times \mathbf{r}_{12})}{r_{12}^3} = d\ell_2 \frac{d\ell_1 \cdot \mathbf{r}_{12}}{r_{12}^3} - \mathbf{r}_{12} \frac{d\ell_1 \cdot d\ell_2}{r_{12}^3}. \quad (489)$$

The first term vanishes after integration around the closed loop C_1 . Indeed, for fixed $\mathbf{r}_2$,

$$\frac{\mathbf{r}_{12}}{r_{12}^3} = -\nabla_{\mathbf{r}_1} \frac{1}{r_{12}}, \quad (490)$$

so

$$\oint_{C_1} d\ell_1 \cdot \frac{\mathbf{r}_{12}}{r_{12}^3} = -\oint_{C_1} d\ell_1 \cdot \nabla_{\mathbf{r}_1} \frac{1}{r_{12}} = 0. \quad (491)$$

The last equality says that the integral of a gradient around a closed path is zero. Therefore

$$\boxed{\mathbf{F}_{2 \rightarrow 1} = -\frac{I_1 I_2}{c^2} \oint_{C_1} \oint_{C_2} \frac{(d\ell_1 \cdot d\ell_2) \mathbf{r}_{12}}{r_{12}^3}.} \quad (492)$$

Significance: In the static limit the force law becomes explicitly compatible with Newton's third law, because interchanging 1 and 2 sends $\mathbf{r}_{12} \mapsto -\mathbf{r}_{12}$.

Thus

$$\boxed{\mathbf{F}_{2 \rightarrow 1} = -\mathbf{F}_{1 \rightarrow 2}.} \quad (493)$$

Significance: Full electrodynamics does not generally obey a particle-only action-reaction law because fields carry momentum, but the magnetostatic closed-loop limit restores the expected equal-and-opposite mechanical forces.

15.7 Parallel Wires as a Local Limit

Physical Motivation

The usual force between two long parallel wires is an idealization. Infinite wires do not exist physically, but two large nearby loops look locally like parallel straight wires if their radius of curvature is much larger than their separation.

For two long parallel wires separated by distance d , the force is proportional to the length L of the wire segment considered. Therefore the meaningful quantity is the force per unit length:

$$\boxed{\frac{F}{L} = \frac{2I_1 I_2}{c^2 d}} \quad (494)$$

with attraction for currents in the same direction and repulsion for currents in opposite directions.

Significance: The sign is not mysterious: the factor $d\ell_1 \cdot d\ell_2$ knows whether the currents are parallel or antiparallel, and the vector $\mathbf{r}_{12}$ points from the source element to the element being pushed.

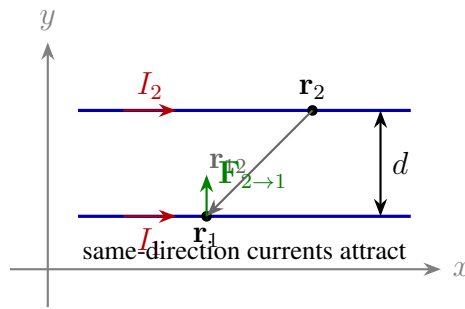


Figure 7: Parallel-wire limit of the loop-force formula. For currents in the same direction, $d\ell_1 \cdot d\ell_2 > 0$, and the force on the lower wire points toward the upper wire.

15.8 Summary

- Magnetostatics means time-independent fields with steady currents, not time-reversal invariant solutions.
- The equations reduce to $\nabla \cdot \mathbf{B} = 0$ and $\nabla \times \mathbf{B} = \frac{4\pi}{c} \mathbf{J}$, with $\nabla \cdot \mathbf{J} = 0$.
- Since $\nabla \cdot \mathbf{B} = 0$, write $\mathbf{B} = \nabla \times \mathbf{A}$.
- Gauge freedom $\mathbf{A} \mapsto \mathbf{A} + \nabla f$ allows the Coulomb gauge $\nabla \cdot \mathbf{A} = 0$.
- In Coulomb gauge, $\mathbf{A}$ satisfies $-\nabla^2 \mathbf{A} = \frac{4\pi}{c} \mathbf{J}$ and therefore $\mathbf{A}(\mathbf{r}) = \frac{1}{c} \int \mathbf{J}(\mathbf{r}') / |\mathbf{r} - \mathbf{r}'| d^3r'$.
- Taking a curl gives the Biot-Savart law $\mathbf{B}(\mathbf{r}) = \frac{1}{c} \int \mathbf{J}(\mathbf{r}') \times (\mathbf{r} - \mathbf{r}') / |\mathbf{r} - \mathbf{r}'|^3 d^3r'$.
- For a thin closed wire, $\mathbf{B}(\mathbf{r}) = \frac{I}{c} \oint d\ell' \times (\mathbf{r} - \mathbf{r}') / |\mathbf{r} - \mathbf{r}'|^3$.
- The force between two current loops is $\mathbf{F}_{2 \rightarrow 1} = -(I_1 I_2 / c^2) \oint \oint (d\ell_1 \cdot d\ell_2) \mathbf{r}_{12} / r_{12}^3$.
- For long parallel wires, $F/L = 2I_1 I_2 / (c^2 d)$, attractive for currents in the same direction.

16 Lecture 15 — Magnetic Multipole Expansion and the Magnetic Dipole Moment

Date: 26/05/2026

Overview

This lecture asks: what does the vector potential of a *localised* current distribution look like at large distances? The answer is a magnetic multipole expansion — the direct analogue of the electrostatic expansion $\phi = Q/r + \mathbf{p} \cdot \mathbf{r}/r^3 + \dots$. The crucial difference is that the monopole term always vanishes in magnetostatics ($\nabla \cdot \mathbf{J} = 0$ forces $\int \mathbf{J} d^3r' = 0$), so the leading term is always a *magnetic dipole*. The results connect directly to the physics of magnetism in materials.

- Monopole term: $\int \mathbf{J} d^3r' = 0$ (current conservation, proved rigorously).
- Magnetic dipole moment: $\boldsymbol{\mu} = \frac{1}{2c} \int \mathbf{r}' \times \mathbf{J}(\mathbf{r}') d^3r'$.
- Dipole vector potential: $\mathbf{A}^{\text{dip}} = \boldsymbol{\mu} \times \mathbf{r}/r^3$.
- For a thin current loop: $\boldsymbol{\mu} = (I/c) \boldsymbol{\Sigma}$, where $\boldsymbol{\Sigma}$ is the (oriented) area vector spanning the loop.
- For a planar loop of area S : $\boldsymbol{\mu} = (IS/c) \hat{n}$.
- For electrons: $\boldsymbol{\mu} = (e/2m_e c) \mathbf{L}$ (orbital gyromagnetic ratio).

16.1 Physical Setup: Why Expand?

Physical Motivation

Magnetism in materials arises from *molecular currents*: electrons orbit nuclei at speeds $v \sim \alpha c \approx 2200$ km/s (fine structure constant $\alpha = e^2/\hbar c \approx 1/137$). These currents are confined to atomic-scale regions of size $d \sim 1$ Å. At macroscopic observation distances $r \gg d$ we cannot — and do not need to — know the full microscopic $\mathbf{J}(\mathbf{r}')$. The multipole expansion replaces all that information with a hierarchy of moments.

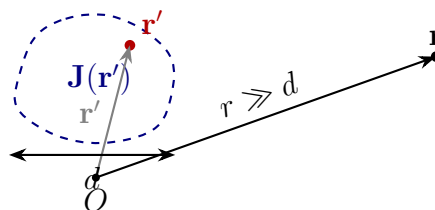


Figure 8: Source currents confined to a ball of radius $\sim d$; observation point $\mathbf{r}$ with $r \gg d$. The exact formula integrates over $\mathbf{r}'$ inside the source ball; the expansion gives a series in $r'/r \ll 1$.

Starting from the exact vector potential derived in Lecture 14,

$$\mathbf{A}(\mathbf{r}) = \frac{1}{c} \int \frac{\mathbf{J}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3r', \quad (495)$$

we expand $1/|\mathbf{r} - \mathbf{r}'|$ as a power series in the small ratio r'/r .

16.2 The Multipole Expansion

Mathematical Setup

Write the denominator as

$$|\mathbf{r} - \mathbf{r}'|^2 = r^2 - 2\mathbf{r} \cdot \mathbf{r}' + r'^2 = r^2(1 + \varepsilon), \quad \varepsilon \equiv \frac{r'^2 - 2\mathbf{r} \cdot \mathbf{r}'}{r^2} = O(r'/r). \quad (496)$$

Expanding $(1 + \varepsilon)^{-1/2}$ to first order in ε ,

$$\begin{aligned} \frac{1}{|\mathbf{r} - \mathbf{r}'|} &= \frac{1}{r}(1 + \varepsilon)^{-1/2} = \frac{1}{r}\left(1 - \frac{\varepsilon}{2} + \dots\right) \\ &= \frac{1}{r}\left(1 + \frac{\mathbf{r} \cdot \mathbf{r}'}{r^2} - \frac{r'^2}{2r^2} + \dots\right) = \frac{1}{r} + \frac{\mathbf{r} \cdot \mathbf{r}'}{r^3} + O\left(\frac{r'^2}{r^3}\right). \end{aligned} \quad (497)$$

Substituting into (495),

$$\boxed{\mathbf{A}(\mathbf{r}) = \underbrace{\frac{1}{cr} \int \mathbf{J}(\mathbf{r}') d^3r'}_{\text{monopole}} + \underbrace{\frac{1}{cr^3} \int (\mathbf{r} \cdot \mathbf{r}') \mathbf{J}(\mathbf{r}') d^3r'}_{\text{dipole}} + \dots} \quad (498)$$

Significance: The structure mirrors the electrostatic multipole expansion. The monopole coefficient is $\int \mathbf{J} d^3r'$; the dipole coefficient involves $\int r'_j \mathbf{J} d^3r'$ (a matrix in source coordinates). As we will show, the monopole always vanishes and the dipole carries all the long-range information.

16.3 No Magnetic Monopole

Physical Motivation

Every element of a steady current loop flows in one direction somewhere and returns in the opposite direction elsewhere — nothing accumulates. Macroscopically: current lines cannot terminate in bulk. Thus there is no “net magnetic charge” to build a monopole field.

Proof that $\int \mathbf{J} d^3r' = 0$

To show a vector is zero it suffices to show its inner product with every constant vector $\mathbf{a}$ is zero. Compute

$$\mathbf{a} \cdot \int \mathbf{J} d^3r' = \int (\mathbf{a} \cdot \mathbf{J}) d^3r'. \quad (499)$$

Use $\nabla' \cdot \mathbf{J} = 0$ and the product rule:

$$\nabla' \cdot [(\mathbf{a} \cdot \mathbf{r}')\mathbf{J}] = (\mathbf{a} \cdot \mathbf{r}') \underbrace{\nabla' \cdot \mathbf{J}}_0 + \mathbf{J} \cdot \underbrace{\nabla'(\mathbf{a} \cdot \mathbf{r}')}_{\mathbf{a}} = \mathbf{a} \cdot \mathbf{J}. \quad (500)$$

Therefore

$$\int (\mathbf{a} \cdot \mathbf{J}) d^3r' = \int \nabla' \cdot [(\mathbf{a} \cdot \mathbf{r}')\mathbf{J}] d^3r' \stackrel{\text{Gauss}}{=} \oint_{\partial V} (\mathbf{a} \cdot \mathbf{r}') \mathbf{J} \cdot d\mathbf{S}' = 0, \quad (501)$$

since $\mathbf{J} = 0$ on a surface taken outside all currents. Since $\mathbf{a}$ was arbitrary,

$$\boxed{\int \mathbf{J}(\mathbf{r}') d^3r' = \mathbf{0}.} \quad (502)$$

Significance: This holds for *every* steady localised current, regardless of geometry. It is not a special symmetry but a direct consequence of current conservation $\nabla \cdot \mathbf{J} = 0$. There is no magnetic monopole.

16.4 The Magnetic Dipole Moment

Mathematical Setup

The leading term is now the dipole:

$$\mathbf{A}^{\text{dip}}(\mathbf{r}) = \frac{1}{cr^3} \int (\mathbf{r} \cdot \mathbf{r}') \mathbf{J}(\mathbf{r}') d^3r'. \quad (503)$$

We simplify the integrand. First we need a lemma.

Mathematical Result 16.1 Current Symmetry Identity

For a steady, localised current ($\nabla \cdot \mathbf{J} = 0$, $\mathbf{J}|_{\partial V} = 0$):

$$\int (r'_i J_j + r'_j J_i) d^3r' = 0 \quad \text{for all } i, j. \quad (504)$$

Proof. Apply the product rule with $\nabla' \cdot \mathbf{J} = 0$:

$$\nabla' \cdot (r'_i r'_j \mathbf{J}) = r'_j J_i + r'_i J_j + r'_i r'_j \underbrace{\nabla' \cdot \mathbf{J}}_0 = r'_j J_i + r'_i J_j. \quad (505)$$

Integrating and applying Gauss's theorem with $\mathbf{J}|_{\partial V} = 0$ gives the result.

Simplifying the dipole integral. Decompose $r'_j J_i$ into symmetric and antisymmetric parts:

$$r'_j J_i = \underbrace{\frac{1}{2}(r'_j J_i + r'_i J_j)}_{\text{integrates to 0 by Lemma}} + \frac{1}{2}(r'_j J_i - r'_i J_j). \quad (506)$$

Hence

$$\int r'_j J_i d^3r' = \frac{1}{2} \int (r'_j J_i - r'_i J_j) d^3r'. \quad (507)$$

The antisymmetric combination is proportional to the cross product. Using the Levi-Civita identity $(\mathbf{r}' \times \mathbf{J})_k = \epsilon_{klm} r'_l J_m$, one can verify:

$$r'_i J_j - r'_j J_i = \epsilon_{ijk} (\mathbf{r}' \times \mathbf{J})_k, \quad (508)$$

so $r'_j J_i - r'_i J_j = -\epsilon_{ijk} (\mathbf{r}' \times \mathbf{J})_k$, giving

$$\int r'_j J_i d^3r' = -\frac{1}{2} \epsilon_{ijk} \int (\mathbf{r}' \times \mathbf{J})_k d^3r'. \quad (509)$$

Define the vector

$$\mathbf{M} \equiv \int \mathbf{r}' \times \mathbf{J}(\mathbf{r}') d^3r'. \quad (510)$$

The i -th component of $\mathbf{A}^{\text{dip}}$ is

$$A_i^{\text{dip}} = \frac{r_j}{cr^3} \int r'_j J_i d^3r' = -\frac{\epsilon_{ijk} r_j M_k}{2cr^3} = -\frac{(\mathbf{r} \times \mathbf{M})_i}{2cr^3} = \frac{(\mathbf{M} \times \mathbf{r})_i}{2cr^3}. \quad (511)$$

Defining

$$\boldsymbol{\mu} \equiv \frac{\mathbf{M}}{2c} = \frac{1}{2c} \int \mathbf{r}' \times \mathbf{J}(\mathbf{r}') d^3r', \quad (512)$$

we obtain the clean result:

Physical Concept 16.1: Magnetic Dipole Moment

The *magnetic dipole moment* of a localised current distribution is

$$\boldsymbol{\mu} = \frac{1}{2c} \int \mathbf{r}' \times \mathbf{J}(\mathbf{r}') d^3 r'. \quad (513)$$

In Gaussian units, $[\boldsymbol{\mu}] = \text{erg/G} = \text{G cm}^3$. The moment is linear in $\mathbf{J}$: moments of separate current loops simply add.

Physical Law 16.2 Dipole Vector Potential

The leading-order vector potential of a steady localised current at distance $r \gg d$ is

$$\mathbf{A}^{\text{dip}}(\mathbf{r}) = \frac{\boldsymbol{\mu} \times \mathbf{r}}{r^3}. \quad (514)$$

Significance: The entire long-range behaviour of any steady current distribution is encoded in a single vector $\boldsymbol{\mu}$. The field falls off as $\mathbf{A}^{\text{dip}} \sim 1/r^2$, and the magnetic field $\mathbf{B} = \nabla \times \mathbf{A}$ will fall off as $1/r^3$ — faster than the electrostatic dipole field, the same power law. This is the fundamental building block of the theory of magnetic materials.

Note:

This is the exact magnetic analogue of the electric dipole potential $\phi^{\text{dip}} = \mathbf{p} \cdot \mathbf{r}/r^3$. The difference is that the “source” is $\boldsymbol{\mu}$ (built from $\mathbf{r}' \times \mathbf{J}$) rather than $\mathbf{p}$ (built from $\mathbf{r}'\rho$), and the potential is a vector (because $\mathbf{A}$ is a vector field).

16.5 Geometric Interpretation: Current Loop

Physical Motivation

For a thin wire carrying steady current I , the volume current element becomes a line element: $\mathbf{J}(\mathbf{r}') d^3 r' \rightarrow I d\ell'$. The magnetic moment reduces to a line integral around the closed loop C :

$$\boldsymbol{\mu} = \frac{1}{2c} \oint_C I \mathbf{r}' \times d\ell' = \frac{I}{2c} \oint_C \mathbf{r}' \times d\ell'. \quad (515)$$

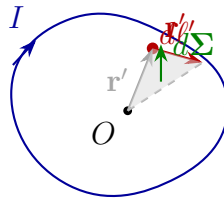


Figure 9: Area element $d\Sigma = \frac{1}{2} \mathbf{r}' \times d\ell'$ is the vector area of the triangle swept from origin O to element $d\ell'$ at $\mathbf{r}'$. Summing all triangles fills the surface spanning the loop.

The quantity $\frac{1}{2} \mathbf{r}' \times d\ell'$ is exactly the vector area of the infinitesimal triangle with two sides $\mathbf{r}'$ and $\mathbf{r}' + d\ell'$:

$$d\Sigma = \frac{1}{2} \mathbf{r}' \times d\ell'. \quad (516)$$

Integrating around the loop fills in a surface $\mathcal{S}$ spanning C :

$$\Sigma \equiv \oint_C \frac{1}{2} \mathbf{r}' \times d\ell' = \iint_{\mathcal{S}} d\mathbf{S}'. \quad (517)$$

Physical Concept 16.2: Magnetic Moment of a Current Loop

For a closed current loop carrying current I ,

$$\boldsymbol{\mu} = \frac{I}{c} \boldsymbol{\Sigma}, \quad (518)$$

where $\boldsymbol{\Sigma} = \iint_{\mathcal{S}} d\mathbf{S}'$ is the oriented area vector of any surface $\mathcal{S}$ bounded by the loop. The direction follows the right-hand rule with respect to the sense of the current.

Significance: $\boldsymbol{\mu}$ depends only on the current and the total area vector — not on the shape of the loop, nor on the choice of origin.

Surface Independence of $\boldsymbol{\Sigma}$

The formula (518) requires $\boldsymbol{\Sigma}$ to be independent of the surface $\mathcal{S}$ chosen to span the loop. Let $\mathcal{S}_1$ and $\mathcal{S}_2$ be two such surfaces, with outward normals oriented so that together they bound a closed volume V . Then for any constant $\mathbf{a}$,

$$\mathbf{a} \cdot (\boldsymbol{\Sigma}_1 - \boldsymbol{\Sigma}_2) = \oint_{\partial V} \mathbf{a} \cdot d\mathbf{S}' = \int_V \nabla' \cdot \mathbf{a} d^3r' = 0, \quad (519)$$

since $\mathbf{a}$ is constant. Because $\mathbf{a}$ is arbitrary, $\boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}_2$.

Planar Loop

For a planar loop all area elements point in the same direction $\hat{n}$ (the right-hand normal), so the integral just sums scalar areas:

Consequence 16.3 Magnetic Moment of a Planar Loop

For a planar loop of area S carrying current I ,

$$\boldsymbol{\mu} = \frac{IS}{c} \hat{n}, \quad (520)$$

where $\hat{n}$ is the right-hand normal to the plane of the loop.

Significance: This is the simplest and most widely used result. A circular loop of radius R has $|\boldsymbol{\mu}| = \pi R^2 I/c$. Two parallel co-axial loops with the same current in the same sense have $|\boldsymbol{\mu}| = 2\pi R^2 I/c$ (moments add).

16.6 Magnetic Moment and Angular Momentum**Physical Motivation**

Inside a material, the molecular currents arise from electrons orbiting nuclei. We model these as N point particles (electrons) with charge e , mass m_e , positions $\mathbf{r}_n(t)$, and velocities $\mathbf{v}_n(t)$. The corresponding current density is

$$\mathbf{J}(\mathbf{r}, t) = e \sum_n \mathbf{v}_n(t) \delta^{(3)}(\mathbf{r} - \mathbf{r}_n(t)). \quad (521)$$

Derivation

Insert (521) into (513):

$$\begin{aligned}
 \boldsymbol{\mu} &= \frac{1}{2c} \int \mathbf{r}' \times \mathbf{J}(\mathbf{r}') d^3r' \\
 &= \frac{e}{2c} \sum_n \int \mathbf{r}' \times \mathbf{v}_n \delta^{(3)}(\mathbf{r}' - \mathbf{r}_n) d^3r' \\
 &= \frac{e}{2c} \sum_n \mathbf{r}_n \times \mathbf{v}_n.
 \end{aligned} \tag{522}$$

The orbital angular momentum of the n -th electron is $\mathbf{L}_n = m_e \mathbf{r}_n \times \mathbf{v}_n$, so $\mathbf{r}_n \times \mathbf{v}_n = \mathbf{L}_n/m_e$. Summing over all electrons,

Physical Law 16.4 Orbital Gyromagnetic Relation

For a system of electrons with total orbital angular momentum $\mathbf{L} = \sum_n \mathbf{L}_n$,

$$\boxed{\boldsymbol{\mu} = \frac{e}{2m_e c} \mathbf{L}.} \tag{523}$$

The ratio $\boldsymbol{\mu}/L = e/(2m_e c)$ is the *gyromagnetic ratio* for orbital motion.

Significance: Any rotating charge generates a magnetic moment proportional to its angular momentum. The orbital gyromagnetic ratio $e/(2m_e c)$ follows purely from classical electrodynamics. In quantum mechanics, spin also carries angular momentum: since orbital and spin angular momenta contribute to $\boldsymbol{\mu}$, and spin behaves like a half-integer angular momentum, the Dirac equation gives

$$\boldsymbol{\mu}_{\text{spin}} = g \frac{e}{2m_e c} \mathbf{S}, \quad g \approx 2, \tag{524}$$

with the Landé g -factor $g \approx 2$ (exact $g = 2$ from Dirac; small corrections from QED). Thus **magnetism is fundamentally a quantum phenomenon**: without spin, completely random orbital orientation gives zero net moment; spin alignment is what creates observable ferromagnetism.

Remark 11. The derivation used instantaneous positions $\mathbf{r}_n(t)$. For atomic electrons orbiting at $v \sim \alpha c$, $\mathbf{r}_n(t)$ oscillates at atomic frequencies. The correct procedure is to time-average: the total $\mathbf{L}$ is conserved and hence time-independent, so (523) holds for the averaged magnetic moment. The connection also justifies replacing the complicated quantum orbital with an effective steady current loop — the central approximation of classical magnetism.

16.7 Summary

- The vector potential of localised currents expands as monopole + dipole + higher multipoles; (498).
- **Monopole vanishes:** $\int \mathbf{J} d^3r' = 0$ because $\nabla \cdot \mathbf{J} = 0$ (steady current loops have no net “current charge”).
- **Magnetic dipole moment:** $\boldsymbol{\mu} = \frac{1}{2c} \int \mathbf{r}' \times \mathbf{J} d^3r'$.
- **Dipole vector potential:** $\mathbf{A}^{\text{dip}} = \boldsymbol{\mu} \times \mathbf{r}/r^3$; falls off as r^{-2} .
- **Current loop:** $\boldsymbol{\mu} = (I/c)\boldsymbol{\Sigma}$; $\boldsymbol{\Sigma}$ is surface-choice independent.
- **Planar loop:** $\boldsymbol{\mu} = (IS/c)\hat{n}$ with right-hand rule.

- **Orbital electrons:** $\boldsymbol{\mu} = (e/2m_e c)\mathbf{L}$.
- Spin contributes $\boldsymbol{\mu}_{\text{spin}} = (ge/2m_e c)\mathbf{S}$ with $g \approx 2$; explaining real ferromagnetism requires quantum mechanics.

17 Lecture 16 – Magnetic Dipoles in External Fields and Magnetic Materials

Date: 31/05/2026

Overview

Lecture 15 introduced the magnetic dipole moment $\boldsymbol{\mu}$ as the leading far-field data of a localized current distribution. This lecture explains why that concept is the right microscopic language for magnetic materials. A piece of matter can be viewed, at sufficiently long wavelengths, as many tiny current loops: orbital electron currents and spin magnetic moments. The external magnetic field does not see the detailed atomic motion directly; it sees the dipole moments those motions produce.

The main results are:

- the magnetic field of a point magnetic dipole is $\mathbf{B}_{\text{dip}}(\mathbf{r}) = [3\hat{\mathbf{r}}(\boldsymbol{\mu} \cdot \hat{\mathbf{r}}) - \boldsymbol{\mu}]/r^3$ away from the dipole;
- the interaction term for a small current loop in a slowly varying external field is $L_{\text{int}} = \boldsymbol{\mu} \cdot \mathbf{B}_{\text{ext}}$;
- therefore the potential energy is $U = -\boldsymbol{\mu} \cdot \mathbf{B}_{\text{ext}}$;
- the force on a dipole is $\mathbf{F} = \nabla(\boldsymbol{\mu} \cdot \mathbf{B})$, so a uniform magnetic field exerts no net force on an ideal dipole;
- the torque is $\boldsymbol{\tau} = \boldsymbol{\mu} \times \mathbf{B}$, so a uniform field can rotate the dipole even when it cannot translate it;
- for orbital motion, combining $\boldsymbol{\mu} = (e/2m_e c)\mathbf{L}$ with $d\mathbf{L}/dt = \boldsymbol{\tau}$ gives Larmor precession;
- magnetic materials are classified by how their microscopic dipoles respond: paramagnets align with the applied field, diamagnets oppose it, ferromagnets retain magnetization, and superconductors expel magnetic field.

17.1 Magnetic Field of a Dipole

Physical Motivation

Last lecture gave the far-zone vector potential of a localized current distribution,

$$\mathbf{A}_{\text{dip}}(\mathbf{r}) = \frac{\boldsymbol{\mu} \times \mathbf{r}}{r^3}. \quad (525)$$

For matter, this is the microscopic building block: an atom or molecule is replaced by a pointlike magnetic dipole when we observe it on length scales much larger than atomic size. To understand how many such dipoles produce magnetization, we first need the magnetic field of one dipole.

Derivation

Use the vector identity

$$\nabla \times (\mathbf{a} \times \mathbf{c}) = \mathbf{a}(\nabla \cdot \mathbf{c}) - \mathbf{c}(\nabla \cdot \mathbf{a}) + (\mathbf{c} \cdot \nabla)\mathbf{a} - (\mathbf{a} \cdot \nabla)\mathbf{c}. \quad (526)$$

Here $\mathbf{a} = \boldsymbol{\mu}$ is constant and $\mathbf{c} = \mathbf{r}/r^3$, so $\nabla \cdot \boldsymbol{\mu} = 0$ and $(\mathbf{r}/r^3 \cdot \nabla)\boldsymbol{\mu} = 0$. Thus

$$\mathbf{B}_{\text{dip}} = \nabla \times \left(\frac{\boldsymbol{\mu} \times \mathbf{r}}{r^3} \right) = \boldsymbol{\mu} \nabla \cdot \left(\frac{\mathbf{r}}{r^3} \right) - (\boldsymbol{\mu} \cdot \nabla) \left(\frac{\mathbf{r}}{r^3} \right). \quad (527)$$

Away from the origin,

$$\nabla \cdot \left(\frac{\mathbf{r}}{r^3} \right) = 0 \quad (r \neq 0), \quad (528)$$

while componentwise

$$\begin{aligned} (\boldsymbol{\mu} \cdot \nabla) \left(\frac{r_i}{r^3} \right) &= \mu_j \partial_j (r_i r^{-3}) \\ &= \mu_j (\delta_{ij} r^{-3} + r_i \partial_j r^{-3}) \\ &= \frac{\mu_i}{r^3} - 3 \frac{r_i (\boldsymbol{\mu} \cdot \mathbf{r})}{r^5}. \end{aligned} \quad (529)$$

Substituting this into (527) gives

$$\mathbf{B}_{\text{dip}}(\mathbf{r}) = \frac{3\mathbf{r}(\boldsymbol{\mu} \cdot \mathbf{r})}{r^5} - \frac{\boldsymbol{\mu}}{r^3} = \frac{3\hat{\mathbf{r}}(\boldsymbol{\mu} \cdot \hat{\mathbf{r}}) - \boldsymbol{\mu}}{r^3}, \quad r \neq 0. \quad (530)$$

Significance: The dipole field decays as $1/r^3$, exactly like the electric dipole field. The formula is meaningful only outside the microscopic source; if one approaches the current loop itself, the point-dipole approximation must be replaced by the actual current distribution.

Remark 12. Distributionally, there is also a contact term at the dipole position. In this lecture we use the far-field expression (530), because magnetic materials are modeled by replacing each small current loop by its field outside atomic scales.

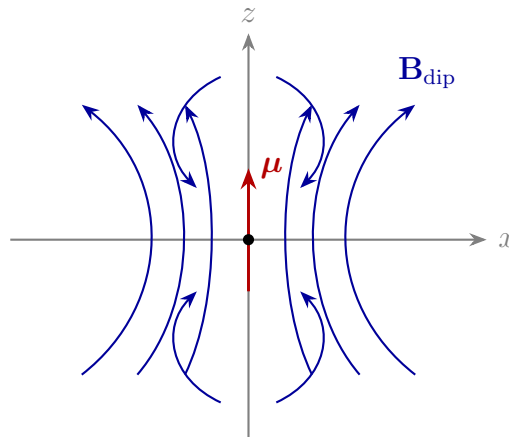


Figure 10: Schematic field pattern of a magnetic dipole. Far from the microscopic current loop, the entire source is represented by the single vector $\boldsymbol{\mu}$.

17.2 Interaction of a Small Current Loop with an External Field

Physical Motivation

To study a material in an applied magnetic field, we need the energy of one microscopic current loop in that field. The applied field is assumed to vary slowly across the loop, because the loop is atomic in size while laboratory fields vary on macroscopic scales. This is the same logic as a multipole expansion: keep the leading term first.

The covariant interaction between matter current and electromagnetic field can be written as

$$S_{\text{int}} = -\frac{1}{c^2} \int J^\mu A_\mu d^4x. \quad (531)$$

In the magnetostatic neutral sector, $J^0 = 0$ and $A^0 = \phi = 0$. With the mostly-minus metric convention used in this course, this reduces to

$$S_{\text{int}} = \frac{1}{c} \int dt \int \mathbf{J}(\mathbf{r}) \cdot \mathbf{A}(\mathbf{r}) d^3r. \quad (532)$$

For a thin current loop carrying current I ,

$$\mathbf{J}(\mathbf{r}) d^3r \longrightarrow I d\ell, \quad (533)$$

so

$$\begin{aligned} S_{\text{int}} &= \frac{I}{c} \int dt \oint_C \mathbf{A} \cdot d\ell \\ &= \frac{I}{c} \int dt \iint_{\Sigma} (\nabla \times \mathbf{A}) \cdot d\mathbf{S} \\ &= \frac{I}{c} \int dt \iint_{\Sigma} \mathbf{B} \cdot d\mathbf{S}. \end{aligned} \quad (534)$$

If $\mathbf{B}$ is nearly constant over the loop,

$$\iint_{\Sigma} \mathbf{B} \cdot d\mathbf{S} \simeq \mathbf{B} \cdot \Sigma, \quad (535)$$

where Σ is the oriented area vector. Since $\boldsymbol{\mu} = (I/c)\Sigma$, we obtain

$$L_{\text{int}} = \boldsymbol{\mu} \cdot \mathbf{B}_{\text{ext}}. \quad (536)$$

Significance: The microscopic loop couples to the external field only through its dipole moment at leading order. This is why $\boldsymbol{\mu}$ is the natural variable for magnetic matter.

Since a mechanical Lagrangian has the structure $L = T - U$, the potential energy is

$$U = -\boldsymbol{\mu} \cdot \mathbf{B}_{\text{ext}}. \quad (537)$$

Significance: The energy is minimized when $\boldsymbol{\mu}$ points along $\mathbf{B}$. A misaligned magnetic moment is therefore not in equilibrium: the field tries to rotate it.

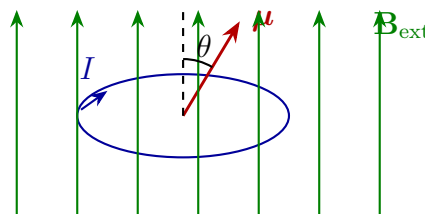


Figure 11: A microscopic current loop in a slowly varying external field. The leading interaction energy is $U = -\boldsymbol{\mu} \cdot \mathbf{B} = -\mu B \cos \theta$.

17.3 Force on a Magnetic Dipole

Conceptual Background

Once the potential energy is known, the force follows from the spatial variation of that energy. We keep $\boldsymbol{\mu}$ fixed while differentiating with respect to the loop's position; physically, the microscopic current loop is moved through an external field whose value changes from point to point.

From (537),

$$\mathbf{F} = -\nabla U = \nabla(\boldsymbol{\mu} \cdot \mathbf{B}). \quad (538)$$

Thus

$$F_i = \partial_i(\mu_j B_j), \quad \mathbf{F} = \nabla(\boldsymbol{\mu} \cdot \mathbf{B}). \quad (539)$$

Significance: A magnetic dipole feels a translational force only if the external field is nonuniform. A uniform field can choose an orientation, but it cannot choose a preferred position.

For constant $\boldsymbol{\mu}$ one often writes

$$\mathbf{F} = (\boldsymbol{\mu} \cdot \nabla)\mathbf{B} \quad (540)$$

when the region is current-free so that $\nabla \times \mathbf{B} = 0$. The form (538) is the direct energy-gradient statement.

Remark 13. This does not contradict the magnetic Lorentz force being perpendicular to the instantaneous velocity of a charge. Holding a steady current loop together and moving it as a composite object involves external constraints and induced electric fields; the effective dipole energy describes the work needed to move the whole object in an inhomogeneous field.

17.4 Torque on a Magnetic Dipole

Physical Motivation

In a uniform field, (539) gives zero net force. But the energy $U = -\mu B \cos \theta$ still depends on orientation, so the dipole must feel a torque. This is exactly what happens to a compass needle: it does not translate in a uniform field, but it rotates.

Loop Derivation

For a current element $d\ell$ in a magnetic field,

$$d\mathbf{F} = \frac{I}{c} d\ell \times \mathbf{B}. \quad (541)$$

The torque element about the loop origin is

$$d\boldsymbol{\tau} = \mathbf{r} \times d\mathbf{F} = \frac{I}{c} \mathbf{r} \times (d\ell \times \mathbf{B}). \quad (542)$$

Using

$$\mathbf{r} \times (d\ell \times \mathbf{B}) = d\ell(\mathbf{r} \cdot \mathbf{B}) - \mathbf{B}(\mathbf{r} \cdot d\ell), \quad (543)$$

and integrating around the loop, one finds that the total derivative terms around the closed path vanish. Keeping the leading term for a uniform $\mathbf{B}$ across the loop, the result is

$$\boldsymbol{\tau} = \boldsymbol{\mu} \times \mathbf{B}. \quad (544)$$

Significance: The torque tries to rotate $\boldsymbol{\mu}$ toward $\mathbf{B}$, precisely the orientation that minimizes $U = -\boldsymbol{\mu} \cdot \mathbf{B}$.

The same result follows immediately from the energy:

$$U(\theta) = -\mu B \cos \theta, \quad \tau_{\perp} = -\frac{\partial U}{\partial \theta} = -\mu B \sin \theta, \quad (545)$$

where the sign indicates rotation toward smaller θ .

17.5 Larmor Precession

Physical Motivation

If a dipole is tied to an angular momentum, the torque does not simply make it fall straight into the field direction. Instead, just like a spinning top in gravity, the angular momentum vector precesses around the applied field.

For orbital electron motion, Lecture 15 gave

$$\boldsymbol{\mu} = \gamma \mathbf{L}, \quad \gamma = \frac{e}{2m_e c}. \quad (546)$$

The rotational equation of motion is

$$\frac{d\mathbf{L}}{dt} = \boldsymbol{\tau}. \quad (547)$$

Using (544),

$$\frac{d\mathbf{L}}{dt} = \boldsymbol{\mu} \times \mathbf{B} = \gamma \mathbf{L} \times \mathbf{B}. \quad (548)$$

Equivalently,

$$\boxed{\frac{d\mathbf{L}}{dt} = \boldsymbol{\Omega}_L \times \mathbf{L}, \quad \boldsymbol{\Omega}_L = -\gamma \mathbf{B} = -\frac{e}{2m_e c} \mathbf{B}.} \quad (549)$$

Significance: The angular momentum keeps its magnitude fixed and rotates around $\mathbf{B}$. The magnetic field creates a preferred axis, not a radial pull.

Indeed,

$$\frac{d}{dt} L^2 = 2\mathbf{L} \cdot \frac{d\mathbf{L}}{dt} = 2\gamma \mathbf{L} \cdot (\mathbf{L} \times \mathbf{B}) = 0, \quad (550)$$

so $|\mathbf{L}|$ is constant. Also, for uniform constant $\mathbf{B}$,

$$\frac{d}{dt} (\mathbf{L} \cdot \mathbf{B}) = \frac{d\mathbf{L}}{dt} \cdot \mathbf{B} = \gamma (\mathbf{L} \times \mathbf{B}) \cdot \mathbf{B} = 0, \quad (551)$$

so the angle between $\mathbf{L}$ and $\mathbf{B}$ is constant. A vector of fixed length and fixed angle relative to $\mathbf{B}$ can only precess around $\mathbf{B}$.

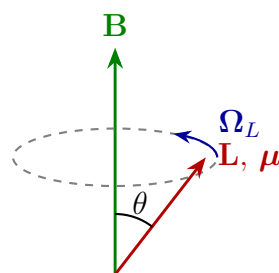


Figure 12: Larmor precession: the torque is perpendicular to $\mathbf{L}$, so it changes the direction of $\mathbf{L}$ while preserving its magnitude and its angle with $\mathbf{B}$.

17.6 From One Dipole to Magnetic Materials

Conceptual Background

The microscopic picture of magnetic matter is a collection of dipoles. The macroscopic question is how those dipoles arrange themselves in response to an external field. This is not fully explainable by classical mechanics: real magnetism, especially spin and ferromagnetism, requires quantum mechanics. The classical dipole model still gives the correct organizing language.

Physical Concept 17.1: Magnetization

The magnetization $\mathbf{M}$ is magnetic dipole moment per unit volume:

$$\mathbf{M}(\mathbf{r}) = \frac{d\boldsymbol{\mu}_{\text{total}}}{dV}. \quad (552)$$

It measures the coarse-grained alignment of microscopic magnetic moments.

Significance: A material produces its own magnetic field once its microscopic dipoles acquire a nonzero average orientation.

Paramagnetism

In a paramagnet, microscopic moments tend to align with the applied field:

$$\mathbf{M} \parallel \mathbf{B}_{\text{ext}}. \quad (553)$$

Significance: The material reinforces the applied field. With the field off, thermal disorder usually randomizes the moments, so the net magnetization disappears.

Diamagnetism

In a diamagnet, the induced magnetic response points opposite the applied field:

$$\mathbf{M} \text{ opposes } \mathbf{B}_{\text{ext}}. \quad (554)$$

Significance: The material partially cancels the applied field. Classically, this is connected to the small induced orbital currents created when the external field perturbs electron motion.

Ferromagnetism

In a ferromagnet, the material can retain magnetization even after the external field is removed:

$$\mathbf{M} \neq 0 \text{ can remain when } \mathbf{B}_{\text{ext}} = 0. \quad (555)$$

Significance: A ferromagnet is a material with memory: microscopic moments remain ordered macroscopically. Explaining why this order persists is a quantum many-body problem, not a purely classical magnetostatic calculation.

Superconductors as Perfect Diamagnets

A superconductor responds even more strongly: in the Meissner state the magnetic field inside is expelled,

$$\mathbf{B}_{\text{inside}} = 0. \quad (556)$$

Significance: A superconductor is the magnetic analogue of an ideal conductor in electrostatics: it rearranges currents so that the interior field vanishes. In magnetism this is the Meissner effect, not merely ordinary perfect conductivity.

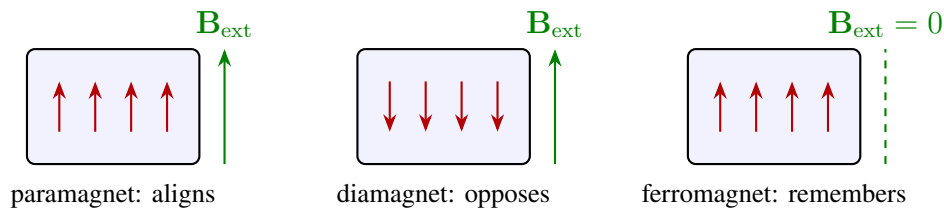


Figure 13: Three schematic magnetic responses. Paramagnets align with the applied field, diamagnets oppose it, and ferromagnets can retain alignment after the applied field is removed.

17.7 Summary

- The dipole vector potential $\mathbf{A}_{\text{dip}} = \boldsymbol{\mu} \times \mathbf{r}/r^3$ gives, for $r \neq 0$, the dipole field $\mathbf{B}_{\text{dip}} = [3\hat{\mathbf{r}}(\boldsymbol{\mu} \cdot \hat{\mathbf{r}}) - \boldsymbol{\mu}]/r^3$.
- For a small current loop in a slowly varying external field, $L_{\text{int}} = \boldsymbol{\mu} \cdot \mathbf{B}$ and $U = -\boldsymbol{\mu} \cdot \mathbf{B}$.
- The force on a magnetic dipole is $\mathbf{F} = \nabla(\boldsymbol{\mu} \cdot \mathbf{B})$; it vanishes in a uniform field.
- The torque is $\boldsymbol{\tau} = \boldsymbol{\mu} \times \mathbf{B}$; a uniform field can rotate a dipole even when it cannot translate it.
- For orbital electron motion, $\boldsymbol{\mu} = (e/2m_e c)\mathbf{L}$, so $d\mathbf{L}/dt = (e/2m_e c)\mathbf{L} \times \mathbf{B}$: Larmor precession.
- Magnetic materials are understood macroscopically through magnetization $\mathbf{M} = d\boldsymbol{\mu}_{\text{total}}/dV$.
- Paramagnets align with the applied field, diamagnets oppose it, ferromagnets retain magnetization, and superconductors expel magnetic field.

18 Lecture 17 – Magnetization Currents, the $\mathbf{H}$ Field, and the Magnetic Scalar Potential

Date: 02/06/2026

Overview

Lecture 16 established that a piece of matter behaves, on macroscopic length scales, as a continuous distribution of magnetic dipoles, summarized by the magnetization $\mathbf{M}(\mathbf{r}) = d\boldsymbol{\mu}_{\text{total}}/dV$. This lecture closes the magnetostatics of matter in three steps:

- identify the microscopic origin of $\mathbf{M}$ (orbital currents and spin) and write the *total* current as free plus magnetization currents;
- start from the dipole vector potential, superpose it over the body, and prove that magnetization is equivalent to a bound *volume* current $\mathbf{J}_M = c \nabla \times \mathbf{M}$ plus a bound *surface* current $\mathbf{K}_M = c \mathbf{M} \times \hat{\mathbf{n}}$;
- insert these into Ampère's law, which forces the definition of the magnetic intensity $\mathbf{H} = \mathbf{B} - 4\pi\mathbf{M}$ sourced only by free currents, and – when free currents are absent – a *magnetic scalar potential* obeying a Poisson equation identical in form to electrostatics.

The recurring theme is that everything magnetic matter does is captured by $\mathbf{M}$: once $\mathbf{M}$ (or equivalently $\mathbf{J}_M$) is given, the field is fully determined.

18.1 Total Current in Magnetic Matter

Physical Motivation

A magnetic field is produced by moving charge. In a laboratory there are two very different kinds of moving charge. The first is the current we deliberately drive through wires – the current in the solenoid we use to magnetize a sample. The second appears only once matter is present: the perpetual microscopic currents bound inside atoms and molecules. In equilibrium these microscopic currents are randomly oriented and average to zero, but an external field can order them and produce a net effect. The macroscopic field therefore responds to the *sum* of these contributions.

Physical Concept 18.1: Free and Magnetization Currents

The total current density measured at macroscopic (classical) resolution splits as

$$\mathbf{J}_{\text{total}} = \mathbf{J}_f + \mathbf{J}_M. \quad (557)$$

$\mathbf{J}_f$ are *free currents*: conduction electrons travelling long distances through wires. $\mathbf{J}_M$ are *magnetization* (or bound) currents: the coarse-grained average of the microscopic currents bound to atoms.

Significance: Only $\mathbf{J}_f$ is under direct experimental control; the whole physics of magnetic matter lives in how $\mathbf{J}_M$ is generated and how it relates to $\mathbf{M}$.

Microscopic Origin of the Magnetization Current

The measured macroscopic current is an average of a microscopic current that itself has two pieces,

$$\mathbf{J}_m = \mathbf{J}_o + \mathbf{J}_s, \quad (558)$$

where $\mathbf{J}_o$ is the genuine *orbital* current produced by the motion of bound charges, and $\mathbf{J}_s$ is an *effective* current standing in for electron spin.

For the orbital part, a small volume ΔV contains a huge number of microscopic current loops; coarse-graining replaces them by their dipole moment per unit volume. Using the dipole-moment definition $\boldsymbol{\mu} = \frac{1}{2c} \int \mathbf{r}' \times \mathbf{J} d^3x'$ from Lecture 15, the orbital magnetization is

$$\mathbf{M}_o(\mathbf{r}) = \frac{1}{2c \Delta V} \int_{\Delta V} \mathbf{r}' \times \mathbf{J}_o(\mathbf{r}') d^3x' = \frac{d\boldsymbol{\mu}_o}{dV}. \quad (559)$$

Significance: The complicated microscopic loop structure is compressed into a single field, the dipole-moment density; this is the only orbital information that survives at macroscopic distances.

The Spin Contribution

Spin is an intrinsic angular momentum of the electron with no classical analogue: it is *not* the circulation of any charge. Nevertheless, because orbital angular momentum $\mathbf{L}$ produces a magnetic moment $\boldsymbol{\mu} = (e/2m_e c)\mathbf{L}$, and spin $\mathbf{S}$ is also an angular momentum, consistency demands that spin produce a magnetic moment too. We therefore model its effect by an effective current whose magnetic moment matches that of spin. Its magnetization is a sum over the spins per unit volume,

$$\mathbf{M}_s(\mathbf{r}) = \sum_k \mathbf{m}_k \delta^{(3)}(\mathbf{r} - \mathbf{r}_k), \quad \mathbf{m}_k \propto \mathbf{S}_k \sim \frac{e}{2m_e c} \mathbf{S}_k, \quad (560)$$

the sum running over all spins inside a unit volume. Although the underlying cause of $\mathbf{M}_s$ is purely quantum, the resulting magnetic effect is measured by ordinary classical instruments.

Note:

The magnitude of spin (the quantum number $\frac{1}{2}$) is Lorentz invariant like a rest mass, but spin is a *vector*: its direction can change, which is exactly what lets an external field reorient it. The total angular momentum – and hence the total current – does transform between frames, while the spin magnitude does not.

Total Magnetization

For our (classical, phenomenological) purposes we never separate the two sources: a classical experiment sees neither the orbital motion around an atom nor the spin directly. We work only with the total magnetization

$$\mathbf{M} = \mathbf{M}_o + \mathbf{M}_s. \quad (561)$$

Significance: The macroscopic theory needs only $\mathbf{M}$; the distinction between “real” orbital currents and “effective” spin currents is a microscopic detail invisible to classical measurement.

18.2 Vector Potential of a Magnetized Body

Physical Motivation

Suppose the magnetization $\mathbf{M}(\mathbf{r})$ is given throughout a body. We want the field it produces. Because the theory is linear, the total field is the superposition of the fields of all the microscopic dipoles, plus (separately) the field of any free currents, which we already know how to handle. Each small volume d^3r' carries a dipole moment $d\boldsymbol{\mu} = \mathbf{M}(\mathbf{r}') d^3r'$, and a dipole at $\mathbf{r}'$ contributes the vector potential $\boldsymbol{\mu} \times (\mathbf{r} - \mathbf{r}')/|\mathbf{r} - \mathbf{r}'|^3$ from Lecture 15.

Mathematical Setup

Summing (integrating) these contributions over the body gives the magnetization vector potential

$$\mathbf{A}_M(\mathbf{r}) = \int d^3r' \frac{\mathbf{M}(\mathbf{r}') \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3}. \quad (562)$$

Significance: This is the exact analogue of replacing a charge distribution by its dipole density; it is the leading (dipole) approximation to the field of matter, valid because classical probes always sit at distances enormous compared with atomic scales.

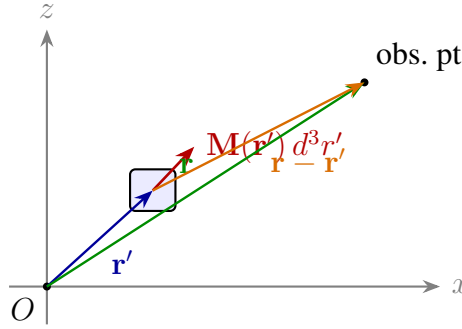


Figure 14: Each volume element at $\mathbf{r}'$ carries a dipole $d\boldsymbol{\mu} = \mathbf{M}(\mathbf{r}') d^3r'$ and contributes to the vector potential at the observation point $\mathbf{r}$ through the separation vector $\mathbf{r} - \mathbf{r}'$.

18.3 Magnetization (Bound) Currents

Mathematical Setup

We rewrite (562) so that it looks like the magnetostatic Poisson solution $\mathbf{A} = \frac{1}{c} \int \mathbf{J}/|\mathbf{r} - \mathbf{r}'| d^3r'$, and read off the effective current. The key observation is that the separation factor is a primed gradient:

$$\nabla' \frac{1}{|\mathbf{r} - \mathbf{r}'|} = \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|^3}, \quad (563)$$

where ∇' differentiates with respect to $\mathbf{r}'$. Apply the product identity $\nabla' \times (f\mathbf{V}) = (\nabla' f) \times \mathbf{V} + f(\nabla' \times \mathbf{V})$ with $f = 1/|\mathbf{r} - \mathbf{r}'|$ and $\mathbf{V} = \mathbf{M}(\mathbf{r}')$:

$$\nabla' \times \left(\frac{\mathbf{M}}{|\mathbf{r} - \mathbf{r}'|} \right) = \left(\nabla' \frac{1}{|\mathbf{r} - \mathbf{r}'|} \right) \times \mathbf{M} + \frac{\nabla' \times \mathbf{M}}{|\mathbf{r} - \mathbf{r}'|}. \quad (564)$$

Substituting (563) and isolating the integrand of (562),

$$\begin{aligned} \frac{\mathbf{M} \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3} &= - \left(\frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|^3} \right) \times \mathbf{M} \\ &= \frac{\nabla' \times \mathbf{M}}{|\mathbf{r} - \mathbf{r}'|} - \nabla' \times \left(\frac{\mathbf{M}}{|\mathbf{r} - \mathbf{r}'|} \right). \end{aligned} \quad (565)$$

Converting the Curl Term to a Surface Integral

For the last term we use the curl form of Gauss's theorem,

$$\int_V (\nabla \times \mathbf{A}) d^3r = \oint_S \hat{\mathbf{n}} \times \mathbf{A} d\sigma = \oint_S d\boldsymbol{\sigma} \times \mathbf{A}, \quad (566)$$

with $\hat{\mathbf{n}}$ the outward normal of the body's surface. Hence

$$\int_V \nabla' \times \left(\frac{\mathbf{M}}{|\mathbf{r} - \mathbf{r}'|} \right) d^3r' = \oint_S \frac{\hat{\mathbf{n}} \times \mathbf{M}}{|\mathbf{r} - \mathbf{r}'|} d\sigma = - \oint_S \frac{\mathbf{M} \times \hat{\mathbf{n}}}{|\mathbf{r} - \mathbf{r}'|} d\sigma. \quad (567)$$

Inserting (565) and (567) into (562),

$$\mathbf{A}_M(\mathbf{r}) = \int_V \frac{\nabla' \times \mathbf{M}}{|\mathbf{r} - \mathbf{r}'|} d^3r' + \oint_S \frac{\mathbf{M} \times \hat{\mathbf{n}}}{|\mathbf{r} - \mathbf{r}'|} d\sigma. \quad (568)$$

Reading Off the Bound Currents

Equation (568) has exactly the structure of the magnetostatic potential generated by a volume current plus a surface current,

$$\mathbf{A}(\mathbf{r}) = \frac{1}{c} \int_V \frac{\mathbf{J}}{|\mathbf{r} - \mathbf{r}'|} d^3r' + \frac{1}{c} \oint_S \frac{\mathbf{K}}{|\mathbf{r} - \mathbf{r}'|} d\sigma. \quad (569)$$

Because the equality must hold for an arbitrary body of arbitrary shape and arbitrary $\mathbf{M}$ (and because, with no boundary, the surface piece drops and the two volume integrals must match term by term), we identify

$$\boxed{\mathbf{J}_M = c \nabla \times \mathbf{M}, \quad \mathbf{K}_M = c \mathbf{M} \times \hat{\mathbf{n}}.} \quad (570)$$

Significance: Magnetization is *exactly* equivalent to a real-looking bound current – a volume current wherever $\mathbf{M}$ curls, and a sheet current $c \mathbf{M} \times \hat{\mathbf{n}}$ wrapping the surface where $\mathbf{M}$ stops abruptly. This is the magnetic counterpart of bound charge $\rho_b = -\nabla \cdot \mathbf{P}$ in dielectrics.

Note:

The surface current is the limit of a thin volume layer: writing a surface current as a volume current times a δ -function collapses one integration and leaves the surface integral, which is why both pieces appear together.

18.4 Ampère's Law in Matter and the Magnetic Intensity $\mathbf{H}$

Conceptual Background

With the bound current known, we return to Maxwell's equations in the presence of matter. In magnetostatics these are $\nabla \cdot \mathbf{B} = 0$ and Ampère's law with the *total* current. Splitting the total current and substituting $\mathbf{J}_M = c \nabla \times \mathbf{M}$ lets us absorb the matter response into a redefined field.

Derivation

Ampère's law with the total current reads

$$\nabla \times \mathbf{B} = \frac{4\pi}{c} \mathbf{J}_{\text{total}} = \frac{4\pi}{c} \mathbf{J}_f + \frac{4\pi}{c} \mathbf{J}_M. \quad (571)$$

Insert $\mathbf{J}_M = c \nabla \times \mathbf{M}$; the factors of c cancel in the last term,

$$\nabla \times \mathbf{B} = \frac{4\pi}{c} \mathbf{J}_f + 4\pi \nabla \times \mathbf{M}. \quad (572)$$

Move the magnetization term to the left,

$$\nabla \times (\mathbf{B} - 4\pi \mathbf{M}) = \frac{4\pi}{c} \mathbf{J}_f. \quad (573)$$

The combination $\mathbf{B} - 4\pi \mathbf{M}$ is sourced only by free currents, so it deserves its own name.

Physical Concept 18.2: Magnetic Intensity $\mathbf{H}$

$$\mathbf{H} = \mathbf{B} - 4\pi\mathbf{M}. \quad (574)$$

Outside matter $\mathbf{M} = 0$, so $\mathbf{H} = \mathbf{B}$ there; only inside a magnetized body do $\mathbf{H}$ and $\mathbf{B}$ differ.

Significance: $\mathbf{H}$ is the magnetic field “with the matter response divided out”; it plays the role $\mathbf{B}$ plays in vacuum but is tied to the controllable free current.

In terms of $\mathbf{H}$, Ampère’s law inside matter becomes

$$\nabla \times \mathbf{H} = \frac{4\pi}{c} \mathbf{J}_f. \quad (575)$$

Significance: The curl of $\mathbf{H}$ knows nothing about the magnetization – it responds to free currents alone, which is precisely why $\mathbf{H}$ is the convenient field for solving boundary problems with given conduction currents.

Taking the divergence of (574) and using $\nabla \cdot \mathbf{B} = 0$ (which is exact, since there are no magnetic monopoles),

$$\nabla \cdot \mathbf{H} = -4\pi \nabla \cdot \mathbf{M}. \quad (576)$$

Significance: Unlike $\mathbf{B}$, the field $\mathbf{H}$ is generally *not* divergence-free: wherever the magnetization has nonzero divergence, $\mathbf{H}$ has an effective source. This is the seed of the magnetic scalar potential.

Remark 14. In general the magnetization is itself a function of the field, $\mathbf{M} = \mathbf{M}(\mathbf{B})$, determined self-consistently by the material. For linear media (paramagnets, diamagnets) the relation is linear; for ferromagnets it is complicated and hysteretic. Throughout this lecture we assume $\mathbf{M}$ (equivalently $\mathbf{J}_M$) is *given* – e.g. a permanent magnet whose moments are fixed and only weakly disturbed by the applied field.

18.5 Magnetic Scalar Potential and the Electrostatic Analogy**Physical Motivation**

Consider a permanent magnet sitting in a region with *no free currents*, $\mathbf{J}_f = 0$. Then (575) gives $\nabla \times \mathbf{H} = 0$, exactly like the electrostatic condition $\nabla \times \mathbf{E} = 0$. A curl-free field is the gradient of a potential, so we expect a *magnetic scalar potential* and an entire electrostatics-style solution method.

The Poisson Equation for ψ_M

Since $\nabla \times \mathbf{H} = 0$, write

$$\mathbf{H} = -\nabla\psi_M. \quad (577)$$

Substituting into (576) gives a Poisson equation,

$$\nabla^2\psi_M = 4\pi \nabla \cdot \mathbf{M} = -4\pi \rho_M, \quad \rho_M \equiv -\nabla \cdot \mathbf{M}. \quad (578)$$

Significance: The magnetic problem becomes formally identical to electrostatics, with an *effective magnetic charge density* $\rho_M = -\nabla \cdot \mathbf{M}$. There are no real magnetic charges; ρ_M is a mathematical bookkeeping device that lets us import every electrostatic technique (images, separation of variables, multipoles).

Explicit Potential from the Single-Dipole Field

To find ψ_M concretely, rewrite the dipole field of Lecture 16 as a gradient. For a moment $\boldsymbol{\mu}$ at the origin, the full field (including the contact term at the source) is

$$\mathbf{B}_{\text{dip}}(\mathbf{r}) = \frac{3\hat{\mathbf{r}}(\boldsymbol{\mu} \cdot \hat{\mathbf{r}}) - \boldsymbol{\mu}}{r^3} + 4\pi \boldsymbol{\mu} \delta^{(3)}(\mathbf{r}). \quad (579)$$

We claim this equals $-\nabla(\boldsymbol{\mu} \cdot \mathbf{r}/r^3) + 4\pi \boldsymbol{\mu} \delta^{(3)}(\mathbf{r})$. Indeed, differentiating the scalar $\boldsymbol{\mu} \cdot \mathbf{r}/r^3 = \mu_j r_j r^{-3}$,

$$\begin{aligned} \partial_i \left(\frac{\mu_j r_j}{r^3} \right) &= \frac{\mu_i}{r^3} + \mu_j r_j \partial_i r^{-3} \\ &= \frac{\mu_i}{r^3} - \frac{3 r_i (\boldsymbol{\mu} \cdot \mathbf{r})}{r^5}, \end{aligned} \quad (580)$$

so that, away from the origin,

$$-\nabla \left(\frac{\boldsymbol{\mu} \cdot \mathbf{r}}{r^3} \right) = \frac{3\hat{\mathbf{r}}(\boldsymbol{\mu} \cdot \hat{\mathbf{r}}) - \boldsymbol{\mu}}{r^3}, \quad (581)$$

which is exactly the $r \neq 0$ part of (579). Hence

$$\boxed{\mathbf{B}_{\text{dip}}(\mathbf{r}) = -\nabla \left(\frac{\boldsymbol{\mu} \cdot \mathbf{r}}{r^3} \right) + 4\pi \boldsymbol{\mu} \delta^{(3)}(\mathbf{r}).} \quad (582)$$

Significance: The dipole field is (almost everywhere) a pure gradient; the δ -function carries the difference between the curl-based and gradient-based representations and is exactly the term that reconstructs $4\pi\mathbf{M}$ in the bulk.

Superposition over the Body

Replace $\boldsymbol{\mu} \rightarrow \mathbf{M}(\mathbf{r}') d^3 r'$ and place each dipole at $\mathbf{r}'$ (so $\mathbf{r} \rightarrow \mathbf{r} - \mathbf{r}'$), then integrate. The first term integrates into a gradient and the δ -function collapses the second integral:

$$\mathbf{B}_M(\mathbf{r}) = -\nabla \psi_M(\mathbf{r}) + 4\pi \mathbf{M}(\mathbf{r}), \quad \psi_M(\mathbf{r}) = \int d^3 r' \frac{\mathbf{M}(\mathbf{r}') \cdot (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3}. \quad (583)$$

Forming $\mathbf{H}_M = \mathbf{B}_M - 4\pi\mathbf{M}$ cancels the contact term and confirms

$$\boxed{\mathbf{H}_M = -\nabla \psi_M, \quad \psi_M(\mathbf{r}) = \int d^3 r' \frac{\mathbf{M}(\mathbf{r}') \cdot (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3}.} \quad (584)$$

Significance: This is fully consistent with the abstract result $\mathbf{H} = -\nabla\psi_M$: starting from the explicit dipole superposition we recover both the scalar potential and the relation $\mathbf{B} = \mathbf{H} + 4\pi\mathbf{M}$. The magnetic problem with no free currents is solved exactly like an electrostatics problem.

The Electrostatic Analogy

The magnetic scalar potential is the literal image of the electric dipole potential. Compare the two theories side by side:

	Electrostatics	Magnetostatics ($\mathbf{J}_f = 0$)
Field from potential	$\mathbf{E} = -\nabla\phi$	$\mathbf{H} = -\nabla\psi_M$
Dipole potential	$\phi = \frac{\mathbf{p} \cdot \mathbf{r}}{r^3}$	$\psi_M = \frac{\boldsymbol{\mu} \cdot \mathbf{r}}{r^3}$
Source density	ρ	$\rho_M = -\nabla \cdot \mathbf{M}$
Poisson equation	$\nabla^2\phi = -4\pi\rho$	$\nabla^2\psi_M = -4\pi\rho_M$

Significance: Polarization-like reasoning carries over completely: $\mathbf{M}$ plays the role of polarization, $-\nabla \cdot \mathbf{M}$ the role of bound charge, and ψ_M the role of the electrostatic potential. The next lecture exploits this to solve concrete magnet geometries with electrostatic methods.

18.6 Summary

- The total current splits as $\mathbf{J}_{\text{total}} = \mathbf{J}_f + \mathbf{J}_M$, with the magnetization current arising microscopically from orbital motion plus an effective spin current; classically only $\mathbf{M} = \mathbf{M}_o + \mathbf{M}_s$ matters.
- Superposing dipole vector potentials gives $\mathbf{A}_M = \int \mathbf{M} \times (\mathbf{r} - \mathbf{r}')/|\mathbf{r} - \mathbf{r}'|^3 d^3r'$.
- A vector identity converts this into bound currents $\mathbf{J}_M = c \nabla \times \mathbf{M}$ (volume) and $\mathbf{K}_M = c \mathbf{M} \times \hat{\mathbf{n}}$ (surface).
- Ampère's law then forces $\mathbf{H} = \mathbf{B} - 4\pi\mathbf{M}$ with $\nabla \times \mathbf{H} = (4\pi/c)\mathbf{J}_f$ and $\nabla \cdot \mathbf{H} = -4\pi \nabla \cdot \mathbf{M}$.
- With no free currents, $\mathbf{H} = -\nabla\psi_M$ and $\nabla^2\psi_M = -4\pi\rho_M$ with effective magnetic charge $\rho_M = -\nabla \cdot \mathbf{M}$.
- Explicitly $\psi_M = \int \mathbf{M}(\mathbf{r}') \cdot (\mathbf{r} - \mathbf{r}')/|\mathbf{r} - \mathbf{r}'|^3 d^3r'$, the exact analogue of the electric dipole potential – magnetostatics in matter reduces to an electrostatics problem.

Homework Solutions

Analytical Electrodynamics: Exercise Set 1

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1. Unit Conversion ($c = 1$)

In the natural unit system where $c = 1$, lengths and times share the same dimension $[L] = [T]$, achieved by the substitution $1 \text{ s} \rightarrow m/c$ (numerically, $1 \text{ s} \approx 3 \times 10^8 \text{ m}$). To convert an SI quantity, multiply by the appropriate power of $1/c$ until all factors of seconds are eliminated.

- (a) **Velocity.** $[v] = \text{m s}^{-1}$ contains one inverse second. Replacing $1 \text{ s}^{-1} \rightarrow c \text{ m}^{-1}$ and setting $c = 1$:

$$v = 22 \text{ m s}^{-1} = \frac{22}{c} \underbrace{(\text{m s}^{-1} \cdot \text{s m}^{-1})}_{=1} \xrightarrow{c=1} \frac{22}{c} \approx 7.34 \times 10^{-8} \quad (\text{dimensionless})$$

- (b) **Momentum.** $[p] = \text{kg m s}^{-1}$. One power of s^{-1} :

$$p = 3.4 \times 10^4 \text{ kg m s}^{-1} = \frac{3.4 \times 10^4}{c} \text{ kg} \approx 1.13 \times 10^{-4} \text{ kg}$$

- (c) **Work (energy).** $[W] = \text{kg m}^2 \text{s}^{-2}$. Two powers of s^{-1} , so divide by c^2 :

$$W = 6 \text{ J} = 6 \text{ kg m}^2 \text{s}^{-2} = \frac{6}{c^2} \text{ kg} \approx 6.68 \times 10^{-17} \text{ kg}$$

- (d) **Power.** $[P] = \text{kg m}^2 \text{s}^{-3}$. Three powers of s^{-1} ; with $1/\text{s}^3 \rightarrow (c/\text{m})^3$ the result picks up three factors of c and three inverse metres:

$$P = 100 \text{ W} = 100 \text{ kg m}^2 \text{s}^{-3} = \frac{100}{c^3} \text{ kg m}^{-1} \approx 3.70 \times 10^{-24} \text{ kg m}^{-1}$$

- (e) **Density.** $[\rho] = \text{kg m}^{-3}$ contains no time factors; the numerical value is unchanged under the $c = 1$ prescription:

$$\rho = 10^3 \text{ kg m}^{-3}$$

- (f) **Reduced Planck constant.** $[\hbar] = \text{kg m}^2 \text{s}^{-1}$. Replacing $1/\text{s} \rightarrow c/\text{m}$ eliminates the single factor of s^{-1} :

$$\hbar = 1.05 \times 10^{-34} \text{ J s} = 1.05 \times 10^{-34} \text{ kg m}^2 \text{s}^{-1} \xrightarrow{1 \text{ s}^{-1} \rightarrow c \text{ m}^{-1}} \frac{1.05 \times 10^{-34}}{c} \text{ kg m} \approx 3.50 \times 10^{-43} \text{ kg m}$$

2. SI to CGS Conversion

In the Gaussian (CGS) system, Coulomb's constant is set to unity ($k_e = 1$, equivalent to $4\pi\epsilon_0 \rightarrow 1$ in SI notation). Charge thereby acquires dimensions $[q]_{\text{Gauss}} = \text{g}^{1/2}\text{cm}^{3/2}\text{s}^{-1}$, and the unit is the statcoulomb (statC).

(a) **Density.**

$$\rho = 10^3 \text{ kg m}^{-3} = 10^3 \left(\frac{10^3 \text{ g}}{10^6 \text{ cm}^3} \right) = 1 \text{ g cm}^{-3}$$

(b) **Reduced Planck constant.** The CGS unit of energy is the erg ($1 \text{ erg} = 1 \text{ g cm}^2 \text{ s}^{-2}$), with $1 \text{ J} = 10^7 \text{ erg}$. Since $\hbar$ has dimensions of action (energy $\times$ time), the conversion factor is the same:

$$\hbar = 1.05 \times 10^{-34} \text{ J s} \times 10^7 \frac{\text{erg}}{\text{J}} = 1.05 \times 10^{-27} \text{ erg s}$$

(c) **Charge.** We derive the Gaussian equivalent of 1 C by equating the Coulomb force for two unit charges separated by 1 m. In SI, substituting $q = 1 \text{ C}$ and $r = 1 \text{ m}$:

$$F_{\text{SI}} = \frac{1}{4\pi\epsilon_0} \frac{q^2}{r^2} = \left(9 \times 10^9 \frac{\text{N m}^2}{\text{C}^2} \right) \frac{(1 \text{ C})^2}{(1 \text{ m})^2} = 9 \times 10^9 \text{ N}$$

Converting to CGS units ($1 \text{ N} = 10^5 \text{ dyn}$, $1 \text{ m} = 10^2 \text{ cm}$):

$$F_{\text{cgs}} = 9 \times 10^{14} \text{ dyn}, \quad r_{\text{cgs}} = 100 \text{ cm}$$

In Gaussian units $k_e = 1$, so Coulomb's law gives $F = q_{\text{cgs}}^2/r^2$:

$$\begin{aligned} 9 \times 10^{14} &= \frac{q_{\text{cgs}}^2}{(100 \text{ cm})^2} \\ q_{\text{cgs}}^2 &= 9 \times 10^{18} \text{ dyn cm}^2 \\ q_{\text{cgs}} &= 3 \times 10^9 \text{ statC} \end{aligned}$$

(d) **Gauss's Law.** In Gaussian units, the charge unit is defined so that $k_e = 1$, i.e., the ϵ_0 in SI is absorbed into the charge rescaling via $4\pi\epsilon_0 \rightarrow 1$. Gauss's law in differential and integral form becomes:

$$(\text{SI}) \quad \nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad \longrightarrow \quad (\text{Gaussian}) \quad \nabla \cdot \vec{E} = 4\pi\rho$$

• **Differential form:** $\nabla \cdot \vec{E} = 4\pi\rho$

• **Integral form:** $\oint_S \vec{E} \cdot d\vec{a} = 4\pi Q_{\text{enc}}$

(e) **Biot-Savart Law.** In Gaussian units $\vec{E}$ and $\vec{B}$ have the same dimensions. This requires a factor of $1/c$ wherever current appears:

$$d\vec{B} = \frac{I d\vec{l} \times \hat{r}}{c r^2}$$

3. Four-Vector Analysis

Given the contravariant components $A^\mu = (A^0, A^1, A^2, A^3) = (0, -1, 3, 6)$ in frame $\mathcal{O}$, with metric signature $\eta_{\mu\nu} = \text{diag}(+1, -1, -1, -1)$.

(a) **Components in $\mathcal{O}'$** (boost $v = 0.8c$ along $+\hat{x}$).

$$\beta = 0.8, \quad \gamma = \frac{1}{\sqrt{1-\beta^2}} = \frac{1}{\sqrt{0.36}} = \frac{5}{3}. \text{ The Lorentz matrix is:}$$

$$\Lambda^{\mu'}_{\nu} = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{5}{3} & -\frac{4}{3} & 0 & 0 \\ -\frac{4}{3} & \frac{5}{3} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Applying $A^{\mu'} = \Lambda^{\mu'}_{\nu} A^{\nu}$:

$$A^{0'} = \gamma(A^0 - \beta A^1) = \frac{5}{3}(0 - (0.8)(-1)) = \frac{5}{3} \cdot \frac{4}{5} = \frac{4}{3}$$

$$A^{1'} = \gamma(A^1 - \beta A^0) = \frac{5}{3}(-1 - 0) = -\frac{5}{3}$$

$$A^{2'} = A^2 = 3$$

$$A^{3'} = A^3 = 6$$

$$A^{\mu'} = \left(\frac{4}{3}, -\frac{5}{3}, 3, 6\right)$$

(b) **Components in $\bar{\mathcal{O}}$** (boost $v = 0.6c$ along $+\hat{x}$).

$$\beta = 0.6, \quad \gamma = \frac{1}{\sqrt{1-\beta^2}} = \frac{1}{0.8} = \frac{5}{4}.$$

$$A^{\bar{0}} = \gamma(A^0 - \beta A^1) = \frac{5}{4}(0 - (0.6)(-1)) = \frac{5}{4} \cdot \frac{3}{5} = \frac{3}{4}$$

$$A^{\bar{1}} = \gamma(A^1 - \beta A^0) = \frac{5}{4}(-1 - 0) = -\frac{5}{4}$$

$$A^{\bar{2}} = 3, \quad A^{\bar{3}} = 6$$

$$A^{\bar{\mu}} = \left(\frac{3}{4}, -\frac{5}{4}, 3, 6\right)$$

(c) **Magnitude in $\mathcal{O}$.**

$$A_{\mu}A^{\mu} = \eta_{\mu\nu}A^{\mu}A^{\nu} = (A^0)^2 - (A^1)^2 - (A^2)^2 - (A^3)^2 = 0 - 1 - 9 - 36 = \boxed{-46}$$

(d) **Magnitude in $\mathcal{O}'$.** The quantity $A_{\mu}A^{\mu}$ is a Lorentz scalar, hence frame-independent.

verification:

$$A_{\mu'}A^{\mu'} = \left(\frac{4}{3}\right)^2 - \left(-\frac{5}{3}\right)^2 - 3^2 - 6^2 = \frac{16}{9} - \frac{25}{9} - 9 - 36 = -\frac{9}{9} - 45 = -1 - 45 = \boxed{-46} \checkmark$$

(e) **Covariant components $A_{\mu'}$ and verification.**

Lowering the index with $\eta_{\mu'\nu'}$:

$$A_{\mu'} = \eta_{\mu'\nu'} A^{\nu'} \implies A_{\mu'} = \left(\frac{4}{3}, \frac{5}{3}, -3, -6\right)$$

We verify via $A_{\mu'} = A_{\nu}(\Lambda^{-1})^{\nu}_{\mu'}$, where $(\Lambda^{-1})^{\nu}_{\mu'}$ is the boost with $\beta \rightarrow -\beta$:

$$(\Lambda^{-1})^{\nu}_{\mu'} = \begin{pmatrix} \gamma & +\gamma\beta & 0 & 0 \\ +\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{5}{3} & \frac{4}{3} & 0 & 0 \\ \frac{4}{3} & \frac{5}{3} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

The covariant 4-vector in $\mathcal{O}$ is $A_{\nu} = (0, +1, -3, -6)$. Contracting:

$$A_{0'} = A_{\nu}(\Lambda^{-1})^{\nu}_{0'} = (0)\gamma + (1)\gamma\beta = \frac{4}{3} \checkmark$$

$$A_{1'} = A_{\nu}(\Lambda^{-1})^{\nu}_{1'} = (0)\gamma\beta + (1)\gamma = \frac{5}{3} \checkmark$$

$$A_{2'} = A_2 = -3 \checkmark$$

$$A_{3'} = A_3 = -6 \checkmark$$

4. Lorentz Transformations

- (a) **General boost in the xy -plane.** Let the boost velocity be v with magnitude $v = \beta c$. Given the angle ϕ from the positive x axis, the velocity components are:

$$\begin{aligned} v_x &= v \cos \phi, & v_y &= v \sin \phi \\ \beta_x &= \beta \cos \phi, & \beta_y &= \beta \sin \phi \end{aligned}$$

Substituting these components into the general boost matrix yields the transformation tensor:

$$\Lambda^\mu{}_\nu = \begin{pmatrix} \gamma & -\gamma\beta \cos \phi & -\gamma\beta \sin \phi & 0 \\ -\gamma\beta \cos \phi & 1 + (\gamma - 1) \cos^2 \phi & (\gamma - 1) \sin \phi \cos \phi & 0 \\ -\gamma\beta \sin \phi & (\gamma - 1) \sin \phi \cos \phi & 1 + (\gamma - 1) \sin^2 \phi & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (585)$$

To construct the transformation without relying on the general boost formula, decompose the physical operation into a sequence of three elementary transformations: a spatial rotation to align the coordinate axes with the boost direction, a standard one-dimensional boost, and an inverse spatial rotation to restore the original coordinate orientation. First, apply a passive coordinate rotation by an angle ϕ around the z -axis. This transforms the coordinate system such that the new x' -axis is parallel to v .

$$R(\phi) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \phi & \sin \phi & 0 \\ 0 & -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

In this rotated frame, the boost occurs strictly along the positive x' -axis with magnitude $v = \beta c$. The standard Lorentz boost matrix is:

$$\Lambda_x(v) = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Finally, rotate the coordinate system back to its original orientation. The inverse rotation matrix is $R^{-1}(\phi) = R(-\phi)$

$$R^{-1}(\phi) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi & 0 \\ 0 & \sin \phi & \cos \phi & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

The full Lorentz transformation Λ is the matrix product $\Lambda = R^{-1}(\phi)\Lambda_x(v)R(\phi)$ which yields equivalent results to [585](#)

- (b) **Successive Lorentz boosts.** We are given two successive boosts:

- $\mathcal{O} \rightarrow \bar{\mathcal{O}} : \quad \vec{v} = 0.6c\hat{x}$

$$\beta_1 = 0.6, \quad \gamma_1 = \frac{1}{\sqrt{1 - 0.6^2}} = 1.25 = \frac{5}{4}$$

- $\bar{\mathcal{O}} \rightarrow \mathcal{O}' : \quad \vec{v} = 0.8c\hat{y}$

$$\beta_2 = 0.8, \quad \gamma_2 = \frac{1}{\sqrt{1 - 0.8^2}} = 1.66... = \frac{5}{3}$$

The corresponding individual transformation matrices are:

$$\Lambda_{\mu}^{\bar{\nu}}(\vec{v}) = \begin{pmatrix} 5/4 & -3/4 & 0 & 0 \\ -3/4 & 5/4 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \Lambda_{\bar{\nu}}^{\alpha'}(\vec{v}') = \begin{pmatrix} 5/3 & 0 & -4/3 & 0 \\ 0 & 1 & 0 & 0 \\ -4/3 & 0 & 5/3 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

The composite transformation $\mathcal{O} \rightarrow \mathcal{O}'$ is the matrix product $\Lambda_{\mu}^{\alpha'} = \Lambda_{\bar{\nu}}^{\alpha'}(\vec{v}')\Lambda_{\mu}^{\bar{\nu}}(\vec{v})$

$$\Lambda_{\mu}^{\alpha'} = \begin{pmatrix} 5/3 & 0 & -4/3 & 0 \\ 0 & 1 & 0 & 0 \\ -4/3 & 0 & 5/3 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 5/4 & -3/4 & 0 & 0 \\ -3/4 & 5/4 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 25/12 & -5/4 & -4/3 & 0 \\ -3/4 & 5/4 & 0 & 0 \\ -5/3 & 1 & 5/3 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

- (c) **Invariance of the spacetime interval.** To prove the result is a valid Lorentz transformation, it must satisfy the invariance of the spacetime metric $\eta = \text{diag}(1, -1, -1, -1)$. This requires $\Lambda^T \eta \Lambda = \eta$. Lets restrict to the non-trivial (ct, x, y) block. we extract a scalar factor of 1/12 for algebraic convenience:

$$\Lambda = \frac{1}{12} \begin{pmatrix} 25 & -15 & -16 \\ -9 & 15 & 0 \\ -20 & 12 & 20 \end{pmatrix}$$

Evaluate $\eta\Lambda$

$$\eta\Lambda = \frac{1}{12} \begin{pmatrix} 25 & -15 & -16 \\ 9 & -15 & 0 \\ 20 & -12 & -20 \end{pmatrix}$$

Now multiply $\Lambda^T(\eta\Lambda)$:

$$\Lambda^T \eta \Lambda = \frac{1}{144} \begin{pmatrix} 25 & -9 & -20 \\ -15 & 15 & 12 \\ -16 & 0 & 20 \end{pmatrix} \begin{pmatrix} 25 & -15 & -16 \\ 9 & -15 & 0 \\ 20 & -12 & -20 \end{pmatrix} = \text{diag}(1, -1, -1) \implies \Delta s'^2 = \Delta s^2 \checkmark$$

- (d) **Reversed Boost Sequence and Physical Interpretation** Evaluate $\Lambda_{\bar{\nu}}^{\alpha'}(\vec{v}')\Lambda_{\mu}^{\bar{\nu}}(\vec{v}')$

$$\begin{pmatrix} 5/4 & -3/4 & 0 & 0 \\ -3/4 & 5/4 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 5/3 & 0 & -4/3 & 0 \\ 0 & 1 & 0 & 0 \\ -4/3 & 0 & 5/3 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 25/12 & -3/4 & -5/3 & 0 \\ -5/4 & 5/4 & 1 & 0 \\ -4/3 & 0 & 5/3 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Comparing this matrix to the result in (b), they are fundamentally unequal $\Lambda(\vec{v}')\Lambda(\vec{v}) \neq \Lambda(\vec{v})\Lambda(\vec{v}')$.

Physical Interpretation:

The Lorentz group $SO(1,3)$ is non-Abelian. Since the commutators of the boost generators satisfy the relation $[K_i, K_j] = -i\epsilon_{ijk}J_k$ (where J_k are the generators of rotations), boosts along non-collinear axes do not commute:

$$\Lambda(\vec{v}')\Lambda(\vec{v}) \neq \Lambda(\vec{v})\Lambda(\vec{v}').$$

While a pure boost is represented by a symmetric matrix, the product of two non-collinear symmetric boost matrices is asymmetric. According to the polar decomposition of the Lorentz group, any such transformation can be uniquely factored into a pure boost followed by a spatial rotation:

$$\Lambda = R(\theta_W)B(\vec{v}_{\text{res}}).$$

This induced rotation is known as the **Wigner rotation**. Physically, this implies that the order of successive boosts determines the final orientation of the reference frame. Even if two observers reach the same final velocity magnitude relative to the laboratory, different boost sequences result in orientationally distinct frames. Their coordinate axes will be physically rotated relative to one another—a purely kinematic effect that serves as the underlying mechanism for **Thomas precession**.¹

Analytical Electrodynamics – Exercise Set 2 Eran Rehani
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19 Conservation of four momentum

We adopt the natural units where $c = 1$. The four-momentum of a particle is defined as $P^\mu = (E, \mathbf{p})$. For any on-shell particle, the invariant mass squared is $P^2 = P_\mu P^\mu = E^2 - |\mathbf{p}|^2 = m^2$.

Single-Photon Annihilation $e^- + e^+ \rightarrow \gamma$

Let p_- and p_+ denote the electron and positron four-momenta, and let k be the photon four-momentum. Four-momentum conservation gives

$$p_- + p_+ = k \quad (586)$$

Squaring,

$$(p_- + p_+)^2 = k^2 \quad (587)$$

For a real photon, $k^2 = 0$. In the CoM frame, $p_- = (E, \mathbf{p})$ and $p_+ = (E, -\mathbf{p})$, so

$$(p_- + p_+)^2 = (2E, \mathbf{0})^2 = 4E^2 > 0. \quad (588)$$

Hence $(p_- + p_+)^2 \neq 0$, in contradiction with $k^2 = 0$. Therefore, **single-photon annihilation is forbidden in vacuum**.

Two-Photon Annihilation $e^- + e^+ \rightarrow \gamma + \gamma$

For the two-photon channel,

$$p_- + p_+ = k_1 + k_2 \quad (589)$$

and after squaring,

$$(p_- + p_+)^2 = (k_1 + k_2)^2 \quad (590)$$

Define the invariant

$$s \equiv (p_- + p_+)^2. \quad (591)$$

In the CoM frame, $s = (2E, \mathbf{0})^2 = 4E^2$.

For massless photons,

$$(k_1 + k_2)^2 = 2k_1 \cdot k_2. \quad (592)$$

In the CoM frame, momentum conservation requires $k_1 = (\omega, \mathbf{k})$ and $k_2 = (\omega, -\mathbf{k})$ with $|\mathbf{k}| = \omega$. Therefore,

$$(k_1 + k_2)^2 = (2\omega, \mathbf{0})^2 = 4\omega^2. \quad (593)$$

Energy conservation implies $2E = 2\omega$, so $\omega = E$. Hence

$$(k_1 + k_2)^2 = 4\omega^2 = 4E^2 = s = (p_- + p_+)^2. \quad (594)$$

Thus the invariant relation is satisfied by physical final states, and **two-photon annihilation is allowed by four-momentum conservation**.

¹This specific phenomenon is the origin of Thomas Precession (Wigner rotation).

20 Relativistic work for a small speed increase

For a particle with rest mass $m \neq 0$, the total relativistic energy is

$$E(v) = \gamma(v)mc^2, \quad \gamma(v) = \frac{1}{\sqrt{1 - v^2/c^2}}. \quad (595)$$

The required work to change the speed from v to $v + \delta v$ (with $\delta v \ll v$) is, to first order,

$$\delta W = \delta E = \frac{dE}{dv} \delta v. \quad (596)$$

Differentiate $E(v)$:

$$\frac{dE}{dv} = mc^2 \frac{d\gamma}{dv} = mc^2 \left(\frac{\gamma^3 v}{c^2} \right) = m\gamma^3 v. \quad (597)$$

Hence

$$\boxed{\delta W \simeq m\gamma^3 v \delta v} \quad (\text{first order in } \delta v). \quad (598)$$

Equivalently, using $\beta = v/c$,

$$\delta W \simeq mc^2 \gamma^3 \beta \delta \beta. \quad (599)$$

To see why reaching light speed is impossible for $m \neq 0$, note that the total energy needed to accelerate from some initial $v_0 < c$ to v is

$$\Delta E = E(v) - E(v_0) = mc^2 (\gamma(v) - \gamma(v_0)). \quad (600)$$

As $v \rightarrow c^-$, we have $\gamma(v) \rightarrow \infty$, therefore

$$\Delta E \rightarrow \infty. \quad (601)$$

So an infinite amount of energy would be required to accelerate a massive particle to the speed of light.

21 Compton Scattering

We restore factors of c throughout this solution. A photon of frequency ν_i propagates along the positive x -axis and scatters off a particle of rest mass m that is initially at rest in the laboratory frame. After the collision the photon emerges at angle θ from the incident direction with frequency ν_f . We use the metric signature $(+---)$.

Four-momenta

A photon of frequency ν carries energy $E = h\nu$ and momentum magnitude $|\mathbf{p}| = h\nu/c$, so its four-momentum is $k^\mu = (h\nu/c)(1, \hat{\mathbf{n}})$ where $\hat{\mathbf{n}}$ is the propagation direction and $k^2 = 0$ (massless). The initial four-momenta are

$$P_i^\mu = (mc, \mathbf{0}), \quad (602)$$

$$k_i^\mu = \frac{h\nu_i}{c}(1, 1, 0, 0), \quad (603)$$

and the final four-momenta are

$$k_f^\mu = \frac{h\nu_f}{c}(1, \cos \theta, \sin \theta, 0), \quad (604)$$

$$P_f^\mu \quad (\text{to be eliminated via the on-shell condition}). \quad (605)$$

Derivation of the Compton formula

Four-momentum conservation gives

$$P_i^\mu + k_i^\mu = P_f^\mu + k_f^\mu \implies P_f^\mu = P_i^\mu + k_i^\mu - k_f^\mu. \quad (606)$$

Squaring both sides and invoking the on-shell condition $P_f^2 = m^2 c^2$:

$$m^2 c^2 = (P_i + k_i - k_f)^2 = P_i^2 + k_i^2 + k_f^2 + 2 P_i \cdot k_i - 2 P_i \cdot k_f - 2 k_i \cdot k_f. \quad (607)$$

Using $P_i^2 = m^2 c^2$ and $k_i^2 = k_f^2 = 0$, this reduces to

$$0 = 2 P_i \cdot k_i - 2 P_i \cdot k_f - 2 k_i \cdot k_f. \quad (608)$$

We evaluate each Minkowski inner product with $P_i = (mc, \mathbf{0})$:

$$P_i \cdot k_i = (mc) \left(\frac{h\nu_i}{c} \right) - \mathbf{0} \cdot \mathbf{0} = mh\nu_i, \quad (609)$$

$$P_i \cdot k_f = (mc) \left(\frac{h\nu_f}{c} \right) = mh\nu_f, \quad (610)$$

$$k_i \cdot k_f = \frac{h\nu_i}{c} \cdot \frac{h\nu_f}{c} - \frac{h\nu_i}{c} \cdot \frac{h\nu_f \cos \theta}{c} = \frac{h^2 \nu_i \nu_f}{c^2} (1 - \cos \theta). \quad (611)$$

Substituting into (608):

$$0 = 2mh\nu_i - 2mh\nu_f - \frac{2h^2 \nu_i \nu_f}{c^2} (1 - \cos \theta). \quad (612)$$

Dividing through by $2mh\nu_i \nu_f$:

$$0 = \frac{1}{\nu_f} - \frac{1}{\nu_i} - \frac{h}{mc^2} (1 - \cos \theta). \quad (613)$$

Rearranging yields the **Compton scattering formula**:

$$\boxed{\frac{1}{\nu_f} = \frac{1}{\nu_i} + \frac{h(1 - \cos \theta)}{mc^2}}. \quad (614)$$

In terms of wavelength $\lambda = c/\nu$, this is the familiar Compton shift $\Delta\lambda = \lambda_f - \lambda_i = \frac{h}{mc} (1 - \cos \theta)$, where $\lambda_C = h/(mc)$ is the Compton wavelength of the scattering particle.

22 Relativistic velocity addition

Frame $\mathcal{O}'$ moves with constant velocity $\mathbf{v} = v \hat{\mathbf{x}}$ relative to inertial frame $\mathcal{O}$, with $\gamma_v = (1 - v^2/c^2)^{-1/2}$ and $\beta_v = v/c$. A body moves with velocity $\mathbf{u}' = u'_{\parallel} \hat{\mathbf{x}} + \mathbf{u}'_{\perp}$ in $\mathcal{O}'$, where $u'_{\parallel}$ is the component along $\hat{\mathbf{x}}$ and $\mathbf{u}'_{\perp}$ is the transverse part.

Four-velocity in $\mathcal{O}'$

The four-velocity of the body in $\mathcal{O}'$ is

$$U'^{\mu} = \gamma_{u'} (c, u'_{\parallel}, \mathbf{u}'_{\perp}), \quad \gamma_{u'} = \frac{1}{\sqrt{1 - |\mathbf{u}'|^2/c^2}}. \quad (615)$$

Lorentz boost to $\mathcal{O}$

A boost with velocity v along $\hat{\mathbf{x}}$ acts on the 0- and 1-components of any four-vector while leaving transverse components unchanged:

$$U^0 = \gamma_v (U'^0 + \beta_v U'^1) = \gamma_v \gamma_{u'} \left(c + \frac{v}{c} u'_{\parallel} \right) = \gamma_v \gamma_{u'} c \left(1 + \frac{v u'_{\parallel}}{c^2} \right), \quad (616)$$

$$U^1 = \gamma_v (U'^1 + \beta_v U'^0) = \gamma_v \gamma_{u'} (u'_{\parallel} + v), \quad (617)$$

$$U_{\perp}^j = U_{\perp}^{'j} = \gamma_{u'} u_{\perp}^{'j}. \quad (618)$$

Since $U^0 = \gamma_u c$ by definition, equation (616) gives

$$\gamma_u = \gamma_v \gamma_{u'} \left(1 + \frac{\mathbf{v} \cdot \mathbf{u}'}{c^2} \right), \quad (619)$$

where we replaced $v u'_{\parallel} = \mathbf{v} \cdot \mathbf{u}'$ since $\mathbf{v}$ is along $\hat{\mathbf{x}}$.

Parallel component

From $U^1 = \gamma_u u_{\parallel}$ and equations (617)–(619):

$$u_{\parallel} = \frac{U^1}{\gamma_u} = \frac{\gamma_v \gamma_{u'} (u'_{\parallel} + v)}{\gamma_v \gamma_{u'} \left(1 + \frac{\mathbf{v} \cdot \mathbf{u}'}{c^2} \right)}. \quad (620)$$

The common factors $\gamma_v \gamma_{u'}$ cancel, yielding

$$\boxed{u_{\parallel} = \frac{u'_{\parallel} + v}{1 + \frac{\mathbf{v} \cdot \mathbf{u}'}{c^2}}}. \quad (621)$$

Transverse components

From $U_{\perp}^j = \gamma_u u_{\perp}^j$ and equations (618)–(619):

$$u_{\perp} = \frac{U_{\perp}^j}{\gamma_u} = \frac{\gamma_{u'} u'_{\perp}}{\gamma_v \gamma_{u'} \left(1 + \frac{\mathbf{v} \cdot \mathbf{u}'}{c^2} \right)}. \quad (622)$$

Again $\gamma_{u'}$ cancels, giving

$$\boxed{u_{\perp} = \frac{u'_{\perp}}{\gamma_v \left(1 + \frac{\mathbf{v} \cdot \mathbf{u}'}{c^2} \right)}}. \quad (623)$$

Summary

The full relativistic velocity addition law for a frame boost $\mathbf{v} = v \hat{\mathbf{x}}$ is therefore

$$u_{\parallel} = \frac{u'_{\parallel} + v}{1 + \frac{\mathbf{v} \cdot \mathbf{u}'}{c^2}}, \quad u_{\perp} = \frac{u'_{\perp}}{\gamma_v \left(1 + \frac{\mathbf{v} \cdot \mathbf{u}'}{c^2} \right)}. \quad (624)$$

Note that in the non-relativistic limit $v, |\mathbf{u}'| \ll c$ the denominators approach unity and we recover the Galilean result $\mathbf{u} = \mathbf{u}' + \mathbf{v}$.

23 Lorentz Transformations Form a Closed Subgroup

Question 23.1. (a) Show that

$$\sum_i (\Lambda^0_i)^2 = \sum_i (\Lambda^i_0)^2 = (\Lambda^0_0)^2 - 1. \quad (625)$$

Hint: Use

$$\eta_{\gamma\delta} = \eta_{\alpha\beta} \Lambda^\alpha_\gamma \Lambda^\beta_\delta \quad (626)$$

and a similar relation, both shown in class.

(b) In class we defined the special subgroup of the Lorentz group by

$$\det \Lambda = 1, \quad \Lambda^0_0 \geq 1. \quad (627)$$

Use the relation proved in part (a) to show that this subgroup is closed, namely that if Λ and $\bar{\Lambda}$ both satisfy these conditions, then

$$\det(\Lambda\bar{\Lambda}) = 1, \quad (\Lambda\bar{\Lambda})^0_0 \geq 1. \quad (628)$$

Hint: Note that $\Lambda^0_i \bar{\Lambda}^i_0$ is a dot product of two vectors.

Solution. We use the mostly negative metric signature

$$\eta = \text{diag}(1, -1, -1, -1). \quad (629)$$

The Lorentz condition is

$$\eta_{\gamma\delta} = \eta_{\alpha\beta} \Lambda^\alpha_\gamma \Lambda^\beta_\delta, \quad (630)$$

or in matrix form $\Lambda^T \eta \Lambda = \eta$.

(a)

Set $\gamma = \delta = 0$. Since $\eta_{00} = 1$, we get

$$1 = \eta_{\alpha\beta} \Lambda^\alpha_0 \Lambda^\beta_0 = (\Lambda^0_0)^2 - \sum_i (\Lambda^i_0)^2. \quad (631)$$

Therefore,

$$\boxed{\sum_i (\Lambda^i_0)^2 = (\Lambda^0_0)^2 - 1}. \quad (632)$$

To obtain the analogous relation for Λ^0_i , use the inverse relation

$$\eta^{\gamma\delta} = \eta^{\alpha\beta} \Lambda^\gamma_\alpha \Lambda^\delta_\beta, \quad (633)$$

which follows from $\Lambda \eta \Lambda^T = \eta$. Setting again $\gamma = \delta = 0$ gives

$$1 = \eta^{\alpha\beta} \Lambda^0_\alpha \Lambda^0_\beta = (\Lambda^0_0)^2 - \sum_i (\Lambda^0_i)^2. \quad (634)$$

Hence,

$$\boxed{\sum_i (\Lambda^0_i)^2 = (\Lambda^0_0)^2 - 1}. \quad (635)$$

Combining the two results,

$$\boxed{\sum_i (\Lambda^0_i)^2 = \sum_i (\Lambda^i_0)^2 = (\Lambda^0_0)^2 - 1}. \quad (636)$$

(b)

Let Λ and $\bar{\Lambda}$ satisfy

$$\det \Lambda = \det \bar{\Lambda} = 1, \quad \Lambda^0_0 \geq 1, \quad \bar{\Lambda}^0_0 \geq 1. \quad (637)$$

First, the determinant condition is immediate:

$$\det(\Lambda \bar{\Lambda}) = \det \Lambda \det \bar{\Lambda} = 1. \quad (638)$$

Now consider the time-time component:

$$(\Lambda \bar{\Lambda})^0_0 = \Lambda^0_\mu \bar{\Lambda}^\mu_0 = \Lambda^0_0 \bar{\Lambda}^0_0 + \Lambda^0_i \bar{\Lambda}^i_0. \quad (639)$$

The second term is a Euclidean dot product of the spatial vectors (Λ^0_i) and $(\bar{\Lambda}^i_0)$, so by Cauchy–Schwarz,

$$\Lambda^0_i \bar{\Lambda}^i_0 \geq -\sqrt{\sum_i (\Lambda^0_i)^2} \sqrt{\sum_i (\bar{\Lambda}^i_0)^2}. \quad (640)$$

Using part (a),

$$(\Lambda \bar{\Lambda})^0_0 \geq \Lambda^0_0 \bar{\Lambda}^0_0 - \sqrt{(\Lambda^0_0)^2 - 1} \sqrt{(\bar{\Lambda}^0_0)^2 - 1}. \quad (641)$$

Write

$$\Lambda^0_0 = \cosh \alpha, \quad \bar{\Lambda}^0_0 = \cosh \beta, \quad \alpha, \beta \geq 0, \quad (642)$$

which is possible because both quantities are at least 1. Then

$$\sqrt{(\Lambda^0_0)^2 - 1} = \sinh \alpha, \quad \sqrt{(\bar{\Lambda}^0_0)^2 - 1} = \sinh \beta, \quad (643)$$

and therefore

$$(\Lambda \bar{\Lambda})^0_0 \geq \cosh \alpha \cosh \beta - \sinh \alpha \sinh \beta = \cosh(\alpha - \beta) \geq 1. \quad (644)$$

So the product also satisfies

$$(\Lambda \bar{\Lambda})^0_0 \geq 1. \quad (645)$$

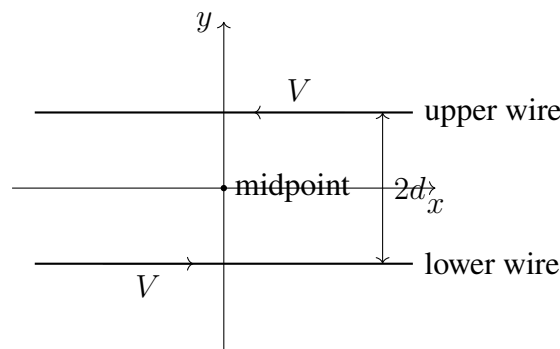
We conclude that

$$\boxed{\det(\Lambda \bar{\Lambda}) = 1, \quad (\Lambda \bar{\Lambda})^0_0 \geq 1}. \quad (646)$$

Hence the proper orthochronous Lorentz transformations form a closed subgroup.

24 Two Parallel Charged Wires

Question 24.1. Two parallel wires, at a distance $2d$, are charged with linear density λ . In a reference frame O , the charges on the lower wire move to the right, and the charges on the upper wire move to the left, each with velocity V .



- (a) Find the electric and magnetic fields at the midpoint between the wires (you may use what you learned in last year's EM course).
- (b) Find the charge and current in each of the wires, in a reference frame O' moving with velocity v to the left relative to O .
- (c) From your result in part (b), find the electric and magnetic fields in reference frame O' , at the midpoint between the wires.
- (d) Write down the electromagnetic tensor $F^{\alpha\beta}$ in reference frame O , at the midpoint between the wires.
- (e) Find the electromagnetic tensor $F'^{\mu\nu}$ in reference frame O' by performing a Lorentz transformation on $F^{\alpha\beta}$, and show that the electric and magnetic fields are the same as in part (c).

Solution. (a)

We work in Gaussian c.g.s. units, so Maxwell's equations give

$$\oint \mathbf{E} \cdot d\mathbf{a} = 4\pi Q_{\text{enc}}, \quad \oint \mathbf{B} \cdot d\boldsymbol{\ell} = \frac{4\pi}{c} I_{\text{enc}}. \quad (647)$$

For an infinite wire with linear charge density λ , a cylindrical Gaussian surface of radius r and length L gives

$$E(2\pi r L) = 4\pi \lambda L \quad \implies \quad E(r) = \frac{2\lambda}{r}. \quad (648)$$

At the midpoint, the lower wire at $y = -d$ contributes

$$\mathbf{E}_{\text{lower}} = \frac{2\lambda}{d} \hat{\mathbf{y}}, \quad (649)$$

while the upper wire at $y = d$ contributes

$$\mathbf{E}_{\text{upper}} = -\frac{2\lambda}{d} \hat{\mathbf{y}}. \quad (650)$$

Therefore,

$$\boxed{\mathbf{E} = 0}. \quad (651)$$

For the magnetic field, a circular Amperian loop of radius r gives

$$B(2\pi r) = \frac{4\pi}{c} I \quad \implies \quad B(r) = \frac{2I}{cr}. \quad (652)$$

Since $I = \lambda V$ for a line charge moving with speed V , the lower wire contributes

$$\mathbf{B}_{\text{lower}} = \frac{2\lambda V}{cd} \hat{\mathbf{z}}, \quad (653)$$

and the upper wire also contributes in the $+\hat{\mathbf{z}}$ direction:

$$\mathbf{B}_{\text{upper}} = \frac{2\lambda V}{cd} \hat{\mathbf{z}}. \quad (654)$$

Hence,

$$\boxed{\mathbf{B} = \frac{4\lambda V}{cd} \hat{\mathbf{z}}}. \quad (655)$$

(b)

Let $\mathcal{O}'$ move with velocity v to the left relative to $\mathcal{O}$, so the boost from $\mathcal{O}$ to $\mathcal{O}'$ is along the x -axis with speed $-v$. Define

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}. \quad (656)$$

For a wire moving along x , it is convenient to use the line-current four-vector

$$J^\mu = (c\lambda, \lambda u), \quad (657)$$

where u is the wire velocity in $\mathcal{O}$. Under the boost to $\mathcal{O}'$,

$$J'^0 = \gamma \left(J^0 + \frac{v}{c} J^1 \right), \quad J'^1 = \gamma \left(J^1 + \frac{v}{c} J^0 \right). \quad (658)$$

Therefore, for a wire with velocity u ,

$$\lambda' = \gamma \lambda \left(1 + \frac{uv}{c^2} \right), \quad I' = \gamma \lambda (u + v). \quad (659)$$

For the lower wire, $u = +V$, so

$$\boxed{\lambda'_{\text{lower}} = \gamma \lambda \left(1 + \frac{Vv}{c^2} \right), \quad I'_{\text{lower}} = \gamma \lambda (V + v).} \quad (660)$$

For the upper wire, $u = -V$, so

$$\boxed{\lambda'_{\text{upper}} = \gamma \lambda \left(1 - \frac{Vv}{c^2} \right), \quad I'_{\text{upper}} = \gamma \lambda (v - V).} \quad (661)$$

(c)

In $\mathcal{O}'$, the electric field of an infinite line charge is still $2\lambda'/r$. At the midpoint,

$$\mathbf{E}'_{\text{lower}} = \frac{2\lambda'_{\text{lower}}}{d} \hat{\mathbf{y}}, \quad \mathbf{E}'_{\text{upper}} = -\frac{2\lambda'_{\text{upper}}}{d} \hat{\mathbf{y}}. \quad (662)$$

Hence

$$\mathbf{E}' = \frac{2}{d} (\lambda'_{\text{lower}} - \lambda'_{\text{upper}}) \hat{\mathbf{y}} = \boxed{\frac{4\gamma\lambda Vv}{c^2 d} \hat{\mathbf{y}}}. \quad (663)$$

For the magnetic field, the lower wire contributes $+\frac{2I'_{\text{lower}}}{cd} \hat{\mathbf{z}}$ and the upper wire contributes $-\frac{2I'_{\text{upper}}}{cd} \hat{\mathbf{z}}$, so

$$\mathbf{B}' = \frac{2}{cd} (I'_{\text{lower}} - I'_{\text{upper}}) \hat{\mathbf{z}} = \boxed{\frac{4\gamma\lambda V}{cd} \hat{\mathbf{z}}}. \quad (664)$$

(d)

In frame $\mathcal{O}$ we found

$$\mathbf{E} = 0, \quad \mathbf{B} = B_z \hat{\mathbf{z}}, \quad B_z = \frac{4\lambda V}{cd}. \quad (665)$$

Using the convention

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & -B_z & B_y \\ E_y & B_z & 0 & -B_x \\ E_z & -B_y & B_x & 0 \end{pmatrix}, \quad (666)$$

The only non-zero components are:

$$F^{12} = -B_z = -\frac{4\lambda V}{cd}, \quad F^{21} = +B_z = +\frac{4\lambda V}{cd}.$$

Thus, we obtain

$$F^{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{4\lambda V}{cd} & 0 \\ 0 & \frac{4\lambda V}{cd} & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (667)$$

(e)

For a boost from $\mathcal{O}$ to $\mathcal{O}'$ with velocity $-v$ along x , the Lorentz matrix is

$$\Lambda^\mu{}_\nu = \begin{pmatrix} \gamma & \gamma\frac{v}{c} & 0 & 0 \\ \gamma\frac{v}{c} & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (668)$$

The transformed tensor is

$$F'^{\mu\nu} = \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta F^{\alpha\beta}. \quad (669)$$

Since the only nonzero entries of $F^{\alpha\beta}$ are $F^{12} = -B_z$ and $F^{21} = B_z$, the relevant transformed components are

$$F'^{02} = \Lambda^0{}_1 \Lambda^2{}_2 F^{12} = -\gamma\frac{v}{c}B_z, \quad (670)$$

$$F'^{12} = \Lambda^1{}_1 \Lambda^2{}_2 F^{12} = -\gamma B_z. \quad (671)$$

Reading off the fields from $F'^{\mu\nu}$,

$$E'_y = -F'^{02} = \gamma\frac{v}{c}B_z, \quad B'_z = -F'^{12} = \gamma B_z. \quad (672)$$

Using $B_z = \frac{4\lambda V}{cd}$, we get

$$\mathbf{E}' = \frac{4\gamma\lambda V v}{c^2 d} \hat{\mathbf{y}}, \quad \mathbf{B}' = \frac{4\gamma\lambda V}{cd} \hat{\mathbf{z}}. \quad (673)$$

This agrees exactly with part (c).

25 Motion in a Uniform Electric Field

Question 25.1. A particle with mass m , charge q , and initial velocity $v_0 \hat{\mathbf{x}}$ enters a uniform electric field $E \hat{\mathbf{y}}$ at time $t = 0$.

- What is the position of the particle as a function of time?
- The particle decays after a time τ measured in its MCRF. What is the decay time in the laboratory frame?

In both parts, to save some writing, one may express the answer using the initial momentum of the particle p_0 .

MCRF denotes the momentarily comoving reference frame of the particle. Since an accelerated particle has no inertial frame in which it is always at rest, we define at each point along the trajectory a momentary inertial frame with the same velocity as the particle. This is the MCRF in which the particle decays after time τ .

Solution. (a)

The equation of motion is

$$\frac{d\mathbf{p}}{dt} = q\mathbf{E}, \quad (674)$$

where $\mathbf{p} = \gamma m \mathbf{v}$ is the relativistic momentum. Since the electric field is along y , the x -component of the momentum is constant:

$$p_x(t) = p_0. \quad (675)$$

The y -component of the momentum changes linearly with time:

$$p_y(t) = qEt. \quad (676)$$

To find the velocity from $\mathbf{p}(t)$, we use the energy–momentum relation.

Start from the definitions of the relativistic momentum and energy:

$$\mathbf{p} = \gamma m \mathbf{v}, \quad \mathcal{E} = \gamma m c^2. \quad (677)$$

Now take the ratio $\mathbf{p}/\mathcal{E}$:

$$\frac{\mathbf{p}}{\mathcal{E}} = \frac{\gamma m \mathbf{v}}{\gamma m c^2} = \frac{\mathbf{v}}{c^2} \implies \mathbf{v} = \frac{c^2}{\mathcal{E}} \mathbf{p}. \quad (678)$$

So it remains to express $\mathcal{E}$ in terms of m and $\mathbf{p}$. Using the invariant energy–momentum relation,

$$\mathcal{E}^2 - (pc)^2 = (mc^2)^2, \quad (679)$$

we get

$$\mathcal{E}^2 = m^2 c^4 + p^2 c^2 \implies \mathcal{E} = \sqrt{m^2 c^4 + p^2 c^2}. \quad (680)$$

Plugging this back into $\mathbf{v} = \frac{c^2}{\mathcal{E}} \mathbf{p}$ gives

$$\mathbf{v} = \frac{c^2 \mathbf{p}}{\sqrt{m^2 c^4 + p^2 c^2}} = \frac{c \mathbf{p}}{\sqrt{m^2 c^2 + p^2}}. \quad (681)$$

The velocity components are

$$v_x(t) = \frac{cp_x}{\sqrt{m^2 c^2 + p_x^2 + p_y^2}}, \quad v_y(t) = \frac{cp_y}{\sqrt{m^2 c^2 + p_x^2 + p_y^2}}. \quad (682)$$

Integrating the velocity gives the position:

$$x(t) = \int_0^t v_x(t') dt' = \int_0^t \frac{cp_0}{\sqrt{m^2 c^2 + p_0^2 + q^2 E^2 t'^2}} dt', \quad (683)$$

$$y(t) = \int_0^t v_y(t') dt' = \int_0^t \frac{cqEt'}{\sqrt{m^2 c^2 + p_0^2 + q^2 E^2 t'^2}} dt'. \quad (684)$$

Performing the integrals,

$$x(t) = \frac{p_0 c}{qE} \operatorname{arcsinh} \left(\frac{qEt}{\sqrt{m^2 c^2 + p_0^2}} \right), \quad (685)$$

$$y(t) = \frac{c}{qE} \left(\sqrt{m^2 c^2 + p_0^2 + q^2 E^2 t^2} - \sqrt{m^2 c^2 + p_0^2} \right). \quad (686)$$

$$\boxed{\mathbf{r}(t) = \frac{p_0 c}{qE} \operatorname{arcsinh} \left(\frac{qEt}{\sqrt{m^2 c^2 + p_0^2}} \right) \hat{\mathbf{x}} + \frac{\sqrt{m^2 c^2 + p_0^2 + q^2 E^2 t^2} - \sqrt{m^2 c^2 + p_0^2}}{qE/c} \hat{\mathbf{y}}.} \quad (687)$$

(b)

The particle decays after a proper time τ measured in its MCRF. Since the particle is accelerated, the Lorentz factor is time-dependent, so we must relate the laboratory time t and the proper time τ by

$$d\tau = \frac{dt}{\gamma(t)}. \quad (688)$$

From part (a),

$$v_x(t) = \frac{cp_0}{\sqrt{m^2c^2 + p_0^2 + q^2E^2t^2}}, \quad v_y(t) = \frac{cqEt}{\sqrt{m^2c^2 + p_0^2 + q^2E^2t^2}}. \quad (689)$$

Therefore,

$$\frac{v(t)^2}{c^2} = \frac{v_x(t)^2 + v_y(t)^2}{c^2} \quad (690)$$

$$= \frac{p_0^2}{m^2c^2 + p_0^2 + q^2E^2t^2} + \frac{q^2E^2t^2}{m^2c^2 + p_0^2 + q^2E^2t^2} \quad (691)$$

$$= \frac{p_0^2 + q^2E^2t^2}{m^2c^2 + p_0^2 + q^2E^2t^2}. \quad (692)$$

Then

$$1 - \frac{v(t)^2}{c^2} = 1 - \frac{p_0^2 + q^2E^2t^2}{m^2c^2 + p_0^2 + q^2E^2t^2} \quad (693)$$

$$= \frac{m^2c^2 + p_0^2 + q^2E^2t^2 - (p_0^2 + q^2E^2t^2)}{m^2c^2 + p_0^2 + q^2E^2t^2} \quad (694)$$

$$= \frac{m^2c^2}{m^2c^2 + p_0^2 + q^2E^2t^2}. \quad (695)$$

Hence the Lorentz factor is

$$\gamma(t) = \frac{1}{\sqrt{1 - \frac{v(t)^2}{c^2}}} = \frac{\sqrt{m^2c^2 + p_0^2 + q^2E^2t^2}}{mc}. \quad (696)$$

Let t_{lab} be the decay time in the laboratory frame. Integrating $d\tau = dt/\gamma(t)$ from $t = 0$ to $t = t_{\text{lab}}$ gives

$$\tau = \int_0^{t_{\text{lab}}} \frac{dt}{\gamma(t)} = \int_0^{t_{\text{lab}}} \frac{mc \, dt}{\sqrt{m^2c^2 + p_0^2 + q^2E^2t^2}}. \quad (697)$$

Using

$$\int \frac{dt}{\sqrt{a^2 + b^2t^2}} = \frac{1}{b} \operatorname{arcsinh} \left(\frac{bt}{a} \right), \quad (698)$$

with $a = \sqrt{m^2c^2 + p_0^2}$ and $b = qE$, we obtain

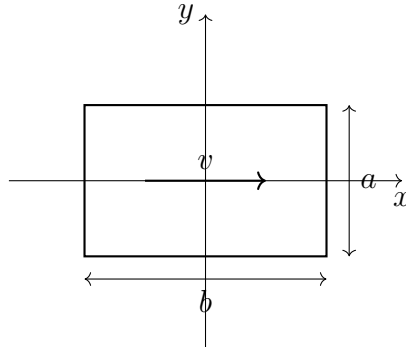
$$\tau = \frac{mc}{qE} \operatorname{arcsinh} \left(\frac{qE t_{\text{lab}}}{\sqrt{m^2c^2 + p_0^2}} \right). \quad (699)$$

Solving for t_{lab} ,

$$\boxed{t_{\text{lab}} = \frac{\sqrt{m^2c^2 + p_0^2}}{qE} \sinh \left(\frac{qE \tau}{mc} \right)}. \quad (700)$$

26 Moving Rectangular Current Loop

Question 26.1. In the MCRF of a rectangular current loop, the current is I , and the lengths of the sides are a and b . The wires through which the current passes have some finite thickness. In the laboratory frame, the loop moves with velocity v , parallel to the side b .



- Find the charge on each wire in the laboratory frame.
- Find the current in each wire in the laboratory frame.
- Is charge conserved in the laboratory frame? If yes, how? If no, why not? You may answer this part qualitatively.

Solution. (a)

In the MCRF, the current is I and the loop is neutral, so the net linear charge density is $\rho_{\text{net}} = 0$.

For the sides of length b (parallel to the direction of motion), the net four-current vector in the MCRF is

$$J_{\text{MCRF}}^{\mu} = (0, \pm I), \quad (701)$$

where the sign depends on the direction of the current in that side. Let's boost this total four-current to the laboratory frame (which moves with velocity $-v$ relative to the MCRF, so the boost parameter is $+v$ along the x -axis):

$$J'^0 = \gamma \left(J^0 + \frac{v}{c^2} J^1 \right) = \gamma \left(0 \pm \frac{v}{c^2} I \right) = \pm \gamma \frac{vI}{c^2}. \quad (702)$$

The length of this side is contracted in the lab frame, so $b' = b/\gamma$. Therefore, the net charge on the wire is exactly

$$Q'_{b,\pm} = J'^0 b' = \left(\pm \gamma \frac{vI}{c^2} \right) \left(\frac{b}{\gamma} \right) = \pm \frac{vIb}{c^2}. \quad (703)$$

For the sides of length a (perpendicular to motion), the current is in the y -direction, so the net four-current vector in the MCRF is

$$J_{\text{MCRF}}^{\mu} = (0, 0, \pm I, 0). \quad (704)$$

Boosting to the laboratory frame, the time component is

$$J'^0 = \gamma(J^0 + 0) = 0, \quad (705)$$

so the net charge on these sides is zero:

$$Q'_a = 0. \quad (706)$$

$$\boxed{Q'_a = 0, \quad Q'_{b,\pm} = \pm \frac{vIb}{c^2}.} \quad (707)$$

(b)

Transforming the spatial component of the four-current for the sides of length b :

$$I'_{b,\pm} = J'^1 = \gamma(J^1 + vJ^0) = \gamma(\pm I + 0) = \pm\gamma I. \quad (708)$$

For the sides of length a , the current is in the y -direction, which is unchanged by a boost along the x -axis:

$$I'_a = J'^2 = J^2 = I. \quad (709)$$

$$\boxed{I'_a = I, \quad I'_{b,\pm} = \pm\gamma I.} \quad (710)$$

(c)

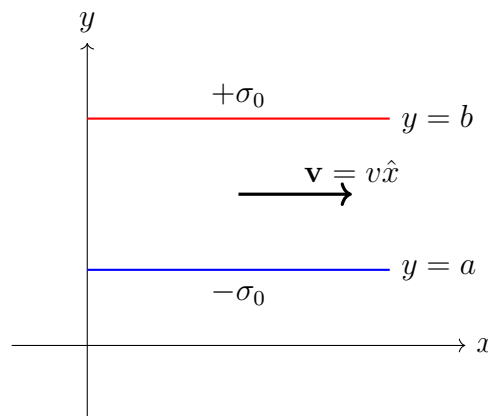
Charge is conserved in the laboratory frame. A Lorentz boost changes the local charge density and current on each segment, but it preserves the continuity equation because $(c\rho, \mathbf{J})$ transforms as a four-vector. If the loop is neutral in its MCRF (ions plus carriers), then the total charge is zero there and stays zero in every inertial frame. The boost only produces equal-and-opposite charge shifts on the two b -sides (from $u \rightarrow \pm u$), so the net charge remains unchanged.

Analytical Electrodynamics - Exercise Set 4 Eran Rehani

207823063 June 2, 2026

27 Moving Capacitor

Question 27.1. Two dielectric plates (capacitor) are at $y = a$ and $y = b$, where $a < b$ are constants. At the reference frame where the capacitor is at rest the surface charges of the two dielectric plates are $\pm\sigma_0$ for the upper and lower infinite plates respectively. Find the electric and magnetic fields of the charged capacitor moving at a constant speed $v\hat{x}$ by



(a) directly solving the Maxwell equations.

(b) solving in a reference frame where the capacitor is at rest and using the Lorentz transformation for the fields.

Compare the results of (a) and (b).

Solution. (a) **Direct solution via Maxwell's equations**

Gauss's law (CGS):

$$\oint \mathbf{E} \cdot d\mathbf{A} = 4\pi Q_{\text{enc}}. \quad (711)$$

For an infinite plate with surface charge density σ , applying a pillbox of face area A gives $2EA = 4\pi\sigma A$, so $E = 2\pi\sigma$ on each side. For a capacitor with $+\sigma$ on the upper plate and $-\sigma$ on the lower plate:

$$\mathbf{E} = -4\pi\sigma \hat{y} \quad (\text{between the plates}), \quad \mathbf{E} = 0 \quad (\text{outside}). \quad (712)$$

Ampère's law (CGS):

$$\oint \mathbf{B} \cdot d\boldsymbol{\ell} = \frac{4\pi}{c} I_{\text{enc}}. \quad (713)$$

For an infinite sheet with surface current $\mathbf{K}$, a rectangular Amperian loop gives the field on either side as

$$\mathbf{B} = \frac{2\pi}{c} \mathbf{K} \times \hat{n}, \quad (714)$$

where $\hat{n}$ points *from the sheet toward the field point*.

In the lab frame, the capacitor moves with $\mathbf{v} = v\hat{x}$. The plates are contracted in x by $1/\gamma$, so the same charge is spread over a smaller area:

$$\sigma = \gamma\sigma_0. \quad (715)$$

By Gauss's law, the electric field is:

$$\boxed{\mathbf{E} = -4\pi\gamma\sigma_0 \hat{y} \quad (a < y < b)}, \quad \mathbf{E} = 0 \quad (y < a \text{ or } y > b). \quad (716)$$

The field points from the positive upper plate ($+\sigma_0$) downward to the negative lower plate ($-\sigma_0$), hence the $-\hat{y}$ direction.

The moving surface charges also constitute surface currents $\mathbf{K} = \sigma\mathbf{v}$:

$$\mathbf{K}_{\text{top}} = +\gamma\sigma_0 v \hat{x}, \quad \mathbf{K}_{\text{bot}} = -\gamma\sigma_0 v \hat{x}. \quad (717)$$

Using (714), the contribution from each plate at a field point *between* the plates is:

$$\mathbf{B}_{\text{top}} = \frac{2\pi}{c} \mathbf{K}_{\text{top}} \times (-\hat{y}) = \frac{2\pi\gamma\sigma_0 v}{c} (\hat{x} \times (-\hat{y})) = -\frac{2\pi\gamma\sigma_0 v}{c} \hat{z}, \quad (718)$$

$$\mathbf{B}_{\text{bot}} = \frac{2\pi}{c} \mathbf{K}_{\text{bot}} \times (+\hat{y}) = \frac{2\pi(-\gamma\sigma_0 v)}{c} (\hat{x} \times \hat{y}) = -\frac{2\pi\gamma\sigma_0 v}{c} \hat{z}. \quad (719)$$

(For the top plate, $\hat{n} = -\hat{y}$ points toward the field point below it; for the bottom plate, $\hat{n} = +\hat{y}$.) Summing both contributions:

$$\boxed{\mathbf{B} = -\frac{4\pi\gamma\sigma_0 v}{c} \hat{z} \quad (a < y < b)}, \quad \mathbf{B} = 0 \quad (y < a \text{ or } y > b). \quad (720)$$

(b) **Lorentz transformation from the rest frame**

Lorentz field transformation (rest frame $\rightarrow$ lab frame): Let S' be the rest frame of the capacitor, moving with velocity $\boldsymbol{\beta} = \mathbf{v}/c = (v/c)\hat{x}$ relative to the lab S . Given fields in S' , the fields in S are

$$\mathbf{E}_{\parallel} = \mathbf{E}'_{\parallel}, \quad (721)$$

$$\mathbf{E}_{\perp} = \gamma(\mathbf{E}'_{\perp} - \boldsymbol{\beta} \times \mathbf{B}'), \quad (722)$$

$$\mathbf{B}_{\parallel} = \mathbf{B}'_{\parallel}, \quad (723)$$

$$\mathbf{B}_{\perp} = \gamma(\mathbf{B}'_{\perp} + \boldsymbol{\beta} \times \mathbf{E}'). \quad (724)$$

(These are the *inverse* transformation, obtained by $\beta \rightarrow -\beta$ relative to the $S \rightarrow S'$ form. We've seen them in Recitation 4.)

In the rest frame S' , the capacitor is static. By Gauss's law with charges $\pm\sigma_0$:

$$\mathbf{E}' = -4\pi\sigma_0 \hat{y} \quad (a < y < b), \quad \mathbf{B}' = 0. \quad (725)$$

The boost direction is $\hat{x}$, so $\mathbf{E}'$ (which is along $\hat{y}$) is entirely transverse: $\mathbf{E}'_{\parallel} = 0$, $\mathbf{E}'_{\perp} = -4\pi\sigma_0 \hat{y}$.

Applying the inverse transformation (721)–(724) with $\mathbf{B}' = 0$:

$$\mathbf{E} = \gamma \mathbf{E}'_{\perp} = -4\pi\gamma\sigma_0 \hat{y}, \quad (726)$$

$$\mathbf{B} = \gamma \beta \times \mathbf{E}' = \gamma \frac{v}{c} \hat{x} \times (-4\pi\sigma_0 \hat{y}) = -\frac{4\pi\gamma\sigma_0 v}{c} \hat{z}. \quad (727)$$

These are the fields between the plates; both vanish outside.

Comparison

Both methods yield identical results:

$$\mathbf{E} = -4\pi\gamma\sigma_0 \hat{y}, \quad \mathbf{B} = -\frac{4\pi\gamma\sigma_0 v}{c} \hat{z}, \quad a < y < b, \quad (728)$$

and zero outside. Note also that $\mathbf{B} = \frac{\mathbf{v}}{c} \times \mathbf{E}$, consistent with question 2.

28 Moving Charge Density

Question 28.1. A charge density $\rho(\mathbf{r})$ is moving at a constant velocity v . Show that the magnetic field is given by (CGS units)

$$\mathbf{B}(\mathbf{r}) = \frac{\mathbf{v}}{c} \times \mathbf{E}(\mathbf{r})$$

where $\mathbf{E}(\mathbf{r})$ is the electric field produced by $\rho(\mathbf{r})$ at rest. Show this by using **one** of the following:

- (a) the transformation of the fields which were studied in class.
- (b) Biot–Savart law (which you studied in course 77102. Please note that here we are using CGS units instead of SI).

You may assume that $\rho(\mathbf{r})$ is constant in time [$\rho(\mathbf{r} - \mathbf{v}t) = \rho(\mathbf{r})$].

Solution. We use approach (a): the Lorentz field transformation. Let S' be the rest frame of ρ and S the lab frame, with S' moving at $\mathbf{v} = v\hat{v}$ relative to S . The assumption $\rho(\mathbf{r} - \mathbf{v}t) = \rho(\mathbf{r})$ means ρ is uniform along $\hat{v}$, so the fields are static in both frames and the steady-state transformation applies.

The inverse transformation ($S' \rightarrow S$) with $\mathbf{B}' = 0$ gives:

$$\mathbf{E}_{\parallel} = \mathbf{E}'_{\parallel}, \quad (729)$$

$$\mathbf{E}_{\perp} = \gamma \mathbf{E}'_{\perp}, \quad (730)$$

$$\mathbf{B} = \gamma \beta \times \mathbf{E}'. \quad (731)$$

Since $\beta \parallel \hat{v}$, the cross product with the parallel component vanishes: $\beta \times \mathbf{E}'_{\parallel} = 0$, so (731) becomes

$$\mathbf{B} = \gamma \beta \times \mathbf{E}'_{\perp}. \quad (732)$$

From (730), $\mathbf{E}'_{\perp} = \mathbf{E}_{\perp}/\gamma$. Substituting:

$$\mathbf{B} = \gamma \beta \times \frac{\mathbf{E}_{\perp}}{\gamma} = \beta \times \mathbf{E}_{\perp} = \beta \times \mathbf{E}, \quad (733)$$

where the last step uses $\beta \times \mathbf{E}_{\parallel} = 0$. Therefore:

$$\mathbf{B}(\mathbf{r}) = \frac{\mathbf{v}}{c} \times \mathbf{E}(\mathbf{r}). \quad (734)$$

Note: $\mathbf{E}(\mathbf{r})$ here is the *lab-frame* electric field, which equals $\mathbf{E}'(\mathbf{r})$ only for the parallel component; the γ from the field transformation cancels the γ from the boost.

29 Applications of Question 2

Question 29.1. Use the results of question 2 to obtain the magnetic field for

- (a) the charged capacitor from question 1.
- (b) an infinite and uniform line of current.
- (c) an infinite and uniform sheet of current.

Solution. (a)

From question 1, the lab-frame electric field of the moving capacitor (between the plates) is $\mathbf{E} = -4\pi\gamma\sigma_0\hat{y}$, $\mathbf{v} = v\hat{x}$. Applying the result of question 2:

$$\mathbf{B} = \frac{\mathbf{v}}{c} \times \mathbf{E} = \frac{v}{c} \hat{x} \times (-4\pi\gamma\sigma_0\hat{y}) = -\frac{4\pi\gamma\sigma_0 v}{c} \hat{z}, \quad \mathbf{B} = 0 \text{ outside}, \quad (735)$$

in agreement with the direct calculation in question 1. ✓

(b)

Place the wire along $\hat{z}$ with linear charge density λ . By Gauss's law, $E(2\pi rL) = 4\pi\lambda L$, giving

$$\mathbf{E} = \frac{2\lambda}{r} \hat{r}. \quad (736)$$

The charges move along the wire, so $\mathbf{v} = v\hat{z}$. Since $\mathbf{E} \perp \mathbf{v}$, question 2 gives

$$\mathbf{B} = \frac{\mathbf{v}}{c} \times \mathbf{E} = \frac{v}{c} \hat{z} \times \frac{2\lambda}{r} \hat{r} = \frac{2\lambda v}{cr} \hat{\phi}, \quad \hat{z} \times \hat{r} = \hat{\phi}. \quad (737)$$

Identifying the current $I = \lambda v$, this is $\mathbf{B} = \frac{2I}{cr} \hat{\phi}$, which is exactly the CGS Biot-Savart result for an infinite line current. ✓

(c)

Let the sheet lie in the xz -plane with surface charge density σ . By Gauss's law,

$$\mathbf{E} = 2\pi\sigma \hat{n}, \quad (738)$$

where $\hat{n}$ is the outward normal (e.g. $+\hat{y}$ above the sheet). The sheet moves with $\mathbf{v} = v\hat{x}$ (in the plane of the sheet), so $\mathbf{E} \perp \mathbf{v}$, and question 2 gives

$$\mathbf{B} = \frac{\mathbf{v}}{c} \times \mathbf{E} = \frac{v}{c} \hat{x} \times 2\pi\sigma \hat{y} = \frac{2\pi\sigma v}{c} \hat{z}. \quad (739)$$

The surface current is $\mathbf{K} = \sigma\mathbf{v} = \sigma v\hat{x}$. By Ampère's law for an infinite current sheet (derived in question 1, eq. (714)): $\mathbf{B} = \frac{2\pi}{c} \mathbf{K} \times \hat{n} = \frac{2\pi\sigma v}{c} (\hat{x} \times \hat{y}) = \frac{2\pi\sigma v}{c} \hat{z}$. ✓

30 Relativistic Particle in Crossed Fields

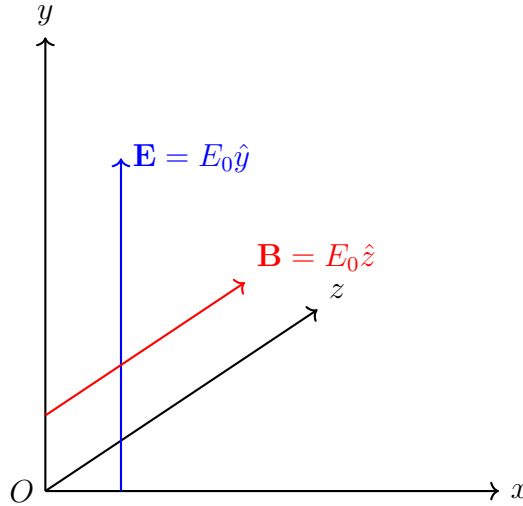
Question 30.1. A relativistic particle with mass m and charge q moves through an electromagnetic field of the form

$$\mathbf{E} = E_0 \hat{y}, \quad \mathbf{B} = E_0 \hat{z},$$

where E_0 is given. The particle starts its motion from rest at $t = 0$. Show that the components p_x and p_y of the momentum satisfy

$$p_x = a_1 + a_2 p_y^2, \quad a_3 p_y^3 + a_4 p_y = a_5 t + a_6,$$

where $\{a_i\}_{i=1}^6$ are constants. Express these constants in terms of the problem parameters.



Solution. Relativistic Lorentz force (CGS):

$$\frac{d\mathbf{p}}{dt} = q \left(\mathbf{E} + \frac{\mathbf{v}}{c} \times \mathbf{B} \right), \quad \frac{d\varepsilon}{dt} = q \mathbf{E} \cdot \mathbf{v}, \quad (740)$$

where $\varepsilon = \gamma m c^2$ is the total energy and $\mathbf{p} = \gamma m \mathbf{v}$ (so $v_i = p_i / (\gamma m)$).

Energy-momentum relation:

$$\varepsilon^2 = (p_x^2 + p_y^2 + p_z^2) c^2 + m^2 c^4. \quad (741)$$

Compute the cross product :

$$\mathbf{v} \times \mathbf{B} = (v_x \hat{x} + v_y \hat{y} + v_z \hat{z}) \times E_0 \hat{z} = E_0 (-v_x \hat{y} + v_y \hat{x}). \quad (742)$$

The total force $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}/c)$ has components:

$$\dot{p}_x = -\frac{qE_0}{c} v_y, \quad (743)$$

$$\dot{p}_y = qE_0 - \frac{qE_0}{c} (-v_x) = qE_0 \left(1 + \frac{v_x}{c} \right), \quad (744)$$

$$\dot{p}_z = 0. \quad (745)$$

Since the particle starts from rest, $p_z(0) = 0$, so $p_z = 0$ for all t .

The power delivered by the fields is:

$$\frac{d\varepsilon}{dt} = q \mathbf{E} \cdot \mathbf{v} = qE_0 v_y. \quad (746)$$

Comparing with (743):

$$\dot{p}_x = -\frac{1}{c} \frac{d\varepsilon}{dt} \implies \frac{d}{dt} \left(p_x + \frac{\varepsilon}{c} \right) = 0. \quad (747)$$

This quantity is conserved. Evaluating at $t = 0$ with $p_x(0) = 0$ and $\varepsilon(0) = mc^2$ (particle at rest):

$$p_x + \frac{\varepsilon}{c} = mc. \quad (748)$$

From (748): $\varepsilon = mc^2 - cp_x$. With $p_z = 0$, substituting into (741):

$$(mc^2 - cp_x)^2 = (p_x^2 + p_y^2)c^2 + m^2c^4. \quad (749)$$

Expand the left side:

$$m^2c^4 - 2mc^3p_x + c^2p_x^2 = c^2p_x^2 + c^2p_y^2 + m^2c^4. \quad (750)$$

The m^2c^4 and $c^2p_x^2$ terms cancel from both sides:

$$-2mc^3p_x = c^2p_y^2. \quad (751)$$

Solving for p_x :

$$\boxed{p_x = -\frac{p_y^2}{2mc}}, \quad (752)$$

so $a_1 = 0$ and $a_2 = -\frac{1}{2mc}$.

Rewrite the factor $1 + v_x/c$ in (744). Since $v_x = cp_x/\varepsilon$:

$$1 + \frac{v_x}{c} = 1 + \frac{cp_x}{\varepsilon} = \frac{\varepsilon + cp_x}{\varepsilon}. \quad (753)$$

From (748), $\varepsilon + cp_x = mc^2$, so:

$$1 + \frac{v_x}{c} = \frac{mc^2}{\varepsilon}. \quad (754)$$

Substitute $\varepsilon = mc^2 - cp_x$ and then use $p_x = -p_y^2/(2mc)$:

$$\varepsilon = mc^2 - c \left(-\frac{p_y^2}{2mc} \right) = mc^2 + \frac{p_y^2}{2m}. \quad (755)$$

Inserting (754) into (744):

$$\dot{p}_y = \frac{qE_0mc^2}{\varepsilon} = \frac{qE_0mc^2}{mc^2 + p_y^2/(2m)}. \quad (756)$$

Multiply numerator and denominator by $2m$:

$$\dot{p}_y = \frac{2qE_0m^2c^2}{2m^2c^2 + p_y^2}. \quad (757)$$

Separate variables:

$$(2m^2c^2 + p_y^2) dp_y = 2qE_0m^2c^2 dt. \quad (758)$$

Integrate both sides from 0 to t (using $p_y(0) = 0$):

$$\int_0^{p_y} (2m^2c^2 + p'^2) dp' = \left[2m^2c^2 p' + \frac{p'^3}{3} \right]_0^{p_y} = 2m^2c^2 p_y + \frac{p_y^3}{3}, \quad (759)$$

$$\int_0^t 2qE_0m^2c^2 dt' = 2qE_0m^2c^2 t. \quad (760)$$

Therefore:

$$\boxed{\frac{p_y^3}{3} + 2m^2c^2 p_y = 2qE_0m^2c^2 t}, \quad (761)$$

so $a_3 = \frac{1}{3}$, $a_4 = 2m^2c^2$, $a_5 = 2qE_0m^2c^2$, $a_6 = 0$.

31 Transformation Law For $F^{\mu\nu}$

Question 31.1. In class, you derived the transformation law of the fields $\vec{E}$ and $\vec{B}$ using the 4-potential A^μ . Derive the transformation law using the field strength tensor $F^{\mu\nu}$, and compare your results to what we saw in class.

Solution. We start with the definition of the field strength tensor:

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$$

Under a Lorentz transformation, the 4-potential transforms as:

$$A'^\mu = \Lambda^\mu_\nu A^\nu$$

Therefore, the field strength tensor transforms as:

$$F'^{\mu\nu} = \partial'^\mu A'^\nu - \partial'^\nu A'^\mu$$

Using the chain rule for differentiation, we can express the derivatives in terms of the original coordinates:

$$\partial'^\mu = \Lambda^\mu_\alpha \partial^\alpha$$

Substituting this into the expression for $F'^{\mu\nu}$, we get:

$$F'^{\mu\nu} = \Lambda^\mu_\alpha \partial^\alpha (\Lambda^\nu_\beta A^\beta) - \Lambda^\nu_\alpha \partial^\alpha (\Lambda^\mu_\beta A^\beta)$$

Expanding this, we have:

$$F'^{\mu\nu} = \Lambda^\mu_\alpha \Lambda^\nu_\beta \partial^\alpha A^\beta - \Lambda^\nu_\alpha \Lambda^\mu_\beta \partial^\alpha A^\beta$$

Since the field strength tensor is antisymmetric, we can write:

$$F'^{\mu\nu} = \Lambda^\mu_\alpha \Lambda^\nu_\beta F^{\alpha\beta}$$

This shows that the field strength tensor transforms as a rank-2 tensor under Lorentz transformations. Comparing this result to the transformation laws for the electric and magnetic fields derived in class, we see that they are consistent, as the components of the field strength tensor correspond to the electric and magnetic fields in the appropriate frames.

32 Electromagnetic field invariants

Question 32.1. Use Lorentz transformations to show that the following quantities are invariant under Lorentz transformations:

1. $\vec{E} \cdot \vec{B}$
2. $\vec{B}^2 - \vec{E}^2$

2

Solution. Dot Product Definition: $\vec{A} \cdot \vec{C} = \sum A_i C_i$ I will use the standard vector transformation equations for a boost with velocity $\vec{v}$. Let $\vec{E}_\parallel$ be the component of $\vec{E}$ parallel to $\vec{v}$, and $\vec{E}_\perp$ be the component perpendicular to $\vec{v}$. Similarly for $\vec{B}$. The formulas are:

$$\vec{E}'_\parallel = \vec{E}_\parallel$$

$$\vec{B}'_\parallel = \vec{B}_\parallel$$

$$\begin{aligned}\vec{E}'_{\perp} &= \gamma(\vec{E}_{\perp} + \vec{v} \times \vec{B}) \\ \vec{B}'_{\perp} &= \gamma\left(\vec{B}_{\perp} - \frac{1}{c^2}\vec{v} \times \vec{E}\right)\end{aligned}$$

Or more simply, decomposing into parallel and perpendicular:

$$\begin{aligned}\vec{E}' &= \vec{E}_{\parallel} + \gamma(\vec{E}_{\perp} + \vec{v} \times \vec{B}) \\ \vec{B}' &= \vec{B}_{\parallel} + \gamma\left(\vec{B}_{\perp} - \frac{1}{c^2}\vec{v} \times \vec{E}\right)\end{aligned}$$

We need to compute:

$$\vec{E}' \cdot \vec{B}' = [\vec{E}_{\parallel} + \gamma(\vec{E}_{\perp} + \vec{v} \times \vec{B})] \cdot [\vec{B}_{\parallel} + \gamma(\vec{B}_{\perp} - \frac{1}{c^2}\vec{v} \times \vec{E})]$$

1. $\vec{E}_{\parallel} \cdot \vec{B}_{\parallel}$
2. $\vec{E}_{\parallel} \cdot [\gamma(\vec{B}_{\perp} - \frac{1}{c^2}\vec{v} \times \vec{E})]$
3. $[\gamma(\vec{E}_{\perp} + \vec{v} \times \vec{B})] \cdot \vec{B}_{\parallel}$
4. $[\gamma(\vec{E}_{\perp} + \vec{v} \times \vec{B})] \cdot [\gamma(\vec{B}_{\perp} - \frac{1}{c^2}\vec{v} \times \vec{E})]$

$$\vec{E}' \cdot \vec{B}' = \vec{E}_{\parallel} \cdot \vec{B}_{\parallel} + \gamma^2[\vec{E}_{\perp} \cdot \vec{B}_{\perp} - \frac{v^2}{c^2}(\vec{E} \cdot \vec{B}) + \frac{1}{c^2}(\vec{v} \cdot \vec{E})(\vec{v} \cdot \vec{B})]$$

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Question 32.2. $A^{\mu}(t, \vec{r})$ is a contravariant vector potential of an electro-magnetic field in 3D spacetime (2 spatial dimensions, one temporal dimension). Here, $A^{\mu} = (A^0, A^1, A^2)$, where $A^0 = \varphi$ is the scalar potential and $\vec{A} = (A^1, A^2)$ is the vector potential in the 2D space. The action of the system is given by

$$S = \iint dt d^2x (\epsilon^{\alpha\mu\nu} A_{\alpha} F_{\mu\nu} + J^{\mu} A_{\mu})$$

where $F_{\mu\nu} = \partial_{\mu} A_{\nu} - \partial_{\nu} A_{\mu}$, $\epsilon^{\alpha\mu\nu}$ is the Levi-Civita symbol, such that $\epsilon^{012} = 1$, and J^{μ} is a contravariant current that satisfies the equation of continuity.

(a) Show that the action is invariant to the gauge transformation

$$A_{\mu} \rightarrow A_{\mu} + \partial_{\mu} f$$

(b) Find a full solution for the electromagnetic fields (E and B) for a system governed by the given action. Express your answer in terms of J^{μ} . Can a non-trivial solution (non-zero field) exist without sources ($J^{\mu} = 0$)?

Solution. (a) First note that $F_{\mu\nu}$ is gauge invariant: under $A_{\mu} \rightarrow A_{\mu} + \partial_{\mu} f$,

$$F_{\mu\nu} \rightarrow \partial_{\mu}(A_{\nu} + \partial_{\nu} f) - \partial_{\nu}(A_{\mu} + \partial_{\mu} f) = F_{\mu\nu} + \partial_{\mu}\partial_{\nu} f - \partial_{\nu}\partial_{\mu} f = F_{\mu\nu}. \quad (762)$$

Thus the transformed action is

$$S = \iint d^3x (\epsilon^{\alpha\mu\nu} (A_{\alpha} + \partial_{\alpha} f) F_{\mu\nu} + J^{\mu} (A_{\mu} + \partial_{\mu} f)) \quad (763)$$

$$= \iint d^3x (\epsilon^{\alpha\mu\nu} A_{\alpha} F_{\mu\nu} + \epsilon^{\alpha\mu\nu} (\partial_{\alpha} f) F_{\mu\nu} + J^{\mu} A_{\mu} + J^{\mu} \partial_{\mu} f) \quad (764)$$

By using $\partial_{\mu}(J^{\mu} f) = (\partial_{\mu} J^{\mu}) f + J^{\mu} (\partial_{\mu} f) \Rightarrow J^{\mu} \partial_{\mu} f = \partial_{\mu}(J^{\mu} f) - (\partial_{\mu} J^{\mu}) f$ Now, Integrate this over all space-time

$$\iint d^3x J^{\mu} \partial_{\mu} f = \iint d^3x \partial_{\mu}(J^{\mu} f) - \iint d^3x (\partial_{\mu} J^{\mu}) f \quad (765)$$

The first term on the right side is a total derivative. By Gauss's theorem, a total derivative integrates to a boundary term at infinity. so this boundary term drops out to zero:

$$\iiint d^3x \partial_\mu (J^\mu f) = 0 \quad (766)$$

That leaves us with the second term

$$\iiint d^3x J^\mu \partial_\mu f = - \iiint d^3x (\partial_\mu J^\mu) f \quad (767)$$

So we get the action

$$S = \iiint d^3x (\epsilon^{\alpha\mu\nu} A_\alpha F_{\mu\nu} + \epsilon^{\alpha\mu\nu} \partial_\alpha f F_{\mu\nu} + J^\mu A_\mu - (\partial_\mu J^\mu) f) \quad (768)$$

Integrate by parts the second term:

$$\iiint d^3x \epsilon^{\alpha\mu\nu} (\partial_\alpha f) F_{\mu\nu} = \iiint d^3x \epsilon^{\alpha\mu\nu} \left[\underbrace{\partial_\alpha (f F_{\mu\nu})}_{\text{Total Derivative}} - f (\partial_\alpha F_{\mu\nu}) \right] \quad (769)$$

Insert into 768

$$S = \iiint d^3x (\epsilon^{\alpha\mu\nu} A_\alpha F_{\mu\nu} - \epsilon^{\alpha\mu\nu} f (\partial_\alpha F_{\mu\nu}) + J^\mu A_\mu - (\partial_\mu J^\mu) f) \quad (770)$$

The remaining two terms vanish. Current conservation gives $\partial_\mu J^\mu = 0$, so $-(\partial_\mu J^\mu) f = 0$. For the other term, note the Bianchi identity:

$$\epsilon^{\alpha\mu\nu} \partial_\alpha F_{\mu\nu} = \epsilon^{\alpha\mu\nu} \partial_\alpha (\partial_\mu A_\nu - \partial_\nu A_\mu) = 0, \quad (771)$$

since $\epsilon^{\alpha\mu\nu}$ is antisymmetric in (α, μ) while $\partial_\alpha \partial_\mu$ is symmetric (and likewise for (α, ν)). Thus,

$$\boxed{S_{\text{transformed}} = \iiint d^3x \epsilon^{\alpha\mu\nu} A_\alpha F_{\mu\nu} + \iiint d^3x J^\mu A_\mu = S_{\text{original}}} \quad (772)$$

(b) To find the field equations, we vary the action with respect to A_α :

$$\delta S = \iiint d^3x (\epsilon^{\alpha\mu\nu} \delta A_\alpha F_{\mu\nu} + \epsilon^{\alpha\mu\nu} A_\alpha \delta F_{\mu\nu} + J^\alpha \delta A_\alpha) \quad (773)$$

Note that $\delta F_{\mu\nu} = \partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu$, so

$$\epsilon^{\alpha\mu\nu} A_\alpha \delta F_{\mu\nu} = \epsilon^{\alpha\mu\nu} A_\alpha (\partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu) = \epsilon^{\alpha\mu\nu} A_\alpha \partial_\mu \delta A_\nu - \epsilon^{\alpha\mu\nu} A_\alpha \partial_\nu \delta A_\mu \quad (774)$$

Integrate by parts the first term:

$$\iiint d^3x \epsilon^{\alpha\mu\nu} A_\alpha \partial_\mu \delta A_\nu = \iiint d^3x \epsilon^{\alpha\mu\nu} \left[\underbrace{\partial_\mu (A_\alpha \delta A_\nu)}_{\text{Total Derivative}} - \delta A_\nu (\partial_\mu A_\alpha) \right] \quad (775)$$

Integrate by parts the second term:

$$\iiint d^3x \epsilon^{\alpha\mu\nu} A_\alpha \partial_\nu \delta A_\mu = \iiint d^3x \epsilon^{\alpha\mu\nu} \left[\underbrace{\partial_\nu (A_\alpha \delta A_\mu)}_{\text{Total Derivative}} - \delta A_\mu (\partial_\nu A_\alpha) \right] \quad (776)$$

Thus (relabelling $\alpha \leftrightarrow \nu$ in the first term and $\alpha \leftrightarrow \mu$ in the second, using $\epsilon^{\nu\mu\alpha} = -\epsilon^{\alpha\mu\nu}$ and $\epsilon^{\mu\alpha\nu} = -\epsilon^{\alpha\mu\nu}$):

$$\begin{aligned} \iint d^3x \epsilon^{\alpha\mu\nu} A_\alpha \delta F_{\mu\nu} &= \iint d^3x \epsilon^{\alpha\mu\nu} [-\delta A_\nu (\partial_\mu A_\alpha) + \delta A_\mu (\partial_\nu A_\alpha)] \\ &= \iint d^3x \epsilon^{\alpha\mu\nu} \delta A_\alpha (\partial_\mu A_\nu - \partial_\nu A_\mu) \\ &= \iint d^3x \epsilon^{\alpha\mu\nu} \delta A_\alpha F_{\mu\nu} \end{aligned} \quad (777)$$

So the total variation is

$$\delta S = \iint d^3x (2\epsilon^{\alpha\mu\nu} F_{\mu\nu} + J^\alpha) \delta A_\alpha \quad (778)$$

Setting $\delta S = 0$ for arbitrary δA_α gives the field equations:

$$\boxed{2\epsilon^{\alpha\mu\nu} F_{\mu\nu} + J^\alpha = 0} \quad (779)$$

The full field tensor in 2+1D is the antisymmetric 3×3 matrix

$$F_{\mu\nu} = \begin{pmatrix} 0 & E_1 & E_2 \\ -E_1 & 0 & B \\ -E_2 & -B & 0 \end{pmatrix},$$

Expanding component by component using $E_i = F_{0i}$, $B = F_{12}$, and $\epsilon^{012} = 1$ (where $\epsilon^{\alpha\mu\nu} F_{\mu\nu}$ counts each antisymmetric pair twice):

$$\alpha = 0 : \quad \epsilon^{0\mu\nu} F_{\mu\nu} = 2F_{12} = 2B \quad \implies \quad B = -\frac{1}{4}J^0 \quad (780)$$

$$\alpha = 1 : \quad \epsilon^{1\mu\nu} F_{\mu\nu} = -2F_{02} = -2E_2 \quad \implies \quad E_2 = \frac{1}{4}J^1 \quad (781)$$

$$\alpha = 2 : \quad \epsilon^{2\mu\nu} F_{\mu\nu} = 2F_{01} = 2E_1 \quad \implies \quad E_1 = -\frac{1}{4}J^2 \quad (782)$$

Substituting the solutions above, reads explicitly

$$F_{\mu\nu} = \frac{1}{4} \begin{pmatrix} 0 & -J^2 & J^1 \\ J^2 & 0 & -J^0 \\ -J^1 & J^0 & 0 \end{pmatrix}.$$

The fields are fully determined by the sources with no propagating modes (no kinetic F^2 term). Without sources ($J^\mu = 0$), every component of $F_{\mu\nu}$ vanishes: no non-trivial sourceless solution exists.

Question 32.3. A relativistic system of currents and fields in 4D space-time (3 spatial dimension and one temporal dimension) is described by the following action:

$$S = -\frac{1}{16\pi c} \int d^4x F_{\mu\nu} F^{\mu\nu} - \frac{1}{c^2} \int d^4x (A_\mu J^\mu - \tilde{A}_\mu \tilde{J}^\mu)$$

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu - \epsilon^{\mu\nu\alpha\beta} \partial_\alpha \tilde{A}_\beta$$

where A^μ , $\tilde{A}^\mu$ are independent 4-potentials and J^μ , $\tilde{J}^\mu$ are distinct 4-currents. We define the electric and the magnetic fields in this system the same way as in the usual EM theory:

$$\begin{aligned} E_i &= F_{0i} \\ B_i &= -\frac{1}{2} \epsilon_{ijk} F_{jk} \end{aligned}$$

where repeated indices are summed over ($i, j, k = 1, 2, 3$).

- (a) Show that the system is invariant to the gauge transformation

$$\begin{aligned} A_\mu &\rightarrow A_\mu + \partial_\mu f \\ \tilde{A}_\mu &\rightarrow \tilde{A}_\mu + \partial_\mu \tilde{f} \end{aligned}$$

if and only if 4-currents are conserved. $f, \tilde{f}$ are arbitrary smooth and independent functions that vanish at infinity.

- (b) Using the given action, write out the field equations for this theory. The answer needs to be in terms of the E and B fields. No additional equations are required; in particular, there is no need to derive the equations of motion for the currents.
- (c) Now we assume that the transformation law of two 4-potentials for reflections is identical (but unspecified). Show that this field theory is not invariant to spatial reflections. There is no need to consider time reflections.

Solution. (a) First, note that $F_{\mu\nu}$ is gauge invariant under the given transformations see 762 (with $A_\mu \rightarrow A_\mu + \partial_\mu f$ and $\tilde{A}_\mu \rightarrow \tilde{A}_\mu + \partial_\mu \tilde{f}$):

$$F^{\mu\nu} \rightarrow \partial^\mu (A^\nu + \partial^\nu f) - \partial^\nu (A^\mu + \partial^\mu f) - \epsilon^{\mu\nu\alpha\beta} \partial_\alpha (\tilde{A}_\beta + \partial_\beta \tilde{f}) = F^{\mu\nu} + \underbrace{(\partial^\mu \partial^\nu - \partial^\nu \partial^\mu) f}_{=0} - \epsilon^{\mu\nu\alpha\beta} \underbrace{\partial_\alpha \partial_\beta \tilde{f}}_{=0} = F^{\mu\nu} \quad (783)$$

The first extra term vanishes because partial derivatives commute; the second vanishes because $\epsilon^{\mu\nu\alpha\beta}$ is antisymmetric in (α, β) while $\partial_\alpha \partial_\beta$ is symmetric. Lowering indices with the constant Minkowski metric, $F_{\mu\nu} = \eta_{\mu\rho} \eta_{\nu\sigma} F^{\rho\sigma}$ is likewise invariant. Hence $F_{\mu\nu} F^{\mu\nu}$ is gauge invariant. Next, consider the transformation of the interaction term:

$$\delta S_{int} = -\frac{1}{c^2} \int d^4x ((\partial_\mu f) J^\mu - (\partial_\mu \tilde{f}) \tilde{J}^\mu) \quad (784)$$

Integrate by parts:

$$\delta S_{int} = -\frac{1}{c^2} \int d^4x (\partial_\mu (f J^\mu) - \partial_\mu (\tilde{f} \tilde{J}^\mu)) + \frac{1}{c^2} \int d^4x (f \partial_\mu J^\mu - \tilde{f} \partial_\mu \tilde{J}^\mu) \quad (785)$$

By Gauss's theorem, the total derivatives integrate to boundary terms. Since f and $\tilde{f}$ vanish at infinity, these boundary terms drop out to zero. Thus,

$$\delta S = \delta S_{int} = \frac{1}{c^2} \int d^4x (f \partial_\mu J^\mu - \tilde{f} \partial_\mu \tilde{J}^\mu) \quad (786)$$

For the system to be gauge invariant ($\delta S = 0$) for arbitrary f and $\tilde{f}$, we must have $\partial_\mu J^\mu = 0$ and $\partial_\mu \tilde{J}^\mu = 0$. Conversely, if the currents are conserved, then $\delta S = 0$ for arbitrary f and $\tilde{f}$, so the system is gauge invariant. Thus, the system is gauge invariant if and only if the currents are conserved.

$$\boxed{\text{Gauge invariance} \iff \partial_\mu J^\mu = 0 \text{ and } \partial_\mu \tilde{J}^\mu = 0} \quad (787)$$

- (b) To find the field equations, we vary the action with respect to A_α and $\tilde{A}_\alpha$:

$$\delta S = -\frac{1}{16\pi c} \int d^4x \delta(F_{\mu\nu} F^{\mu\nu}) - \frac{1}{c^2} \int d^4x (\delta A_\mu J^\mu - \delta \tilde{A}_\mu \tilde{J}^\mu) \quad (788)$$

Using the antisymmetry of $F_{\mu\nu}$ and integrating by parts (where boundary terms vanish), we obtain:

$$\delta_A S = \int d^4x \left(\frac{1}{4\pi c} \partial_\mu F^{\mu\nu} - \frac{1}{c^2} J^\nu \right) \delta A_\nu = 0 \quad (789)$$

This yields the first field equation:

$$\partial_\mu F^{\mu\nu} = \frac{4\pi}{c} J^\nu \quad (790)$$

Now, varying with respect to $\tilde{A}_\alpha$:

$$\delta_{\tilde{A}} S = -\frac{1}{8\pi c} \int d^4x F_{\mu\nu} (-\epsilon^{\mu\nu\alpha\beta} \partial_\alpha \delta \tilde{A}_\beta) + \frac{1}{c^2} \int d^4x \tilde{J}^\beta \delta \tilde{A}_\beta \quad (791)$$

Integrating by parts again yields:

$$\delta_{\tilde{A}} S = \int d^4x \left(-\frac{1}{8\pi c} \epsilon^{\mu\nu\alpha\beta} \partial_\alpha F_{\mu\nu} + \frac{1}{c^2} \tilde{J}^\beta \right) \delta \tilde{A}_\beta = 0 \quad (792)$$

This yields the second field equation:

$$\epsilon^{\mu\nu\alpha\beta} \partial_\alpha F_{\mu\nu} = \frac{8\pi}{c} \tilde{J}^\beta \quad (793)$$

Expressing these equations in terms of E and B fields by using $E_i = F_{0i}$ and $B_i = -\frac{1}{2} \epsilon_{ijk} F_{jk}$, we have:

$$\text{For } \nu = 0 : \quad \partial_i E_i = \frac{4\pi}{c} J^0 \quad (794)$$

$$\text{For } \nu = i : \quad \epsilon_{ijk} \partial_j B_k - \partial_0 E_i = \frac{4\pi}{c} J^i \quad (795)$$

$$\text{For } \beta = 0 : \quad \partial_i B_i = \frac{4\pi}{c} \tilde{J}^0 \quad (796)$$

$$\text{For } \beta = k : \quad \epsilon_{ijk} \partial_i E_j + \partial_0 B_k = -\frac{4\pi}{c} \tilde{J}^k \quad (797)$$

- (c) Since A^μ and $\tilde{A}^\mu$ transform identically under $\mathcal{P}$, so do J^μ and $\tilde{J}^\mu$: both are polar 4-vectors, so $\tilde{J}^0 \rightarrow +\tilde{J}^0$ under reflections. To establish the transformation of $\vec{B}$: under $\mathcal{P}$, spatial derivatives flip ($\partial_i \rightarrow -\partial_i$) and the covariant spatial components of a 4-vector satisfy $A_i = -A^i \rightarrow -(-A^i) = -A_i$, so

$$F_{jk} = \partial_j A_k - \partial_k A_j \xrightarrow{\mathcal{P}} (-\partial_j)(-A_k) - (-\partial_k)(-A_j) = \partial_j A_k - \partial_k A_j = F_{jk}. \quad (798)$$

Since $\epsilon_{ijk} = \epsilon^{0ijk}$ is invariant under $\mathcal{P}$ (the factor $\det(\mathcal{P}) = -1$ and three spatial sign flips combine as $(-1)(-1)^3 = +1$), it follows that $B_k = -\frac{1}{2} \epsilon_{ijk} F_{jk} \rightarrow +B_k$, confirming $\vec{B}$ is axial. Therefore $\partial_i B_i \rightarrow -\partial_i B_i$ under $\mathcal{P}$. The $\beta = 0$ field equation $\partial_i B_i = \frac{4\pi}{c} \tilde{J}^0$ maps to $-\partial_i B_i = \frac{4\pi}{c} \tilde{J}^0$, which is inconsistent with the original. Hence the theory is not invariant under spatial reflections.

Analytical Electrodynamics – Homework Set 7 June 2, 2026 Eran Rehani (ID: 207823063)

Question 32.4. Find the Green function of a grounded sphere of radius a (both inside and outside the sphere).

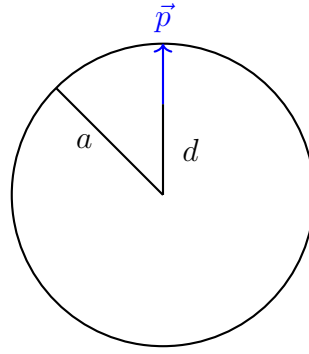
Hint: No need for eigenfunction expansions here – you already saw the solution in course 77102.

Solution. The Green function satisfies $\nabla_{\mathbf{r}}^2 G(\mathbf{r}, \mathbf{r}') = -4\pi \delta^{(3)}(\mathbf{r} - \mathbf{r}')$ with $G = 0$ on $|\mathbf{r}| = a$. Place an image (unit) charge of strength $-a/r'$ at the inverse point $\mathbf{r}'' = \frac{a^2}{r'^2} \mathbf{r}'$. For any source location $r' \neq a$ (both $r' < a$ and $r' > a$) the image lies outside the source domain, so the Green function is

$$G(\mathbf{r}, \mathbf{r}') = \frac{1}{|\mathbf{r} - \mathbf{r}'|} - \frac{a}{r'} \frac{1}{|\mathbf{r} - \mathbf{r}''|}. \quad (799)$$

Verification: $G = 0$ on $|\mathbf{r}| = a$ using the identity $|\mathbf{r} - \mathbf{r}''| = \frac{a}{r'} |\mathbf{r} - \mathbf{r}'|$ (for $|\mathbf{r}| = a$).

Question 32.5. An electric dipole of magnitude p is placed inside a grounded sphere of radius a , at a distance d , and pointing outwards from the center of the sphere (see figure).



1. Show that the charge density of an electric dipole $\mathbf{p}$ can be written as $\rho(\mathbf{x}') = -\mathbf{p} \cdot \nabla_{\mathbf{x}'} \delta(\mathbf{x}' - \mathbf{x}_0)$, where $\mathbf{x}_0$ is the location of dipole.
2. Integrating in parts and using the Green function you found in question 1, find the potential $\Phi(\mathbf{x})$ at any point inside the sphere. When you integrate in parts, make sure you explain why the uv term can be discarded!

Solution. (a) A dipole consists of charges $+q$ at $\mathbf{x}_0 + \epsilon$ and $-q$ at $\mathbf{x}_0$, with $\mathbf{p} = q\epsilon$ held fixed as $\epsilon \rightarrow 0$:

$$\rho(\mathbf{x}') = q\delta(\mathbf{x}' - \mathbf{x}_0 - \epsilon) - q\delta(\mathbf{x}' - \mathbf{x}_0). \quad (800)$$

Taylor-expand the shifted delta using $f(\mathbf{y} - \epsilon) = f(\mathbf{y}) - \epsilon \cdot \nabla f(\mathbf{y}) + O(\epsilon^2)$, where the sign follows from $\frac{\partial}{\partial \epsilon_i} \delta(\cdot - \epsilon) \Big|_{\epsilon=0} = -\partial_{x'_i} \delta$:

$$= q \left[\delta(\mathbf{x}' - \mathbf{x}_0) - \epsilon \cdot \nabla_{\mathbf{x}'} \delta(\mathbf{x}' - \mathbf{x}_0) + O(\epsilon^2) \right] - q\delta(\mathbf{x}' - \mathbf{x}_0) \quad (801)$$

$$= \underbrace{q \delta(\mathbf{x}' - \mathbf{x}_0) - q \delta(\mathbf{x}' - \mathbf{x}_0)}_{=0} - q\epsilon \cdot \nabla_{\mathbf{x}'} \delta(\mathbf{x}' - \mathbf{x}_0) + O(\epsilon^2) \quad (802)$$

$$= -\mathbf{p} \cdot \nabla_{\mathbf{x}'} \delta(\mathbf{x}' - \mathbf{x}_0), \quad (803)$$

where the last step uses $q\epsilon = \mathbf{p}$ and drops $O(\epsilon^2)$ in the point-dipole limit.

(b) The potential is $\Phi(\mathbf{x}) = \int G(\mathbf{x}, \mathbf{x}') \rho(\mathbf{x}') d^3x'$. Substituting ρ :

$$\Phi(\mathbf{x}) = - \int G(\mathbf{x}, \mathbf{x}') \mathbf{p} \cdot \nabla_{\mathbf{x}'} \delta(\mathbf{x}' - \mathbf{x}_0) d^3x'. \quad (804)$$

Integrate by parts ($u = G$, $dv = \mathbf{p} \cdot \nabla_{\mathbf{x}'} \delta d^3x'$):

$$\int G \mathbf{p} \cdot \nabla_{\mathbf{x}'} \delta d^3x' = \underbrace{\oint_{\partial V} G (\mathbf{p} \cdot d\mathbf{S}') \delta(\mathbf{x}' - \mathbf{x}_0)}_{=0, \text{ since } G=0 \text{ on sphere}} - \int \delta(\mathbf{x}' - \mathbf{x}_0) \mathbf{p} \cdot \nabla_{\mathbf{x}'} G(\mathbf{x}, \mathbf{x}') d^3x'. \quad (805)$$

The surface term vanishes because $G = 0$ on the grounded sphere (Dirichlet BC). Applying the delta function:

$$\Phi(\mathbf{x}) = \mathbf{p} \cdot \nabla_{\mathbf{x}'} G(\mathbf{x}, \mathbf{x}') \Big|_{\mathbf{x}'=\mathbf{x}_0}. \quad (806)$$

Question 32.6. The sphere of radius R is charged with the surface charge density $\sigma = \sigma_0 \cos \theta$. Find the potential ϕ of the electric field, and the electric field $\mathbf{E}$, using the multipole expansion (i.e., Legendre polynomials) in spherical coordinates.

Hint: Note the azimuthal symmetry.

Solution. By azimuthal symmetry the potential is independent of ϕ , so the multipole expansion reduces to a sum over Legendre polynomials:

$$\Phi(r, \theta) = \sum_{l=0}^{\infty} \left(A_l r^l + \frac{B_l}{r^{l+1}} \right) P_l(\cos \theta). \quad (807)$$

Since $\sigma_0 \cos \theta = \sigma_0 P_1(\cos \theta)$, only $l = 1$ contributes, so

$$\Phi(r, \theta) = \begin{cases} A_1 r \cos \theta, & r < R, \\ \frac{B_1}{r^2} \cos \theta, & r > R. \end{cases} \quad (808)$$

The boundary conditions at $r = R$ are:

$$\Phi(R^-, \theta) = \Phi(R^+, \theta) \implies A_1 R = \frac{B_1}{R^2}, \quad (809)$$

$$(E_r^{\text{out}} - E_r^{\text{in}})|_R = 4\pi\sigma_0 \cos \theta \implies \frac{2B_1}{R^3} + A_1 = 4\pi\sigma_0. \quad (810)$$

Substituting $A_1 = B_1/R^3$ from (809) into (810) gives $\frac{3B_1}{R^3} = 4\pi\sigma_0$, so

$$A_1 = \frac{4\pi\sigma_0}{3}, \quad B_1 = \frac{4\pi\sigma_0 R^3}{3}. \quad (811)$$

Thus the potential is

$$\Phi(r, \theta) = \begin{cases} \frac{4\pi\sigma_0}{3} r \cos \theta, & r < R, \\ \frac{4\pi\sigma_0 R^3}{3r^2} \cos \theta, & r > R. \end{cases} \quad (812)$$

The electric field $\mathbf{E} = -\nabla\Phi$ (using $\nabla(r \cos \theta) = \hat{z} = \cos \theta \hat{r} - \sin \theta \hat{\theta}$):

$$\mathbf{E}(r, \theta) = \begin{cases} -\frac{4\pi\sigma_0}{3} (\cos \theta \hat{r} - \sin \theta \hat{\theta}), & r < R, \\ \frac{4\pi\sigma_0 R^3}{3r^3} (2 \cos \theta \hat{r} + \sin \theta \hat{\theta}), & r > R. \end{cases} \quad (813)$$

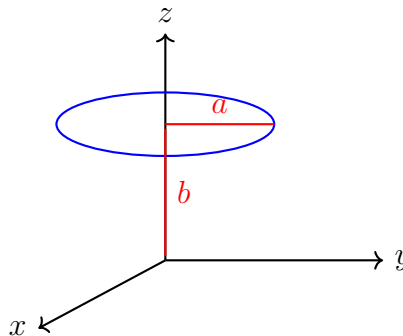
Note: the interior field is uniform, $\mathbf{E}_{\text{in}} = -\frac{4\pi\sigma_0}{3} \hat{z}$, consistent with a uniformly polarized sphere.

Question 32.7. (a) Consider a set of N point charges $\{q_n, \mathbf{r}_n\}_{n=1}^N$ that are all inside a ball of radius R around the origin ($|\mathbf{r}_n| < R$). Show that if $|\mathbf{r}| > R$, the potential can be written as

$$\Phi(\mathbf{r}) = \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{4\pi}{2l+1} \frac{Q_{lm}}{r^{l+1}} Y_{lm}(\theta, \phi), \quad (814)$$

Find the expression for Q_{lm} in terms of the point charges.

(b) Consider a thin ring of total charge q and radius a in the plane $z = b$, centered around the z -axis. Calculate Q_{lm} for this configuration.



Solution. (a) The Coulomb potential of the N charges is

$$\Phi(\mathbf{r}) = \sum_{n=1}^N \frac{q_n}{|\mathbf{r} - \mathbf{r}_n|}. \quad (815)$$

Since $|\mathbf{r}_n| < R < |\mathbf{r}|$ we have $r_n < r$ for every n , so the expansion

$$\frac{1}{|\mathbf{r} - \mathbf{r}_n|} = \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{4\pi}{2l+1} \frac{r_n^l}{r^{l+1}} Y_{lm}^*(\theta_n, \phi_n) Y_{lm}(\theta, \phi) \quad (816)$$

is valid for all n . Substituting and rearranging:

$$\begin{aligned} \Phi(\mathbf{r}) &= \sum_{n=1}^N q_n \sum_{l,m} \frac{4\pi}{2l+1} \frac{r_n^l}{r^{l+1}} Y_{lm}^*(\theta_n, \phi_n) Y_{lm}(\theta, \phi) \\ &= \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{4\pi}{2l+1} \frac{Y_{lm}(\theta, \phi)}{r^{l+1}} \underbrace{\left[\sum_{n=1}^N q_n r_n^l Y_{lm}^*(\theta_n, \phi_n) \right]}_{Q_{lm}}. \end{aligned}$$

Matching with the given form, the multipole moments are

$$Q_{lm} = \sum_{n=1}^N q_n r_n^l Y_{lm}^*(\theta_n, \phi_n). \quad (817)$$

(b) Let $r_0 = \sqrt{a^2 + b^2}$ and $\cos \theta_0 = b/r_0$. Every point on the ring has the same $r' = r_0$ and $\theta' = \theta_0$; only ϕ' varies. The linear charge density is $\lambda = q/(2\pi a)$, and the arc-length element is $a d\phi'$, so from the result of (a):

$$Q_{lm} = \int_{\text{ring}} \lambda r'^l Y_{lm}^*(\theta', \phi') a d\phi' = \frac{q}{2\pi a} \cdot a \cdot r_0^l \int_0^{2\pi} Y_{lm}^*(\theta_0, \phi') d\phi'.$$

Using $Y_{lm}^*(\theta_0, \phi') = N_{lm} P_l^m(\cos \theta_0) e^{-im\phi'}$ and $\int_0^{2\pi} e^{-im\phi'} d\phi' = 2\pi \delta_{m,0}$, the integral selects $m = 0$ only:

$$Q_{lm} = \frac{q r_0^l}{2\pi} \cdot 2\pi \delta_{m,0} Y_{l0}^*(\theta_0) = q r_0^l Y_{l0}(\theta_0) \delta_{m,0}.$$

Substituting $Y_{l0}(\theta_0) = \sqrt{\frac{2l+1}{4\pi}} P_l(\cos \theta_0)$:

$$Q_{lm} = q (a^2 + b^2)^{l/2} \sqrt{\frac{2l+1}{4\pi}} P_l\left(\frac{b}{\sqrt{a^2 + b^2}}\right) \delta_{m,0}. \quad (818)$$

Only $m = 0$ moments are non-zero, as expected from azimuthal symmetry.